Shifted Mills Constants and Cubic Pisot Trace Dynamics

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27 October 2026

Abstract

We develop a unified treatment of least prime-representing constants on affine ternary exponent schedules and of recurring composite floors for cubic Pisot numbers. The positive-shift argument reduces algebraic candidates to three units and eliminates them by exact local certificates and an infinite prime continuation. Negative shifts admit fixed-point and multiplier extensions, but leave a genuinely different even-multiplier problem. On cubic coefficient lines we obtain simultaneous composite-floor returns for density-one sets, including an improved logarithmic exceptional exponent $2/3$ and prime-relative bounds with exponent $5/3$, while explicit thin families require adaptive projective witnesses. Finite-field trace-zero geometry, cyclotomic norm bounds, and a two-sign three-adic depth theorem for arbitrary irreducible totally real cubics provide exact tests for hypothetical classical Mills candidates. We also establish precise limitations of finite congruence covers, skipped-level intervals, translated prime patterns, and phase-blind Fourier estimates. The transcendence of the classical Mills minimum and of the residual constant with exponents $2 \cdot 3^k - 1$ is not proved. Conditional reductions, proved restricted families, finite computations, and unresolved arithmetic inputs are kept separate throughout.

Contents

1 Introduction and the structure of the argument **3**

I Affine schedules and their least prime representing constants **5**

2 Scaling consequences local minima and normalized tails **5**

3 Positive shifts and elimination of the exceptional unit candidates **10**

4 Full prefix positive extensions and cube rescaling **18**

5 Negative shifts and integral fixed point rigidity **27**

6 The multiplier thirteen and a residual branch exclusion **36**

7 Odd multiplier extensions **46**

II Prime cells and finite to infinite continuation **54**

8 Prime cells height bounds and Pell families **54**

9 Finite prime forests and explicit continuation criteria 66

10 Shifted prime patterns and compactness criteria 76

III Coefficient density and obstructions to fixed congruence covers 85

11 Nonunit cubic families and fixed modulus limitations 85

12 Density on fixed norm lines 94

13 Prime coefficients and simultaneous composite floors 103

14 Signature controlled density and quantitative exceptional sets 110

IV Thin families and adaptive projective witnesses 117

15 The thin prime power family 117

16 A prime seed outlier and certified returns 123

17 Adaptive returns and ancestral minimality 130

18 Projective return criteria and finite horizon limitations 144

19 Coset coverage and output prime order bounds 157

V Actual cubic candidates and analytic prime interfaces 171

20 Both residue signs and all norm depth 171

21 Three analytic routes and their exact missing hypotheses 183

22 Final status and the remaining arithmetic problem 193

A Source coverage and reproducibility 193

1 Introduction and the structure of the argument

For a specified positive exponent schedule $C = (C_k)$, write

$$\mathcal{W}(C) = \{A > 1 : \left\lfloor A^{C_k} \right\rfloor \text{ is prime for every specified } k\}, \qquad \xi(C) = \min \mathcal{W}(C),$$

whenever the minimum exists. The starting index is part of the schedule. In particular, removing a finite prefix is not a harmless notational change. Our central distinction is between a minimum of this entire admissible set, a least completion of one feasible prime prefix, and an arbitrary cubic Pisot number satisfying some of the resulting arithmetic constraints. These three objects are not interchangeable.

A cubic Pisot number has one real conjugate $\beta > 1$ and two conjugates inside the unit disk. Integer traces then approximate its large powers. When the small-conjugate contribution has the appropriate sign, the trace is exactly the floor. The exponent schedules 3^k and $2 \cdot 3^k - 1$ therefore turn a prime-floor question into a highly sparse recurrence question. For a single auxiliary prime the resulting trace orbit is finite, so a zero on a recurrent phase yields infinitely many composite floors. The difficulty is not this implication, but ensuring a suitable prime and a suitable recurrent phase for every fixed candidate.

1.1 The main conclusions

The following roadmap separates the results that close restricted cases from the statements still needed for the principal problems.

1. For positive affine shifts, the exceptional unit classes left by an earlier fixed-point calculation are removed by explicit divisibility certificates and a bound on the actual infinite minimum (Sections 3 and 4). Exact scaling then yields further equivalent schedules without treating them as independent results.

2. For negative shifts, the integral fixed-point method, the multiplier thirteen argument, and the additional odd multiplier certificates give restricted-family results using the structural reductions cited below (Sections 5, 6, and 7). They do not eliminate every candidate for $\xi(2 \cdot 3^k - 1)$.

3. On each fixed coefficient line, recurrent divisors exist for a set of coefficients of density one. Prime-relative estimates and the explicit logarithmic exceptional-set saving are proved in Sections 12–14. Density one does not mean that the exceptional set is empty, nor that a prescribed minimum is covered.

4. Thin prime-power families and prime-leading examples require pointwise certificates. Sections 15–19 give exact families, projective return tests, and limitations of bounded witness strategies. A finite collection of successful residues is never promoted to a universal covering theorem.

5. Hypothetical classical Mills candidates satisfy stronger order and three-adic constraints in Sections 19–20. Section 21 identifies what three analytic routes do supply and what each still lacks. No contradiction for every candidate is obtained.

1.2 Strengthenings obtained by combining the arguments

The combined treatment yields stronger conclusions than a sequential reading of the source papers alone.

- Scalar returns in cubic finite algebras enlarge the usable Frobenius residues from one to four classes modulo 9. This improves the fixed-line logarithmic exceptional exponents from 1/6 to 2/3 for integers and from 7/6 to 5/3 for prime coefficients (Theorem 14.1).
- The depth argument extends to both residue signs and to every irreducible totally real cubic satisfying the stated congruences, without assuming it is Pisot. A single cyclotomic argument then controls both output signs (Theorem 20.9 and Corollary 20.10). Multiplicative independence also makes the conditional shadow bounds strict.
- The prime-31 family extends from one negative clock to every even multiplier not divisible by 5 (Corollary 11.6). A different small-prime witness synchronizes both clocks for the prime-seed outlier and an entire coefficient progression (Theorem 16.6).

These are strengthened restricted-family and necessary-condition results. They do not change the unresolved status of the two main constants.

1.3 Proof dependencies and interpretation

The analytic-to-Pisot reductions use the cited work of Saito and the explicitly identified structural lemmas in the source corpus. The original source references are retained so that these dependencies are visible rather than hidden inside a new title. Exact certificates establish finite algebraic assertions; their periodic-return proofs explain when those assertions imply infinitely many composite floors. A finite search alone does not establish an infinite prime chain. Likewise, a theorem for an auxiliary sequence of allowed integers is not a counterexample for the actual primes.

This article combines the mathematical content of twenty source packages. The sections are arranged by their role in the argument rather than by delivery order. Repeated abstracts and title matter have been removed; proofs, relevant finite tables, and verification procedures are retained. The density proof is replaced by its stronger projective version, and the positive-only depth proof by a two-sign totally real version; neither earlier proof is duplicated as a separate result. The common convention is that a local symbol has the meaning assigned in its section. Some local results are intentionally stated under their own notation to avoid silently changing hypotheses. A source map at the end records their provenance. Earlier "remaining" cases are interpreted in light of the later sections and the common final status, not as contradictory claims about the final theorem list. No assertion of external peer review or literature priority is made.

Part I

Affine schedules and their least prime representing constants

2 Scaling consequences local minima and normalized tails

2.1 Scaling, prime-output consequences, and prefix minima

The results in this section consolidate the first archive's consequence and proof-transfer notes. They distinguish an exact change of variables, a reuse of the local-minimum proof, and a continuity statement that does not preserve transcendence in its limit. All prescribed starting indices are part of the definitions. For a positive clock $C = (C_k)_{k\ge 1}$, write

$$\mathcal{W}(C) = \{A > 1 : \lfloor A^{C_k} \rfloor \text{ is prime for all } k \ge 1\}.$$

Its least element, when it exists, is denoted by $\xi(C)$.

Proposition 2.1 (Exact scaling)**.** *For any real $d > 0$,*

$$\mathcal{W}(dC) = \{A > 1 : A^d \in \mathcal{W}(C)\}, \qquad \xi(dC) = \xi(C)^{1/d},$$

whenever one, and hence both, of these minima exists. For rational $d > 0$, algebraicity of the two minima is equivalent. In particular, if $\alpha = \xi((3^k - 1)_{k\ge 1})$, then

$$\xi((m(3^k - 1)/2)_{k\ge 1}) = \alpha^{2/m} \qquad (m \ge 1).$$

These are all positive rational scalings of $3^k - 1$ that are integral at every prescribed index.

Proof. The map $A \mapsto A^d$ is an increasing bijection on $(1, \infty)$, and $(A^d)^{C_k} = A^{dC_k}$ term by term. A positive rational power of a positive real number is algebraic if and only if that number is algebraic. Finally, $\gcd_{k\ge 1}(3^k - 1) = 2$: necessity follows from the first exponent 2, and every exponent is even. □

Thus parameter hypotheses should be stated, where possible, for the primitive pair $(R, S) = (r/d, s/d)$, $d = \gcd(r, s)$. For example, $3 \nmid S$ is equivalent to $v_3(s) \le v_3(r)$; it does not require $3 \nmid s$. This is a strengthening of the literal unnormalized statements, but not a new independent transcendence mechanism.

2.1.1 A uniform consequence for all nonzero affine shifts

The next theorem extends the first archive's output consequences from $3^k - 1$ to every transcendental affine-clock base. Primality is not needed for its proof; it applies in particular to each transcendental minimum proved in the subsequent sections.

Theorem 2.2 (Laurent limits and escape from fixed curves)**.** *Let $b \ge 2$ and $r \ge 1$ be integers, let $\sigma \in \mathbb{Z} \setminus \{0\}$, and suppose $C_k = rb^k + \sigma > 0$ for $k \ge 1$. Fix a transcendental real number $A > 1$ and put $p_k = \lfloor A^{C_k} \rfloor$. For every fixed $h \ge 1$,*

$$\frac{p_{k+h}}{p_k^{b^h}} \longrightarrow L_h := A^{\sigma(1-b^h)}.$$

For any fixed integers $a_0, \dots, a_m$, put $U = \sum_{j=0}^{m} a_j b^j$ and $V = \sum_{j=0}^{m} a_j$. Then

$$\lim_{k\to\infty} \prod_{j=0}^{m} p_{k+j}^{a_j} = \begin{cases} +\infty, & U > 0, \\ 0, & U < 0, \\ A^{\sigma V}, & U = 0. \end{cases}$$

The finite nonzero limits, as all fixed integral Laurent monomials are allowed, are exactly

$$\{A^{\sigma(b-1)n} : n \in \mathbb{Z}\}.$$

Only 1 in this set is algebraic. Moreover, for every nonzero $F \in \overline{\mathbb{Q}}[X, Y]$ and every fixed $h \geq 1$,

$$F(p_k, p_{k+h}) \neq 0 \qquad \text{for all sufficiently large } k.$$

Consequently no fixed polynomial recurrence with algebraic coefficients can hold for these pairs on an infinite subsequence.

Proof. For every fixed finite window, $p_{k+j} = A^{rb^{k+j}+\sigma}(1 + o(1))$, and hence

$$\prod_{j=0}^{m} p_{k+j}^{a_j} = A^{rb^k U + \sigma V}(1 + o(1)).$$

This proves the three alternatives, including negative exponents. If $U = 0$, reduction modulo $b - 1$ gives $V \equiv 0 \pmod{b-1}$. Conversely $(a_0, a_1) = (bn, -n)$ gives $U = 0$ and $V = (b-1)n$. A nonzero integral power of a transcendental positive real number is transcendental, proving the arithmetic assertion. The ratio formula is the special case $(a_0, a_h) = (-b^h, 1)$.

Write $F = \sum c_{ij} X^i Y^j$, put $d = b^h$, and let $D = \max\{i + dj : c_{ij} \neq 0\}$. Since $p_k \to \infty$,

$$\frac{F(p_k, p_{k+h})}{p_k^D} \longrightarrow \sum_{i+dj=D} c_{ij} L_h^j.$$

The right side is a nonzero polynomial evaluated at the transcendental number L_h: for each j, the equation $i + dj = D$ determines at most one i, so distinct terms do not cancel as formal powers of L_h. The limit is therefore nonzero. □

For the ternary clock this yields exactly the limit group $\{A^{2\sigma n} : n \in \mathbb{Z}\}$. For α and $p_k = \lfloor \alpha^{3^k - 1} \rfloor$, in particular,

$$\frac{p_{k+1}}{p_k^3} \to \alpha^2, \qquad \frac{p_{k+1} - p_k^3}{p_k^3} \to \alpha^2 - 1,$$

and the exact telescoping product is

$$\alpha = p_1^{1/3} \prod_{k\geq 1} \left(\frac{p_{k+1}}{p_k^3} \right)^{1/3^{k+1}}.$$

Its finite products equal $p_n^{1/3^n}$. The curve-escape conclusion means that every infinite subset of the pair sequence is Zariski dense over $\overline{\mathbb{Q}}$; it does not say that individual integer pairs are algebraically independent. Nor does the two-coordinate proof automatically extend to arbitrary polynomials in three or more output coordinates.

2.1.2 Every feasible-prefix minimum for the clock $3^k - 1$

A *local left minimum* of $\mathcal{W}(C)$ is a member A for which $(A - \eta, A) \cap \mathcal{W}(C) = \varnothing$ for some $\eta > 0$. The following result uses more than transcendence of the global minimum. Its explicit inherited inputs are the local Type B reduction of [6, section "Local Type B and factorial perturbations"], and the negative-fixed-point norm exclusion of [7]. The localization in the former is the Saito prime-chain replacement argument with every fixed finite prefix retained. These are unpublished corpus results, not additional external hypotheses about primes.

Theorem 2.3 (Local and prefix minima)**.** *Every local left minimum of $\mathcal{W}((3^k - 1)_{k\geq 1})$ is transcendental. These points are exactly the least completions of feasible finite prime prefixes. They form a countable dense subset of the admissible set.*

Proof. Put $C_k = 3^k - 1$. For each m, a later proper multiple of C_m occurs among the C_k, by taking $k - m$ a positive multiple of $\operatorname{ord}_{C_m}(3)$. Thus every admissible A satisfies $A^{C_m} \notin \mathbb{N}$: otherwise the prime at level m would become a composite proper prime power at that later level.

Suppose an algebraic admissible A were a local left minimum. The stated local reduction applies because $C_{k+1}/C_k > 3$, these ratios tend to 3, and the recurring divisibility just verified preserves the initial endpoints. It gives a cubic Pisot number $\beta = A^g$, $g \mid 2$, and, for all sufficiently large k,

$$n_k = (3^k - 1)/g, \qquad T_k = \operatorname{Tr}(\beta^{n_k}) = \lfloor A^{C_k} \rfloor \text{ prime}, \qquad 0 \leq \beta^{n_k} - T_k \leq T_k^{-17/40}.$$

Here the degree bound is $3 \leq \deg \beta \leq 1 + 40/17 < 4$. The norm-exclusion lemma of [7] gives $3 \nmid \mathrm{N}(\beta)$.

For every k divisible by 12, $208 \mid 3^k - 1$, so $104 \mid n_k$. Each residue of a conjugate of β in an algebraic closure of $\mathbb{F}_3$ is nonzero and has degree at most three over $\mathbb{F}_3$. Its multiplicative order therefore divides one of $2, 8, 26$, hence divides 104. Reducing the trace at these exponents modulo a prime above 3 gives $T_k \equiv 1 + 1 + 1 = 0 \pmod 3$. The traces tend to infinity, contradicting eventual primality.

For the remaining topological assertion, a feasible prefix defines a nonempty cylinder $I = [a, b)$, the finite intersection of its prime cells. Let $A = \inf(\mathcal{W}(C) \cap I)$. A sequence of completions tending to A from the right, and right continuity of each fixed floor, show that $A \in \mathcal{W}(C)$. A chosen completion gives $A < b$. Also $A \neq a$, since a is a lower prime-cell endpoint and integral powers were excluded. Thus $(a, A) \cap \mathcal{W}(C) = \varnothing$, so this least completion is a local left minimum. Conversely, a sufficiently long prefix cylinder about any local left minimum has width smaller than its left gap; its least completion is therefore that point. There are countably many finite prefixes, and successive prefix cylinders at any admissible point have widths tending to zero. Their minima converge to that point, proving density. □

This theorem and Proposition 2.1 also transfer local left minima and feasible-prefix minima throughout the rational scaling class of $3^k - 1$. They make no claim that every admissible base is transcendental.

2.2 Continuity at the unshifted clock, without arithmetic transfer

Extend ξ to positive real exponent clocks and put

$$F(t) = \xi((3^k - t)_{k\geq 1}) \quad (-1 \leq t \leq 1), \qquad M = F(0).$$

All these minima exist by Saito's existence theorem [4, Theorem 1.3], since the consecutive ratios are at least 5/2. The proof below needs only a uniform sufficiently-large prime-gap threshold, not its numerical value.

Lemma 2.4 (Uniform large-prime continuation)**.** *There is a prime threshold P_0 such that, for every positive clock with consecutive ratios in $[5/2, 4]$, every prime $p \geq P_0$ at any one level admits an infinite prime continuation inside its closed cell, whose subsequent closed cells lie strictly inside their parents.*

Proof. The Baker–Harman–Pintz prime-interval theorem [1] supplies constants $K, X_0 > 0$ such that every real $x \geq X_0$ has a prime q with $x < q \leq x + Kx^{21/40}$ (enlarging K, X_0 if necessary). Let $c \in [5/2, 4]$. The mean value theorem gives

$$(p+1)^c - p^c \geq cp^{c-1}, \qquad \frac{cp^{c-1}}{p^{21c/40}} = cp^{19c/40-1} \geq \frac{5}{2}p^{3/16}.$$

Thus a single sufficiently large P_0 ensures, uniformly in c, $p^c + Kp^{21c/40} + 1 < (p+1)^c$. The prime just supplied satisfies $p^c < q$ and $q + 1 < (p+1)^c$. Iteration gives nested compact root cells, and any point in their intersection has the required floors: the strict upper containment at the next level removes every open right-endpoint ambiguity. The same threshold remains valid because the primes strictly increase. □

Theorem 2.5 (Two-sided continuity and normalized tails)**.** *One has $F(t) \to M$ as $t \to 0$. Consequently*

$$\xi((3^{k+h} - 1)_{k\geq 1})^{3^h} \longrightarrow M, \qquad \xi((3^{k+h} + 1)_{k\geq 1})^{3^h} \longrightarrow M \qquad (h \to \infty).$$

The corresponding unnormalized minima tend to 1. For each fixed finite set of output indices, the two normalized families eventually reproduce the prime outputs of M at those indices, when evaluated on their own perturbed clocks.

Proof. The ratios $(3^{k+1} - t)/(3^k - t)$ lie in $[5/2, 4]$ for $|t| \leq 1$. A fixed sufficiently large first prime and Lemma 2.4 give a uniform upper bound on $F(t)$; also $F(t) \geq 2^{1/4} > 1$.

Take a sequence $t_n \to 0$ for which $F(t_n) \to A$. If $A^{3^j} = m \in \mathbb{N}$ for some j, then $m \geq 2$ and

$$F(t_n)^{3^{j+2}-t_n} \longrightarrow m^9.$$

Its floor, for all sufficiently large n, is either $m^9 - 1$ or m^9. Both are composite; indeed $m^9 - 1 = (m^3 - 1)(m^6 + m^3 + 1)$. This contradicts admissibility. Therefore each A^{3^j} is nonintegral, and continuity at each fixed index gives $\lfloor A^{3^j} \rfloor$ prime. Thus $A \geq M$, proving $\liminf_{t\to 0} F(t) \geq M$.

Write $q_j = \lfloor M^{3^j} \rfloor$. No M^{3^j} is integral, because otherwise the next required power would be composite. Moreover the closed classical prime cells strictly nest:

$$q_{j+1} > q_j^3, \qquad q_{j+1} + 1 < (q_j + 1)^3.$$

The lower equality is composite; the only possible upper equality would give $q_{j+1} = (q_j+1)^3 - 1$, also composite. Given $\varepsilon > 0$, choose n with $q_n \geq P_0$ and $(q_n + 1)^{1/3^n} < M + \varepsilon$. Its closed cell lies strictly inside every earlier prime cell. By continuity of finitely many endpoints, this containment persists with exponents $3^j - t$ whenever $|t|$ is sufficiently small. Continue from the same prime q_n by Lemma 2.4. The resulting admissible base preserves the earlier prefix, so

$$F(t) \leq (q_n + 1)^{1/(3^n - t)}.$$

Taking the upper limit and then $\varepsilon \downarrow 0$ proves continuity.

Exact scaling gives $\xi(3^{k+h}-1)^{3^h} = F(3^{-h})$ and $\xi(3^{k+h}+1)^{3^h} = F(-3^{-h})$. These converge to $M > 1$, and taking 3^h-th roots gives the raw limits 1. Finally, $F(t)^{3^j-t} \to M^{3^j}$ at every fixed j; nonintegrality of the latter makes the finitely many floors stable. □

The ninth power in the lower-limit proof is essential to this simple argument: $2^3 - 1 = 7$ is prime. The displayed limits do not give an identity between a full minimum and a tail minimum. Although the shifted approximants here are transcendental by the shifted-clock results, convergence of transcendental numbers does not determine the arithmetic nature of M.

2.3 Classical exponential-value consequences

The following bound is the theorem of Ramachandra and Srinivasan [47], included with its short six-exponentials proof. Its use here is an application of an existing theorem, not a new transcendence theorem for individual exponential values.

Theorem 2.6 (Sparse algebraic exponential values)**.** *Let $A > 1$ be transcendental and fix $z \in \mathbb{C} \setminus \{0\}$. In any interval of $N \geq 1$ consecutive positive integer indices, at most*

$$\frac{1 + \sqrt{16N - 15}}{2}$$

of the numbers $\exp(zA^n)$ are algebraic. For real A, the same exceptional-index bound holds simultaneously for algebraicity of $\sin(A^n)$ and $\cos(A^n)$.

Proof. Let $E = \{n \geq 1 : \exp(zA^n) \in \overline{\mathbb{Q}}\}$. For a fixed positive difference d, there are at most two indices n with $n, n+d \in E$. Otherwise three such distinct indices n_1, n_2, n_3 give three $\mathbb{Q}$-linearly independent numbers zA^{n_i} and two $\mathbb{Q}$-linearly independent numbers $1, A^d$. Their six exponentials would all be algebraic, contrary to the six-exponentials theorem. The independence follows directly from transcendence of A. If an interval of length N contains m elements of E, its $\binom{m}{2}$ pairwise positive differences have at most $2(N-1)$ occurrences in total. Solving this quadratic inequality gives the bound. For $z = i$, put $u = e^{ix}$ with $x = A^n$ real. Algebraicity of u, $\cos x$, and $\sin x$ are equivalent: u satisfies $u^2 - 2\cos(x)u + 1 = 0$ and $u^2 - 2i\sin(x)u - 1 = 0$, while the reverse implications follow from $\cos x = (u + u^{-1})/2$ and $\sin x = (u - u^{-1})/(2i)$. □

For every fixed real $b > 0$, $b \neq 1$, one may take $z = \log b$; no algebraicity assumption on b is needed. A separate, weaker-input consequence is that for any irrational real A, at most two distinct primes p can have p^A algebraic. Indeed, the logarithms of three distinct primes are $\mathbb{Q}$-linearly independent, and applying six exponentials with $1, A$ would contradict algebraicity of all six values. More generally the multiplicative group

$$\{b \in \overline{\mathbb{Q}} \cap \mathbb{R}_{>0} : b^A \in \overline{\mathbb{Q}}\}$$

has $\mathbb{Q}$-rank at most two; it need not have only two elements. None of these assertions identifies e^α, $\log \alpha$, or $\alpha^{\sqrt{2}}$ individually, or gives algebraic independence of an infinite family of exponential values.

3 Positive shifts and elimination of the exceptional unit candidates

The next argument completes the positive unit elimination that was only partial in the initial notes. Its finite-ring certificates must be combined with a bound for the actual infinite minimum; divisibility tests alone do not finish the argument.

3.1 Statement and dependencies

For a positive sequence $C = (C_k)_{k\ge 1}$ define

$$\mathcal{W}(C) = \{A > 1 : \lfloor A^{C_k} \rfloor \text{ is prime for every } k \ge 1\}.$$

When this set has a least element, denote it by $\xi(C)$. Saito's existence theorem [4, Theorem 1.3] applies if $C_{k+1}/C_k \ge 2$. For $C_k = r3^k + s$ this condition follows from $s \le 3r$.

Theorem 3.1. *Let $r, s \ge 1$ be integers satisfying $3 \nmid s$ and $s \le 3r$. Then*

$$\xi\big((r3^k + s)_{k\ge 1}\big)$$

is transcendental.

Theorem 3.1 removes the exceptional divisibility classes in the positive-shift theorem of [7]. It concerns the actual least element of $\mathcal{W}(C)$, with the schedule starting at $k = 1$. We use Saito's algebraic-minimum reduction [4, Proposition 3.1], based on the prime-interval exponent 21/40 of Baker, Harman and Pintz [1], and the theoretical ternary trace results of [7]. The latter is an unpublished research manuscript; the present argument is not an independent replacement for all of its theoretical lemmas. We isolate exactly the assertions used below. The finite candidate computation is also reproduced by a separate unit-only enumeration, rather than relying on the original enumeration of all cubic Pisot numbers below 9.

The additional argument has two parts. Finite modular certificates make any surviving normalized coefficient $R = r/s$ very large. A separate infinite construction makes the *global* minimum small for all such R. Neither part alone removes every R.

3.2 Reduction to three algebraic candidates

A Pisot number is a real algebraic integer $\beta > 1$ whose other conjugates have absolute value less than 1. All field traces below are over $\mathbb{Q}$.

Proposition 3.2 (Inherited algebraic-minimum and trace inputs)**.** *Under the assumptions of Theorem 3.1, suppose that $\xi = \xi(r3^k + s)$ is algebraic. There is an integer $g \mid 2s$ such that $\beta = \xi^g$ is a cubic Pisot number and*

$$n_k = \frac{r3^k + s}{g} \in \mathbb{N}$$

for every sufficiently large k. Write x_2, x_3 for its other conjugates and $\lambda = s/g$. Then:

(i) $T_k = \mathrm{Tr}(\beta^{n_k}) = \lfloor \xi^{r3^k+s} \rfloor$ *is prime for all sufficiently large k.*

(ii) *For every fixed $\varepsilon > 0$,*

$$|x_2^{n_k} + x_3^{n_k}| = O\big(\beta^{-(17/40-\varepsilon)n_k}\big).$$

(iii) $\beta^\lambda < 3$, *hence* $\beta < 9$.

(iv) *If* β *is a unit, then the possible pairs* (λ, β) *and the necessary eventual exponent residues are precisely among the five rows of Table 1.*

Items (i) and (ii) follow from Saito's Proposition 3.1, after excluding integer powers and the quadratic alternative as in the Mills-shadow corollary of [7]. For clarity, $3 \nmid s$ makes every C_m coprime to 3, and $C_m \mid C_{m+h}$ whenever $3^h \equiv 1 \pmod{C_m}$. An integer power ξ^{C_m} would therefore give composite outputs at infinitely many later indices. Likewise, a putative sequence of prime integers C_k/g would repeat one of its prime divisors along an affine return, contrary to its strict growth. The surviving degree is 3, since $1 + (3(1 - 21/40) - 1)^{-1} < 4$. Finally, g divides every sufficiently late C_k and therefore divides $C_{k+1} - 3C_k = -2s$.

Items (iii) and (iv) are the positive-fixed-point consequence and the necessary finite table of [7]. Its affine trace-return theorem forces each relevant twisted 3-adic limit to equal ± 1. Transporting that algebraic identity to $\mathbb{C}$ gives $\beta^\lambda < 1 + 1 + 1 = 3$. The exact unit-only check of (iv) is described in Section 3.7. All Saito numbering here refers to arXiv version 3; in particular, the reduction is Proposition *3.1*, not 3.2.

Lemma 3.3 (Unit reduction)**.** *The number* β *in Proposition 3.2 has* $|\,\mathrm{N}(\beta)| = 1$.

Proof. Take $\varepsilon = 1/40$ in Proposition 3.2(ii), so that the approximation exponent is $\kappa = 2/5$. The approximation-magnitude lemma of [7] gives

$$|\,\mathrm{N}(\beta)| \le \beta^{1-2\kappa} = \beta^{1/5} < 9^{1/5} < 2.$$

The norm is a nonzero integer, proving the claim. This use of the magnitude lemma is valid for both totally real and nonreal cubic Pisot numbers. In the nonreal case its proof uses the polynomial lower bound for a nonzero linear form in logarithms [2]; it does not assume a uniform lower bound on an oscillating cosine. □

Let δ and τ denote the unique real roots greater than 1 of

$$f_\delta(X) = X^3 - X^2 - 1, \qquad f_\tau(X) = X^3 - X^2 - X - 1,$$

respectively. Their relevant powers have minimal polynomials

$$f_{\delta^2}(X) = X^3 - X^2 - 2X - 1, \qquad f_{\tau^2}(X) = X^3 - 3X^2 - X - 1.$$

λ	β	Minimal polynomial of β	ξ^s	Necessary exponent residue
1	δ	$X^3 - X^2 - 1$	δ	$n_k \equiv 1 \pmod 8$
1	τ	$X^3 - X^2 - X - 1$	τ	$n_k \equiv 1 \pmod{13}$
1	δ^2	$X^3 - X^2 - 2X - 1$	δ^2	$n_k \equiv 1 \pmod 4$
1/2	τ^2	$X^3 - 3X^2 - X - 1$	τ	$n_k \equiv 7 \pmod{13}$
2	δ	$X^3 - X^2 - 1$	δ^2	$n_k \equiv 2 \pmod 8$

Table 1: The reduced unit table. Each listed residue is a necessary condition, not an assertion that the corresponding trace is prime.

Proposition 3.4 (Normalized candidate reduction)**.** *If* $\xi(r3^k + s)$ *is algebraic under the hypotheses of Theorem 3.1, then* $s \mid r$. *Put*

$$R = r/s, \qquad \eta = \xi(r3^k + s)^s.$$

Then

$$\eta = \xi(R3^k + 1) \in \{\delta, \delta^2, \tau\},$$

and $\mathrm{Tr}_{\mathbb{Q}(\eta)/\mathbb{Q}}(\eta^{R3^k+1})$ *is prime for every sufficiently large* k*. Moreover,* $8 \mid R$ *for the* δ *branch,* $4 \mid R$ *for the* δ^2 *branch, and* $13 \mid R$ *for the* τ *branch.*

Proof. For $\lambda = 1$ we have $g = s$, so the integrality of n_k and $3 \nmid s$ give $s \mid r$. Table 1 then gives $8 \mid R$, $13 \mid R$, or $4 \mid R$, respectively. For $\lambda = 1/2$ we have $g = 2s$, so again $s \mid r$, and $n_k = (R3^k + 1)/2 \equiv 7 \pmod{13}$ implies $13 \mid R$. For $\lambda = 2$ we have $g = s/2$; the last residue condition says

$$\frac{2r3^k}{s} \equiv 0 \pmod{8}.$$

Consequently $4s \mid r3^k$, and since $3 \nmid 4s$, we get $4s \mid r$. These arguments establish the stated integrality and divisibility conditions in every row.

The possible values of $\eta = \beta^\lambda$ are displayed in the table. A Pisot number and any positive integer power generate the same field: an embedding fixing that power must send the Pisot number to a conjugate of the same absolute value, and only the dominant conjugate has that absolute value. Thus the trace field is unchanged in the τ^2 and δ^2 rows, and

$$\mathrm{Tr}_{\mathbb{Q}(\beta)/\mathbb{Q}}(\beta^{n_k}) = \mathrm{Tr}_{\mathbb{Q}(\eta)/\mathbb{Q}}(\eta^{R3^k+1}).$$

Lastly, for any positive integer d,

$$\mathcal{W}(dC) = \{A > 1 : A^d \in \mathcal{W}(C)\}, \qquad \xi(dC)^d = \xi(C).$$

Applying this exact minimum-preserving identity with $d = s$ proves $\eta = \xi(R3^k + 1)$, not just $\eta \in \mathcal{W}(R3^k + 1)$. □

3.3 Exact large divisors from finite rings

For a cubic unit with minimal polynomial $f(X) = X^3 - aX^2 + bX - 1$, put $T_n = \mathrm{Tr}(\eta^n)$. Newton's identities give

$$T_0 = 3, \quad T_1 = a, \quad T_2 = a^2 - 2b, \qquad T_n = aT_{n-1} - bT_{n-2} + T_{n-3} \quad (n \geq 3). \tag{1}$$

Lemma 3.5 (Finite-ring obstruction)**.** *Let* η *be a cubic Pisot unit, and let* p, t *be primes with* $t \neq 3$*. Suppose*

(a) $\mathrm{ord}_t(3) = t - 1$*;*

(b) $X^t = 1$ *in* $\mathbb{F}_p[X]/(f)$*;*

(c) *for some* $n \not\equiv 1 \pmod{t}$*, one has* $T_n \equiv 0 \pmod{p}$*.*

If T_{R3^k+1} *is prime for every sufficiently large* k*, where* $R \in \mathbb{N}$*, then* $t \mid R$*.*

Proof. If $t \nmid R$, then $R3^k$ runs periodically through every nonzero residue modulo t, so that $R3^k + 1 \equiv n \pmod{t}$ infinitely often. Condition (b) makes the power traces periodic modulo p with period dividing t. Hence $p \mid T_{R3^k+1}$ for infinitely many k. Since η is Pisot, $T_m = \eta^m + o(1) \longrightarrow +\infty$ as $m \longrightarrow \infty$. These divisible terms eventually exceed p, contradicting their primality. Neither irreducibility of f modulo p nor exact order t of X in the quotient ring is required. □

p	t	n for δ	n for δ^2
2	7	3	3
5	31	4	2
41	1723	42	21
101	10303	58	29
1217	1482307	1218	609
1373	1886503	86	43
2729	7450171	2730	1365

Table 2: Certificates for δ and δ^2.

p	t	n for τ
5	31	11
1193	1424443	1620
1373	1886503	78
1973	3894703	331

Table 3: Certificates for τ.

Every row of Tables 2 and 3 satisfies Lemma 3.5; in each row $t = p^2 + p + 1$. Section 3.9 gives a complete exact verifier. It checks primality by trial division, the order of 3 using every prime divisor of $t - 1$, quotient-ring powering, and the zero trace both by powering and by the independent recurrence (1). A separate companion-matrix implementation reproduces these checks.

Proposition 3.6. *If* $\mathrm{Tr}(\eta^{R3^k+1})$ *is eventually prime for* $\eta \in \{\delta, \delta^2, \tau\}$ *and* $R \in \mathbb{N}$*, then*

$$\begin{aligned} 802547251619991615029086812 43 \mid R \qquad & \text{if } \eta \in \{\delta, \delta^2\}, \\ 324443153856491628397 \mid R \qquad & \text{if } \eta = \tau. \end{aligned}$$

In particular, $R > 10^{20}$ *in all three cases.*

Proof. Apply Lemma 3.5 to every row for the given candidate. Within each table the moduli t are distinct primes, so their product divides R. Their exact products are the displayed integers. □

These certificates assert necessary divisibility, not compositeness for every R. In particular, multiples of all listed moduli survive this finite sieve. The next section supplies the separate global bound that eliminates this escape.

3.4 An explicit prime interval

We give a deliberately conservative consequence of the explicit estimates in Cully-Hugill [3]. The large threshold keeps the verification elementary and leaves ample room below the divisors in Proposition 3.6.

Lemma 3.7. *For every real* $u \geq X_0 := \exp(10^{18})$*, there is a prime* p *with*

$$u < p \leq u + u^{2/3}.$$

Proof. First let x be a half-odd-integer at least X_0, and put

$$y = \log x, \qquad \alpha = \frac{17}{50}, \qquad T = x^{\alpha}, \qquad h = \left\lfloor x^{2/3}/2 \right\rfloor, \qquad w = \log \frac{x}{T^{8/3}(\log T)^2}.$$

Thus $x + h$ is also a half-odd-integer, $50 < T < x$, and $x^{2/3}/3 \le h \le x^{2/3}/2$.

Write $\theta(x) = \sum_{p \le x} \log p$. Theorem 2, Table 2, and equations (2), (9), (15), and (19)–(21) of [3], with $\sigma_1 = 3/5$, imply

$$\theta(x+h) - \theta(x) > h\left(1 - F(x) - \frac{KG(x,h)}{Th} + \frac{E(x)}{h}\right), \tag{2}$$

where $K = 3.4747$,

$$G(x,h) = (x+h)\log(x+h)\log\log(x+h) + x\log x \log\log x,$$
$$E(x) = 0.999\sqrt{x} + x^{1/3} - a_1\sqrt{x+h} - a_2(x+h)^{1/3},$$

with $a_1, a_2 < 2$. Taking $C_1 = 17.418$, $C_2 = 5.272$, and

$$v = \frac{1}{57.54(\alpha y)^{2/3}(\log(\alpha y))^{1/3}},$$

one may use the upper estimate

$$F(x) \le \frac{\alpha}{\pi} y e^{-(1-\alpha-3/5)y} + \frac{2C_1\alpha^3 y^4}{w} e^{-vw} + 2C_2\alpha^2 y^2 e^{-vy}. \tag{3}$$

Here the negative subtraction involving $\exp((\sigma_1 - 1)w)$ has been discarded, and the source's zero-free width has been replaced by the smaller width v.

For $y \ge 10^{18}$, elementary inequalities give

$$w = \frac{7}{75}y - 2\log(\alpha y) \ge \frac{y}{12}, \quad 0 < w < y, \quad \log y \le y^{1/10}.$$

Consequently

$$v \ge \frac{1}{60y^{7/10}}, \qquad vw \ge \frac{y^{3/10}}{720} =: D(y), \qquad vy \ge D(y).$$

Since $24C_1\alpha^3 < 20$ and $2C_2\alpha^2 < 2$,

$$F(x) < ye^{-3y/50} + (20y^3 + 2y^2)e^{-D(y)} < \frac{1}{100}. \tag{4}$$

For completeness, at $y = 10^{18}$ one has $D(y) > 300$ and $\log(22y^3) < 130$. These follow from $\log 10 < 7/3$, $\log 22 < 4$, and $10^{2/5} > 5/2$; the last inequality follows on taking fifth powers. For $1 \le j \le 3$, the logarithmic derivative of $y^j e^{-D(y)}$ with respect to $\log y$ is $j - (3/10)D(y) < 0$. The first term of (4) also decreases. This proves the bound throughout the asserted range, not just at the left endpoint. The auxiliary bound $\log y \le y^{1/10}$ and the estimate for w follow by the same elementary endpoint and monotonicity checks.

Because $x + h < 2x$, we have $G(x,h) < 9xy\log y$. Hence

$$\frac{KG(x,h)}{Th} < 108y\log y\, e^{-y/150} < \frac{1}{100}.$$

This expression decreases on $y \ge 10^{18}$, and its value at the left endpoint is below $1/100$ without any delicate numerical estimate. Also $E(x) > -6\sqrt{x}$, so

$$\frac{E(x)}{h} > -18e^{-y/6} > -\frac{1}{100}.$$

The bracket in (2) is therefore greater than 97/100, proving the existence of a prime in $(x, x+h]$.

For an arbitrary real $u \geq X_0$, choose the least half-odd-integer $x \geq u$. Then $0 \leq x-u < 1$, and the prime just obtained satisfies

$$u < p \leq x + \tfrac{1}{2}x^{2/3} \leq u + 1 + \tfrac{1}{2}(u+1)^{2/3} < u + u^{2/3}.$$

The last inequality holds already for $u \geq 8$, using $(u+1)^{2/3} \leq u^{2/3} + 1$. □

3.5 A bound for the actual infinite minimum

Proposition 3.8. *For every integer $R \geq 10^{19}$,*

$$\xi(R3^k + 1) < \frac{4}{3}.$$

Proof. Put $C_k = R3^k + 1$, $B = 4/3$, and

$$c_{k+1} = \frac{C_{k+1}}{C_k} = 3 - \frac{2}{C_k} \geq \frac{5}{2}.$$

Bertrand's postulate gives a prime p_1 with $X_0 < p_1 < 2X_0$; one can apply the integer version to $\lfloor X_0 \rfloor$. The upper root endpoint satisfies

$$\frac{\log(p_1+1)}{C_1} < \frac{10^{18} + \log 3}{3R+1} < \frac{1}{20} < \log\frac{4}{3}.$$

For the last inequality, $\log(1+t) > t/(1+t)$ at $t = 1/3$ gives $\log(4/3) > 1/4$.

Suppose a prime p_k has been chosen with $(p_k+1)^{1/C_k} < B$, and set $c = c_{k+1}$ and $x = p_k^c$. Then $x \geq X_0$ and

$$\frac{cp_k^{c-1}}{x^{2/3}} = cp_k^{-2/(3C_k)} > \frac{5}{2}B^{-2/3} > 1. \tag{5}$$

Taylor's theorem gives

$$(p_k+1)^c \geq p_k^c + cp_k^{c-1} + \frac{c(c-1)}{2}p_k^{c-2} > p_k^c + cp_k^{c-1} + 1,$$

since $c \geq 5/2$ and $p_k \geq 2$. Applying Lemma 3.7 at x therefore supplies a prime p_{k+1} with

$$p_k^c < p_{k+1} \leq x + x^{2/3}, \qquad p_{k+1} + 1 < (p_k+1)^c. \tag{6}$$

The second inequality, including its strictness and the extra 1, is what is needed for the floor endpoints.

Thus the compact intervals

$$I_k = [p_k^{1/C_k}, (p_k+1)^{1/C_k}]$$

are nested; each child has its upper endpoint strictly below its parent's upper endpoint. Their intersection is nonempty. If A lies in this intersection, then for every k the containment $A \in I_{k+1}$ and (6) imply

$$p_k \leq A^{C_k} < p_k + 1.$$

Hence $A \in \mathcal{W}(C)$, and

$$\xi(C) \leq A \leq (p_1+1)^{1/C_1} < B.$$

This constructs an entire infinite prime chain. It makes no assumption about extending an arbitrary prime prefix below X_0. □

3.6 Completion of the transcendence proof

Proof of Theorem 3.1. Suppose $\xi(r3^k + s)$ were algebraic. By Proposition 3.4, $R = r/s \in \mathbb{N}$ and

$$\eta = \xi(r3^k + s)^s = \xi(R3^k + 1) \in \{\delta, \delta^2, \tau\},$$

with eventually prime traces along $R3^k + 1$. Proposition 3.6 gives $R > 10^{20}$, so Proposition 3.8 yields $\eta < 4/3$.

On the other hand, $f_\delta(4/3) = -11/27 < 0$, and f_δ is strictly increasing on $(1, \infty)$, so $\delta > 4/3$. Also $\delta^2 > \delta$, while $f_\tau(\delta) = -\delta < 0$ and f_τ is strictly increasing on $(1, \infty)$, whence $\tau > \delta$. Every possible η is therefore greater than $4/3$, a contradiction. □

3.7 The finite unit enumeration

Here are the exact checks underlying Table 1. They reproduce the necessary unit rows of [7] and can be carried out without floating-point root tests. Write the minimal polynomial of a cubic Pisot unit below 9 as

$$X^3 - aX^2 + bX - c.$$

The root bounds give the safe exhaustive ranges

$$0 \le a \le 10, \quad |b| \le 20, \quad c \in \{-1, 1\}.$$

Reject polynomials having a root at 1 or -1; for these monic cubics with unit constant term, this is the full rational-root test. For negative discriminant, a root in $(1, 9)$ and unit norm force the nonreal pair to have modulus $\beta^{-1/2} < 1$. For positive discriminant, use exact Sturm counts to require one root in $(1, 9)$ and two in $(-1, 1)$. Exactly 144 cubic Pisot units remain.

Since $g \mid 2s$, the fixed point is a positive half-integer. For $\lambda = \ell/2$, $1 \le \ell \le 7$, the positive-fixed-point bound is $\beta^\ell < 9$. Rational bisection tests this inequality exactly. The numbers of candidates are

$$144,\ 12,\ 4,\ 2,\ 2,\ 1,\ 1.$$

There are no candidates for $\lambda \ge 4$: Siegel's smallest-Pisot theorem [5] gives $\beta \ge \rho$, where $\rho^3 - \rho - 1 = 0$ and $\rho > 3^{1/4}$. The impossible values $\lambda = 3/2, 3$ may harmlessly be included in the finite check.

To test the twisted-limit condition, fix $J \in \{12, 22, 30\}$ and $K = J + 3$. Let n_0 be the least nonnegative residue of $\ell/2$ modulo 3^K, and consider

$$n_t = n_0 + t3^K, \qquad 0 \le t < 104.$$

These integers cover every residue modulo 104. Compute $\mathrm{Tr}(\beta^{n_t})$ in the exact ring $(\mathbb{Z}/3^J\mathbb{Z})[X]/(f)$ by binary powering, and retain the residue n_t mod 104 only when the trace equals 1 or -1 modulo 3^J. The principal-unit estimate from [7] shows that these computed traces equal the corresponding twisted limits modulo 3^J. Indeed, each root lies in a local field of ramification index at most 3; for its principal-unit factor u, cubing gives $v_3(u^{3^K} - 1) \ge K$. The Teichmüller orders divide $\mathrm{lcm}(2, 8, 26) = 104$.

For all three precisions, the same five rows remain, with exactly the residue sets displayed in Table 1. A genuine limit ± 1 must pass this test. Conversely, passing a finite-precision test is *not* used to prove a limit identity or eventual primality. The program `unit_table_independent.py`, supplied with this draft's certificates, performs the enumeration using exact Sturm counts and rational arithmetic and records the complete residue sets. The full logical direction needed here is exclusion: every unlisted candidate fails an exact necessary congruence.

3.8 Scope and remaining questions

The hypothesis $3 \nmid s$ is used in the algebraic-minimum reduction, in affine returns, and in the 3-adic fixed-point analysis. The hypothesis $s \leq 3r$ guarantees the consecutive-ratio condition from the first prescribed exponent onward. We do not remove either hypothesis here, although exact common-factor scaling can transport an individual result to an equivalent schedule.

The global minimum is essential in Proposition 3.8. Producing a single prime-representing $A < 4/3$ bounds the least element but need not bound any other local minimum. Thus this proof does not establish transcendence of all local minima or all prime-representing constants for the same schedule. Likewise, the trace argument starts from the eventual floor–trace identity of an algebraic minimum. It does not identify arbitrary Pisot floors with traces; the small-conjugate sum may have either sign.

For the classical Mills schedule $C_k = 3^k$, the fixed point of the affine exponent dynamics is 0, not positive. The principal-unit factors then disappear from the transferred identity, so the positive-fixed-point candidate bound used here is unavailable. No unconditional transcendence statement about the classical Mills constant follows from Theorem 3.1. The negative-shift families left open in [7] also require separate arguments. Earlier finite-certificate work in [6] motivates the modular step, but its negative-shift cutoffs are not used in the present proof.

3.9 An exact verifier for the large divisors

The following Python 3 program uses only the standard library. All arithmetic affecting acceptance is integral. Its final assertions check both product values and the inequality used in the proof. The trace verification is performed in two distinct ways.

```
from math import isqrt, prod

def prime(n):
    return n >= 2 and all(n % d for d in range(2, isqrt(n) + 1))

def prime_factors(n):
    out, d = [], 2
    while d * d <= n:
        if n % d == 0:
            out.append(d)
            while n % d == 0:
                n //= d
        d += 1
    if n > 1:
        out.append(n)
    return out

# f = X^3 - a X^2 + b X - 1; tuples are coefficients of 1,X,X^2.
def mul(u, v, a, b, p):
    z = [0] * 5
    for i in range(3):
        for j in range(3):
            z[i+j] += u[i] * v[j]
    for i in (4, 3):
        z[i-1] += a * z[i]
        z[i-2] -= b * z[i]
        z[i-3] += z[i]
    return tuple(z[i] % p for i in range(3))

def xp(n, a, b, p):
    v, x = (1, 0, 0), (0, 1, 0)
    while n:
        if n & 1:
            v = mul(v, x, a, b, p)
        x = mul(x, x, a, b, p)
        n //= 2
    return v
```

```
def trace_power(n, a, b, p):
    x, y, z = xp(n, a, b, p)
    return (3*x + a*y + (a*a - 2*b)*z) % p

def trace_recurrence(n, a, b, p):
    v = [3 % p, a % p, (a*a - 2*b) % p]
    for j in range(3, n + 1):
        v.append((a*v[-1] - b*v[-2] + v[-3]) % p)
    return v[n]

DATA = {
    'delta': (1, 0, [
        (2,7,3), (5,31,4), (41,1723,42), (101,10303,58),
        (1217,1482307,1218), (1373,1886503,86),
        (2729,7450171,2730)]),
    'delta_square': (1, -2, [
        (2,7,3), (5,31,2), (41,1723,21), (101,10303,29),
        (1217,1482307,609), (1373,1886503,43),
        (2729,7450171,1365)]),
    'tau': (1, -1, [
        (5,31,11), (1193,1424443,1620),
        (1373,1886503,78), (1973,3894703,331)])
}

products = {}
for name, (a, b, rows) in DATA.items():
    moduli = []
    for p, t, n in rows:
        assert prime(p) and prime(t)
        assert t == p*p + p + 1 and t != 3
        assert pow(3, t-1, t) == 1
        assert all(pow(3, (t-1)//q, t) != 1
                   for q in prime_factors(t-1))
        assert xp(t, a, b, p) == (1, 0, 0)
        assert 0 < n < t and n != 1
        assert trace_power(n, a, b, p) == 0
        assert trace_recurrence(n, a, b, p) == 0
        moduli.append(t)
    assert len(moduli) == len(set(moduli))
    products[name] = prod(moduli)
    print(name, products[name])

assert products['delta'] == 80254725161999161502908681243
assert products['delta_square'] == products['delta']
assert products['tau'] == 324443153856491628397
assert all(value > 10**20 for value in products.values())
print('ALL EXACT CERTIFICATES PASSED')
```

4 Full prefix positive extensions and cube rescaling

The preceding elimination concerns its stated starting index. We now extend the full-prefix range and isolate the exact cube rescaling that yields the clock $3^k + 3$. This keeps tail minima separate from minima with every initial constraint.

4.1 Statement, indexing, and dependencies

For a positive clock $C = (C_i)$ define

$$\mathcal{W}(C) = \{A > 1 : \left\lfloor A^{C_i} \right\rfloor \text{ is prime at every prescribed index}\}.$$

If this set has a least element, write $\xi(C) = \min \mathcal{W}(C)$. The starting index is part of the definition.

Theorem 4.1 (Full-clock extension)**.** *Let r, s be integers satisfying*

$$r \geq s \geq 1, \qquad 3 \nmid s, \qquad 3 \nmid (r+s).$$

Then the minimum

$$B = \xi\big((r3^j + s)_{j\geq 0}\big)$$

exists and is transcendental. Consequently

$$\xi\big((r3^k + 3s)_{k\geq 1}\big) = B^{1/3}$$

exists and is transcendental, as does the minimum for every positive integer scaling of either clock.

Corollary 4.2. *For every integer $q \geq 1$,*

$$\xi\big((q(3^k + 3))_{k\geq 1}\big)$$

exists and is transcendental.

The full clock $(3^j+1)_{j\geq 0}$ is $2, 4, 10, 28, \ldots$; it is not the shorter clock $4, 10, 28, \ldots$. *Transcendence of a tail's minimum does not, by itself, imply transcendence of the full clock's minimum.* No such inference is made here. Only after applying Saito's theorem to the full minimum do we use an eventual statement about its trace orbit.

The inputs are Saito's Theorem 1.3 and Proposition 3.1 in arXiv version 3 [4], the prime-interval exponent 21/40 of Baker–Harman–Pintz [1], and the positive trace theory of corpus manuscript [064] [7]. The unit table, large-divisor certificates, and explicit prime-interval lemma are inherited from [19]; the finite checks are reproduced with this note. The two preceding manuscripts are unpublished research manuscripts. This note proves the new full-prefix bridge and infinite-chain adaptation; it does not independently reprove all their trace or analytic lemmas.

For later use, the increasing bijection $A \mapsto A^d$ gives, for any positive integer d,

$$\mathcal{W}(dC) = \{A > 1 : A^d \in \mathcal{W}(C)\}, \qquad \xi(dC) = \xi(C)^{1/d}. \tag{7}$$

This is an exact identity of global minima whenever they exist.

4.2 The full-clock algebraic reduction

A Pisot number is a real algebraic integer greater than 1 whose other conjugates have absolute value less than 1. All traces and norms below are over $\mathbb{Q}$.

Proposition 4.3. *Under the hypotheses of Theorem 4.1, the full minimum B exists. If B is algebraic, there is $g \mid 2s$ such that $\beta = B^g$ is a cubic Pisot number. For every sufficiently large j,*

$$n_j = \frac{r3^j + s}{g} \in \mathbb{N}, \qquad \mathrm{Tr}(\beta^{n_j}) = \left\lfloor B^{r3^j+s} \right\rfloor \text{ is prime.}$$

If x_2, x_3 are the other conjugates of β, then, for every fixed $0 < \kappa < 17/40$,

$$|x_2^{n_j} + x_3^{n_j}| = O(\beta^{-\kappa n_j}).$$

Proof. Put $D_j = r3^j + s$ for $j \geq 0$. The consecutive ratios are

$$c_j := \frac{D_{j+1}}{D_j} = 3 - \frac{2s}{D_j} \geq 2, \qquad c_j \longrightarrow 3.$$

Thus Saito's Theorem 1.3 gives existence of the minimum for this *full* clock, with $C_{j+1} = D_j$ in his indexing.

Every D_m is coprime to 3: for $m \geq 1$ this follows from $3 \nmid s$, and for $m = 0$ it is exactly the additional condition $3 \nmid (r + s)$. Let $h = \mathrm{ord}_{D_m}(3)$. Then

$$D_{m+h} \equiv D_m \equiv 0 \pmod{D_m}, \qquad D_{m+h} > D_m.$$

If B^{D_m} were an integer, its prime floor would make it a prime p, and the later output $B^{D_{m+h}} = p^{D_{m+h}/D_m}$ would be composite. Hence

$$B^{D_m} \notin \mathbb{N} \quad \text{for every } m \geq 0. \tag{8}$$

This verifies the initial boundary as well as every later one.

Take $\theta = 21/40$ and $\varepsilon_0 = 1/10$ in Saito's Proposition 3.1. Since $D_j \geq 4s$ for $j \geq 1$, we have $c_j \geq 5/2$ there. The selected set

$$I = \{j \geq 0 : c_j \geq 40/19 + 1/10\}$$

therefore contains every $j \geq 1$, possibly also $j = 0$. The proposition gives $\beta = B^g$ Pisot, with

$$2 \leq \deg \beta \leq 1 + (3 \cdot 19/40 - 1)^{-1} = 1 + 40/17 < 4,$$

and $g \mid D_j$ at all sufficiently large indices. Taking consecutive indices shows $g \mid D_{j+1} - 3D_j = -2s$, so $g \mid 2s$ and $3 \nmid g$.

If the degree were 2, Saito's quadratic alternative would require $n_j = D_j/g$ to be an odd prime for every sufficiently large j. Choose one such prime $p = n_j > g$ and put $h = \mathrm{ord}_p(3)$; this order exists since $3 \nmid D_j$. The congruence $D_{j+h} \equiv D_j \pmod{p}$ and $p \nmid g$ imply $p \mid n_{j+h}$, while $n_{j+h} > p$. This contradicts primality. Thus the degree is 3, and Saito's trace identity gives the asserted prime traces.

Finally Saito's fractional-part estimate is

$$\{B^{D_j}\} \leq \frac{2}{c_j} \left\lfloor B^{D_j} \right\rfloor^{-(1-\theta)c_j+1} \quad \text{for all sufficiently large } j \in I.$$

Since $(1 - \theta)c_j - 1 \to 17/40$, this implies the stated estimate for every fixed $\kappa < 17/40$, using the trace identity and $\left\lfloor B^{D_j} \right\rfloor \sim \beta^{n_j}$. We do not assert the limiting exponent $17/40$ itself. □

Saito's proposition is used on the original minimum, not on a tail minimum. Its finite-prefix preservation is part of the primary proof [4, Lemma 5.2, equations (5.5)–(5.14)]; the selected set need not include the first coordinate. The numbering here is that of version 3: the reduction is Proposition *3.1*, not 3.2.

4.3 A short proof for the clock $3^k + 3$

We isolate the part of [064] needed for this family.

Lemma 4.4 (Inherited positive trace rigidity)**.** *Let β be a cubic Pisot number and let $r, s, g \in \mathbb{N}$ with $3 \nmid gs$ and $n_j = (r3^j + s)/g \in \mathbb{N}$ for all sufficiently large j. If $\mathrm{Tr}(\beta^{n_j})$ is eventually prime, then $\beta^{s/g} < 3$. Moreover, if $4s \nmid r$ and $13s \nmid r$, no such eventually prime trace orbit exists.*

This is the positive-fixed-point part (iv) of the ternary trace-rigidity theorem in [7]. It is a theorem about eventual *traces*, not about transferring minima between clocks. Its proof uses affine finite-field return, twisted 3-adic limits, Galois transfer, and a finite candidate calculation.

Proof of Corollary 4.2, independently of the large-height bound. Let $E_j = 3^j + 1$ for $j \geq 0$, and let $\eta = \xi((E_j)_{j\geq 0})$. Proposition 4.3, with $r = s = 1$, gives existence. If η were algebraic, the same proposition would give a cubic Pisot number $\beta = \eta^g$, $g \mid 2$, with eventually prime traces at $(3^j + 1)/g$. Lemma 4.4 rules this out, since neither 4 nor 13 divides 1.

In this case the initial boundary check is especially simple: $\eta^2 = p \in \mathbb{N}$ would force the next output $\eta^4 = p^2$. The selected set omits just this first coordinate, because $c_0 = 2$ and $c_1 = 5/2$; it nevertheless belongs to the minimum problem throughout the proof. Since $q(3^k + 3) = 3qE_{k-1}$, equation (7) gives

$$\xi((q(3^k + 3))_{k\geq 1}) = \eta^{1/(3q)}.$$

An algebraic value on the left would make η algebraic, proving the corollary. No explicit large prime-gap threshold is used here. □

4.4 Inherited unit candidates and their normalization

For the broader theorem we state the further inherited inputs precisely. The approximation-magnitude lemma of [7] says that a cubic Pisot number satisfying $|x_2^n + x_3^n| \leq A_0\beta^{-\kappa n}$ for infinitely many n, with $0 < \kappa < 1/2$, satisfies

$$|\,\mathrm{N}(\beta)| \leq \beta^{1-2\kappa}. \tag{9}$$

For nonreal conjugates, its proof uses a polynomial lower bound from linear forms in logarithms [2], not a uniform lower bound on an oscillating cosine.

Apply this to Proposition 4.3 with $\kappa = 2/5$. Lemma 4.4 and $g \leq 2s$ give $\beta < 9$, whence

$$|\,\mathrm{N}(\beta)| \leq \beta^{1/5} < 2.$$

The nonzero integral norm is therefore ± 1.

Write $\lambda = s/g$. The exact unit-only table of [19], using the necessary twisted-limit identities of [7], leaves only Table 4. Define $\delta, \tau > 1$ by

$$\delta^3 - \delta^2 - 1 = 0, \qquad \tau^3 - \tau^2 - \tau - 1 = 0.$$

λ	β	Minimal polynomial	B^s	Necessary eventual residue
1	δ	$X^3 - X^2 - 1$	δ	$n_j \equiv 1 \pmod 8$
1	τ	$X^3 - X^2 - X - 1$	τ	$n_j \equiv 1 \pmod{13}$
1	δ^2	$X^3 - X^2 - 2X - 1$	δ^2	$n_j \equiv 1 \pmod 4$
1/2	τ^2	$X^3 - 3X^2 - X - 1$	τ	$n_j \equiv 7 \pmod{13}$
2	δ	$X^3 - X^2 - 1$	δ^2	$n_j \equiv 2 \pmod 8$

Table 4: Necessary unit candidates; no row asserts eventual primality.

The hypotheses of this inherited table are exactly those now available: a cubic Pisot unit $\beta < 9$, $g \mid 2s$, $3 \nmid gs$, the bound $\beta^{s/g} < 3$, and eventual prime traces along $(r3^j + s)/g$. Its finite verification is described in Section 4.8.1.

Proposition 4.5. *If the full minimum B is algebraic, then $s \mid r$ and, with $R = r/s$ and $\eta = B^s$,*

$$\boxed{\eta = \xi((R3^j + 1)_{j\geq 0}) \in \{\delta, \delta^2, \tau\}.} \tag{10}$$

Furthermore $\mathrm{Tr}(\eta^{R3^j+1})$ *is prime for all sufficiently large j.*

Proof. In the rows $\lambda = 1$ and $1/2$, integrality of n_j and $3 \nmid s$ imply $s \mid r$. In the row $\lambda = 2$, the residue condition gives

$$\frac{2r3^j}{s} = n_j - 2 \equiv 0 \pmod 8.$$

Thus $4s \mid r3^j$, and $3 \nmid 4s$ implies $4s \mid r$. Every row therefore has integral $R = r/s$, and the possible values of $\eta = \beta^\lambda$ are those displayed.

A Pisot number and any positive integer power generate the same field: an embedding fixing that power must send the Pisot number to a conjugate of the same absolute value, and no other conjugate has that value. This handles both the τ^2 and δ^2 rows and gives

$$\mathrm{Tr}_{\mathbb{Q}(\beta)/\mathbb{Q}}(\beta^{n_j}) = \mathrm{Tr}_{\mathbb{Q}(\eta)/\mathbb{Q}}(\eta^{R3^j+1}).$$

Finally, apply the exact scaling identity (7) to $D_j = s(R3^j + 1)$ at *every* $j \geq 0$. This proves the equality of minima in (10), with initial exponent $R+1$; no shorter-clock minimum appears. □

4.5 Exact divisibility obstructions

For $\eta \in \{\delta, \delta^2, \tau\}$ write its minimal polynomial as $f(X) = X^3 - aX^2 + bX - 1$, and put $T_n = \mathrm{Tr}(\eta^n)$. Newton's identities give

$$T_0 = 3, \quad T_1 = a, \quad T_2 = a^2 - 2b, \qquad T_n = aT_{n-1} - bT_{n-2} + T_{n-3} \quad (n \geq 3). \tag{11}$$

Lemma 4.6. *Suppose p, t are primes, $t \neq 3$,* $\mathrm{ord}_t(3) = t-1$, *$X^t = 1$ in $\mathbb{F}_p[X]/(f)$, and $T_n \equiv 0 \pmod p$ for some $n \not\equiv 1 \pmod t$. If T_{R3^j+1} is eventually prime, then $t \mid R$.*

Proof. If $t \nmid R$, the residues $R3^j$ run periodically through all nonzero classes modulo t, so $R3^j + 1 \equiv n \pmod t$ infinitely often. The quotient-ring identity makes the traces periodic modulo p with period dividing t. Thus these trace values are divisible by p. They tend to infinity because $T_m = \eta^m + o(1)$, so they eventually exceed p, contradicting primality. Neither irreducibility of f modulo p nor exact order t of X in the quotient is required. □

p	t	n for δ	n for δ^2
2	7	3	3
5	31	4	2
41	1723	42	21
101	10303	58	29
1217	1482307	1218	609
1373	1886503	86	43
2729	7450171	2730	1365

p	t	n for τ
5	31	11
1193	1424443	1620
1373	1886503	78
1973	3894703	331

Table 5: Exact inherited finite-ring certificates. Each row has $t = p^2 + p + 1$.

Every row of Table 5 satisfies Lemma 4.6. The supplied programs check primality, the primitive-root condition, quotient-ring powering, and the zero trace by two methods; a separate

companion-matrix implementation reproduces the identities. The distinct prime t's in each branch have products

$$802547251619991615029086812 43 \mid R \qquad (\eta = \delta \text{ or } \delta^2), \tag{12}$$

$$324443153856491628397 \mid R \qquad (\eta = \tau). \tag{13}$$

In particular, $R > 10^{20}$ in every branch. These certificates use only eventual traces, so adding the initial coordinate does not alter their conclusions. Multiples of these products survive the finite sieve; the global bound below eliminates them separately.

4.6 A full-prefix infinite prime chain

Lemma 4.7 (Inherited explicit prime-interval bound)**.** *For every real $u \geq X_{\mathrm{gap}} := \exp(10^{18})$, there is a prime*

$$u < p \leq u + u^{2/3}.$$

This is the explicit lemma proved in [19, Section 4] from Cully-Hugill [3], using Theorem 2, Table 2, and equations (2), (9), (15), (19)–(21) of that paper. The substituted parameters are

$$T = x^{17/50}, \quad \sigma_1 = 3/5, \quad K = 3.4747, \quad C_1 = 17.418, \quad C_2 = 5.272,$$

with the zero-free width

$$\frac{1}{57.54(\log T)^{2/3}(\log\log T)^{1/3}}.$$

The derivation treats half-odd-integer endpoints first and then arbitrary real u. This explicit lemma is inherited as stated; no numerical constant is inferred from an asymptotic O-bound.

Proposition 4.8 (The initial exponent is $R+1$)**.** *For every integer $R \geq 10^{19}$,*

$$\xi\big((R3^j + 1)_{j\geq 0}\big) < \frac{4}{3}.$$

Proof. Put $C_j = R3^j + 1$ for $j \geq 0$ and $B_0 = 4/3$. Bertrand's postulate, applied to $\lfloor X_{\mathrm{gap}} \rfloor$, gives a prime p_0 with $X_{\mathrm{gap}} < p_0 < 2X_{\mathrm{gap}}$. As $p_0 + 1 < 3X_{\mathrm{gap}}$, we have

$$\frac{\log(p_0+1)}{C_0} < \frac{10^{18} + \log 3}{R+1} < \frac{1}{5} < \frac{1}{4} < \log\frac{4}{3}. \tag{14}$$

Here $C_0 = R + 1$, not $3R + 1$. The last inequality follows from $\log(1+t) > t/(1+t)$ with $t = 1/3$.

Suppose p_j has been chosen with $(p_j+1)^{1/C_j} < B_0$. Set

$$c = \frac{C_{j+1}}{C_j} = 3 - \frac{2}{C_j} \geq \frac{5}{2}, \qquad u = p_j^c \geq X_{\mathrm{gap}}.$$

Since $p_j^{1/C_j} < B_0$, the first-order width satisfies

$$\frac{cp_j^{c-1}}{u^{2/3}} = cp_j^{-2/(3C_j)} > \frac{5}{2}B_0^{-2/3} > 1. \tag{15}$$

Taylor's theorem, with increasing second derivative for $c > 2$, gives

$$(p_j+1)^c \geq p_j^c + cp_j^{c-1} + \frac{c(c-1)}{2}p_j^{c-2} > u + cp_j^{c-1} + 1 > u + u^{2/3} + 1.$$

The remainder exceeds 1 since $c \geq 5/2$ and $p_j \geq 2$. By Lemma 4.7, choose a prime p_{j+1} with

$$p_j^c < p_{j+1} \leq u + u^{2/3}, \qquad p_{j+1} + 1 < (p_j + 1)^c. \tag{16}$$

Its upper root endpoint remains below B_0, so this constructs an entire infinite sequence.

The compact intervals

$$I_j = [p_j^{1/C_j}, (p_j + 1)^{1/C_j}]$$

are nested, with each child upper endpoint strictly below its parent's. Their intersection is nonempty. For any point A in the intersection, use $A \in I_{j+1}$ and (16) to obtain

$$p_j \leq A^{C_j} < p_j + 1 \quad \text{for every } j \geq 0.$$

Thus every floor is prime, including the first floor at exponent $R+1$, and $A \leq (p_0+1)^{1/C_0} < B_0$ by (14). Saito's existence theorem supplies the minimum, so $\xi((C_j)_{j\geq 0}) \leq A < B_0$ as asserted. This constructs a full infinite chain; it does not presume that an arbitrarily chosen finite prime prefix extends. □

4.7 Completion and scope

Proof of Theorem 4.1. Existence and the algebraic reduction are established in Proposition 4.3. If B were algebraic, Proposition 4.5 would give

$$\eta = B^s = \xi((R3^j + 1)_{j\geq 0}) \in \{\delta, \delta^2, \tau\}, \qquad R = r/s \in \mathbb{N}.$$

Equations (12)–(13) force $R > 10^{20}$, so Proposition 4.8 gives $\eta < 4/3$. But $f_\delta(4/3) = -11/27 < 0$ and f_δ is strictly increasing on $(1, \infty)$, so $\delta > 4/3$. Also $\delta^2 > \delta$ and $f_\tau(\delta) = -\delta < 0$, whence $\tau > \delta$. Every candidate exceeds $4/3$, a contradiction.

Finally, $r3^k + 3s = 3D_{k-1}$ for $k \geq 1$. Equation (7) proves the translated statement and all positive integer scalings, preserving global minima exactly. □

The condition $3 \nmid (r + s)$ is retained in Theorem 4.1: it excludes an integral power at the newly imposed coordinate. Without that condition the following necessary-only statement is valid.

Proposition 4.9 (The only possible algebraic initial boundary)**.** *Let $r \geq s \geq 1$ and $3 \nmid s$, without assuming $3 \nmid (r + s)$. The full minimum $B = \xi((r3^j + s)_{j\geq 0})$ exists. If B is algebraic, then for some prime p,*

$$B^{r+s} = p, \qquad B = p^{1/(r+s)}, \qquad \xi((r3^k + 3s)_{k\geq 1}) = p^{1/[3(r+s)]}.$$

This is a necessary condition only: it asserts neither the existence of such an algebraic minimum nor the admissibility or minimality of any specified prime radical.

Proof. The ratio bound still gives existence. For every $j \geq 1$, the exponent $r3^j + s$ is coprime to 3, so the return argument in Proposition 4.3 excludes an integral B^{r3^j+s}. If an algebraic B also had $B^{r+s} \notin \mathbb{N}$, all the hypotheses of Saito's reduction would hold. The remainder of the full-clock candidate, divisibility, and global-bound argument above would then give the same contradiction: none of those later steps uses $3 \nmid (r + s)$. Consequently B^{r+s} must be an integer, hence a prime because its floor is prime. The last identity follows from (7). □

The condition $r \geq s$ supplies the ratio bound from the first coordinate. No claim is made about every admissible base, every local minimum, the classical Mills constant, or the unresolved negative-shift families.

Relation to the earlier corpus. The earlier additive-clock theorem [18, Section 6, Theorem 6.1] proves transcendence of the full minima for $(ba^k \pm k)_{k\geq 1}$, with $a \geq 3$ and $b \geq 1$. Its perturbation is the varying term $\pm k$, not a constant shift. A stronger antecedent relevant here is [6, Section 27, Theorem 27.2]: for $C_n = P(a^n) + Q(n)$, with P of positive leading coefficient and positive degree, $P(0) + Q$ not the zero polynomial, and $L = a^{\deg P} > 40/19$, every sufficiently late tail minimum is transcendental. Constant shifts are included. Taking $a = 3$, $P(X) = rX$, and $Q(X) = s$ gives $L = 3$ and applies to sufficiently late tails of the present clocks; taking $P(X) = X$, $Q(X) = 3$ applies to sufficiently late tails of $3^k + 3$.

The distinction is the prescribed starting coordinate. At $L = 3$, that proof uses the fact that the tail minima tend to 1 as the starting index tends to infinity. It does not give the corresponding claim for an arbitrary fixed full prefix. The same theorem's stronger conclusion for every tail left-gap point (a local left minimum), and its full-clock extension under recurring divisibility, are stated for $L \geq 4$. Moreover, the finite-prefix dichotomy in [6, Section 27, Theorem 27.1] requires transcendence of *every* left-gap point of the underlying tail set, not merely of its global minimum. Thus the already established $L = 3$ tail-minimum result alone does not justify a finite-prefix transfer here. The present argument instead applies Saito directly to the specified full minimum and keeps its initial coordinate throughout. It is a full-prefix extension of the existing framework; no historical priority claim is made.

4.8 Reproducing the exact finite checks

The accompanying `certificates/` directory contains four portable Python 3 programs. Both `unit_table_independent.py` and `positive_prefix_certificate.py`, which imports the unit-table module, require SymPy. The two large-divisor programs use only the standard library. All acceptance tests use exact integer or rational arithmetic. Run from the corresponding extracted source archive's directory:

```
python3 certificates/large_divisor_certificate.py
python3 certificates/independent_matrix_audit.py
python3 certificates/unit_table_independent.py
python3 certificates/positive_prefix_certificate.py
```

The programs assert their claimed conclusions. Expected terminal lines are

```
ALL EXACT CERTIFICATES PASSED
ALL INDEPENDENT MATRIX CHECKS PASSED
ALL EXACT UNIT TABLE CHECKS PASSED
PASS: all full-prefix algebraic candidates excluded by exact finite residues
```

Rerun logs are supplied alongside them. The companion-matrix verifier reads only the literal certificate records from the first program and implements its powering checks separately.

For the large divisors, primality of all p, t is checked by deterministic trial division. The test $\mathrm{ord}_t(3) = t - 1$ uses every prime divisor of $t - 1$. Quotient-ring multiplication uses $X^3 = aX^2 - bX + 1$; the zero trace is checked both by powering and by (11). These finite identities prove the implications of Lemma 4.6; they do not certify eventual primality for any surviving class.

4.8.1 The necessary unit table

Write a cubic Pisot unit below 9 as a root of $X^3 - aX^2 + bX - c$. Safe exhaustive bounds are

$$0 \leq a \leq 10, \qquad |b| \leq 20, \qquad c \in \{-1, 1\}.$$

The rational-root test rejects roots ± 1. Exact Sturm counts handle the totally real case; in the nonreal case the unit norm and the unique root $\beta \in (1, 9)$ give the other roots modulus $\beta^{-1/2} < 1$. Exactly 144 cubic Pisot units remain.

As $g \mid 2s$, write $\lambda = \ell/2$ with $\ell \geq 1$. The bound $\beta^\lambda < 3$ becomes $\beta^\ell < 9$ and is tested by rational bisection. For $\ell = 1, \ldots, 7$ the respective candidate counts are

$$144,\ 12,\ 4,\ 2,\ 2,\ 1,\ 1.$$

There are no candidates with $\lambda \geq 4$, by Siegel's smallest-Pisot theorem [5]: its smallest value ρ, satisfying $\rho^3 - \rho - 1 = 0$, exceeds $3^{1/4}$. The impossible values $\lambda = 3/2, 3$ can harmlessly be tested too.

For each $J \in \{12, 22, 30\}$ set $K = J + 3$. Let $n_0 \equiv \ell/2 \pmod{3^K}$ be the least nonnegative representative and test $n_t = n_0 + t3^K$, $0 \leq t < 104$. These cover every residue modulo 104. Binary powering in $(\mathbb{Z}/3^J\mathbb{Z})[X]/(f)$ computes the trace exactly; retain a residue only if that trace equals 1 or -1 modulo 3^J. The principal-unit estimate and twisted-limit theorem of [7] make this a necessary test: the Teichmüller orders divide $104 = \mathrm{lcm}(2, 8, 26)$, and principal-unit powers at 3^K are sufficiently close to 1 modulo 3^J. Every genuine twisted limit ± 1 passes, whereas passing is not claimed sufficient. All three precisions give exactly the five necessary rows of Table 4.

For the small family, the additional program intersects these necessary residues with the full orbits

$$\left\{ \frac{3^j + 1}{g} \bmod 104 : j \geq 0 \right\}, \qquad g = 1, 2.$$

Their periods are 6 and 12, respectively. At precision $J = 22$, all 12 bounded unit candidates for $g = 1$ and all 144 for $g = 2$ are excluded. Equivalently, the first three relevant rows would require 3^j divisible by $8, 4$, or 13; the $g = 2$ row would require $(3^j + 1)/2 \equiv 7 \pmod{13}$, hence $3^j \equiv 0 \pmod{13}$. This separately checks the finite part of the small-family argument. It does not independently reprove the inherited trace or gap lemmas.

4.8.2 The portable large-divisor verifier

For convenience, the complete first verifier is included below; the separate unit and matrix programs are supplied as companion files.

```
#!/usr/bin/env python3
"""Exact finite certificates for trace(eta^(R*3^k+1)) eventual primality.
No nonstandard dependencies; all primality/order checks are deterministic.
Polynomials are x^3-a*x^2+b*x-1.
"""
from math import isqrt, prod

def prime(n):
    if n<2:return False
    return all(n%d for d in range(2,isqrt(n)+1))

def prime_factors(n):
    ans=[];d=2
    while d*d<=n:
        if n%d==0:
            ans.append(d)
            while n%d==0:n//=d
        d+=1
    if n>1:ans.append(n)
    return ans

def mul(u,v,a,b,p):
    z=[0]*5
    for i in range(3):
        for j in range(3):z[i+j]+=u[i]*v[j]
    for i in (4,3):
        z[i-1]+=a*z[i]
        z[i-2]-=b*z[i]
        z[i-3]+=z[i]
```

```
    return tuple(z[i]%p for i in range(3))

def xp(n,a,b,p):
    v=(1,0,0);x=(0,1,0)
    while n:
        if n&1:v=mul(v,x,a,b,p)
        x=mul(x,x,a,b,p);n//=2
    return v

def trace_pow(n,a,b,p):
    x,y,z=xp(n,a,b,p)
    return (3*x+a*y+(a*a-2*b)*z)%p

def trace_recurrence(n,a,b,p):
    v=[3%p,a%p,(a*a-2*b)%p]
    for j in range(3,n+1):v.append((a*v[-1]-b*v[-2]+v[-3])%p)
    return v[n]

DATA={
'delta':(1,0,[(2,7,3),(5,31,4),(41,1723,42),(101,10303,58),(1217,1482307,1218),(1373,1886503,86)
     ,(2729,7450171,2730)]),
'delta_square':(1,-2,[(2,7,3),(5,31,2),(41,1723,21),(101,10303,29),(1217,1482307,609),(1373,1886503,43)
     ,(2729,7450171,1365)]),
 'tau':(1,-1,[(5,31,11),(1193,1424443,1620),(1373,1886503,78),(1973,3894703,331)])}

def verify():
    for name,(a,b,rows) in DATA.items():
        ts=[]
        for p,t,n in rows:
            assert prime(p) and prime(t)
            assert t==p*p+p+1
            assert pow(3,t-1,t)==1
            factors=prime_factors(t-1)
            assert all(pow(3,(t-1)//ell,t)!=1 for ell in factors)
            assert xp(t,a,b,p)==(1,0,0)
            assert 0<n<t and n!=1
            assert trace_pow(n,a,b,p)==0
            assert trace_recurrence(n,a,b,p)==0
            ts.append(t)
            print(name,'p',p,'t',t,'zero exponent',n,'factors(t-1)',factors)
        assert len(ts)==len(set(ts))
        L=prod(ts)
        print('NECESSARY DIVISOR OF R:',name,L)
    assert prod(row[1] for row in DATA['delta'][2])==802547251619991615029086812 43
    assert prod(row[1] for row in DATA['tau'][2])==324443153856491628397
    print('ALL EXACT CERTIFICATES PASSED')

if __name__=='__main__':verify()
```

5 Negative shifts and integral fixed point rigidity

The negative shift changes the fixed-point argument. The results below establish a structural reduction and explicit families; the multiplier-thirteen and other odd-multiplier cases are subsequently strengthened in Sections 6 and 7. Statements of what remains at this stage are superseded only by those explicitly later theorems.

5.1 Main results and scope

For an integer sequence $C = (C_k)_{k\geq 1}$ with $C_{k+1} \geq 2C_k > 0$, write

$$\xi(C) = \min\{A > 1 : \left\lfloor A^{C_k} \right\rfloor \text{ is prime for every } k \geq 1\}.$$

Existence is covered by Saito's general existence theorem [4, Theorem 1.3]. Throughout this note,

$$r, s \in \mathbb{N}, \qquad 3 \nmid s, \qquad s < 3r, \qquad C_k = r3^k - s. \tag{17}$$

Thus $C_k > 0$, $C_{k+1} = 3C_k + 2s$, and the consecutive ratios tend to 3 from above. We abbreviate $\xi(C)$ to $\xi(r3^k - s)$.

Theorem 5.1 (Extension to even shifts)**.** *Assume* (17) *and put* $d = \gcd(r, s)$*. If* r/d *is odd and* $13 \nmid r/d$*, then* $\xi(r3^k - s)$ *is transcendental.*

The odd–odd case is Theorem B of [7]. The added cases have r/d odd and s/d even; for example, $\xi(3^k - 2)$ is covered. The hypothesis $3 \nmid s$ has *not* been removed.

Theorem 5.2 (A uniform cutoff for the shift minus one)**.** *For every integer* $r \geq 12\,230\,000\,000\,000$*, the number* $\xi(r3^k - 1)$ *is transcendental.*

Theorem 5.2 includes even r and odd multiples of 13. Its proof uses a finite computation only after obtaining a global bound on *every* hypothetical algebraic minimum in the stated range. It does not infer a theorem for arbitrary r from a finite prime prefix or a bounded search for Pisot numbers.

Section 5.9 gives a further sufficient condition for general $-s$: with $R = r/d$, $S = s/d$ and the parity gcd $D \in \{1, 2\}$, transcendence holds if $(9R - S)/D \geq 2\,924\,000\,000\,000\,000$. In particular, the still-unresolved *normalized* parameter pairs form a finite set; this finiteness is not a proof that the set is empty.

The new structural result is Lemma 5.4. It gives $\mathrm{Tr}(\beta^{-s}) = \pm 1$ without an upper bound on β and without assuming that β is a unit or is non-totally-real. A positive-shift argument cannot simply be reused: at a negative fixed point the terms in the limiting identity are inverse powers of the conjugates. The proof below treats this sign directly.

5.2 Inputs and normalization

We use the following prior results, with their provenance made explicit. The initial reduction is Saito's [4, Proposition 3.1]; its specialization and the subsequent 3-adic arguments are from [7]. The numbering of Saito here refers to arXiv version 3. In particular, the general reduction is Proposition 3.1 in that version, rather than the Proposition 3.2 citation used in [7].

(I1) *Cubic Pisot shadow.* If $\xi = \xi(C)$ is algebraic, the Mills-shadow corollary in [7, Section 2] provides $g \mid D$, where $D = \lim_m \gcd(C_m, C_{m+1}, \ldots)$, and a cubic Pisot number $\beta = \xi^g$. For all sufficiently large k, putting $n_k = C_k/g$ gives

$$T_k = \mathrm{Tr}(\beta^{n_k}) = \left\lfloor \xi^{C_k} \right\rceil \text{ prime}, \qquad 0 \leq \beta^{n_k} - T_k \leq T_k^{-17/40}. \tag{18}$$

Here $D \mid 2s$, and $3 \nmid D$.

(I2) *Norm bounds.* The approximation-magnitude lemma in [7, Section 3.7] gives

$$|\,\mathrm{N}(\beta)| \leq \beta^{3/20}. \tag{19}$$

The negative-fixed-point norm-exclusion lemma in [7, Section 4, lemma titled $3 \nmid \mathrm{N}(\beta)$ at negative fixed points] gives $3 \nmid \mathrm{N}(\beta)$ when (18) holds and $g \leq 2s$. The magnitude lemma uses the Pisot power-field property and, for a nonreal conjugate pair, a lower bound for a nonzero linear form in logarithms.

(I3) *Actual twisted limit identities.* Write the 3-adic unit conjugates as $\beta_i = \omega_i u_i$, with $\omega_i \in \mu_{104}$ and u_i a principal unit. The affine-return lemma and the limit-identity proposition in [7, Sections 3.4–3.5] imply that, in each phase class of k,

$$\sum_i \omega_i^{n_k} u_i^{-s/g} = \varepsilon_k, \qquad \varepsilon_k \in \{1, -1\}. \tag{20}$$

When $3 \nmid \mathrm{N}(\beta)$, all three conjugates occur. The phases are those of the actual orbit; the coefficients are not freely chosen roots of unity.

(I4) *Odd primitive coefficients.* Theorem B of [7] settles coprime odd r, s with $13 \nmid r$. For coprime odd r, s with $13 \mid r$, the theorem in [7, Section 7.2], together with its immediately following general-s remark, implies that an algebraic minimum ξ itself is a cubic Pisot number, with nonreal conjugates, 3 inert in $\mathbb{Q}(\xi)$, and $\mathrm{Tr}(\xi^{-s}) = \pm 1$.

We use these as prior results; their full proofs are not replaced by the new argument in this note. In contrast, all the additional structural steps and the implications of the finite certificates are proved below.

Lemma 5.3 (Normalization)**.** *For $d \in \mathbb{N}$, one has $\xi(dC) = \xi(C)^{1/d}$. If $R = r/d$, $S = s/d$ with $d = \gcd(r, s)$, then the asymptotic gcd of $R3^k - S$ equals 1 when R, S have opposite parity, and equals 2 when both are odd.*

Proof. The first assertion follows from $\left\lfloor A^{dC_k} \right\rfloor = \left\lfloor (A^d)^{C_k} \right\rfloor$ and monotonicity of $A \mapsto A^d$. For the second, the asymptotic gcd divides $2S$ and every $R3^k - S$. No odd prime can divide it: such a prime would divide S and $R3^k$, and cannot be 3 because $3 \nmid S$. This contradicts $\gcd(R, S) = 1$. Opposite parity therefore gives gcd 1. If both are odd, all exponents are even, but they are not all divisible by 4: multiplication of the odd residue $R3^k$ by 3 interchanges its two residues modulo 4. Thus the gcd is exactly 2. □

5.3 Inverse trace rigidity at integral negative fixed points

Lemma 5.4 (Uniform inverse trace)**.** *Let β be a cubic Pisot number with $3 \nmid \mathrm{N}(\beta)$, and let $r, s \in \mathbb{N}$ with $3 \nmid s$. Suppose that $\mathrm{Tr}(\beta^{r3^k - s})$ is prime for every sufficiently large k. Then*

$$\mathrm{Tr}(\beta^{-s}) = \pm 1.$$

Moreover, in every phase class of the orbit, the three actual twists $\omega_i^{r3^k}$ are all one common sign ± 1.

Proof. Work inside a fixed algebraic closure of $\mathbb{Q}_3$ and identify the algebraic numbers appearing there with an algebraic closure of $\mathbb{Q}$. All three β_i are 3-adic units. Put

$$a_i = \beta_i^{-s}, \qquad q = a_1 + a_2 + a_3 \in \mathbb{Q}, \qquad F = \mathbb{Q}(\mu_{104}).$$

Although q need not initially be an integer, it is rational. The a_i are pairwise distinct. Indeed, for a Pisot number $\mathbb{Q}(\beta^s) = \mathbb{Q}(\beta)$: otherwise two embeddings over $\mathbb{Q}(\beta^s)$ would send β to conjugates of the same modulus $\beta > 1$, contrary to the Pisot property. The same degree statement holds for β^{-s}.

By input (I3), the identity at the integral fixed point $-s$ takes the form

$$\zeta_1 a_1 + \zeta_2 a_2 + \zeta_3 a_3 = \varepsilon, \qquad \zeta_i = \omega_i^{r3^k}, \qquad \varepsilon \in \{1, -1\}. \tag{21}$$

This follows directly from $\omega_i^{n_k} u_i^{-s} = \omega_i^{n_k + s} \beta_i^{-s}$. We prove the conclusion first in the generic Galois case, then treat the exceptional cubic field.

A cyclic normal-basis argument over F. Let M be the splitting field of β and put $E = MF$. Unless M is a cyclic cubic subfield of F, the group $\mathrm{Gal}(E/F)$ contains an element τ cyclically permuting the β_i. If $\mathrm{Gal}(M/\mathbb{Q}) = S_3$, the intersection $M \cap F$ is abelian and Galois over $\mathbb{Q}$, hence is at most quadratic; its fixing subgroup contains A_3. If $\mathrm{Gal}(M/\mathbb{Q}) = C_3$, the intersection is $\mathbb{Q}$ unless $M \subseteq F$. These assertions also show that the required three-cycle extends while fixing F.

Apply $1, \tau, \tau^2$ to (21) and add. Since τ fixes the ζ_i, this gives

$$(\zeta_1 + \zeta_2 + \zeta_3)q = 3\varepsilon, \tag{22}$$

so $q \neq 0$. Consider the relation space

$$\mathcal{R} = \{(v_1, v_2, v_3) \in F^3 : \textstyle\sum_i v_i a_i = 0\}.$$

It is stable under cyclic permutation. The field F contains no primitive cube root of unity; for example, 3 is unramified in $\mathbb{Q}(\mu_{104})$, whereas it ramifies in $\mathbb{Q}(\mu_3)$. Consequently $X^2 + X + 1$ is irreducible over F. The permutation representation decomposes as

$$F^3 = U \oplus V, \qquad U = F(1,1,1), \qquad V = \{(v_i) : \textstyle\sum_i v_i = 0\},$$

where U and V are nonisomorphic simple modules for the three-cycle. Equivalently, the projections onto these summands are polynomials in that cycle and its restriction to V has irreducible minimal polynomial $X^2 + X + 1$. Every invariant subspace is therefore one of $0, U, V, F^3$. Equation (22) excludes $U \subseteq \mathcal{R}$, and $a_1 \neq a_2$ excludes $V \subseteq \mathcal{R}$. Hence $\mathcal{R} = 0$.

Subtracting $(\varepsilon/q)(a_1 + a_2 + a_3) = \varepsilon$ from (21) now gives $\zeta_i = \varepsilon/q$ for every i. This common value is a rational root of unity, hence is 1 or -1. Thus $q = \pm 1$, with the asserted common signed twists.

The exceptional cyclic field of conductor 13. The group

$$\mathrm{Gal}(F/\mathbb{Q}) \simeq (\mathbb{Z}/104\mathbb{Z})^\times \simeq C_2 \times C_2 \times C_{12}$$

has a unique index-3 subgroup. Its cubic fixed field K is the real cubic subfield of $\mathbb{Q}(\mu_{13})$, fixed by $H = \{1, 5, 8, 12\} \subseteq (\mathbb{Z}/13\mathbb{Z})^\times$. The class of 3 has order 3 and is not in H, so 3 is unramified and inert in K.

We must allow the possibility that the integral generator β has nonmaximal index at 3. Its residue in $\mathbb{F}_{27}$ either belongs to $\mathbb{F}_3$ or generates $\mathbb{F}_{27}$. In the former case all three conjugates have the same nonzero residue e, and $\mathrm{Tr}(\beta^n) \equiv 3e^n = 0 \pmod 3$ for every positive n. Since these traces tend to infinity, this contradicts eventual primality. Thus the residue generates $\mathbb{F}_{27}$, and, after labeling,

$$\omega_i = \omega^{3^{i-1}}, \qquad \mathrm{ord}(\omega) \in \{13, 26\}.$$

If $13 \mid r$, each $\omega_i^{r3^k}$ is the same sign, and (21) immediately proves the conclusion.

Suppose instead that $13 \nmid r$. Put $\eta = \omega^{r3^k}$. The three successive phase identities have cyclically permuted coefficient rows, because $\eta^{27} = \eta$. Adding them yields

$$(\eta + \eta^3 + \eta^9)q = \varepsilon_0 + \varepsilon_1 + \varepsilon_2. \tag{23}$$

The right side is an odd, nonzero integer. The Gaussian period on the left is irrational. For η of order 13, a rational value would give a polynomial $X^9 + X^3 + X - c$ over $\mathbb{Q}$ annihilating η, impossible because its degree is 12. For order 26, replace η by $-\eta$, of order 13; all three exponents are odd, so the same argument applies with a common minus sign. Equation (23) first implies $q \neq 0$ and then contradicts $q \in \mathbb{Q}$. This exhausts the exceptional case and proves the lemma. □

5.4 The extended negative shift theorem

Proof of Theorem 5.1. By Lemma 5.3, it is enough to take $\gcd(r,s)=1$ with r odd. For odd s, apply input (I4). Suppose s is even and that the minimum is algebraic. Lemma 5.3 gives $D=1$, so input (I1) provides $\beta=\xi$ itself as a cubic Pisot number. Input (I2) gives $3\nmid \mathrm{N}(\beta)$, and Lemma 5.4 shows that the three twists $\omega_i^{r3^k}$ are one common sign. Since $r3^k$ is odd, inspect the factorization of the minimal polynomial modulo 3.

(i) If it has three linear factors, allowing repetitions, every ω_i is a sign. Equality of their odd powers makes all three signs equal. Every trace is then 0 modulo 3, impossible.

(ii) If it has a linear factor and an irreducible quadratic factor, the latter's Teichmüller roots are ω,ω^3, of order 4 or 8. Equality of their $r3^k$-th powers requires $\omega^{2r3^k}=1$, impossible for odd $r3^k$.

(iii) If it is irreducible, its roots have Teichmüller lifts ω,ω^3,ω^9, of order 13 or 26. Equality requires $13\mid r$, contrary to the hypothesis.

These are all cubic factorization types with nonzero constant term over $\mathbb{F}_3$. The contradiction proves the theorem. □

Corollary 5.5 (A uniform necessary algebraic form)**.** *Assume* (17) *and write* $d=\gcd(r,s)$, $R=r/d$, $S=s/d$. *If* $\xi(r3^k-s)$ *is algebraic, then* $\alpha=\xi(r3^k-s)^d=\xi(R3^k-S)$ *itself is a cubic Pisot number and*

$$3\nmid \mathrm{N}(\alpha),\qquad \mathrm{Tr}(\alpha^{-S})=\pm 1,\qquad |\mathrm{N}(\alpha)|\le \alpha^{3/20}. \tag{24}$$

If R *is odd, then* $13\mid R$, *the prime* 3 *is inert in* $\mathbb{Q}(\alpha)$, *and the two small conjugates are nonreal.*

Proof. For opposite parity, the asymptotic gcd is 1, so the assertions follow from inputs (I1)–(I3) and Lemma 5.4. If R is odd and S even, the preceding modulo-3 classification forces an irreducible reduction and $13\mid R$. Irreducibility modulo 3 gives inertness. Total reality would make each small conjugate's inverse even power greater than 1, so $\mathrm{Tr}(\alpha^{-S})>2$, a contradiction.

If R,S are both odd, input (I4) gives the conclusion except possibly the displayed norm bound. Its degree descent uses α^S and α^2 in the same cubic field and $\gcd(S,2)=1$; thus it is a statement about α, not merely α^2. Apply (19) to $\beta=\alpha^2$ and take square roots to obtain the bound for α. □

In particular, for $S=1$, a hypothetical algebraic minimum has minimal polynomial

$$f(X)=X^3-aX^2+bX-c,\qquad b/c=\mathrm{Tr}(\alpha^{-1})=\pm 1. \tag{25}$$

The condition $b=\pm c$ is uniform, not an extrapolation from the $\beta<9$ table in [7].

5.5 Finite ring certificates and the global cutoff

5.5.1 The certificate implication

For integers A,B put

$$F_{A,B}(X)=X^3-AX^2+BX-1,\qquad M_{A,B}=\begin{pmatrix}0&0&1\\1&0&-B\\0&1&A\end{pmatrix}.$$

This is the matrix of multiplication by X in the basis $1, X, X^2$. Its power traces are the Newton traces of $F_{A,B}$.

Lemma 5.6 (A modular certificate)**.** *Suppose p, q are primes, d is a nonnegative integral power of 3, $t = dq$, and $n \geq 0$ satisfy*

$$\operatorname{ord}_q(3) = q - 1, \qquad M_{A,B}^t = I \pmod{p}, \tag{26}$$

$$\operatorname{Tr}(M_{A,B}^n) = 0 \pmod{p}, \qquad d \mid n+1, \qquad q \nmid n+1. \tag{27}$$

If the absolute values of the traces at exponents $r3^k - 1$ tend to infinity and are eventually prime, then $q \mid r$.

Proof. Assume $q \nmid r$. Since 3 generates $(\mathbb{Z}/q\mathbb{Z})^\times$, there are infinitely many k, in one residue class modulo $q-1$, with $r3^k \equiv n+1 \pmod{q}$. For sufficiently large such k, $r3^k \equiv 0 \equiv n+1 \pmod{d}$ as well. As $q \neq 3$, the Chinese remainder theorem gives $r3^k - 1 \equiv n \pmod{t}$. Equations (26)–(27) make the corresponding trace divisible by p. Its absolute value eventually exceeds p, contradicting primality. □

Proposition 5.7 (Exact finite certificate)**.** *Set $R_0 = 400\,300\,000\,000\,000$. For every integer pair*

$$-102 \leq A \leq 102, \qquad B \in \{-1, 1\},$$

other than $(-3, 1)$, $(-1, -1)$ and $(1, 1)$, the supplementary certificate file contains tuples satisfying (26)–(27) whose q-values have least common multiple greater than R_0. Thus no irreducible polynomial in this range can have eventually prime absolute traces along $r3^k - 1$ with $1 \leq r \leq R_0$ and traces tending in absolute value to infinity.

Exact verification and deduction. There are 410 pairs and 2 202 recorded tuples. The independent verifier described in Section 5.7 checks every condition by integer arithmetic and modular matrix multiplication. Exactly 407 rows have the required least common multiple; the smallest of these least common multiples is $404\,048\,787\,244\,321 > R_0$. The three exceptions factor as

$$\begin{aligned} F_{-3,1} &= (X+1)(X^2+2X-1), \\ F_{-1,-1} &= (X-1)(X+1)^2, \\ F_{1,1} &= (X-1)(X^2+1). \end{aligned}$$

They cannot be minimal polynomials of cubic numbers. For every other row, Lemma 5.6 forces each certified q to divide r, hence forces their least common multiple to divide r. This is impossible for $1 \leq r \leq R_0$. □

5.5.2 Why the finite certificate covers all large parameters

Put

$$E = e^{32.76}, \qquad x_0 = e^{3E}, \qquad M = 2e^E + 1, \qquad r_* = 12\,230\,000\,000\,000. \tag{28}$$

Saito's Type C theorem gives the following bound for a hypothetical algebraic minimum:

$$\xi(r3^k - 1) \leq M^{1/(3r-1)}. \tag{29}$$

Indeed, [4, Theorem 2.6 and its application in Section 2.3] gives $\xi^g \leq (2x_0^{1/c_2} + 1)^{g/c_1}$, with $c_1 = 3r - 1$ and $c_2 = (9r-1)/(3r-1) > 3$; taking the g-th root gives (29). This is a global prime-chain bound, not a finite-prefix estimate.

Proof of Theorem 5.2. For $r \geq R_0$, Saito's established large-r result already applies [4, Theorem 1.9(C)]. Let $r_* \leq r < R_0$, and suppose that $\alpha = \xi(r3^k - 1)$ is algebraic. Corollary 5.5 makes α itself cubic Pisot with (24). The rational estimates in Section 5.8 prove

$$(3r_* - 1)\log 100 > \log M.$$

Therefore (29) gives $\alpha < 100$. Its norm is a nonzero integer and

$$|\,\mathrm{N}(\alpha)| \leq \alpha^{3/20} < 100^{3/20} < 2, \qquad \text{since } 100^3 < 2^{20}.$$

Thus $c = \pm 1$ in (25), and $b = \pm 1$. If x_2, x_3 are the small conjugates, then $-1 < \alpha + x_2 + x_3 < 102$, so the integral coefficient satisfies $0 \leq a \leq 101$.

For $c = 1$, the polynomial already has the form $F_{a,b}$. For $c = -1$, put $\gamma = -\alpha$; its polynomial is $F_{-a,b}$, and

$$\mathrm{Tr}(\gamma^n) = (-1)^n \,\mathrm{Tr}(\alpha^n).$$

Consequently all zero-trace certificates transfer, regardless of exponent parity. Both signs are covered by Proposition 5.7. The traces of α^{r3^k-1} are the eventual prime floors from input (I1); for the odd–odd descent the field is unchanged, so these traces are unchanged as well. They tend to infinity. The finite certificate now contradicts $r < R_0$. □

The unrounded sufficient threshold in this argument is

$$\frac{\log M}{3\log 100} + \frac{1}{3}.$$

The stated integer cutoff is a conservative rationally certified rounding, not an optimized endpoint.

5.6 What remains open

Neither theorem settles all parameters in (17). For $s = 1$, the cases not settled by the present conclusions can still include even $r < r_*$ and odd multiples of 13 below r_*. This section alone does not settle $\xi(2 \cdot 3^k - 1)$ or $\xi(13 \cdot 3^k - 1)$; the latter is settled in Section 6, while the former remains open. The next proposition explains one limitation of simply extending a fixed modular sieve.

Let α_a be the dominant root of $F_a = X^3 - aX^2 - X - 1$ for $a \geq 1$. These are cubic Pisot units with nonreal conjugates and $\mathrm{Tr}(\alpha_a^{-1}) = -1$; see [7, Section 7.1]. Put $T_n(a) = \mathrm{Tr}(\alpha_a^n)$. The residual families in [7, Section 7] use $a \equiv 2 \pmod 3$ for even r, and $a \equiv 1 \pmod 3$ for odd r divisible by 13. They satisfy the cited 3-adic necessary conditions; this does not assert prime traces or the full shadow condition.

Proposition 5.8 (A finite fixed-prime sieve cannot cover the residual family)**.** *Fix an even r, or an odd r divisible by 13. For every finite set of primes $\mathcal{P}$, infinitely many parameters a in the corresponding residual family have*

$$p \nmid T_{r3^k-1}(a) \qquad \textit{for all } p \in \mathcal{P} \textit{ and all } k \geq 1.$$

Proof. The exact specialization $F_{-1} = (X-1)(X+1)^2$ gives, by Newton traces,

$$T_n(a) \equiv 1 + 2(-1)^n \pmod{a+1}. \tag{30}$$

For even r, choose infinitely many positive $a \equiv -1 \pmod{3\prod_{p\in\mathcal{P}} p}$. Then $a \equiv 2 \pmod 3$ and $r3^k - 1$ is odd, so every tested trace is -1 modulo every tested prime.

For odd $13 \mid r$, use the Chinese remainder theorem to choose $a \equiv 1 \pmod 3$ and $a \equiv -1 \pmod p$ for each $p \in \mathcal{P} \setminus \{3\}$. The exponents are even, so the traces are 3 modulo those primes. Modulo 3 the polynomial is $X^3 - X^2 - X - 1$, irreducible, and $X^{13} = 1$ in its residue field. Hence $13 \mid r$ gives $T_{r3^k-1}(a) \equiv T_{-1}(a) = -1 \pmod 3$ as well. All required divisibilities fail. □

This is a limitation of finite *fixed-prime divisibility covers*, not a counterexample to transcendence. Parameter-dependent primes, finer use of the real shadow, or additional global bounds remain possible. The finite-ring certificate above is compatible with this barrier because it is applied only after bounding the candidate coefficients.

5.7 Reproducible finite verification

The proof-critical supplementary files are

- `finite_modular_scan_signed102.json`, containing all 410 rows and 2 202 tuples;
- `bounds_verify_certificates.py`, an independent verifier;
- `bounds_roundoff.py`, containing the rational numerical checks.
- `cutoff_audit_checks.py`, checking completeness of the signed grid and the small Pisot gap used in Section 5.9.

For each row, the verifier performs the following exhaustive operations. It tests p and q by trial division; checks $t = dq$ and that repeated division of d by 3 ends at 1; factors $q - 1$ and checks

$$3^{q-1} \equiv 1 \pmod q, \qquad 3^{(q-1)/\ell} \not\equiv 1 \pmod q \quad (\ell \text{ prime},\ \ell \mid q-1);$$

and uses binary matrix powering modulo p to check $M_{A,B}^t = I$ and $\operatorname{Tr}(M_{A,B}^n) = 0$. It then checks the two conditions on $n + 1$, recomputes the least common multiple of the q-values, and verifies that the only failed rows are the three explicitly reducible rows of Proposition 5.7. The additional check verifies that the row keys are exactly the full 410-element signed grid, with no duplicates or omissions. No floating-point root approximation or probable-prime test is used. The matrix implementation is separate from the generating program's quotient-ring representation.

The SHA-256 digest of the proof-critical JSON data is

```
c97b6693c8057e73c90641f5f30d6ad0
2167f3e83b82b4c428c342866f1840a1.
```

The line break and final punctuation are not part of the digest. The verifier's terminal certificate is

```
INDEPENDENT MATRIX CHECK PASSED 410 rows 2202 certificates;
only reducible failures [(-3, 1), (-1, -1), (1, 1)].
```

These are exact computational checks, not a claim of external peer review.

As a nonessential cross-check, a separate exact Sturm enumeration of a safe superset of cubic Pisot units below 100 produced 10 277 candidates. Necessary twisted-limit tests at precisions 3^{12} and 3^{22} left the same 394 odd-residue candidates, all with $|b| = 1$, and all mapped to certified signed rows. The proof above does not depend on inferring any exact limit identity from such finite-precision agreement.

5.8 Rational certification of the cutoff

Set $x = 819/25$ and $N = 250$. The Taylor sum with geometric tail bound

$$e^x \le \sum_{j=0}^{N} \frac{x^j}{j!} + \frac{x^{N+1}}{(N+1)!}\frac{1}{1 - x/(N+2)} < 168\,845\,000\,000\,000$$

is verified over $\mathbb{Q}$. For $0 < z < 1$, the positive expansion

$$\log\frac{1+z}{1-z} = 2\sum_{j=0}^{\infty}\frac{z^{2j+1}}{2j+1}$$

gives lower bounds by truncation. Taking 20 terms at $z = 1/3$ and $z = 1/9$, and using $\log 100 = 6\log 2 + 2\log(5/4)$, proves $\log 100 > 460517/100000 = 4.60517$. Since $E = e^{32.76} > 2$, one has $2 + e^{-E} < e$, and therefore $\log M = E + \log(2 + e^{-E}) < E + 1$. Finally the exact positive difference is

$$(3r_* - 1)\frac{460517}{100000} - 168\,845\,000\,000\,001 = \frac{11\,868\,729\,999\,439\,483}{100000} > 0.$$

Combining these strict inequalities gives $(3r_* - 1)\log 100 > \log M$, as required.

5.9 A sufficient bound for general negative shifts

Theorem 5.9. *Let $r, s \in \mathbb{N}$, $3 \nmid s$, and $s < 3r$. Put $d = \gcd(r, s)$, $R = r/d$, $S = s/d$, and*

$$D = \begin{cases} 2, & R, S \text{ both odd}, \\ 1, & R, S \text{ of opposite parity}. \end{cases}$$

If

$$\frac{9R - S}{D} \ge Q_0 := 2\,924\,000\,000\,000\,000, \tag{31}$$

then $\xi(r3^k - s)$ is transcendental. In particular, all normalized $R \ge 975\,000\,000\,000\,000$ are covered; for opposite parity, $R \ge 487\,500\,000\,000\,000$ suffices.

Proof. By normalization, take the coprime schedule $R3^k - S$ and write $A = \xi(R3^k - S)$. Set

$$H = 3R - S > 0, \quad J = 9R - S, \quad N = H/D \in \mathbb{N}, \quad c_2 = J/H, \quad L = \log x_0 = 3e^{32.76}.$$

Suppose A is algebraic. The reduction gives a cubic Pisot number A^g with $g \mid D$, so A^D and A^H are cubic Pisot numbers by the power-field property. This observation does not require the degree descent in Corollary 5.5.

The exact rational checks give

$$Q_0\frac{693147}{1000000} > 12 \cdot 168\,845\,000\,000\,000, \qquad \log 2 > \frac{693147}{1000000}.$$

Together with Section 5.8, condition (31) therefore implies

$$\frac{J}{D} > \frac{4L}{\log 2}, \qquad x_0^{1/c_2} = \exp(LH/J) < 2^{N/4}.$$

Let κ be the smallest Pisot number, the real root of $X^3 - X - 1$ [5]; one has $\kappa > k_0 := 331/250$.

For $N = 4, 5, 6, 7, 8, 9, 10$, respectively choose the prime

$$p = 2, 3, 3, 5, 5, 5, 7.$$

Exact rational inequalities give $p^4 \geq 2^N$ and $p + 1 < k_0^N < \kappa^N$. For $N \geq 11$, Bertrand's postulate gives $2^{N/4} < p < 2 \cdot 2^{N/4}$. The rational inequalities

$$3^4 2^{11} < k_0^{44}, \qquad 2 < k_0^4$$

imply $p + 1 < 3 \cdot 2^{N/4} < \kappa^N$ for every such N. In either case $p > x_0^{1/c_2}$. Saito's nested prime-chain construction [4, Lemma 9.1] produces a prime-representing number with initial prime p, so minimality gives

$$A^D \leq (p + 1)^{D/H} = (p + 1)^{1/N} < \kappa,$$

contradicting the Pisot property.

It remains to handle $1 \leq N \leq 3$, where a crude initial-prime bound would not suffice. Here $p = 2 > x_0^{1/c_2}$ and $c_2 > 100$. Choose the first prime child q using the explicit prime interval:

$$2^{c_2} \leq q \leq 2^{c_2} + 3 \cdot 2^{2c_2/3}.$$

Continue this same child by the nested construction, rather than replacing the original problem by an unrelated tail. It gives

$$2 \leq A^H \leq (q + 1)^{1/c_2} < \frac{21}{10}.$$

For the last strict inequality, for every $c \geq 100$,

$$1 + 3 \cdot 2^{-c/3} + 2^{-c} < 2 < (21/20)^c.$$

But there is no cubic Pisot number in $[2, 21/10]$. To verify this small gap exhaustively, the root bounds leave only

$$f = X^3 - aX^2 + bX - c, \qquad 0 \leq a \leq 4, \quad |b| \leq 5, \quad c \in \{-2, -1, 1, 2\}.$$

The supplementary rational Sturm verifier checks all 220 triples. It first rejects rational roots (possible integral roots divide c), then counts roots in $(2, 21/10)$ and $(-1, 1)$. For negative discriminant, a root $\beta \in (2, 21/10)$ can be Pisot only for $c > 0$, and then its conjugate-pair modulus is $\sqrt{c/\beta} < 1$; for positive discriminant it must have two roots in $(-1, 1)$. No irreducible polynomial passes either test. The endpoints cannot be roots of an irreducible monic cubic. An independent Fraction-only implementation and a separate symbolic Sturm implementation both verify the empty list. This contradicts the fact that A^H is cubic Pisot.

Finally $J = 9R - S > 6R$ and $D \leq 2$, giving the stated simpler bounds on R by direct comparison with Q_0. □

Since $1 \leq S < 3R$, a bound on the remaining normalized R leaves only finitely many coprime schedule pairs (R, S) within the stated scope. This is not a finite enumeration of all cubic Pisot candidates. Arbitrarily large common scalings d are still possible, but scaling preserves algebraicity. The theorem does not enumerate or resolve that finite residual set.

6 The multiplier thirteen and a residual branch exclusion

This section closes the multiplier-thirteen case left by the basic negative theorem. Its final branch exclusion is only one part of the even-multiplier residual problem, not a proof for all candidates.

6.1 Main results and their scope

For a sequence $C = (C_k)_{k\ge 1}$ of positive integers, put

$$\mathcal{W}(C) = \{A > 1 : \ \left\lfloor A^{C_k} \right\rfloor \text{ is prime for every } k \ge 1\}.$$

If $C_{k+1} \ge 2C_k$, this set is nonempty and has a least element [4, Theorem 1.3]; denote it by $\xi(C)$. Our principal schedule is

$$n_k = 13 \cdot 3^k - 1 \qquad (k \ge 1), \qquad n_1 = 38, \quad n_2 = 116. \tag{32}$$

Its consecutive ratios tend to 3 from above.

Theorem 6.1. *The number $\xi((13 \cdot 3^k - 1)_{k\ge 1})$ is transcendental.*

The proof uses the structural theorem for odd multiples of 13 in [7, Section 7.2, Theorem `D13`], whose precise consequences are recorded in Section 6.2. That theorem is a result of the author's earlier, unpublished manuscript; it is an explicit dependency, not an assertion proved anew in this paper. The underlying algebraic-minimum reduction is Saito's [4, Proposition 3.1]. Our citation numbers for Saito refer to arXiv version 3; the corresponding reference in [064] is numbered Proposition 3.2 there.

The added steps are an analytic bound from an initial prime floor and a complete modular exclusion. The analytic bound applies to every hypothetical algebraic minimum, not merely to a numerical search range. The modular exclusion proves infinitely recurring divisibility, not merely compositeness in a finite initial segment. No prime-gap conjecture is assumed. The additional finite computation is independent of numerical root approximations and probabilistic primality tests.

We also obtain the following result, which does not require the structural theorem for multiples of 13.

Theorem 6.2. *Let $\beta > 1$ be a cubic Pisot number with minimal polynomial*

$$f(X) = X^3 - aX^2 + cX - c, \qquad a, c \in \mathbb{Z}, \quad c > 0.$$

If $c \le \beta^{3/20}$, then $\lfloor \beta^5 \rfloor$ is composite.

The exponent 5 is the first exponent of $2 \cdot 3^k - 1$. Thus Theorem 6.2 excludes the entire positive-norm branch $b = c$ in the algebraic-minimum problem for this schedule, including nonunits. Section 6.6 specifies the remaining branches. We make no claim here that $\xi(2 \cdot 3^k - 1)$, the classical Mills constant, or all negative-shift minima have been resolved.

6.2 The structural reduction and its residue-field consequence

A Pisot number is a real algebraic integer $\beta > 1$ all of whose other conjugates have modulus strictly less than one. Write its cubic minimal polynomial and its power traces as

$$f(X) = X^3 - aX^2 + bX - c, \qquad T_n = \mathrm{Tr}_{\mathbb{Q}(\beta)/\mathbb{Q}}(\beta^n). \tag{33}$$

Newton's identities give

$$\begin{aligned} T_0 = 3, \qquad T_1 = a, \qquad T_2 = a^2 - 2b, \\ T_n = aT_{n-1} - bT_{n-2} + cT_{n-3} \qquad (n \ge 3). \end{aligned} \tag{34}$$

Proposition 6.3 (Structural input from [7]). *If $\xi = \xi((n_k)_{k\geq 1})$ is algebraic, then $\beta = \xi$ itself is a cubic Pisot number with nonreal conjugates $u, v = \overline{u}$. For its minimal polynomial (33), one has*

$$b = \varepsilon c, \qquad \varepsilon \in \{-1, 1\}, \qquad 0 < c \leq \beta^{3/20}, \qquad 3 \nmid c. \tag{35}$$

The polynomial f is irreducible modulo 3, and

$$T_{n_k} = \lfloor \beta^{n_k} \rfloor \text{ is prime for every sufficiently large } k. \tag{36}$$

Independently of the eventual identity, the definition of ξ gives

$$\lfloor \beta^{n_k} \rfloor \text{ is prime for every } k \geq 1. \tag{37}$$

Explanation of the cited reduction and the mod 3 bridge. The Mills-shadow corollary in [7, Section 2], based on Saito's [4, Proposition 3.1], first produces the cubic Pisot number $B = \xi^2$, with eventual prime traces at exponents $n_k/2$. The theorem for odd multiples of 13 in [7, Section 7.2] proves that ξ itself is cubic Pisot, with nonreal conjugates, and that

$$\mathbb{Q}(\xi) = \mathbb{Q}(B), \qquad \mathrm{Tr}(\xi^{-1}) = \pm 1, \qquad \xi \geq |\,\mathrm{N}(\xi)|^{20/3}.$$

Since $\mathrm{Tr}(\xi^{-1}) = b/c$ and $c = \beta u \overline{u} > 0$, these statements give the coefficient and norm assertions in (35). Equality of the fields gives

$$\mathrm{Tr}_{\mathbb{Q}(B)/\mathbb{Q}}(B^{n_k/2}) = \mathrm{Tr}_{\mathbb{Q}(\xi)/\mathbb{Q}}(\xi^{n_k}),$$

so the eventual prime-trace identity transfers to (36) with $\beta = \xi$, not merely with $\beta = \xi^2$.

There is one residue-field detail that is essential for the finite coefficient cover. In the proof of the cited theorem, the minimal polynomial of $B = \xi^2$ is shown to be irreducible modulo 3. Consequently 3 is inert in $K = \mathbb{Q}(B) = \mathbb{Q}(\xi)$, its residue field is $\mathbb{F}_{27}$, and the residue $\overline{B}$ generates $\mathbb{F}_{27}$ over $\mathbb{F}_3$. The algebraic integer ξ has a residue $\overline{\xi}$ in this same field and $\overline{\xi}^2 = \overline{B}$. Hence

$$\mathbb{F}_3(\overline{B}) \subseteq \mathbb{F}_3(\overline{\xi}) \subseteq \mathbb{F}_{27},$$

and both inclusions are equalities. The reduction of the monic cubic minimal polynomial of ξ annihilates an element of residue degree 3, so it is irreducible. In particular its constant coefficient is nonzero modulo 3, giving $3 \nmid c$. This uses the stronger square-generator statement from the cited proof: inertness alone would not justify irreducibility modulo 3 for an arbitrary integral generator. □

Theorem 6.1 uses only this input from [064] and Saito, together with the arguments below. In particular it does not depend on the more general inverse-trace theorem in [8], which is used only to describe the $r = 2$ application in Section 6.6.

6.3 Parity and the second prime floor

We first isolate the parity obstruction. It will apply to both schedules used below.

Lemma 6.4. *Let β be a cubic Pisot number as in (33), with $b \equiv c \pmod 2$. Suppose that, for $r \in \{2, 13, 39\}$, the traces T_{r3^k-1} are prime for all sufficiently large k. Then a is odd, and T_n is odd for every $n \geq 0$.*

Proof. The Pisot property gives $T_n = \beta^n + o(1)$, so these traces tend to infinity. If a and c are even, then b is even, and (34) makes every T_n, $n \geq 1$, even. This contradicts eventual primality.

If a is even and c is odd, then b is odd, and the polynomial modulo 2 is $X^3 + X + 1$. It is irreducible over $\mathbb{F}_2$, because it has no root there. Its nonzero roots belong to $\mathbb{F}_8^\times$, a group of order 7. Thus its power traces have period dividing 7; the residues for $n = 0, \dots, 6$ are

$$(T_n \bmod 2)_{n=0}^{6} = (1, 0, 0, 1, 0, 1, 1).$$

For $r = 13$, $r3^k - 1 \equiv 4 \pmod 7$ whenever $k \equiv 2 \pmod 6$. For $r = 2$, the same residue occurs whenever $k \equiv 3 \pmod 6$, and for $r = 39$ it occurs whenever $k \equiv 1 \pmod 6$. In every case infinitely many arbitrarily large required traces are even, again impossible.

Therefore a is odd. The initial values T_0, T_1, T_2 are all odd. Since $a - b + c \equiv 1 \pmod 2$, recurrence (34) now proves by induction that every trace is odd. □

Suppose henceforth that the hypothetical algebraic minimum of Proposition 6.3 exists. Put

$$q = uv = \frac{c}{\beta}, \qquad s = u + v = a - \beta. \tag{38}$$

Vieta's identity $b = \beta s + q$, together with $b = \varepsilon c$, gives

$$s = q\left(\varepsilon - \frac{1}{\beta}\right), \qquad 0 < q \le \beta^{-17/20} < 1. \tag{39}$$

Choose $\theta \in (0, \pi)$ so that $u = \sqrt{q}\, e^{i\theta}$ and $v = \sqrt{q}\, e^{-i\theta}$. Then

$$\cos\theta = \frac{\sqrt{q}}{2}\left(\varepsilon - \frac{1}{\beta}\right), \qquad S_n := u^n + v^n = 2q^{n/2}\cos(n\theta), \qquad \beta^n = T_n - S_n. \tag{40}$$

Proposition 6.5. *The algebraic minimum in Proposition 6.3 would satisfy*

$$\beta < 8192.$$

Proof. Suppose instead that $\beta \ge 8192 = 2^{13}$. The norm bound gives

$$0 < q \le \beta^{-17/20} \le 2^{-221/20} < \frac{1}{2048}.$$

As $45^2 = 2025 < 2048$, this implies $\sqrt{q} < 1/45$ and hence

$$t := |\cos\theta| \le \frac{\sqrt{q}}{2}\left(1 + \frac{1}{\beta}\right) < \frac{1 + 1/8192}{90} < \frac{1}{89}.$$

Set $\delta = |\theta - \pi/2| = \arcsin t$. By integrating the increasing function $(1 - x^2)^{-1/2}$ on $[0, t]$, we obtain

$$\delta \le \frac{t}{\sqrt{1 - t^2}} < \frac{1}{88},$$

where $\sqrt{1 - 1/89^2} > 88/89$ was used. Consequently

$$116\delta < \frac{116}{88} = \frac{29}{22} < \frac{3}{2} < \frac{\pi}{2}.$$

The exponent 116 is divisible by 4, so $\cos(116\theta) = \cos(116\delta) > 0$. Formula (40) therefore gives the exact inequalities

$$0 < S_{116} = 2q^{58}\cos(116\delta) < 2\left(\frac{1}{2048}\right)^{58} < 1.$$

Since T_{116} is an integer,

$$\left\lfloor \beta^{116} \right\rfloor = T_{116} - 1. \tag{41}$$

Lemma 6.4, applied to the eventual prime traces from (36), makes this floor even. It exceeds 2, since $\beta \ge 8192$. But $116 = n_2$ is an actual required prime exponent, contradicting (37). □

Remark 6.6. The proof does not assume the eventual identity $\lfloor \beta^{n_k} \rfloor = T_{n_k}$ at $k = 2$. Instead, it proves the opposite rounding identity (41) directly under the hypothesis $\beta \geq 8192$. Eventual prime traces justify the parity lemma; the definition of the minimum justifies primality of this particular early floor. These are different uses of the reduction.

Corollary 6.7. *The minimal polynomial of the hypothetical algebraic minimum belongs to the finite set*

$$\mathcal{G} = \left\{ X^3 - aX^2 + bX - c : \begin{array}{l} 1 \leq a \leq 8191, \quad a \text{ odd}, \\ c \in \{1, 2\}, \quad b \in \{-c, c\}, \\ f \bmod 3 \text{ irreducible} \end{array} \right\}. \tag{42}$$

This set contains exactly 8192 *polynomials.*

Proof. From Proposition 6.5,

$$1 \leq c \leq \beta^{3/20} < 8192^{3/20} = 2^{39/20} < 4.$$

Since $3 \nmid c$, necessarily $c = 1$ or $c = 2$. If $b = -c$, then (39) gives $s < 0$ and $a < \beta < 8192$. If $b = c$, it gives $0 < s = q(1 - 1/\beta) < 1$, so $a < \beta + 1 < 8193$. In both cases oddness forces $a \leq 8191$. Moreover $|s| < 2$, because both small conjugates have modulus less than one. Hence $a = \beta + s > \beta - 2 > -1$. As an odd integer, $a \geq 1$. Irreducibility modulo 3 was established in Proposition 6.3.

For a cubic over a field, reducibility is equivalent to having a root. Testing $0, 1, 2$ over $\mathbb{F}_3$ gives the residue classes and counts in Table 6. Their sum is 8192. □

c	b	Allowed $a \pmod 3$	Odd a in $[1, 8191]$
1	-1	$0, 1$	2731
1	1	2	1365
2	-2	1	1366
2	2	$0, 2$	2730
		Total	8192

Table 6: The complete coefficient set after irreducibility modulo 3.

No further root test, norm cutoff, or test of the Pisot property is used to shrink this grid. Eliminating this deliberately larger set suffices.

6.4 Finite certificates for infinitely recurring divisors

For any monic cubic $f = X^3 - aX^2 + bX - c$ with integer coefficients, let $T_n(f)$ denote the integer sequence defined by (34). It agrees with the sum of the nth powers of the three roots, counted with multiplicity. Over $\mathbb{F}_p$, put

$$R_p = \mathbb{F}_p[X]/(f), \qquad x = X \bmod f.$$

The matrix for multiplication by x in the ordered basis $1, x, x^2$ is

$$M_f = \begin{pmatrix} 0 & 0 & c \\ 1 & 0 & -b \\ 0 & 1 & a \end{pmatrix} \pmod p, \qquad \operatorname{tr}(M_f^n) \equiv T_n(f) \pmod p. \tag{43}$$

The trace identity follows, for example, from triangularizing the matrix over an algebraic closure, with algebraic multiplicities. It does not require $f \bmod p$ to be squarefree.

Lemma 6.8 (Return certificate)**.** *Let p be a prime. Suppose there are positive integers E, U, k, h and an integer $v \geq 0$ such that*

$$\begin{gathered} E = 3^v U, \quad \gcd(U, 3) = 1, \quad k \geq v, \quad k \geq 1, \\ M_f^E = I \pmod{p}, \qquad 3^h \equiv 1 \pmod{U}, \\ n \equiv 13 \cdot 3^k - 1 \pmod{E}, \qquad \operatorname{tr}(M_f^n) \equiv 0 \pmod{p}, \quad 0 \leq n < E. \end{gathered} \tag{44}$$

Then

$$p \mid T_{13 \cdot 3^{k+jh}-1}(f) \qquad \text{for every } j \geq 0. \tag{45}$$

Proof. For every $j \geq 0$,

$$(13 \cdot 3^{k+jh} - 1) - (13 \cdot 3^k - 1) = 13 \cdot 3^k (3^{jh} - 1).$$

Its right-hand side is divisible by 3^v, since $k \geq v$, and by U, since $3^h \equiv 1 \pmod{U}$. The two factors 3^v and U are coprime, so it is divisible by E. The identity $M_f^E = I$ therefore identifies the corresponding matrix powers with M_f^n. Taking the trace proves (45). □

Remark 6.9. The hypothesis $M_f^E = I$ is checked in the full matrix algebra. It allows repeated factors modulo p and needs no exclusion of primes dividing the discriminant. Nor is it necessary that $p \nmid 13$. The witness prime 13 is as valid as the other primes in the cover.

Theorem 6.10 (Exact finite exclusion)**.** *For every polynomial $f \in \mathcal{G}$, there are a prime*

$$p \in \{5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47\}$$

and positive integers k, h satisfying (45). Consequently no polynomial in $\mathcal{G}$ is the minimal polynomial of a Pisot number whose traces on (32) are eventually prime.

Proof. The accompanying exact data file `r13_scan.json` contains 230 certificates. Each stores

$$(p, a_0, b, c, k, E, U, n, h), \qquad a_0 \equiv a \pmod{p},$$

with h stored under the field name `period`. For every certificate the verifier checks primality of p by trial division, the coefficient conditions, and all identities in (44). Exponentiation is repeated squaring using integer arithmetic modulo p or modulo U.

The file also assigns each polynomial of $\mathcal{G}$ to one certificate with exactly the same reduced coefficients modulo its witness prime. The verifier independently constructs $\mathcal{G}$ from (42), checks the coefficient match for every assignment, rejects duplicate triples, and checks equality of the expected and assigned sets. Thus it verifies a complete cover, not just the validity of a listed collection of local witnesses. Section 6.7 gives a complete matrix verifier. Its output is

```
PASS: 230 independent matrix certificates cover all 8192 candidates.
```

Lemma 6.8 proves the first assertion. If f were the minimal polynomial of a Pisot number $\beta > 1$, then $T_n(f) = \beta^n + o(1) \to \infty$. Along the infinite progression in (45), all sufficiently late traces are larger than p and divisible by p. They are therefore composite, contradicting eventual primality. □

Witness prime	5	7	11	13	17	19	23
Remaining	3549	1911	1113	664	391	230	112
Witness prime	29	31	37	41	43	47	
Remaining	77	33	18	8	1	0	

Table 7: Successive coverage counts in the exact modular exclusion.

6.4.1 Discovery and verification

For clarity, the search procedure and the mathematical verification have separate roles. The search uses quotient-ring coefficient triples and iterates

$$z_k = x^{13\cdot 3^k}, \qquad z_{k+1} = z_k^3.$$

If $z_k = z_0 + z_1 x + z_2 x^2$, then $p \nmid c$ gives

$$\mathrm{Tr}_{R_p/\mathbb{F}_p}(x^{-1} z_k) = \frac{b}{c} z_0 + 3z_1 + a z_2.$$

Here $b/c = \pm 1$. A zero of this expression suggests a certificate. The search chooses

$$E = p^2 \,\mathrm{lcm}(p-1, p^2-1, p^3-1), \qquad U = E/3^{v_3(E)}, \qquad h = \varphi(U).$$

Neither formula is trusted as an unverified premise: the verification checks $M_f^E = I$ and $3^h \equiv 1 \pmod{U}$ directly for every witness. Similarly, stopping the search at a finite number of indices causes no logical problem, because every candidate receives a verified infinite-return certificate. The largest seed index used is 81.

Three separately implemented checks of the same finite cover are included in the supplement:

(i) `r13_verify.py` uses 3×3 matrix multiplication.

(ii) `audit_r13_certificates.py` uses polynomial multiplication followed by explicit long division. It also checks the first two returned exponent residues and raises explicit exceptions on failure.

(iii) `r13_verify_direct_cycles.py` enumerates the triple-state cycles of recurrence (34) modulo each witness prime, without matrix or quotient-ring exponentiation. It checks the zero trace and the affine return against those cycle periods, and builds the coefficient grid from Table 6.

All three checks pass. The recurrence verification enumerates 662665 trace states in total, with largest period 50652. These are exact reproducibility checks, not formal verification in a proof assistant or external peer review.

6.4.2 Two explicit certificates

For $(a, b, c) = (1857, 2, 2)$, one certificate is

$$p = 5, \quad a_0 = 2, \quad k = 2, \quad E = 18600 = 3 \cdot 6200, \quad U = 6200, \quad n = 116, \quad h = 2400.$$

The matrix (43) satisfies

$$M_f^{18600} = I \pmod 5, \qquad \mathrm{tr}(M_f^{116}) = 0 \pmod 5, \qquad 3^{2400} \equiv 1 \pmod{6200}.$$

Thus 5 divides the trace at every index $k = 2 + 2400j$. For $(a, b, c) = (935, 1, 1)$, the same p, E, U, h work with $k = 5$ and $n = 3158$. These examples illustrate why a certificate excludes eventual prime traces even if its seed is small.

6.5 Proof of the transcendence theorem

Proof of Theorem 6.1. Suppose that $\xi = \xi((13 \cdot 3^k - 1)_{k\geq 1})$ is algebraic. By Proposition 6.3, $\beta = \xi$ is a cubic Pisot number with the structural properties (35), irreducible minimal polynomial modulo 3, and eventual prime traces $T_{13\cdot 3^k-1}$.

Lemma 6.4 forces a to be odd. Proposition 6.5 uses the actual prime floor at exponent 116 to force $\beta < 8192$. Corollary 6.7 then puts the minimal polynomial in the exhaustive finite set $\mathcal{G}$. Theorem 6.10 supplies a fixed prime dividing infinitely many of the required traces. Those traces tend to infinity and so cannot all be prime from any point onward. This contradicts Proposition 6.3 and proves transcendence. □

The proof has two distinct finite ingredients. The exponent 116 is used analytically to bound all possible algebraic minima. The finite cover is then used arithmetically to exclude that entire bounded set by infinite progressions of divisors. Neither ingredient is a substitute for the other.

Corollary 6.11. *The number $\xi((39 \cdot 3^k - 1)_{k\geq 1})$ is transcendental.*

Proof. The structural theorem for odd multiples of 13 in [7, Section 7.2] applies also to 39. If this minimum were algebraic, it would itself be a cubic Pisot number with nonreal conjugates with $b = \pm c$, $0 < c \leq \beta^{3/20}$, and eventual prime traces on $39 \cdot 3^k - 1$. The same square-generator argument as in Proposition 6.3 gives irreducibility modulo 3. Lemma 6.4 makes a odd.

Here $116 = 39 \cdot 3^1 - 1$ is the *first* required exponent. Thus the proof of Proposition 6.5 applies unchanged: if $\beta \geq 8192$, the actual floor at exponent 116 is an even integer greater than 2. Hence $\beta < 8192$, and Corollary 6.7 places its polynomial in the same set $\mathcal{G}$.

For that polynomial, Theorem 6.10 gives a prime divisor of $T_{13\cdot 3^{k_0+jh}-1}$ for every $j \geq 0$. For all sufficiently large j, put $k = k_0 + jh - 1 \geq 1$. Then

$$13 \cdot 3^{k_0+jh} - 1 = 39 \cdot 3^k - 1.$$

The resulting traces on the 39 schedule tend to infinity and are divisible by the same fixed prime, contradicting eventual primality. □

This is not a deduction from the transcendence of the minimum for $r = 13$: deleting an initial floor condition changes the minimum. The proof instead reuses the global bound and the infinite-return certificates. It does not settle every schedule $13 \cdot 3^h \cdot 3^k - 1$, because for $h \geq 2$ the exponent 116 is absent and the same initial-floor bound is not available.

6.6 A branch exclusion for the exponents $2 \cdot 3^k - 1$

We now prove Theorem 6.2. Its hypotheses do not require any prime-trace assumption or irreducibility modulo 3.

Proof of Theorem 6.2. Let u, v be the small conjugates and put $q = uv = c/\beta$ and $s = u + v = a - \beta$. As in (39),

$$s = q(1 - 1/\beta) > 0, \qquad 0 < q \leq \beta^{-17/20} < 1.$$

Set $z = s^2/q = q(1 - 1/\beta)^2$. Then $0 < z < 1$, so $s^2 - 4q = q(z - 4) < 0$. In particular u, v are necessarily nonreal; there is no omitted totally real case.

We first show $\beta > 3/2$. Otherwise the norm bound forces $c = 1$, and Vieta's identity gives

$$a = g(\beta), \qquad g(x) = x + \frac{1}{x} - \frac{1}{x^2}.$$

For $x > 1$, $g'(x) = 1 - x^{-2} + 2x^{-3} > 0$, and therefore

$$1 < a \le g(3/2) = \frac{31}{18} < 2,$$

contradicting integrality. If $c = 1$, this gives $q = 1/\beta < 2/3$. If $c \ge 2$, the norm bound gives $\beta \ge c^{20/3}$ and hence

$$q = \frac{c}{\beta} \le c^{-17/3} \le 2^{-17/3} < \frac{1}{2}.$$

Thus $q < 2/3$ in every case.

The two-root Newton identity gives

$$S_5 = u^5 + v^5 = s^5 - 5qs^3 + 5q^2 s = sq^2(z^2 - 5z + 5) > 0.$$

Indeed $z^2 - 5z + 5 > 1$ for $0 < z < 1$. Since the conjugate pair has modulus $\sqrt{q}$,

$$|S_5| \le 2q^{5/2} < 2(2/3)^{5/2} < 1;$$

the last inequality follows on squaring from $128/243 < 1$. Consequently

$$\left\lfloor \beta^5 \right\rfloor = T_5 - 1. \tag{46}$$

Newton's identities factor the integer on the right exactly:

$$\begin{aligned} T_5 - 1 &= (a-1)Q(a,c), \\ Q(a,c) &= a^4 + a^3 + a^2 + a + 1 - 5a^2c + 5c^2. \end{aligned} \tag{47}$$

For verification, expanding the product yields $a^5 - 5a^3c + 5a^2c + 5ac^2 - 5c^2 - 1$, the fifth power trace minus one for $b = c$.

Since $a = \beta + s > \beta > 3/2$, the integer a is at least 2. If $a = 2$, then $\beta < 2$ and $c \le \beta^{3/20} < 2$, so $c = 1$ and (47) equals 16. If $a \ge 3$, then

$$c \le \beta^{3/20} < a^{3/20} \le a/2,$$

where the last inequality follows from $a^{17} \ge 3^{17} > 2^{20}$. Therefore

$$Q(a,c) \ge a^4 - \frac{3}{2}a^3 + a^2 + a + 1 > 1.$$

Both factors $a - 1$ and $Q(a,c)$ are integers greater than one. Equations (46)–(47) prove compositeness. □

Corollary 6.12. *If $\alpha = \xi((2 \cdot 3^k - 1)_{k \ge 1})$ is algebraic, then, using the structural reduction of [7] and the inverse-trace lemma of [8, Section 3], its cubic minimal polynomial $X^3 - aX^2 + bX - c$ must belong to one of the two sign branches*

$$\begin{aligned} &c > 0, \quad b = -c, \quad \textit{with a nonreal small conjugate pair;} \\ &c < 0, \quad b = c, \quad \textit{with two real small conjugates of opposite signs.} \end{aligned}$$

In both branches a is odd, $3 \nmid c$, and $|c| \le \alpha^{3/20}$.

Proof. For $2 \cdot 3^k - 1$, the eventual gcd of the exponents is 1, so the Mills-shadow reduction gives α itself as cubic Pisot. The negative-fixed-point norm exclusion and magnitude bound of [7] give $3 \nmid c$ and $|c| \le \alpha^{3/20}$. The inverse-trace lemma of [8] gives $\mathrm{Tr}(\alpha^{-1}) = \pm 1$, hence $b = \pm c$. The same reduction gives eventual floor–trace equality and eventual prime traces,

so Lemma 6.4 makes a odd. The case $c > 0, b = c$ is excluded by Theorem 6.2, because $5 = 2 \cdot 3^1 - 1$ is an actual required prime exponent.

It remains to inspect the signs of the small conjugates u, v. Write $q = c/\alpha$ and $s = q(\varepsilon - 1/\alpha)$, with $\varepsilon = b/c$. If $c > 0$ and $b = -c$, then

$$\frac{s^2}{q} = q(1 + 1/\alpha)^2 < 4,$$

since $q \le \alpha^{-17/20} < 1$ and $1 + 1/\alpha < 2$. Thus the small roots are nonreal. If $c < 0$, their product is negative, so they are real with opposite signs. When $b = -c$, one has $s > 0$, and the positive root has larger absolute value. For every odd n, therefore, $u^n + v^n > 0$. For all sufficiently large odd n this sum is less than one, so $\lfloor \alpha^n \rfloor = T_n - 1$, contradicting eventual floor–trace equality on the entirely odd schedule $2 \cdot 3^k - 1$. Hence $b = c$ when $c < 0$. In that case $s < 0$, so the negative small root has larger absolute value, as claimed. □

This corollary is a branch reduction, not a transcendence theorem for $r = 2$. Theorem 6.2 uses primality at the first required floor. It does not rule out eventual prime traces for every polynomial in its branch by itself. Conversely, the $r = 13$ modular cover rules out eventual prime traces but is justified only after its global coefficient bound has been proved.

6.7 Certificate data and a complete verifier

The supplementary file `r13_scan.json` is part of the computer-assisted proof of Theorem 6.10. Its SHA-256 digest, written on two consecutive lines for readability, is

```
b14b061fb261ebff87c6fe27412bfd5d
bc0d917f45bbc4b9f3de96133845d957.
```

The two strings are to be concatenated, without a line break or the sentence-ending period. The data include 230 local certificates, 8192 candidate-to-certificate assignments, and an empty remainder. The digest fixes the exact evidence being referred to; the verifier's arithmetic and set-equality checks establish its mathematical content.

The following Python 3 program uses only the standard library. Run it in the same directory as `r13_scan.json`. All failed checks raise an exception; no assertion can be disabled by an optimized interpreter. An independent long-division verifier and the sequential recurrence verifier are also supplied.

```
import json
from math import isqrt
from pathlib import Path

D = json.loads(Path("r13_scan.json").read_text())
def check(ok):
    if not ok:
        raise ValueError("Certificate verification failed")

def prime(p):
    return p >= 2 and all(p % d for d in range(2, isqrt(p)+1))

I = tuple(tuple(int(i == j) for j in range(3)) for i in range(3))
def mul(A, B, p):
    return tuple(tuple(sum(A[i][t]*B[t][j] for t in range(3)) % p
                       for j in range(3)) for i in range(3))

def power(A, n, p):
    B = I
    while n:
        if n & 1:
```

```
            B = mul(B, A, p)
        A = mul(A, A, p)
        n //= 2
    return B

check(D["r"] == 13 and D["a_max"] == 8191)
check(D["remaining"] == [])
certs = D["certificates"]
check(len(certs) == 230)
for w in certs:
    p,a,b,c,k,E,U,n,h = [w[t] for t in
        ("p","a_mod","b","c","k","E","U","n","period")]
    check(prime(p) and 0 <= a < p and c in (1,2))
    check(b in (-c,c) and c % p != 0)
    check(k >= 1 and E >= 1 and U >= 1 and h >= 1)
    check(0 <= n < E)
    U0, v = E, 0
    while U0 % 3 == 0:
        U0 //= 3
        v += 1
    check(U == U0 and k >= v and pow(3,h,U) == 1 % U)
    check(n == (13*pow(3,k,E)-1) % E)
    M = ((0,0,c % p), (1,0,(-b) % p), (0,1,a))
    check(power(M,E,p) == I)
    B = power(M,n,p)
    check(sum(B[i][i] for i in range(3)) % p == 0)

expected = {(a,b,c) for a in range(1,8192,2)
            for c in (1,2) for b in (-c,c)
            if all((t**3-a*t*t+b*t-c) % 3 for t in range(3))}
actual = set()
for row in D["excluded"]:
    a,b,c = [row[t] for t in ("a","b","c")]
    key = (a,b,c)
    check(key not in actual)
    actual.add(key)
    j = row["certificate"]
    check(isinstance(j,int) and 0 <= j < len(certs))
    w = certs[j]
    check(b == w["b"] and c == w["c"])
    check(a % w["p"] == w["a_mod"])
check(len(expected) == 8192 == D["initial_count"])
check(actual == expected)
print("PASS: 230 independent matrix certificates cover all 8192 candidates.")
```

7 Odd multiplier extensions

The same certificate mechanism extends to further odd multipliers. The finite coefficient grids are exhaustive only under the height and structural hypotheses proved here; they are not a finite enumeration of every cubic Pisot candidate.

7.1 The results and their dependencies

For a sequence $C = (C_k)_{k\ge1}$ of positive integers, write

$$\mathcal{W}(C) = \{A > 1 : \ \left\lfloor A^{C_k} \right\rfloor \text{ is prime for every } k \ge 1\}, \qquad \xi(C) = \min \mathcal{W}(C).$$

Existence of this minimum when $C_{k+1} \ge 2C_k$ is supplied by [4, Theorem 1.3]. All schedules in this paper satisfy that inequality. We abbreviate

$$n_k(r) = r3^k - 1 \quad (k \ge 1), \qquad \xi_r = \xi((n_k(r))_{k\ge1}). \tag{48}$$

Theorem 7.1. *The numbers ξ_{65}, ξ_{91} and ξ_{195} are transcendental.*

Our explicit structural dependency is the theorem for odd multiples of 13 in [7, Section 7.2, Theorem D13]. That theorem is proved in the author's earlier unpublished manuscript; we do not claim to reprove it here. Its consequences, including the residue-field detail needed by the computation, are recorded in Proposition 7.3. Subject to that cited input, the new analytic and finite-cover arguments below are exact and unconditional: no conjecture about primes, floating-point root test, or extrapolation from a finite prime prefix is used.

The analytic part is uniform, rather than a search cutoff chosen for convenience. Define, for odd $r \geq 1$,

$$j(r) = \begin{cases} 1, & r \equiv 3 \pmod 4, \\ 2, & r \equiv 1 \pmod 4, \end{cases} \qquad N(r) = r3^{j(r)} - 1. \tag{49}$$

Then $4 \mid N(r)$ and $N(r) \geq 8$.

Theorem 7.2. *If r is odd and ξ_r is algebraic, then the structural reduction from [7] either already gives a contradiction, when $13 \nmid r$, or gives $\beta = \xi_r$ with positive integral norm c satisfying*

$$\beta < \left(\frac{N(r)}{2}\right)^{40/17}, \qquad c < \left(\frac{N(r)}{2}\right)^{6/17}.$$

The latter bound alone does not exclude the remaining coefficients for every odd multiple of 13.

The first two new cases are settled by finite covers of entire necessary coefficient grids, not just of numerically identified Pisot roots. The third case reuses a cover only after proving that the same grid still contains every hypothetical algebraic minimum.

7.2 The structural input and the trace sequence

A Pisot number is a real algebraic integer $\beta > 1$ whose other conjugates have absolute value less than one. In the cubic case put

$$f(X) = X^3 - aX^2 + bX - c, \qquad T_n = \mathrm{Tr}_{\mathbb{Q}(\beta)/\mathbb{Q}}(\beta^n). \tag{50}$$

Newton's identities give the integer recurrence

$$\begin{aligned} T_0 = 3, \quad T_1 = a, \quad T_2 = a^2 - 2b, \\ T_n = aT_{n-1} - bT_{n-2} + cT_{n-3} \qquad (n \geq 3). \end{aligned} \tag{51}$$

Proposition 7.3 (Structural input from [7])**.** *Let r be odd with $13 \mid r$. If ξ_r is algebraic, then $\beta = \xi_r$ itself is a cubic Pisot number, with nonreal conjugates $u, v = \overline{u}$. Its minimal polynomial (50) satisfies*

$$b = \varepsilon c, \quad \varepsilon \in \{-1, 1\}, \qquad 0 < c \leq \beta^{3/20}, \qquad 3 \nmid c, \qquad f \text{ irreducible over } \mathbb{F}_3. \tag{52}$$

Furthermore,

$$T_{n_k(r)} = \left\lfloor \beta^{n_k(r)} \right\rfloor \text{ is prime} \quad \text{for all sufficiently large } k, \tag{53}$$

$$\left\lfloor \beta^{n_k(r)} \right\rfloor \text{ is prime} \quad \text{for every } k \geq 1. \tag{54}$$

The equality in (53) is not asserted at every initial index.

Explanation of the cited input. The algebraic-minimum reduction in [7, Section 2], based on [4, Proposition 3.1], first gives the cubic Pisot number $B = \xi_r^2$ with eventual prime traces at

exponents $n_k(r)/2$. Theorem D13 proves that ξ_r itself is cubic Pisot with nonreal conjugates and

$$\mathbb{Q}(\xi_r) = \mathbb{Q}(B), \qquad \mathrm{Tr}(\xi_r^{-1}) = \pm 1, \qquad \xi_r \geq |\mathrm{N}_{\mathbb{Q}(\xi_r)/\mathbb{Q}}(\xi_r)|^{20/3}.$$

Since $c = \beta u\overline{u} > 0$ and $\mathrm{Tr}(\beta^{-1}) = b/c$, these give the coefficient and norm assertions. Equality of the fields gives

$$\mathrm{Tr}_{\mathbb{Q}(B)/\mathbb{Q}}(B^{n_k(r)/2}) = \mathrm{Tr}_{\mathbb{Q}(\beta)/\mathbb{Q}}(\beta^{n_k(r)}),$$

which transfers the eventual prime-trace statement to $\beta = \xi_r$. The initial floor condition (54) follows separately from the definition of the minimum.

For completeness, irreducibility modulo 3 requires more than the statement that 3 is inert. The proof of D13 shows that the minimal polynomial of $B = \beta^2$ is irreducible modulo 3. Thus 3 is inert in $K = \mathbb{Q}(B) = \mathbb{Q}(\beta)$, with residue field $\mathbb{F}_{27}$, and the residue $\overline{B}$ generates that field over $\mathbb{F}_3$. The integral element β has a residue in the same field, and

$$\mathbb{F}_3(\overline{B}) \subseteq \mathbb{F}_3(\overline{\beta}) \subseteq \mathbb{F}_{27}, \qquad \overline{\beta}^{\,2} = \overline{B}.$$

Therefore $\overline{\beta}$ has degree three over $\mathbb{F}_3$. Its monic cubic polynomial $\overline{f}$ is irreducible; in particular $3 \nmid c$. This square-generator argument avoids an unjustified inference from inertness for an arbitrary integral generator. □

For $13 \nmid r$, the transcendence result already follows from [7, Theorem B]. All further arguments concern the residual case of Proposition 7.3.

7.3 Scheduled parity and the uniform height bound

Lemma 7.4. *Let $r \geq 1$ and let β be a cubic Pisot number with $b \equiv c \pmod 2$. If $T_{n_k(r)}$ is eventually prime, then every scheduled trace $T_{n_k(r)}$, $k \geq 1$, is odd. If $7 \nmid r$, then a is odd. If $7 \mid r$, only the simultaneous conditions $a \equiv c \equiv 0 \pmod 2$ are excluded by this parity argument.*

Proof. The Pisot property gives $T_n = \beta^n + o(1)$, so eventually prime scheduled traces eventually exceed two. There are three cases. If c is even, the roots modulo two have multiset $\{0, 0, a\}$, and $T_n \equiv a^n \pmod 2$ for $n \geq 1$. If c and a are both odd, then $\overline{f} = (X+1)^3$ over $\mathbb{F}_2$, and every trace is odd. If c is odd and a is even, then $\overline{f} = X^3 + X + 1$, whose roots are nonzero elements of $\mathbb{F}_8$ with order seven. In this case

$$(T_n \bmod 2)_{n=0}^{6} = (1, 0, 0, 1, 0, 1, 1), \qquad T_{n+7} \equiv T_n \pmod 2. \tag{55}$$

The sequence $r3^k - 1$ modulo seven is periodic from its first index, with period dividing six. Consequently the parity of the scheduled traces is purely periodic in all three cases. An eventually odd, purely periodic sequence is odd from the first index.

If $7 \nmid r$, the powers of three run through all nonzero residues modulo seven, and multiplication by r permutes them. Thus the scheduled exponents include residues $1, 2, 4$ infinitely often; (55) excludes the third case. The first case excludes even a as well. If $7 \mid r$, all scheduled exponents are 6 modulo seven, and the third case has odd scheduled traces. It must not be excluded. □

In particular, $r = 91$ does not force a to be odd. The conclusion needed for the initial floor is only oddness on the actual schedule.

Proposition 7.5. *Suppose r is odd, and β satisfies Proposition 7.3. With $N = N(r)$ from (49),*

$$\beta < (N/2)^{40/17}, \qquad c < (N/2)^{6/17}.$$

Proof. Put $q = uv = c/\beta$. Vieta's identities and $b = \varepsilon c$ give

$$u + v = q\left(\varepsilon - \frac{1}{\beta}\right), \qquad 0 < q \le \beta^{-17/20} < 1. \tag{56}$$

Write $u = \sqrt{q}\, e^{i\theta}$ and $v = \sqrt{q}\, e^{-i\theta}$ with $0 < \theta < \pi$. Then

$$\cos\theta = \frac{\sqrt{q}}{2}\left(\varepsilon - \frac{1}{\beta}\right), \qquad S_n = u^n + v^n = 2q^{n/2}\cos(n\theta), \qquad \beta^n = T_n - S_n. \tag{57}$$

Suppose $\beta \ge (N/2)^{40/17}$. Since $N \ge 8$, we have $\beta > 4$. For $t = |\cos\theta|$, (56) yields

$$0 < t < \frac{5}{8}\beta^{-17/40} \le \frac{5}{4N} \le \frac{5}{32} < \frac{1}{2}.$$

The strict positivity follows from $q > 0$, $\beta > 1$ and $\varepsilon \in \{-1, 1\}$. With $\delta = |\theta - \pi/2| = \arcsin t$,

$$\delta \le \frac{t}{\sqrt{1-t^2}} < \frac{6}{5}t < \frac{3}{4}\beta^{-17/40} \le \frac{3}{2N}.$$

Here $2/\sqrt{3} < 6/5$ follows by squaring from $100 < 108$. It follows that $N\delta < 3/2 < \pi/2$. Since $4 \mid N$, $\cos(N\theta) = \cos(N\delta) > 0$. Moreover $q \le (2/N)^2$, and therefore

$$0 < S_N \le 2q^{N/2} \le 2(2/N)^N < 1.$$

The integer T_N consequently satisfies

$$\left\lfloor \beta^N \right\rfloor = T_N - 1.$$

By Lemma 7.4, T_N is odd. This floor is even and, since $\beta > 4$, greater than two. But $N = n_{j(r)}(r)$ is an actual required exponent, so (54) says that this same floor is prime. This contradiction proves the bound on β. Finally $c \le \beta^{3/20} < (N/2)^{6/17}$. □

Proposition 7.5, together with the cited result for $13 \nmid r$, proves Theorem 7.2. Notice that no initial floor/trace equality was assumed: the sign and size of S_N determined the initial floor directly.

7.4 The two exhaustive coefficient grids

For $r = 65$ and $r = 91$, the parameters are

r	$j(r)$	$N(r)$	B
65	2	584	2^{20}
91	1	272	2^{17}

The following integer comparisons, checked by both verifiers, imply $\beta < B$ and the required norm cutoffs:

$$\begin{aligned} 292^{40} &< (2^{20})^{17}, \quad 292^6 < 8^{17}, \\ 136^{40} &< (2^{17})^{17}, \quad 136^6 < 6^{17}. \end{aligned} \tag{58}$$

For example, $292^2 = 85\,264 < 2^{17}$ proves the first inequality; $136^5 = 46\,525\,874\,176 < 2^{36}$ proves the third after raising to the eighth power. The norm cutoff also follows from $c < 2^3 = 8$ in the first case and $c < 2^{51/20} < 6$, with $2^{51} < 6^{20}$, in the second. As c is a positive integer not divisible by three, the norm sets are

$$C_{65} = \{1, 2, 4, 5, 7\}, \qquad C_{91} = \{1, 2, 4, 5\}.$$

There is a safe coefficient bound without approximating any roots. From (56), $a = \beta + u + v > \beta - 2 > -1$, hence $a \geq 0$. If $\varepsilon = -1$ then $a < \beta < B$; if $\varepsilon = 1$ then $0 < u + v < 1$, so $a < \beta + 1 < B + 1$. Thus in all cases $a \leq B$. When a is odd and B even this improves to $a \leq B - 1$.

Define $\mathcal{G}_{65}$ and $\mathcal{G}_{91}$ to be the sets of integer triples (a, b, c) with $b \in \{-c, c\}$, with $X^3 - aX^2 + bX - c$ irreducible over $\mathbb{F}_3$, and with the following additional conditions:

$$\begin{aligned} &\mathcal{G}_{65} : 1 \leq a \leq 2^{20} - 1, \quad a \text{ odd}, \quad c \in C_{65}; \\ &\mathcal{G}_{91} : 0 \leq a \leq 2^{17}, \quad c \in C_{91}, \quad \text{not both } a, c \text{ even}. \end{aligned} \tag{59}$$

Every algebraic minimum for the corresponding schedule yields a triple in its grid. In particular, the even-a, odd-c branch and both endpoints of the $r = 91$ interval are retained whenever the modulo-three condition permits them. We do not impose any numerical Pisot, discriminant, or additional norm filter.

For a cubic over $\mathbb{F}_3$, irreducibility is exactly the absence of roots in $\{0, 1, 2\}$. Direct enumeration, or counting the admissible a classes modulo six, gives Table 8.

Table 8: Exact grid sizes separated by the coefficients b, c.

		Number of admissible a	
c	b	$\mathcal{G}_{65}$	$\mathcal{G}_{91}$
1	-1	349526	87382
1	1	174762	43691
2	-2	174763	21846
2	2	349525	43690
4	-4	349526	43691
4	4	174762	21845
5	-5	174763	43691
5	5	349525	87382
7	-7	349526	—
7	7	174762	—
Total		2621440	393218

7.5 Exact certificates for infinitely recurring divisibility

For any integer coefficients a, b, c, define T_n by (51); equivalently it is the sum of the nth powers of the roots of f, counted with multiplicity. The companion matrix

$$M(a, b, c) = \begin{pmatrix} 0 & 0 & c \\ 1 & 0 & -b \\ 0 & 1 & a \end{pmatrix} \quad \text{satisfies} \quad \operatorname{tr}(M^n) = T_n \qquad (n \geq 0). \tag{60}$$

This remains true after reduction modulo any prime, including when f becomes reducible or inseparable.

Lemma 7.6 (Recurrent-divisor certificate)**.** *Fix $r \geq 1$ and integers b, c. Let*

$$(p, a_0, b, c, k, E, U, n, h)$$

be a tuple of integers satisfying the following tests:

(i) *p is prime, $p \nmid c$, and $0 \le a_0 < p$;*

(ii) *$E = 3^v U > 0$ with $v \ge 0$, $3 \nmid U$, $k \ge \max(v, 1)$, $h \ge 1$, and $3^h \equiv 1 \pmod{U}$;*

(iii) *$0 \le n < E$ and $n \equiv r3^k - 1 \pmod{E}$;*

(iv) *modulo p, $M(a_0, b, c)^E = I$ and $\operatorname{tr}(M(a_0, b, c)^n) = 0$.*

Then every integer $a \equiv a_0 \pmod{p}$ satisfies

$$p \mid T_{r3^{k+th}-1} \qquad \text{for every integer } t \ge 0.$$

Proof. The factor 3^v divides 3^k, while U divides $3^{th} - 1$. Since 3^v and U are coprime,

$$E \mid r3^k(3^{th} - 1).$$

Thus $r3^{k+th} - 1 \equiv n \pmod{E}$, and the matrix identities in (iv) give the claimed trace divisibility. Reduction of a to a_0 does not change the matrix modulo p. □

Proposition 7.7 (Complete finite covers)**.** *Every triple in $\mathcal{G}_{65}$ or $\mathcal{G}_{91}$ admits a certificate from Lemma 7.6, for its corresponding r. The supplied exact certificate files have the following sizes:*

r	*grid size*	*certificates*	$\max p$	*uncovered*
65	2 621 440	1 447	107	0
91	393 218	774	89	0

Finite verification and its meaning. The data consist of the tuples of Lemma 7.6, not one row for each coefficient triple. For each fixed (b, c) a tuple covers exactly the grid elements satisfying $a \equiv a_0 \pmod{p}$. The matrix verifier independently reconstructs (59), tests every tuple, and removes these arithmetic progressions. It verifies the initial counts in Table 8, all per-prime survivor counts, and an empty final survivor list in each case. Primality of every witness prime is checked by trial division; all other calculations are exact integer or finite-field operations.

A separately authored verifier changes both central implementations: it checks identities using generic polynomial long division in $\mathbb{F}_p[X]/(f)$ instead of matrices, and reconstructs and subtracts coefficient sets using Python sets instead of bytearrays. It also checks the seed exponent and two explicit returns for every tuple. It verifies the same two empty final sets. The certificate files, verifier sources, commands, and their identifying hashes are specified in Section 7.7. Thus the computational assertion is an exhaustive finite check with independently checkable witnesses, not an assumption that a divisor observed once must recur. Lemma 7.6 proves the required infinite recurrence. □

The discovery program chooses

$$E = p^2 \operatorname{lcm}(p - 1, p^2 - 1, p^3 - 1), \qquad U = E/3^{v_3(E)}, \qquad h = \varphi(U). \tag{61}$$

The proof does not depend on a claim that this is a minimal period. The verifiers test $M^E = I$ and $3^h \equiv 1 \pmod{U}$ directly. Thus reducible or repeated factors modulo p cause no gap; primes dividing r are also harmless. A prime dividing c is not used for a unit-return certificate.

7.6 Transcendence and the limits of the transfer

Proof of Theorem 7.1. Suppose first that ξ_r is algebraic for $r = 65$ or $r = 91$. Proposition 7.3 gives $\beta = \xi_r$ and eventual prime traces. Proposition 7.5 and Section 7.4 place its coefficients in $\mathcal{G}_r$. Proposition 7.7 and Lemma 7.6 supply a fixed prime p dividing infinitely many of its

scheduled traces. But $T_n = \beta^n + u^n + v^n \longrightarrow +\infty$. At all sufficiently large certified indices, $T_n > p$ and is divisible by p, hence composite. This contradicts eventual primality.

Now let $r = 195 = 3 \cdot 65$ and suppose ξ_{195} algebraic. The structural input still applies. Since $7 \nmid 195$, a is odd. Its first exponent is

$$N(195) = 3 \cdot 195 - 1 = 584 = N(65).$$

Therefore the same height and norm estimates put its coefficients in the identical grid $\mathcal{G}_{65}$. A certificate for that grid provides divisibility at exponents $65 \cdot 3^{k+th} - 1$. These equal

$$195 \cdot 3^{k+th-1} - 1,$$

and $k + th - 1 \geq 1$ for all sufficiently large t. Thus infinitely many actual $r = 195$ traces are divisible by the same prime and eventually exceed it. This again contradicts eventual primality. □

Corollary 7.8. *For every odd integer r with $1 \leq r \leq 100$, ξ_r is transcendental.*

Proof. Theorem B of [7] covers $13 \nmid r$. The only positive odd multiples of 13 at most 100 are $13, 39, 65, 91$. The first two are proved in [10]; the last two are Theorem 7.1. □

The $r = 195$ proof is not a transfer of transcendence merely because a schedule is a tail of another. Deleting an initial prime-floor condition changes the minimization problem. We instead reestablished the same finite coefficient bound and then transferred infinite recurring-divisor witnesses. For example, the exponent 272 used for $r = 91$ is absent from the $r = 273$ schedule, so the present proof does not automatically settle $r = 273$.

The uniform bound leaves a finite problem for each residual odd multiplier, but this paper proves complete covers only for the stated grids. No result for all odd multiples of 13 is asserted. The constant $\xi_2 = \xi((2 \cdot 3^k - 1)_{k \geq 1})$ remains unresolved by this argument, as does the transcendence of the classical Mills constant with exponents 3^k.

7.7 Reproducible finite verification

All files in Table 9 are supplied together in the accompanying `certificates` directory. Only the two JSON files and either verifier are needed to check the finite-cover assertion. The verifiers use the Python standard library; SymPy is needed only to rerun discovery. The data encode exactly the tests in Lemma 7.6 and the grids in (59).

Table 9: Reproduction files and their roles.

File	Role
`odd65_91_r65.json`	1,447 local certificates and grid metadata
`odd65_91_r91.json`	774 local certificates and grid metadata
`odd65_91_scan.py`	Exact quotient-ring discovery program
`odd65_91_verify.py`	Matrix and bytearray cover verifier
`audit_odd65_91_verify.py`	Long-division and set cover verifier
`odd65_91_verification.json`	Matrix verifier terminal results
`audit_odd65_91_verification.json`	Independent audit terminal results
`odd65_91_certificates.sha256`	Frozen certificate digests

The following commands verify the existing frozen data without rerunning discovery. Run them without Python's optimization flag, so that the matrix verifier's assertion checks remain enabled.

```
cd certificates
python3 odd65_91_verify.py
python3 audit_odd65_91_verify.py
sha256sum -c odd65_91_certificates.sha256
```

Both verifiers report 2 621 440 candidates, 1 447 certificates, and zero survivors for $r = 65$; 393 218 candidates, 774 certificates, and zero survivors for $r = 91$. The largest seed indices are respectively 290 and 230.

For clarity, the complete logical structure of the finite check is:

```
for r in (65, 91):
    G = exact_grid(r)  # stated bounds, parity, no root in F_3
    assert all_pair_counts(G) == supplied_initial_counts
    for (p, a0, b, c, k, E, U, n, h) in certificates[r]:
        verify_all_recurrent_divisor_certificate_conditions()
        G[b,c] = {a for a in G[b,c] if a % p != a0}
    assert every coefficient set G[b,c] is empty
```

Here `exact_grid` uses only integer inequalities and the three evaluations $x^3 - ax^2 + bx - c \pmod 3$ for $x = 0, 1, 2$. Matrix powers are computed by binary exponentiation modulo p. The alternative verifier computes powers by polynomial multiplication and long division by f; for a residue $z_0 + z_1X + z_2X^2$, its trace is $3z_0 + az_1 + (a^2 - 2b)z_2$ modulo p. These tests completely specify the finite proposition independently of the method used to discover the witnesses.

To rediscover the covers, work in a separate copy of the certificates folder, because discovery rewrites the JSON and timing metadata:

```
python3 odd65_91_scan.py 65 1000
python3 odd65_91_scan.py 91 1000
```

The upper search limit 1000 is not a mathematical hypothesis: the scans stop once coverage is complete, at witness primes 107 and 89. Their search updates X^{r3^k} by cubing in $\mathbb{F}_p[X]/(f)$ and uses the identity

$$\mathrm{Tr}\Big(X^{-1}(z_0 + z_1X + z_2X^2)\Big) = (b/c)z_0 + 3z_1 + az_2.$$

The verification does not reuse these discovery formulas.

Frozen SHA256 digests. Line breaks below are typographical only; each digest is one 64-character hexadecimal string.

```
odd65_91_r65.json
73a813b489d134e6e3c7d085c768ebc5715ea88dc98f6ebdc2e00221c13d96a6
odd65_91_r91.json
56aa4b60938822f0d00c4df7f5f9e8c4c45d14792599a9feefb3e71c9206b2ce
odd65_91_scan.py
7aa615056aa2119c53e0fbacf3749ee1b24677f0050c74371a427903aa605984
odd65_91_verify.py
ef5b9c5a084d468da95024a0ebc5e2e57409efd537d045b25c0b0ed151aa4f77
audit_odd65_91_verify.py
fbd28dece660e180d94b0108503eee0eb4630012b6cfd86cd10e56bf927e315a
```

Part II

Prime cells and finite to infinite continuation

8 Prime cells height bounds and Pell families

We now separate finite prime prefixes from their infinite continuations. The prime-cell formalism supports the height and Pell-family arguments below and provides the precise interface needed for the subsequent prime-forest criteria.

8.1 Statements and dependencies

For positive exponents $n_k \to \infty$, define

$$\mathcal{W}(n_k) = \{A > 1 : \lfloor A^{n_k} \rfloor \in \mathbb{P} \text{ for every } k \geq 1\}.$$

When a least element exists, write it as $\xi(n_k)$. The two schedules in this report are

$$n_k = 2 \cdot 3^k - 1 = 5, 17, 53, \ldots \quad (k \geq 1), \qquad M_r = 3^r \quad (r \geq 1). \tag{62}$$

The prime-representing sets for both schedules are nonempty and have least elements by [4, Theorem 1.3]. Their arithmetic problems must nevertheless be kept separate: a statement about 3^r does not automatically transfer to $2 \cdot 3^k - 1$.

A *cubic Pisot number* is a real algebraic integer $\beta > 1$ of degree three whose other conjugates have modulus less than one. Write

$$T_n = \mathrm{Tr}_{\mathbb{Q}(\beta)/\mathbb{Q}}(\beta^n), \qquad S_n = T_n - \beta^n.$$

Thus the sign of S_n determines which of T_n and $T_n - 1$ is the floor once $|S_n| < 1$. We reserve $V_r = T_{3^r}$ for the classical schedule in Section 8.5.

The shifted structural reduction used in earlier work [7, 8, 10] leaves the following two families:

$$\begin{aligned} f_+(X) &= X^3 - aX^2 - CX - C \quad \text{with a nonreal conjugate pair,} \\ f_-(X) &= X^3 - aX^2 - CX + C \quad \text{with three real roots,} \\ a &\text{ odd}, \quad C \in \mathbb{Z}_{\geq 1}, \quad C \leq \beta^{3/20}. \end{aligned} \tag{63}$$

Here β is required to be cubic Pisot. Our first theorem is stated directly under these hypotheses; its proof does not reprove the earlier reduction, and does not assume that every polynomial with coefficients as displayed is a Pisot polynomial.

Put $\theta = 21/40$. For a real number $X \geq 8$, let (P_X) mean

$$[x - x^\theta, x] \cap \mathbb{P} \neq \varnothing \qquad \text{for every real } x \geq X. \tag{P_X}$$

The Baker–Harman–Pintz theorem supplies such a threshold [1, Theorem 1].

Theorem 8.1. *Suppose β satisfies (63), every $\left\lfloor \beta^{2 \cdot 3^k - 1} \right\rfloor$ is prime, and β is the global minimum of $\mathcal{W}(2 \cdot 3^k - 1)$. For every $X \geq 8$ satisfying (P_X),*

$$\beta < \max\{4, (X+2)^{1/17}\}.$$

The prime competitor used in the proof has a complete infinite continuation and retains the first prime floor.

Theorem 8.2. *Define*

$$\begin{pmatrix} m_0 \\ y_0 \end{pmatrix} = \begin{pmatrix} 3 \\ 7 \end{pmatrix}, \qquad \begin{pmatrix} m_{t+1} \\ y_{t+1} \end{pmatrix} = \begin{pmatrix} 577 & 204 \\ 1632 & 577 \end{pmatrix} \begin{pmatrix} m_t \\ y_t \end{pmatrix} - \begin{pmatrix} 468 \\ 1326 \end{pmatrix}, \qquad a_t = 2m_t - 1. \tag{64}$$

For every $t \geq 0$, the real root β_t of $X^3 - a_t X^2 - X - 1$ is a cubic Pisot unit and has infinitely many composite values $\left\lfloor \beta_t^{2 \cdot 3^k - 1} \right\rfloor$. The five primes $5, 19, 29, 31, 59$ suffice to certify this assertion for the entire ray. No minimum or conjugate-phase assumption is needed. The same conclusion holds for the totally real reflected family in Corollary 8.8.

For the classical schedule, let $\mathcal{S}$ consist of totally real cubic Pisot numbers with conjugates

$$\beta, -\rho, \gamma, \qquad 0 < \sigma = |\gamma| < \rho < 1, \qquad \rho \leq \min\{\sigma^{17/23}, \beta^{-17/40}\}. \tag{65}$$

Saito's reduction says that a hypothetical algebraic classical Mills constant has a positive integral power in this locus [4, Theorem 1.7].

Theorem 8.3. *The locus $\mathcal{S}$ has the exact coefficient description in Theorem 8.10. Among its polynomials $X^3 - aX^2 - BX - c$ with $1 \leq a \leq 1000$, there are exactly $32,232$ coefficient triples. Every one has a prime $p \leq 59$ dividing $V_r = \mathrm{Tr}(\beta^{3^r})$ on an infinite arithmetic progression of indices. Consequently every such β has infinitely many composite classical floors $\lfloor \beta^{3^r} \rfloor$.*

Theorem 8.3 is a computer-assisted finite theorem; its exhaustive coefficient bounds, exact comparisons, and certificate semantics are given below. It does not bound the trace of an arbitrary member of $\mathcal{S}$. The norm bound, redundancy of one inequality in (65), and fixed-norm density-one results are already in manuscript [031] [6]; they are background, not new results of this report. The bounded theorem was obtained in the preceding round and is presented here with an independent integer-only verification and a uniform floor-to-trace onset.

8.2 Exact prime cells and minimum criteria

8.2.1 Viability includes the infinite continuation

Assume throughout this subsection that n_k are positive and tend to infinity. A prime prefix $p_1, \ldots, p_k$ determines the half-open cell

$$J_k = [L_k, U_k), \qquad L_k = \max_{j \leq k} p_j^{1/n_j}, \quad U_k = \min_{j \leq k} (p_j + 1)^{1/n_j}. \tag{66}$$

A prime q is a *viable next label* if some member of $\mathcal{W}(n_k)$ lies in

$$J_k \cap [q^{1/n_{k+1}}, (q+1)^{1/n_{k+1}}).$$

The first label is viable if its cell contains a member of $\mathcal{W}(n_k)$. A nonempty finite intersection alone is not a viability certificate. An element $A \in \mathcal{W}(n_k)$ is a *local left minimum* if some $\varepsilon > 0$ satisfies $(A - \varepsilon, A) \cap \mathcal{W}(n_k) = \varnothing$.

Proposition 8.4. *An all-prime base is the global minimum if and only if its first label and its next label after every prefix are the least viable choices. If additionally $A^{n_j} \notin \mathbb{Z}$ for every j, then A is a local left minimum if and only if its next label is the least viable choice after every sufficiently long prefix.*

Proof. Two distinct bases have a first different label: for $1 < A < B$, $B^{n_k} - A^{n_k} \to \infty$. At that first difference, the smaller base has the smaller label. This proves both global implications.

The cell containing a fixed $A > 1$ has width tending to zero, by the mean value theorem (or directly from its endpoints). Infinitely many smaller viable siblings therefore produce smaller all-prime bases sharing successively longer prefixes and converging to A; a local left gap is impossible. Conversely, if all choices after a prefix of length K are least viable, nonintegrality puts A in the interior of that entire finite prefix. Any smaller all-prime base in a sufficiently small left neighborhood has a first different label after K, contrary to leastness. Equal labels forever would force the bases to be equal. □

8.2.2 The affine boundary child cannot be omitted

For the shifted schedule, put $n = n_k$, $m = n_{k+1} = 3n + 2$ and $d = m/n$. Suppose for the moment that the current prefix cell is the full p-cell. Since $\gcd(n, m) = 1$ and $n \geq 5$, the number p^d is not an integer. The integer

$$q_0 = \left\lfloor p^d \right\rfloor \tag{67}$$

can be prime, and its cell can meet the parent in the truncated cell

$$[p^{1/n}, (q_0 + 1)^{1/m}). \tag{68}$$

Indeed, $q_0 < p^d < q_0 + 1 < (p + 1)^d$. Labels below q_0 cannot meet the parent. The child (68) has no analogue for the integer cubic ratio: there the boundary value is the composite integer p^3. Thus "least prime strictly above p^d" alone is not an exact minimum criterion for the affine clock.

Lemma 8.5. *Let β be a cubic Pisot number with all shifted floors prime. Suppose on the scheduled tail that $p_k = T_{n_k}$ and $e_{n_k} = \beta^{n_k} - T_{n_k} > 0$. Then, for every sufficiently large k, the full current prime cell equals the entire prefix cell J_k, and its successor has closure strictly inside it.*

Proof. Let $\rho < 1$ be the largest modulus of the two small conjugates. The algebraic integer $\beta^n - T_n$ has nonzero integral norm. To see nonvanishing, a rational positive power of β would have the same value under every embedding, contradicting the small conjugates. Its other two conjugates have modulus below $T_n + 1$. Consequently, for large scheduled n,

$$\tfrac{1}{4}\beta^{-2n} < (T_n + 1)^{-2} < e_n \leq 2\rho^n. \tag{69}$$

For $\delta_n = \beta - T_n^{1/n}$, the mean value theorem yields

$$\delta_n \geq \frac{\beta}{4n}\beta^{-3n}, \qquad \delta_m \leq \frac{4\beta}{m}(\rho/\beta)^m.$$

Since $m = 3n + 2$, their ratio is at most $16(n/m)\beta^{-2}\rho^m$, which tends to zero. Hence the lower endpoints strictly increase eventually. The upper distances satisfy

$$(T_n + 1)^{1/n} - \beta \sim \frac{\beta}{n\beta^n},$$

and their successive ratio tends to zero, so the upper endpoints strictly decrease. Every fixed earlier prefix contains β in its interior, again by nonintegrality of Pisot powers. The shrinking current cells are eventually contained in that whole earlier prefix. This proves the assertion without a lower bound on a lacunary cosine. □

Corollary 8.6. *Under Lemma 8.5's hypotheses, β is a local left minimum if and only if, for every sufficiently large k, both of the following hold:*

(i) $(p_k^{n_{k+1}/n_k}, p_{k+1})$ *contains no prime;*

(ii) the boundary label $q_0 = \left\lfloor p_k^{n_{k+1}/n_k} \right\rfloor$ *is composite, or its truncated cell* (68) *has no infinite all-prime continuation.*

Proof. Every sufficiently large full prime child extends indefinitely by Lemma 8.7 below. A prime strictly between $p_k^{n_{k+1}/n_k}$ and p_{k+1} has a full cell inside the parent: its upper endpoint is bounded by $p_{k+1}^{1/n_{k+1}}$. Lemma 8.5 makes q_0 a strictly smaller child for large k. These full children and the one boundary child are all possible smaller labels. Apply Proposition 8.4. □

Global leastness requires the earlier least viable choices as well. In particular, eventual prime-free windows by themselves prove neither global leastness nor the viability failure of a boundary child.

8.3 The first affine window and a height theorem

8.3.1 Exact fifth-power cancellation

Assume (63), all scheduled floors prime, and $\beta \geq 4$. Put $h = C/\beta$. Then

$$h \leq \beta^{-17/20} < \tfrac{1}{3}. \tag{70}$$

The sums and products of the two small roots are

$$(s_+, q_+) = (-h(1+\beta^{-1}), h), \qquad (s_-, q_-) = (-h(1-\beta^{-1}), -h).$$

The two-root Newton identity $S_5 = s^5 - 5qs^3 + 5q^2 s$ gives

$$\begin{aligned} e_{5,+} &= h^3(1+\beta^{-1})(5 - 5z_+ + z_+^2), \quad z_+ = h(1+\beta^{-1})^2, \\ e_{5,-} &= h^3(1-\beta^{-1})(5 + 5z_- + z_-^2), \quad z_- = h(1-\beta^{-1})^2. \end{aligned} \tag{71}$$

Here $e_n = -S_n$. Since $z_+ < 25/48$, the first bracket exceeds $5 - 125/48 > 2$. In the second formula the product of the outside factor $1-\beta^{-1}$ and the bracket exceeds $5(3/4) > 2$. Therefore

$$e_5 > 2C^3\beta^{-3} \geq 2\beta^{-3}. \tag{72}$$

Also $e_5 < 1$. For the nonreal pair use $|S_5| \leq 2h^{5/2} < 1$. For the real pair put $\delta = h(1-\beta^{-1}) < h$; then

$$e_5 = \delta^5 + 5h\delta^3 + 5h^2\delta < h^5 + 5h^4 + 5h^3 < 61/243 < 1.$$

For exponent 17, the nonreal modulus estimate again gives $|S_{17}| < 1$. In the real case the larger small modulus is below $\sqrt{h} + h < 11/12$, so $|S_{17}| < 2(11/12)^{17} < 1$. All traces are odd: Newton's recurrence modulo 2 has odd initial traces, an odd leading coefficient, and equal lower coefficients. If $S_{17} > 0$, the actual floor would be $T_{17} - 1$, an even integer greater than two, contradicting primality. No S_n can vanish, since that would make β^n integral. We have therefore proved

$$p = T_5 = \beta^5 - e_5, \qquad P = T_{17} = \beta^{17} - e_{17}, \qquad 0 < e_5, e_{17} < 1. \tag{73}$$

These are the actual initial floors, not just tail traces.

The mean value theorem now gives

$$\begin{aligned} \beta^{17} - (\beta^5 - e_5)^{17/5} &\geq \frac{17}{5}(\beta^5 - 1)^{12/5} e_5 \\ &> \frac{34}{5}\beta^9(1-\beta^{-5})^{12/5} > 6\beta^9. \end{aligned}$$

The last inequality follows from Bernoulli's inequality already at $\beta = 4$. Subtracting $e_{17} < 1$ proves

$$\boxed{T_{17} - T_5^{17/5} > 5\beta^9.} \tag{74}$$

The relevant exponent relative to β^{17} is $9/17$, not $19/34$. The cancellation in (71) is essential.

8.3.2 A full infinite continuation

Lemma 8.7. *Assume* (P_X). *If $r \geq 2$ is prime, $d \geq 3$, and $r^d \geq X$, there is a prime r' such that*

$$r^d < r', \qquad r' + 1 < (r+1)^d.$$

Consequently every sufficiently large full prime cell extends indefinitely along the shifted schedule.

Proof. Put $y = r^d$ and $z = y + 2y^\theta$. Apply (P_X) at z. Since $z \leq 3y$ and $3^\theta < 2$, the supplied prime satisfies

$$y < z - z^\theta \leq r' \leq z = y + 2y^\theta.$$

For $r \geq 2$ and $d \geq 3$,

$$dr^{(1-\theta)d-1} > 3 > 2 + r^{-\theta d},$$

because $(1-\theta)d - 1 \geq 17/40 > 0$. Thus convexity gives

$$(r+1)^d - r^d \geq dr^{d-1} > 2r^{\theta d} + 1.$$

This proves strict inclusion of the child closure inside the parent. Iterate. The compact cell closures are nested, their diameters tend to zero, and strict one-step upper inclusion places the limit inside every original half-open prime cell. Hence the limit is an actual all-prime base, not merely a limit satisfying weak endpoint inequalities. □

Proof of Theorem 8.1. Suppose $\beta \geq 4$ and $\beta^{17} \geq X + 2$. In (73), take $x = P - 1 > \beta^{17} - 2 \geq X$. The prime q supplied by (P_X) satisfies $q \leq P - 1 < P$. Moreover (74) gives

$$q - p^{17/5} > 5\beta^9 - 1 - \beta^{357/40} > 0,$$

since $357/40 < 9$. Also

$$q > \beta^{17} - 2 - \beta^{357/40} > \tfrac{1}{2}\beta^{17}, \qquad q^3 > X.$$

For the middle inequality, divide by β^{17} and use $2 \cdot 4^{-17} + 4^{-8} < 1/2$.

The full q-cell at exponent 17 is strictly inside the first p-cell, because

$$p^{17/5} < q, \qquad q + 1 \leq P < \beta^{17} < (p+1)^{17/5}.$$

Lemma 8.7 then constructs every subsequent child using the same X, since all later ratios are greater than 3 and all later prime labels are at least q. Its limit B has the original first prime floor and all later floors prime, while

$$B^{17} < q + 1 \leq P < \beta^{17}.$$

Thus $B < \beta$, contradicting global minimality. This proves the strict bound. □

Applying the prime theorem at P, rather than $P - 1$, would not suffice: it could return the already known prime P. Likewise a finite prime prefix or a tail constant would not suffice without the complete nesting argument above.

8.3.3 What the height theorem does and does not provide

Theorem 8.1 is sensitive to global minimality: it changes the second prime floor, so it does not exclude every large algebraic local minimum. BHP's original theorem states that its threshold can in principle be made effective, but does not print a numerical value at that theorem [1, p. 532]. No numerical X is established by the present argument. It therefore does not supply an executable finite coefficient bound by itself.

An enormous explicit upper bound for the global least base already follows from Saito's construction [4, Section 2.3 and Lemma 9.1]. For example, his recorded sufficient prime-interval threshold allows

$$X_0 = \exp(3\exp(32.76)), \qquad \xi(2\cdot 3^k - 1) < (2X_0^{5/17} + 1)^{1/5}.$$

Thus finiteness in principle is not a new conclusion. The exponent $2/3$ in that explicit interval estimate is too large to force a prime into (74); substituting that estimate into the thin window would be invalid.

There is also a limitation on repeating the argument later. For fixed C and a fixed scheduled odd exponent n, the two-root power-sum identity gives

$$e_n \sim nC^{(n+1)/2}\beta^{-(n+1)/2} \qquad (\beta \to \infty)$$

within either family. For the transition $n \mapsto 3n+2$ the window consequently has order

$$(3n+2)C^{(n+1)/2}\beta^{3(n+1)/2}, \qquad \eta_n = \frac{3n+3}{6n+4} = \frac{1}{2} + \frac{1}{6n+4}$$

relative to $x = \beta^{3n+2}$. Although $\eta_5 = 9/17 > 21/40$, $\eta_{17} = 27/53 < 21/40$. This is a fixed-index, large-height comparison, not a fixed-β, large-index asymptotic.

8.4 An infinite Pell family with composite shifted floors

8.4.1 The family and its surviving primitive character

The recurrence (64) preserves

$$8m_t^2 - 13m_t + 16 = y_t^2. \tag{75}$$

Indeed, with $x = 16m - 13$, it is multiplication of $x + y\sqrt{32}$ by $577 + 102\sqrt{32}$, a unit of norm one, on $x^2 - 32y^2 = -343$. Both coordinates increase from $(3, 7)$, and $m_t \equiv 3 \pmod{24}$ and oddness of y_t are invariant. Hence

$$a_t \equiv 5 \pmod{48}, \qquad a_t = 5, 5381, 6208949, 7165121045, \ldots.$$

For $a \geq 5$, the only possible rational roots of $X^3 - aX^2 - X - 1$ are ± 1, and neither is a root. Its discriminant is

$$\Delta = -4m(8m^2 - 13m + 16) = -4my^2 < 0. \tag{76}$$

The real root lies in $(a, a+1)$; the other two roots have common modulus $\beta^{-1/2} < 1$. Thus every member is a cubic Pisot unit.

This ray is useful because a natural quadratic-character obstruction does not exclude it. Modulo $a + 1 = 2m$,

$$X^3 - aX^2 - X - 1 \equiv (X+1)(X^2 - 1), \qquad T_n \equiv -1 \pmod{2m} \quad (n \text{ odd}).$$

For scheduled $n \equiv 1 \pmod 4$, Newton's recurrence also gives $T_n \equiv a \equiv 1 \pmod 4$. Since $m \equiv 3 \pmod 8$, reciprocity therefore gives, for positive such traces,

$$\left(\frac{-m}{T_n}\right) = \left(\frac{T_n}{m}\right) = \left(\frac{-1}{m}\right) = -1. \tag{77}$$

The symbol in (77) is the primitive quadratic-field character attached to the square class $-m$; the coprimality needed here follows from $T_n \equiv -1 \pmod m$. The earlier projective prime-return test requires this negative sign for eventual prime traces in the complex unit branch $a \equiv 2 \pmod 3$ [10]. Thus the ray survives that particular necessary test. This says nothing about primes dividing the discarded square factor y^2: the full polynomial-discriminant symbol can be zero there. It also says nothing by itself about primality or minimality.

8.4.2 Seven exact return certificates

For a prime p and coefficient residue a_0, work in

$$R = \mathbb{F}_p[X]/(X^3 - a_0X^2 - X - 1), \qquad x = X \bmod f.$$

The element x is a unit with $x^{-1} = x^2 - a_0x - 1$. Its algebra trace is

$$\tau(v_0 + v_1x + v_2x^2) = 3v_0 + a_0v_1 + (a_0^2 + 2)v_2. \tag{78}$$

For a seed k_0, put $z = x^{2\cdot 3^{k_0}}$. The two finite identities

$$\tau(x^{-1}z) = 0, \qquad z^{3^H} = z \tag{79}$$

imply $p \mid T_{2\cdot 3^{k_0+jH}-1}$ for every $j \geq 0$. This remains valid for non-squarefree reductions: multiplication trace in the quotient algebra gives the usual power sum with multiplicities.

Table 10: A complete Pell-index cover for Theorem 8.2

p	condition on t	$a_0 \pmod p$	k_0	H	$z = (z_0, z_1, z_2)$
5	$0 \pmod 3$	0	1	2	$(1, 2, 1)$
5	$1 \pmod 3$	1	10	30	$(4, 2, 3)$
19	$0 \pmod 5$	5	4	126	$(1, 11, 5)$
19	$2 \pmod 5$	15	9	126	$(18, 1, 1)$
29	$3 \pmod 5$	14	7	66	$(22, 6, 21)$
31	$11 \pmod{15}$	23	8	330	$(23, 13, 2)$
59	$4 \pmod 5$	23	6	28	$(0, 10, 50)$

All entries in Table 10 satisfy (79). For completeness, the coefficient cycles from the initial Pell state modulo the five primes are

$$\begin{aligned}
p = 5 &: 0, 1, 4; \\
p = 19 &: 5, 4, 15, 18, 11; \\
p = 29 &: 5, 16, 20, 14, 17; \\
p = 31 &: 5, 18, 21, 29, 20, 11, 19, 22, 4, 30, 13, 23, 17, 27, 10; \\
p = 59 &: 5, 12, 25, 34, 23.
\end{aligned}$$

The full (m, y) states return with periods $3, 5, 5, 15, 5$, respectively; one need not infer a state period from an accidentally repeated coefficient. These small cycles and the seven quotient-ring identities are finite exact certificates. A separate matrix implementation checks

$$M = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & a_0 \end{pmatrix}, \qquad \operatorname{tr}(M^{2\cdot 3^{k_0}-1}) = 0, \quad M^{2\cdot 3^{k_0}(3^H-1)} = I \pmod p. \tag{80}$$

Binary exponentiation makes these checks short even when $H = 330$; no enormous integer trace is factored.

Proof of Theorem 8.2. The first two rows cover $t \not\equiv 2 \pmod 3$. Of the remaining indices, the rows for $19, 29, 59$ cover the four residues $0, 2, 3, 4 \pmod 5$. The only uncovered condition is $t \equiv 2 \pmod 3$, $t \equiv 1 \pmod 5$, namely $t \equiv 11 \pmod{15}$, and the row for 31 covers it. Thus a fixed prime from the table divides infinitely many scheduled traces for every fixed t. Those traces tend to infinity and are eventually larger than that prime, so they are composite.

To pass from traces to floors without a phase assumption, note that all traces are odd: $T_0 = 3$, $T_1 = a_t$, $T_2 = a_t^2 + 2$ are odd and $T_n = a_t T_{n-1} + T_{n-2} + T_{n-3}$ preserves oddness. For $n \geq 5$,

$$|S_n| \leq 2\beta_t^{-n/2} < 1, \qquad \lfloor \beta_t^n \rfloor = \begin{cases} T_n, & S_n \leq 0, \\ T_n - 1, & S_n > 0. \end{cases}$$

At each sufficiently large certified return, the first option exceeds its fixed prime divisor and is composite. The second is an even integer greater than two and is composite as well. Hence infinitely many scheduled floors are composite, regardless of the sign of S_n. □

8.4.3 The totally real reflected family

Corollary 8.8. *Start at* $(m_0, y_0) = (165, 469)$ *and iterate*

$$\begin{pmatrix} m_{t+1} \\ y_{t+1} \end{pmatrix} = \begin{pmatrix} 577 & 204 \\ 1632 & 577 \end{pmatrix} \begin{pmatrix} m_t \\ y_t \end{pmatrix} + \begin{pmatrix} 468 \\ 1326 \end{pmatrix}, \qquad a_t = 2m_t + 1.$$

For every $t \geq 0$, *the Pisot root of* $X^3 - a_t X^2 - X + 1$ *has infinitely many composite shifted floors. The same five primes as in Theorem 8.2 suffice; the coefficients begin* $331, 382699, \ldots$.

Proof. Extend the complex recurrence (64) to all integer indices; its affine matrix is integrally invertible. Its index -1 state is $(-165, 469)$. Reversing the recurrence and changing (m, y) to $(-m, y)$ gives the displayed real recurrence, and hence

$$a_t^{\mathrm{R}} = -a_{-1-t}^{\mathrm{C}}.$$

Writing $f_a^{\mathrm{C}}(X) = X^3 - aX^2 - X - 1$, the identity $X^3 - aX^2 - X + 1 = -f_{-a}^{\mathrm{C}}(-X)$ shows that every odd real trace is the negative of the corresponding complex trace. The modular state cycles in Table 10 hold also at negative indices, since their maps are invertible. Reflect each Pell-index class $r \pmod P$ to $-1 - r \pmod P$ and negate its coefficient residue. The same seven return identities therefore give a complete real cover modulo 15.

For $a \geq 331$, polynomial signs place the roots in $(-1, 0), (0, 1), (a, a + 1)$, proving irreducibility and the Pisot property. Writing the small roots as $-\rho, \sigma$ gives $0 < \sigma < \rho < 1$, since $\beta > a$. At every positive odd n, $0 < \beta^n - T_n = \rho^n - \sigma^n < 1$, so the floor equals the trace. Every recurring divisor thus supplies infinitely many composite floors. The reflected seven identities are also checked directly by the independent polynomial and matrix verifiers, without relying on the reflection argument. □

8.4.4 Why a finite fixed-prime cover cannot solve every Pell ray

The following obstruction is structural, not the outcome of a capped search. For $s \in \{1, -1\}$, start at $(m_0, y_0) = (0, 4)$ and set

$$\begin{pmatrix} m_{t+1} \\ y_{t+1} \end{pmatrix} = \begin{pmatrix} 577 & 204 \\ 1632 & 577 \end{pmatrix} \begin{pmatrix} m_t \\ y_t \end{pmatrix} - s \begin{pmatrix} 468 \\ 1326 \end{pmatrix}, \qquad a_t = 2m_t - s. \tag{81}$$

For $s = 1$ this is another complex ray with positive-index coefficients $695, 801311, \ldots$; for $s = -1$ it is a real ray with coefficients $2569, 2965345, \ldots$. The index-zero polynomials are reducible and are discarded. Every positive-index member is an irreducible cubic Pisot unit of the indicated type. The Pell identities are respectively $8m^2 - 13m + 16 = y^2$ and $8m^2 + 13m + 16 = y^2$.

Proposition 8.9. *For either ray (81) and every finite set of primes S, there are infinitely many positive indices t such that*

$$T_n \equiv -s \pmod{p} \qquad (p \in S, n \text{ any positive odd integer}).$$

In particular, no finite set of fixed recurring prime divisors covers all members of either ray.

Proof. The affine Pell map has linear determinant $577^2 - 204 \cdot 1632 = 1$. Modulo each prime it is a permutation of the finite state space, so its orbit from $(0, 4)$ is purely periodic. Let L be a common multiple of these periods for $p \in S$. At every positive multiple t of L, one has $m_t \equiv 0 \pmod{p}$ and $a_t \equiv -s \pmod{p}$. Therefore

$$X^3 - a_t X^2 - X - s \equiv (X + s)(X^2 - 1) \pmod{p}.$$

Its odd power sum, with multiplicities, is $-s + 1 - 1 = -s$. This is nonzero modulo every prime, proving the assertion. □

For the real ray this obstruction occurs even inside the full shrinking-conjugate region. If $-\rho$ is the larger small root,

$$\rho < \beta^{-1/2} + \beta^{-1} < \beta^{-17/40} \quad (\beta > 2569),$$

since $1 + \beta^{-1/2} < 17/16 < 3/2 < \beta^{3/40}$. It follows neither that any trace is prime nor that any member is minimal. Adaptive primes, noncongruence arguments, and global competitors remain possible. Theorem 8.2 and Corollary 8.8 solve their two specified rays, not all Pell rays or all unit candidates.

8.5 An exact bounded theorem for the classical schedule

8.5.1 The coefficient locus in both directions

Use the convention

$$f(X) = X^3 - aX^2 - BX - c, \qquad q = |c| \geq 1, \tag{82}$$

so c is the signed norm. Put

$$D = a^2B^2 + 4B^3 - 4a^3c - 27c^2 - 18aBc. \tag{83}$$

Theorem 8.10. *The polynomial (82) has roots $\beta > 1, -\rho, \gamma$, with $0 < |\gamma| < \rho < 1$, if and only if a, B, q are positive integers and one of the following holds:*

(i) $c = -q$ *and* $a \geq B + q$;

(ii) $c = q$, $D > 0$, $B \leq a + q$, *and* $B < 2a + 3$.

It is then irreducible, and $a < \beta < a + 2$. Among these polynomials, membership in $\mathcal{S}$ is equivalent to

$$\rho^{40}\beta^{17} < 1. \tag{84}$$

Writing $\delta = \beta - a$ and $z = \beta^{1/40}$, an equivalent test using only the large root is

$$\begin{aligned} \delta &= B/\beta + c/\beta^2, \qquad \rho = \frac{\delta + \sqrt{\delta^2 - 4c/\beta}}{2}, \\ z^{17}(z^{40} - a) &< 2, \qquad c + z^6 + az^{23} - z^{63} > 0. \end{aligned} \tag{85}$$

Proof. For negative norm, the stated inequality gives $f(-1) < 0 < f(0)$ and $f(1) < 0$, placing one root in each of $(-1, 0), (0, 1), (1, \infty)$. Also $f(a) = -aB + q < 0$, so $\beta > a$ and the negative small root has larger modulus. Conversely that configuration gives $B = \beta(\rho - \gamma) + \rho\gamma > 0$ and $f(-1) < 0$, hence $a \geq B + q$ by integrality.

For positive norm the two small roots must be negative. The negative critical point $u = (a - \sqrt{a^2 + 3B})/3$ lies in $(-1, 0)$ exactly when $B < 2a + 3$. Positive discriminant gives a positive local maximum at u. The additional inequality $B \leq a + q$ is equivalent to $f(-1) < 0$; together with $f(0) < 0$, these locate both negative roots in $(-1, 0)$. The positive root exceeds one and a. Conversely the same signs and critical-point location follow from the root configuration. The formula $B = \beta(\rho + \sigma) - \rho\sigma$ is positive.

There is no missing zero-trace case. In the positive-norm configuration, $a = \beta - \rho - \sigma > -1$, so $a \geq 0$. If $a = 0$, then $\beta < 2$, whence $q = 1$ and $B \leq 1$, giving $D = 4B^3 - 27 < 0$, a contradiction. In the negative-norm case $a > 0$ directly. Reducibility of a monic cubic would give an integer root; neither small nonzero root is integral. If β were that integer root, the remaining monic integral quadratic would have nonzero constant term of absolute value below one, also impossible. Finally $\beta - a = \rho - \gamma \in (0, 2)$.

We recall the norm/redundancy argument from [6]. If $\rho \leq \beta^{-17/40}$, then

$$1 \leq q = \beta\rho\sigma < \beta\rho^2 \leq \beta^{3/20}, \qquad \rho^{40} \leq \beta^{-17} \leq q^{17}\beta^{-17} = \rho^{17}\sigma^{17}.$$

Thus $\rho \leq \sigma^{17/23}$ follows. Equality $\rho^{40}\beta^{17} = 1$ is impossible: in the totally real splitting field, apply an automorphism sending β to its negative conjugate in $\beta^{17}(-\rho)^{40} = 1$. The transformed left side is negative. This proves (84).

The two small roots give the displayed quadratic formula for ρ. For $R = \beta^{-17/40}$, the larger quadratic root is below R if and only if $R > \delta/2$ and $R^2 - \delta R + c/\beta > 0$. Substituting $\beta = z^{40}$ gives the two inequalities in (85). Both are needed to select the correct side of the quadratic. □

8.5.2 Exhaustiveness and exact arithmetic

Every member of $\mathcal{S}$ satisfies the necessary bounds

$$q^{20} < (a + 2)^3, \qquad B^{40} < 2^{40}(a + 2)^{23}. \tag{86}$$

The first follows from $q < \beta^{3/20}$ and $\beta < a + 2$. For positive norm, $B < 2\beta\rho < 2\beta^{23/40}$; for negative norm the sharper $B < \beta\rho < \beta^{23/40}$ holds. Hence (86) bounds *every* possible B and q for a specified trace a, not just parameters chosen for a scan.

The independent verifier enumerates all triples satisfying (86) with $1 \leq a \leq 1000$, then applies the exact integer root-locus conditions of Theorem 8.10. It isolates $\beta \in (a, a + 2)$ using integer polynomial signs. For ρ it uses

$$g(t) = -f(-t) = t^3 + at^2 - Bt + c.$$

When $c < 0$, g has one positive root in $(0, 1)$. When $c > 0$, exact bisection of $g'(t) = 3t^2 + 2at - B$ first identifies its positive minimum; the larger zero is then isolated on the increasing branch. This avoids confusing the two positive roots of g.

With denominator $d = 2^{32}$, each bracket $b_l/d < \beta < b_h/d$, $r_l/d < \rho < r_h/d$ is classified by one of

$$r_h^{40} b_h^{17} < d^{57}, \qquad r_l^{40} b_l^{17} > d^{57}. \tag{87}$$

The first certifies membership and the second nonmembership. Every one of the $166,052$ root-locus candidates is decided at this precision; there are no unclassified boundary cases.

The implementation permits adaptive refinement, although none is needed here. All arithmetic is integral, including the root isolation; floating-point proposals are not an input to this independent verification.

Table 11: Exact counts in the bounded classical locus

Stage or signed norm	Number of triples
Coefficient bounds with $1 \le a \le 1000$	265,126
Exact negative-dominant real root locus	166,052
Shrinking locus $\mathcal{S}$	32,232
$c = -1$ in $\mathcal{S}$	20,018
$c = 1$ in $\mathcal{S}$	4,759
$c = -2$ in $\mathcal{S}$	6,936
$c = 2$ in $\mathcal{S}$	519

8.5.3 A recurrent divisor is a finite certificate of infinitely many floors

For a prime p, work in the finite ring

$$R_p = \mathbb{F}_p[X]/(X^3 - aX^2 - BX - c), \qquad u_0 = X, \quad u_{r+1} = u_r^3.$$

The linear algebra trace is

$$\tau(v_0 + v_1X + v_2X^2) = 3v_0 + av_1 + (a^2 + 2B)v_2 \pmod{p}. \tag{88}$$

Therefore $\tau(u_r) \equiv V_r \pmod{p}$. Iterate until an exact polynomial state repeats. If a zero trace occurs on the eventual cycle, it occurs on an infinite arithmetic progression of indices. A transient zero alone is not enough. This argument works also when $p \mid cD$; neither invertibility nor squarefreeness is assumed.

This verification is independent of the original symmetric Newton map, whose state transformation is

$$(T, U, C) \longmapsto (T^3 - 3TU + 3C,\ U^3 - 3TUC + 3C^2,\ C^3).$$

The independent implementation uses direct polynomial multiplication and reduction, rather than importing that graph calculation.

Table 12: Recurring-divisor certificates for all 32,232 triples

p	Newly certified	Remaining	p	Newly certified	Remaining
2	18,105	14,127	29	34	35
3	4,711	9,416	31	15	20
5	5,196	4,220	37	11	9
7	2,583	1,637	41	4	5
11	834	803	43	3	2
13	411	392	47	1	1
17	165	227	53	0	1
19	73	154	59	1	0
23	85	69			

The complete accepted coefficient set, not merely its cardinality, agrees with the original enumeration. Table 12 records the successive removals, read down the left half and then the right half. The final two certificates are particularly compact:

$$\begin{aligned}(a,B,c)=(917,63,1),\quad p=47:\quad 47\mid V_{8+88j},\\(a,B,c)=(845,65,1),\quad p=59:\quad 59\mid V_{3+28j},\end{aligned}\qquad j\ge 0.$$

Both cycles begin at index 1. Another illustrative case is $(a,B,c)=(211,7,-1)$, for which both 7 and 31 divide V_{2+4j} for all $j\ge 0$. Thus nonreturn of a selected prime seed, as in the earlier example in [031], does not prevent a different prime from giving infinitely recurring divisibility.

Lemma 8.11. *Every $\beta\in\mathcal{S}$ satisfies $\beta>2$, and*

$$\left\lfloor \beta^{3^r}\right\rfloor = V_r \qquad (r\ge 1).$$

For $r\ge 2$ this integer is at least 512.

Proof. For negative norm, $a\ge B+q\ge 2$ and $\beta>a$. For positive norm, if $\beta\le 2$, then $a=1$ and $\rho+\sigma=\beta-1$, implying

$$1\le q=\beta\rho\sigma\le\frac{\beta(\beta-1)^2}{4}\le\frac{1}{2},$$

a contradiction. For $M=3^r$, oddness of M and negative dominance give

$$0<\beta^M-V_r=\rho^M-\gamma^M<2\rho^M<2\cdot 2^{-51/40}<1\quad(r\ge 1).$$

The floor identity follows. For $r\ge 2$, $\beta^{3^r}>2^9=512$. □

Proof of Theorem 8.3. The coefficient bounds and exact tests above enumerate the locus exhaustively for the stated trace range. The associated finite data provide one recurrent-divisor certificate for each of its 32,232 triples. The finite-ring cycle argument proves divisibility on the recorded infinite progression. By Lemma 8.11, every certified return with $r\ge 2$ is a floor equal to $V_r\ge 512$, and hence is strictly larger than its divisor $p\le 59$. It is therefore composite. Every progression has infinitely many such indices, proving the theorem. □

8.6 Reproducibility and the remaining problems

8.6.1 Finite evidence and its exact role

The accompanying supplement contains the following proof artifacts. They are ordinary Python 3 programs and JSON data; the theorem statements do not rely on exploratory scans or probabilistic primality.

- `minimality_verify.py` checks the fifth-shadow identities, rational inequalities, and leading cancellation terms used in Section 8.3. The analytic prime theorem and the infinite continuation are proved or cited in the text, not inferred from this finite program.
- `pell_cover_certificate.json`, `pell_cover_real_certificate.json`, and `pell_cover_verify.py` contain and verify the seven matrix certificates for each ray, the Pell invariants, and both covers of all 15 coefficient-index residue classes. `discriminant_pell_cover_audit.py` independently checks the 14 return identities by polynomial multiplication.

- `totreal_independent_verify.py` performs the integer-only root isolation and quotient-ring cycle verification in Section 8.5. Its output `totreal_independent_verify.json` records one recurrent-divisor certificate for every accepted triple. The corresponding log records the counts in Tables 11 and 12.

The SHA-256 digest of the independent totally real verifier is

```
6a514e149ec38f5203daa92bd925a54eaa2955746676bdca013f501cdb7a1bfb
```

Running the Pell verifier and then the totally real verifier reproduces the two finite certificate checks. The larger JSON file is supplemental data rather than a printed code appendix. The mathematical proofs explain why each finite identity entails an infinite conclusion.

8.6.2 What remains open after these results

For the shifted clock, the exact global criterion is least viable choice after *every* prefix; the exact local criterion must also keep the truncated boundary child. The first-window theorem turns a usable numerical prime-gap threshold into a numerical height bound, but no such value is certified in this report. An independent small admissible base with a proved infinite continuation could also supply a useful bound. A long finite all-prime prefix is not a substitute.

The Pell theorem and its real reflection remove two specified unbounded families without minimality or phase assumptions. Proposition 8.9 shows why one cannot extrapolate its five-prime cover to every boundary-based ray. It leaves open adaptive recurring-divisor arguments, phase arguments, and comparisons with smaller admissible bases.

For the classical schedule, Saito's Problem 1.1 asks for infinitely many composite cubic Pisot floors at powers 3^r [9]. His theorem for iterated reversible recurrences does not cover the inner recurrence 3^r. Theorem 8.3 settles the specified bounded part of the shrinking locus, but every trace coefficient $a > 1000$ remains outside its scope. The earlier fixed-norm density-one theorem in [031] is stronger asymptotic background, yet density one does not remove a possibly nonempty exceptional family. No finite trace bound for a hypothetical algebraic classical Mills shadow is proved here.

Finally, universal compositeness of cubic Pisot floors is a sufficient route to transcendence of the least Mills constant, not an established equivalent formulation. An algebraic prime-generating branch need not be the least branch. Keeping that distinction is precisely why the global-minimum argument of Section 8.3 and the unconditional family exclusions are stated separately.

9 Finite prime forests and explicit continuation criteria

The preceding cells give genuine finite forests, but a large forest alone does not prove infinite survival. This section records the quantitative gap-budget conditions that would provide that continuation and identifies the missing effective estimate.

9.1 The finite and infinite questions

Put

$$n_k = 2 \cdot 3^k - 1 \quad (k \geq 1), \qquad n_1 = 5, \quad n_2 = 17, \quad n_3 = 53, \quad n_{k+1} = 3n_k + 2. \tag{89}$$

Write ξ_2 for the least member of

$$\mathcal{W}_2 = \{A > 1 : \ \lfloor A^{n_k} \rfloor \in \mathbb{P} \text{ for every } k \geq 1\}.$$

The existence of this minimum follows from [4, Theorem 1.3], since $n_{k+1} \geq 2n_k$. We seek a small member of $\mathcal{W}_2$, not merely a long finite sequence of prime floors.

For a prime p, the half-open interval

$$[p^{1/n}, (p+1)^{1/n})$$

is the prime cell at exponent n. Its closed envelope is denoted $J_n(p) = [p^{1/n}, (p+1)^{1/n}]$. A cell is *reachable* if its closed envelope is nested in a chain of cells at the preceding required exponents. In our finite constructions the containments in earlier half-open cells are strict.

Let $p_1, p_2, \ldots$ be the consecutive primes. Throughout the paper,

$$M_2(X) = \sum_{p_j \leq X} (p_{j+1} - p_j)^2. \tag{90}$$

The gap crossing X is included. This convention is essential in the interval-counting argument below.

Theorem 9.1 (Unconditional finite forests)**.** *The schedule* (89) *has the following finite forests.*

(a) *Starting with the first prime floor* 31*, there are* 1151 *strictly internal full cells at exponent* 17*, and at least*

$$N_0 = 149\,564\,695\,915$$

distinct reachable full cells at exponent 53*. All their bases lie in* $[31^{1/5}, 131064^{1/17}] \subset (1, 2)$*.*

(b) *A forest with* 4332 *first prime labels, from* 241361 *through* 295703*, has at least*

$$N_{\mathrm{B}} = 107\,204\,570\,624\,222$$

reachable full cells at exponent 17*, all with bases in* $[241361^{1/5}, 295704^{1/5}] \subset (1, 25/2)$*.*

(c) *Restricting the second forest to its* 1865 *first prime labels at least* 271927 *leaves at least*

$$N_{\mathrm{O}} = 52\,537\,179\,616\,066$$

full cells, with base lower endpoint $271927^{1/5}$*. Separately, the second forest contains at least*

$$N_{\mathrm{U}} = 90\,210\,603\,204\,803$$

full cells whose prime labels at exponent 17 *lie in* $[2^{61}, 2^{62})$*.*

This theorem uses the published finite bound 1476 for prime gaps in its verified range [14]. It does not assert that any of its cells has descendants at every subsequent level.

Theorem 9.2 (Explicit conditional continuations)**.** *Each row below is a sufficient hypothesis for its stated conclusion. Here the hypothesis is* $M_2(X) \leq CX^{\tau}$ *for every* $X \geq X_0$*.*

C	τ	X_0	*Conclusion*
$6 \cdot 10^{15}$	$31/25$	2^{161}	$\xi_2 < 2$
$2 \cdot 10^{9}$	$4/3$	2^{161}	$\xi_2 < 2$
$15 \cdot 10^{19}$	$31/25$	$(25/2)^{53}$	$\xi_2 < 25/2$
$15 \cdot 10^{11}$	$4/3$	$(25/2)^{53}$	$\xi_2 < 25/2$

None of these numerical moment hypotheses is established in this paper.

Theorem 9.3 (Conditional long-gap alternatives)**.** *Put* $X_0 = (25/2)^{53}$ *and* $G(X) = X^{2/3}/64$. *Either of the estimates*

$$\sum_{\substack{p_j \le X \\ p_{j+1}-p_j \ge G(X)}} (p_{j+1} - p_j) \le 10^{13} X^{2/3},$$
$$\sum_{\substack{p_j \le X \\ p_{j+1}-p_j \ge G(X)}} (p_{j+1} - p_j)^2 \le 7.5 \cdot 10^{11} X^{4/3},$$

if valid for every $X \ge X_0$, *implies* $\xi_2 < 25/2$. *These are alternative sufficient hypotheses, not claimed equivalent to one another or to the full-moment hypotheses. Neither is proved here.*

The finite forests and the implications in Theorems 9.2 and 9.3 are the results proved here. An unspecified asymptotic constant, a sufficiently-large threshold, or a finite prime prefix is not a substitute for the displayed hypotheses. The later sections identify concrete issues in applying known prime-gap results at these particular scales.

9.2 Certified finite forests

9.2.1 A finite prime-gap counting principle

Oliveira e Silva, Herzog and Pardi computed the consecutive-prime gaps below $4 \cdot 10^{18}$, with maximum 1476 [14, Section 2.2 and Table 8]. The upper endpoint of a gap must lie within the computed range when invoking this datum. Suppose an integer interval $[A, B]$ lies in a range for which all enclosing consecutive-prime gaps are covered. Subdivide $(A, B]$ into $\lfloor (B - A)/1476 \rfloor$ disjoint blocks of length 1476, ignoring a final remainder. Every such block $(u, u + 1476]$ contains a prime: the last prime at or below u has its successor no farther than 1476 away. Hence

$$\#([A, B] \cap \mathbb{P}) \ge \left\lfloor \frac{B - A}{1476} \right\rfloor. \tag{91}$$

This is a conservative count, not an enumeration of all primes in these very large intervals.

If $m > n$ are consecutive required exponents, define

$$A_{p;n,m} = \left\lfloor p^{m/n} \right\rfloor + 1, \qquad B_{p;n,m} = \left\lfloor (p+1)^{m/n} \right\rfloor - 1. \tag{92}$$

A prime q between these endpoints supplies a full child cell. All endpoint evaluations below use integer roots: for instance $\left\lfloor p^{m/n} \right\rfloor$ is the unique integer z satisfying $z^n \le p^m < (z+1)^n$. The supplied data also verify the strict upper inequality $(B_{p;n,m} + 1)^n < (p+1)^m$ for every retained row. The intervals for distinct parent primes are pairwise disjoint.

9.2.2 The seed with first floor 31

Let $a_0 = 31^{1/5}$. Exact arithmetic gives

$$\left\lceil 31^{17/5} \right\rceil = 117661.$$

There are exactly 1152 primes between 117661 and $2^{17} - 1$; the first is 117671 and the last is 131071. Their closed exponent-17 cells lie in $[a_0, 2]$. The last cell has upper endpoint 2 and is not contained in the half-open first cell $[31^{1/5}, 2)$, so it is discarded. The remaining 1151 primes end at 131063, and all retained closed cells lie in

$$[a_0, b_0], \qquad b_0 = 131064^{1/17} < 2. \tag{93}$$

For every retained prime q, apply (92) with $(n, m) = (17, 53)$. The smallest and largest integer endpoints across these rows are

$$6435135249309836 \quad \text{and} \quad 9005485420906690.$$

They are below 2^{53}. Bertrand's postulate supplies a prime above any such endpoint and below $2^{54} < 4 \cdot 10^{18}$, so both endpoints of every enclosing prime gap lie in the verified finite range. Formula (91) now gives the exact certified lower count

$$\sum_{\substack{q \in \mathbb{P} \\ 117661 \le q \le 131063}} \left\lfloor \frac{B_{q;17,53} - A_{q;17,53}}{1476} \right\rfloor = N_0 = 149564695915. \tag{94}$$

These are distinct reachable cells. No individual primality claims about 149 billion listed integers are being made; the count is a consequence of the verified maximum-gap result and exact endpoints.

9.2.3 The larger and restricted forests

Put $H = 399 \cdot 10^{16} = 3.99 \cdot 10^{18}$, and let $\mathcal{P}$ be the primes in

$$\left\lceil (2 \cdot 10^{18})^{5/17} \right\rceil \le p \le \left\lfloor H^{5/17} \right\rfloor - 1. \tag{95}$$

The integer limits are 241355 and 295714. There are 4332 such primes, with first 241361 and last 295703. For $p \in \mathcal{P}$, set $A_p = A_{p;5,17}$ and $B_p = B_{p;5,17}$. The endpoints range from

$$2000175614492238064 \quad \text{to} \quad 3989463330471287939,$$

and every row satisfies $2 \cdot 10^{18} < A_p < B_p < H$.

To justify the rightmost enclosing prime, use the corrected Cully-Hugill–Lee interval theorem [12, Theorem 1 and Table 2]. At $x = 4 \cdot 10^{18}$ its parameter $\Delta = 39097000$ supplies a prime in $(x(1 - 1/\Delta), x]$. The exact inequality

$$4 \cdot 10^{18}(39097000 - 1) > H \cdot 39097000 \tag{96}$$

places this prime above H. Since $4 \cdot 10^{18}$ is composite, the prime is strictly below the finite computation's upper limit. Thus the gap bound 1476 applies to every block used here. Consequently

$$\sum_{p \in \mathcal{P}} \left\lfloor \frac{B_p - A_p}{1476} \right\rfloor = N_{\mathrm{B}} = 107204570624222. \tag{97}$$

All bases lie between $241361^{1/5}$ and $295704^{1/5}$; the inequality $32 \cdot 295704 < 25^5$ proves that the latter is below $25/2$.

For the fixed-window improvement, retain only first primes

$$p \ge \left\lceil (3 \cdot 10^{18})^{5/17} \right\rceil = 271924.$$

This subset $\mathcal{P}_{\mathrm{O}}$ has 1865 members, from 271927 through 295703, and

$$\sum_{p \in \mathcal{P}_{\mathrm{O}}} \left\lfloor \frac{B_p - A_p}{1476} \right\rfloor = N_{\mathrm{O}} = 52537179616066. \tag{98}$$

Its larger lower base, $271927^{1/5}$, improves the later moment budget more than the lost nodes diminish it.

For a different restriction, clip every large-forest integer interval to a dyadic band. Define

$$L_{p,j} = \max(A_p, 2^j), \qquad U_{p,j} = \min(B_p, 2^{j+1} - 1), \qquad j \in \{60, 61\}.$$

Counting blocks only when $L_{p,j} < U_{p,j}$ gives

$$\sum_{p \in \mathcal{P}} \max\left(0, \left\lfloor \frac{U_{p,j} - L_{p,j}}{1476} \right\rfloor\right) = \begin{cases} 16993967419419, & j = 60, \\ 90210603204803, & j = 61. \end{cases} \tag{99}$$

The latter is N_{U}. These counts can lose short blocks at the band boundary, so they are not inferred by partitioning the already rounded total (97); they are recomputed from clipped endpoints. Equations (94), (97), (98) and (99), with their exact endpoint tests, prove Theorem 9.1.

9.3 A gap budget for a whole generation

Lemma 9.4 (Clipped-gap Cauchy inequality)**.** *Let $I_1, \dots, I_N$ be real intervals in $[2, X]$ with disjoint interiors. Let L be their total length, and let R count the primes in their interiors. Then*

$$R \geq \frac{L^2}{M_2(X)} - N. \tag{100}$$

In particular, if all intervals have length at least ℓ, then $M_2(X) \leq N\ell^2/2$ implies $R \geq N$.

Proof. Split each interval at its interior primes. There are $R + N$ resulting segments, with total length L. Each segment lies in a gap between consecutive primes. If one full gap meets several intervals, the corresponding clipped segments have disjoint interiors; the sum of their squared lengths is no greater than the square of the full gap. All gaps used have left endpoint at most X, including a possible gap crossing X. Thus the sum of all segment-length squares is at most $M_2(X)$. Cauchy–Schwarz gives $L^2 \leq (R + N)M_2(X)$, proving the result. □

At exponent n, write $d = (3n + 2)/n$. For each reachable prime parent p consider the real interval

$$I_p = [p^d, (p + 1)^d - 1]. \tag{101}$$

The intervals belonging to distinct parents are disjoint. We use only primes q in their *interiors*. Then

$$p^d < q, \qquad q + 1 < (p + 1)^d,$$

so $J_{3n+2}(q)$ is strictly inside the parent prime cell. This endpoint convention gives strict nesting directly, without needing to rule out integral values of the real endpoints.

Lemma 9.5 (Passage to a base)**.** *Suppose a finite reachable frontier consists of full cells in a fixed interval $[a, b_*]$, where $1 < a < b_* < b$. If at each subsequent level one can retain a positive fixed number of cells, each strictly inside a retained parent, then $\mathcal{W}_2$ contains a number below b, and therefore $\xi_2 < b$.*

Proof. The retained cells form a finitely branching forest with finitely many roots and nodes at every depth. König's lemma gives an infinite branch. Its nested closed cells lie in $[a, b_*]$. Their diameters tend to zero, since the mean value theorem gives, for $p \geq a^n$,

$$(p + 1)^{1/n} - p^{1/n} \leq \frac{1}{na^{n-1}}.$$

Their unique common point lies in the interior of every earlier cell, because the next cell is strictly internal. It has all the required prime floors and belongs to $[a, b_*] \subset (1, b)$. □

9.4 Continuation in a fixed base window

Theorem 9.6 (Fixed-window criterion)**.** *Suppose a reachable frontier at exponent n_* has at least N_* full cells in $[a, b_*]$, where $1 < a < b_* < b$. Let $C > 0$ and $\tau > 0$. The three conditions*

$$M_2(X) \le CX^\tau \qquad (X \ge b^{3n_*+2}), \tag{102}$$

$$a^4 \ge b^{3\tau}, \tag{103}$$

$$C \le \frac{2N_* a^{4n_*+4}}{b^{\tau(3n_*+2)}} \tag{104}$$

imply $\xi_2 < b$.

Proof. Retain N_* parents at the starting level. At any later level n, let $N \ge N_*$ retained parents have bases in the same window. Their child intervals (101) lie below $X = b^{3n+2}$. Convexity and $p \ge a^n$ give

$$|I_p| \ge dp^{d-1} - 1 \ge 3a^{2n+2} - 1 \ge 2a^{2n+2}. \tag{105}$$

Lemma 9.4 therefore gives

$$R \ge \frac{4N^2 a^{4n+4}}{Cb^{\tau(3n+2)}} - N.$$

This is at least N whenever $C \le 2N_* a^{4n+4}/b^{\tau(3n+2)}$. Condition (103) makes that right-hand side nondecreasing in n, and (104) verifies it initially. We may retain N_* children and repeat indefinitely. Lemma 9.5 completes the proof. □

For the seed 31, use $(a, b, n_*, N_*) = (31^{1/5}, 2, 53, N_0)$ and $\tau = 31/25$. The exact integer checks are

$$\begin{gathered}(6 \cdot 10^{15})^{25} 2^{31\cdot 161} < (2N_0)^{25} 31^{1080}, \\ 31^{20} > 2^{93}.\end{gathered} \tag{106}$$

They prove the first implication in Theorem 9.2. The coefficient ceiling in (104) is approximately $6.3834568534 \cdot 10^{15}$; this decimal is descriptive, not a proof step. More generally this forest works for

$$\tau < \frac{4 \log 31}{15 \log 2} = 1.3211190161\ldots, \qquad C \le \frac{2N_0 31^{216/5}}{2^{161\tau}}.$$

For the larger restricted forest, use

$$(a, b, n_*, N_*) = (271927^{1/5}, 25/2, 17, N_{\mathrm{O}}).$$

The checks for $C = 15 \cdot 10^{19}$ and $\tau = 31/25$ are

$$\begin{gathered}(15 \cdot 10^{19})^{25} 25^{31\cdot 53} < (2N_{\mathrm{O}})^{25} 271927^{360} 2^{31\cdot 53}, \\ 271927^{20} 2^{93} > 25^{93}.\end{gathered} \tag{107}$$

Thus the third implication in Theorem 9.2 holds. The precise coefficient ceiling is approximately $1.5441314472 \cdot 10^{20}$, and $\log_{10}((25/2)^{53}) = 58.13623\ldots$. The hypothesis is needed only at the scheduled values $b^{n_{k+1}}$, but the all-X form is convenient and sufficient.

9.5 Dyadic concentration at exponent four thirds

The exponent restriction of the fixed-window criterion is not intrinsic. Concentrating the children into a dyadic band gives a different continuation theorem at the critical exponent 4/3.

Theorem 9.7 (Dyadic criterion)**.** *Suppose at exponent $n_* \ge 17$ a reachable frontier contains at least N_* full cells with prime labels in $[Y_*, 2Y_*)$, where $Y_* \ge 4$ is a power of two. Suppose all these cells have bases in $[a, b_*] \subset (1, b)$ and $N_* \ge 60C$. If*

$$M_2(X) \le CX^{4/3} \qquad (X \ge X_0)$$

and $(2Y_* + 1)^{(3n_*+2)/n_*} \ge X_0$, then $\xi_2 < b$.

Proof. At any stage let $N \ge N_*$ retained parent labels lie in $[Y, 2Y)$, and set $d = (3n + 2)/n$ and $X = (2Y + 1)^d$. Since $n \ge 17$, we have $3 < d \le 53/17 < 13/4$. Every child interval has length at least $2Y^{d-1}$. Moreover

$$\frac{4Y^{2d-2}}{X^{4/3}} = \frac{4Y^{2d/3-2}}{(2 + 1/Y)^{4d/3}} > \frac{1}{10}. \tag{108}$$

Indeed $Y^{2d/3-2} \ge 1$, $2 + 1/Y \le 9/4$, and $4d/3 < 9/2$, while $(9/4)^{9/2} = (3/2)^9 < 40$. The moment hypothesis and Lemma 9.4 imply

$$R > \frac{N^2}{10C} - N \ge 5N. \tag{109}$$

The child labels lie between Y^d and $(2Y + 1)^d$. Their ratio is less than $(9/4)^{10/3} < 16$; the last inequality is equivalent to $3^{20} < 2^{32}$. This range meets at most five dyadic bands, so one band contains at least $R/5 > N$ child labels. Retain N_* in that band and call its lower endpoint Y'. Because it contains a label exceeding Y^d, $Y' > Y^d/2$. For the next exponent ratio $d' > 3$, its moment endpoint satisfies

$$X' = (2Y' + 1)^{d'} > Y^{3d} > (2Y + 1)^d = X,$$

since $Y^3 > 2Y + 1$. Thus successive moment endpoints increase, and all remain above the stated onset. The process can continue forever; Lemma 9.5 gives the desired base. □

For the seed-31 forest, all exponent-53 labels lie in $[2^{52}, 2^{53})$. Since

$$N_0 > 60(2 \cdot 10^9), \qquad (2^{53} + 1)^{161/53} > 2^{161},$$

Theorem 9.7 proves the second implication in Theorem 9.2.

For the large forest, use the directly counted upper band from (99), with $Y_* = 2^{61}$ and $n_* = 17$. The exact capacity and onset checks are

$$N_{\mathrm{U}} = 90210603204803 > 60(15 \cdot 10^{11}), \qquad (2^{62} + 1)^{53/17} > (25/2)^{53}. \tag{110}$$

The latter follows from the exact integer comparison $25^{17} < 2^{79}$. This proves the fourth implication in Theorem 9.2. The proof does not require a prime between every two consecutive cubes, nor does it require every parent interval to contain a prime.

9.6 Only the long gaps need be controlled

For $X_0 = (25/2)^{53}$ and $G(X) = X^{2/3}/64$, put

$$T_j(X) = \sum_{\substack{p_i \le X \\ p_{i+1}-p_i \ge G(X)}} (p_{i+1} - p_i)^j \qquad (j = 1, 2).$$

As in M_2, the full gap crossing X is included if it meets the size cutoff. We prove Theorem 9.3 using the upper-band frontier of size $N = N_{\mathrm{U}}$; at every subsequent level retain this same number of nodes in one dyadic band.

Use $\ell = 2Y^{d-1}$, $X = (2Y+1)^d$, and let L be the total length of the child intervals. By (108),

$$L \ge N\ell, \qquad \ell^2 > X^{4/3}/10, \qquad G(X) < \ell/12.$$

The last comparison follows from $10 \cdot 12^2 < 64^2$. The range of children meets at most five dyadic bands, as before.

First-moment alternative. Assume $T_1(X) \le 10^{13} X^{2/3}$ and suppose that fewer than $5N$ prime children occur in the interiors. Splitting the child intervals at these primes produces fewer than $6N$ clipped segments. Those contained in full gaps shorter than $G(X)$ have total length less than $6NG(X) < N\ell/2 \le L/2$. The remaining clipped segments have total length at most $T_1(X)$: even when a long full gap meets several intervals, the clipped lengths are disjoint and sum to at most that full gap's length. Since $N = 90210603204803 > 8 \cdot 10^{13}$ and $\ell > X^{2/3}/\sqrt{10} > X^{2/3}/4$,

$$T_1(X) \le 10^{13} X^{2/3} < N\ell/2 \le L/2.$$

The short and long parts cannot add up to L, a contradiction. Thus there are at least $5N$ children, and one dyadic band retains at least N of them. The moment endpoints stay above X_0 and increase by the proof of Theorem 9.7; strict nesting and Lemma 9.5 finish the construction. □

Second-moment alternative. Assume $T_2(X) \le CX^{4/3}$, where $C = 7.5 \cdot 10^{11}$. In the clipped-gap proof, the squares of the segments lying in short full gaps sum to at most $G(X)L$. Those from long gaps have square sum at most $T_2(X)$. The exact capacity $N \ge 120C$ gives

$$T_2(X) < 10C\ell^2 \le \frac{N\ell^2}{12} \le \frac{L^2}{12N}, \qquad G(X)L < \frac{L^2}{12N}.$$

Cauchy–Schwarz now gives

$$R + N \ge \frac{L^2}{T_2(X) + G(X)L} \ge 6N.$$

Hence $R \ge 5N$, and the same dyadic retention proves the conclusion. □

These hypotheses avoid controlling all the smaller prime gaps. They are not inferred merely from the existence of numerical zero-density constants: an actual large-gap estimate at the stated onset is still required.

9.7 What the available prime-gap theorems supply

9.7.1 An asymptotic exponent is not a numerical initialization

Stadlmann proves, for every $\varepsilon > 0$,

$$M_2(X) \ll_\varepsilon X^{1.23+\varepsilon} \qquad [15, \text{Theorem 1}]. \tag{111}$$

The convention in that theorem is the same as (90), so there is no missing crossing-gap term. Taking $\varepsilon = 0.01$ supplies the exponent $31/25$, but does not supply the particular coefficient and onset in Theorem 9.2. A bound with unknown constant and unknown starting point cannot be applied to the first unknown generation of a prescribed finite forest.

Nor is a direct numerical substitution in the proof automatic. For example, [15, Lemma 21, p. 43] uses

$$D = \left\lfloor \frac{\log(x^\beta / x^{1/\log\log x})}{\log 2} \right\rfloor, \qquad \beta \in [0.01, 0.15],$$

in a decomposition whose intended large-x regime requires $x^{1/\log\log x} \le x^\beta$. At the two fixed-window onsets,

x	$\log\log x$	$1/\log\log x$
2^{161}	$4.71489\ldots$	$0.21209\ldots$
$(25/2)^{53}$	$4.89682\ldots$	$0.20421\ldots$

so this ordering fails for every allowed β. Increasing the external coefficient C does not repair that parameter-ordering condition.

A second issue appears in [15, Lemma 20 and equation (5.15)]. The sieve application has weights supported on $d < x^{0.0001}$, and a remainder involves

$$\frac{x}{\tau}(\log x)^{2-\log\log\log x/(8K_2)}, \qquad K_2 = 10000,$$

which is subsequently absorbed into an arbitrary negative power of $\log x$ in the sufficiently-large regime. At the displayed onsets $x^{0.0001} < 1.014$, so that support does not even contain an integer $d \ge 2$. The literal asymptotic absorption therefore is not a quantitative error estimate at our initialization scales. Other small-parameter choices, large-value constants and prime-counting errors also require numerical control.

These observations rule out claiming a direct specialization of (111) from its displayed asymptotic bounds. They do not disprove any moment hypothesis in this paper, and are not lower bounds on the true moment. No claim of inherent ineffectivity is made: extracting or replacing the necessary constants is a separate analytic problem.

9.7.2 A corrected pointwise bound is quantitatively too weak

The corrected version of Cully-Hugill–Lee [12] gives intervals $(x(1-1/\Delta), x]$ containing a prime. At log-thresholds $60, 90, 105$, its parameters are respectively

$$7.69184 \cdot 10^{10}, \qquad 1.84304 \cdot 10^{11}, \qquad D_* = 1.91886 \cdot 10^{11}.$$

At $X = 2^{161}$ these rows, together with Bertrand's postulate for the smaller range, give a uniform bound $G = X/(D_* - 1)$ for gaps whose left endpoints are at most X. Summing $g^2 \le Gg$ over consecutive gaps gives

$$M_2(X) \le G(X + G). \tag{112}$$

At this first unverified scale, the right-hand side of (112), divided by $6 \cdot 10^{15} X^{31/25}$, lies strictly between $5.927 \cdot 10^9$ and $5.928 \cdot 10^9$. The portable verification checks this comparison using rational powers. Thus this particular pointwise-to-moment deduction does not meet the required budget. It says nothing against a sharper mean-square estimate.

9.7.3 Other routes retain a quantitative gap

Maynard's proof of the earlier exponent $5/4 + \varepsilon$ [13] is another possible starting point for an explicit argument, but does not state a constant and onset that establish a row of Theorem 9.2. The $+\varepsilon$ cannot simply be discarded. Its Lemma 5.5 also uses a zero-free-region saving to absorb a power $(\log x_1)^{246}$, without the numerical threshold needed here.

There are genuinely explicit estimates for tails of large gaps. For example, [11, Theorem 3.3] bounds the first moment of gaps larger than a parameter N, with explicit constants, but assumes $x \geq \exp(4000)$ and $N \geq 8x^{1/4+\alpha} \log^3 x$. This onset is far beyond either prescribed frontier scale, and that first-moment result is not, by itself, one of our second-moment hypotheses. These distinctions matter: a numerical zero-density estimate or an asymptotic exceptional-set bound is not automatically an explicit branching theorem at a chosen seed.

9.8 Arithmetic scope and the remaining step

The forest construction is independent of an algebraic-minimum classification. It proves finite prime-floor realizations and conditional upper bounds for the least infinite base. The earlier necessary-form and finite-certificate results [8, Corollary 4.1 and Proposition 5.2] exclude an algebraic $\xi_2 < 100$, subject to their stated structural inputs. Indeed the norm bound gives $|c| \leq \xi_2^{3/20} < 2$ and hence $c = \pm 1$, while the inverse-trace relation gives $b = \pm 1$. The Pisot property puts the trace coefficient in $0 \leq a \leq 101$. Changing the sign of the algebraic number when $c = -1$ reduces both norm signs to the signed coefficient grid in the cited proposition; its recurrent-divisor certificates exclude the schedule $r = 2$. Thus, *with those earlier results as an explicit dependency*, any large-forest hypothesis of Theorem 9.2 or 9.3 would imply transcendence of ξ_2. Neither the structural reduction nor that earlier finite certificate is reproved here, and neither is needed for the forest theorems.

In particular, none of the arguments above proves unconditionally that $\xi_2 < 2$, that $\xi_2 < 25/2$, or that ξ_2 is transcendental. They also do not resolve the classical Mills constant, whose schedule is different. The outstanding step is now a quantitative analytic statement: prove one of the displayed global moment budgets, or a suitable clipped-gap budget on precisely the child intervals needed by the forest.

A bound merely counting empty parents would need additional multiplicity control: a nonempty parent could have only one child, which might die in the next generation. Lemma 9.4 controls the total number of children, and the two continuation arguments preserve a whole frontier. This is why the conditional implications give an actual infinite branch rather than only arbitrarily long, unrelated prefixes.

9.9 Portable reproduction data

The accompanying `certificates` directory contains the files in Table 13. The finite source bound 1476 is a cited external computation [14]; these programs do not re-sieve all primes below $4 \cdot 10^{18}$. They verify the much smaller first-label lists, every exact integer endpoint, all block-count sums, the dyadic clipping, and the integer comparisons used in the continuation criteria.

Table 13: Finite reproduction files.

File	Contents or role
`branching_seed_verify.py`	Sieve and integer-root computation
`branching_seed_verify.json`	1152 exponent-17 label rows
`branching_seed_broad_verify.json`	4332 large-forest endpoint rows
`branching_seed_optimized_verify.json`	1865 restricted-forest rows
`audit_branching_seed_verify.py`	Trial-division and integer-root check
`audit_branching_broad_verify.py`	Large forest and dyadic checks
`audit_branching_seed_verification.json`	Seed and moment-comparison results
`audit_branching_broad_verification.json`	Large-forest results
`audit_branching_optimized_verification.json`	Restricted forest and upper-band counts

The first program uses Python's standard library only. The two checking programs additionally use SymPy's exact `integer_nthroot`; their primality lists use trial division rather than the first program's sieve. Run without Python's optimization flag, so assertion checks remain enabled:

```
cd certificates
python3 branching_seed_verify.py
python3 audit_branching_seed_verify.py
python3 audit_branching_broad_verify.py
```

The programs record passing terminal results and reproduce the exact counts in Theorem 9.1. The four data sums can also be reconstructed directly from (94), (97), (98) and (99); no floating-point approximation enters those definitions.

Frozen endpoint-data SHA256 digests. Each line below is a single 64-character digest; any wrapping is only typographical.

```
branching_seed_verify.json
aeb167c9de52d80287700cc6bcbcbb168581b2fa9d1bbfe96f360391cffeea51
branching_seed_broad_verify.json
6489814babe0e5514f6e72deca4baf2017b28d0426237153dcdec774bb03cda2
branching_seed_optimized_verify.json
5dbcb247af22d3b86c51ee08a43e9cb7649650c4ab969bfa257c4910455e0378
```

10 Shifted prime patterns and compactness criteria

Two different bridges are useful here: prime-pattern theorems give translated finite configurations, whereas compactness converts uniformly bounded full-prefix feasibility into one infinite base. The uniform bound and the absence of forbidden boundary limits are essential.

10.1 The question and the scope

A Pisot number is a real algebraic integer $\beta > 1$ whose other conjugates have modulus less than one. If degree one is permitted, every integer $p + 1 > 1$ is a Pisot number vacuously. Its powers of exponent at least two are composite, however, and it is not the degree-three object appearing in cubic algebraic-candidate reductions. The useful question is whether $p + 1$ can be a *coefficient or trace* of a genuine cubic Pisot number, and what prime-pattern theorems then imply. Here p always denotes a positive rational prime, not a Pisot number.

For a Pisot number, write $T_n = \mathrm{Tr}(\beta^n)$ and $S_n = T_n - \beta^n$. The integer T_n is the sum of all conjugate powers; $S_n \to 0$ exponentially. Our two clocks retain their original indices:

$$E_k^+ = 3^k, \qquad E_k^- = 2 \cdot 3^k - 1, \qquad k \geq 1. \tag{113}$$

Both consist of odd exponents. A prime integer part, a finite pattern among several different Pisot numbers, and prime floors on the whole power orbit of one fixed number are different conclusions.

The external inputs are used only in their stated forms: Green–Tao gives finite arithmetic progressions in a positive-relative-density prime set [24, Theorems 1.1–1.2]; Maynard's linear-forms conclusion gives arbitrarily many, not necessarily all, prime values in a sufficiently large admissible fixed tuple [13, p. 2]; Lê–Wolf permits shifted-prime polynomial steps with a free additive translation [25, Theorem 4]. We prove the algebraic deductions and the compactness statement below. No historical priority claim is made.

10.2 Two actual cubic realizations of prime plus one

Proposition 10.1. *For every prime $p \geq 2$, the polynomial*

$$F_p(X) = X^3 - (p+1)X^2 + X - 1$$

has a cubic Pisot unit root α_p satisfying

$$p < \alpha_p < p+1, \qquad \mathrm{Tr}(\alpha_p) = p+1, \qquad \lfloor \alpha_p \rfloor = p.$$

Furthermore $\left\lfloor \alpha_p^2 \right\rfloor = (p+1)^2 - 2$, which is composite for every odd prime p.

Proof. Put $a = p+1 \geq 3$. We have $F_p(p) = -p^2 + p - 1 < 0$ and $F_p(p+1) = p > 0$. The only possible rational roots are ± 1, and neither is a root. The discriminant is

$$D(a) = -4a^3 + a^2 + 18a - 31.$$

It is negative at $a = 3$, and $D'(a) = -12a^2 + 2a + 18 < 0$ for $a \geq 3$. Thus there is one real root α_p and a nonreal conjugate pair. Their product is $1/\alpha_p$, so each has modulus $\alpha_p^{-1/2} < 1$. The constant term gives norm one.

Set $\epsilon = a - \alpha_p$. The equation gives $\epsilon = (\alpha_p - 1)/\alpha_p^2$, hence $0 < \epsilon < 1/\alpha_p$ and

$$\alpha_p^2 - (a^2 - 2) = \frac{2}{\alpha_p} - \epsilon^2 \in (0, 1).$$

For odd p, a is even, so $a^2 - 2 > 2$ is even. □

In the entire family $X^3 - aX^2 + X - 1$, $a \geq 3$, the same proof gives

$$\lfloor \alpha_a \rfloor \text{ is prime} \quad \Longleftrightarrow \quad a - 1 \text{ is prime.}$$

This exact equivalence realizes the shifted-prime idea without any prime-pattern theorem. Its square calculation does not settle either clock in (113), which never uses exponent two.

Lemma 10.2. *Let $C \geq 1$ and $a > 2C$ be integers. The polynomial*

$$f_{a,C,-1}(X) = X^3 - aX^2 - CX + C$$

is irreducible and has roots $-w, v, \beta$ with $0 < v < w < 1$ and $a < \beta < a + 1$. Thus β is totally real cubic Pisot, and for every positive odd n,

$$0 < \beta^n - T_n = w^n - v^n < 1, \qquad \lfloor \beta^n \rfloor = T_n. \tag{114}$$

Proof. The values at $-1, 0, 1$ have signs $-, +, -$, since $f(-1) = 2C - a - 1$ and $f(1) = 1 - a$. Also $f(a) = C(1-a) < 0$ and $f(a+1) = (a+1)^2 - Ca > 0$. There is one root in each of $(-1, 0), (0, 1), (a, a+1)$. None is integral, so a monic integral cubic cannot have a rational root and is irreducible. The two small roots sum to $a - \beta < 0$, proving $w > v$ and (114). □

For the signed polynomial $f_{a,C,s} = X^3 - aX^2 - CX - sC$, $s \in \{1, -1\}$, Newton's identities give

$$\begin{aligned} &T_0 = 3, \quad T_1 = a, \quad T_2 = a^2 + 2C, \\ &T_n = aT_{n-1} + CT_{n-2} + sCT_{n-3} \qquad (n \geq 3). \end{aligned} \tag{115}$$

These identities define integral trace polynomials even at parameters where the polynomial is reducible.

Proposition 10.3. *For every prime $p \geq 11$, the dominant root β_p of*

$$X^3 - (p+1)X^2 - 4X + 4$$

is a totally real cubic Pisot number of trace $p+1$ and norm -4. Every floor $\left\lfloor \beta_p^n \right\rfloor$ at a positive odd exponent is composite. In particular this holds at every index of both clocks (113). Moreover,

$$p + 1 < \beta_p < p + 2, \qquad \frac{4(p+1)}{(p+2)^2} < \beta_p - (p+1) < \frac{4}{p+1}.$$

Proof. Apply Lemma 10.2 with $a = p + 1$ and $C = 4$. Since a and C are even, (115) makes every T_n, $n \geq 1$, even. Equation (114) identifies the odd floors with these traces, which exceed two. Finally, $\beta_p - a = 4(\beta_p - 1)/\beta_p^2$; the function $(x-1)/x^2$ decreases for $x > 2$, giving the displayed bounds. □

Euclid's infinitude of primes already makes this an infinite family. It lies in the elementary even-coefficient region, not in the remaining odd-a locus. More generally, in Lemma 10.2 with a even, an even C gives the same parity obstruction. If C is odd, the trace sequence modulo two has period seven and initial residues $(1, 0, 0, 1, 0, 1, 1)$. Consequently its classical-clock traces are even when k is even, and its negative-clock traces are even when $k \equiv 0, 3, 5 \pmod 6$. This follows directly from (115) modulo two.

10.3 A shifted-prime trace does not determine its floor

Lemma 10.4. *Suppose $|S_n| < 1$ and $T_n = q + 1$ for an odd prime q. Then*

$$\lfloor \beta^n \rfloor = \begin{cases} q, & S_n > 0, \\ q + 1, & S_n \leq 0. \end{cases}$$

Proof. Use $\beta^n = T_n - S_n$ and the strict bound on $|S_n|$. □

The sign distinction occurs at a classical scheduled exponent, not only in abstract examples. Let β be the real root of

$$X^3 - 6X^2 - 2X - 2.$$

Its discriminant is -2092, and its unique real root lies in $(6, 7)$. It is irreducible, since a rational root would be integral and real. The nonreal conjugates have modulus $\sqrt{2/\beta} < 1$. Thus β is cubic Pisot and $T_3 = 258 = 257 + 1$, but

$$\left\lfloor \beta^3 \right\rfloor = 257 \quad \text{is prime.} \tag{116}$$

For an exact check, set $l = 6363/1000$ and $u = 1591/250$. Then

$$f(l) = -\frac{28939853}{10^9} < 0, \quad f(u) = \frac{221571}{15625000} > 0, \quad 257 < l^3 < u^3 < 258.$$

Trial division up to $\sqrt{257} < 17$ proves the primality assertion.

There are two immediate limitations in the signed family. If a is odd, (115) makes *every* positive trace odd: the coefficient sum modulo two is $a + C + sC \equiv 1$. Thus $T_n = q + 1$ with odd prime q cannot occur anywhere in this odd-a locus. Also

$$T_n \equiv a^n \pmod{C} \qquad (n \geq 1). \tag{117}$$

If $\gcd(a, C) = 1$, putting $p + 1$ into the norm parameter C does not make any prime divisor of C divide a trace. Neither a composite norm nor a shifted-prime trace alone supplies the desired floor conclusion.

10.4 Prime coefficient patterns with certified composite powers

We now use a family in which both a prime first floor and composite scheduled floors are rigorously controlled. For

$$a = 43 + 186t, \quad t \geq 0, \qquad f_a(X) = X^3 - aX^2 - 4X + 4, \tag{118}$$

let β_a be its dominant root. Lemma 10.2 gives $\lfloor \beta_a \rfloor = a$ and exact odd-power traces.

Proposition 10.5. *Every member of* (118) *satisfies*

$$\begin{aligned} 31 \mid \left\lfloor \beta_a^{3^k} \right\rfloor, &\qquad k \equiv 1 \pmod 4, \\ 31 \mid \left\lfloor \beta_a^{2\cdot 3^k - 1} \right\rfloor, &\qquad k \equiv 3 \pmod 4. \end{aligned}$$

All these indicated terms, with $k \geq 1$, are composite.

Proof. Since $a \equiv 12 \pmod{31}$, $f_a(X) = (X + 6)^2(X + 7)$ modulo 31. Hence, for odd n,

$$T_n \equiv 2(-6)^n + (-7)^n = -6^n(2 - 4^n) \pmod{31}.$$

The element 4 has order five modulo 31 and $4^3 = 2$, so the trace vanishes exactly when $n \equiv 3 \pmod 5$. The powers of 3 modulo five have period four, giving the two index classes. The actual floors are T_n, and $n \geq 3$, $\beta_a > 43$ make them larger than 31. □

This is a direct repeated-reduction calculation; it needs no claim about ramification in the cubic number field. It is also independent of whether a itself is prime.

10.4.1 Arithmetic progressions of prime integer parts

The prime number theorem in the fixed progression $43 \pmod{186}$ gives

$$\frac{\#\{p \leq x : p \equiv 43 \pmod{186}\}}{\pi(x)} \longrightarrow \frac{1}{\varphi(186)} = \frac{1}{60}. \tag{119}$$

For the needed fixed-modulus assertion one may also use the Bombieri–Vinogradov estimate recalled in [13, §1]. Infinitude alone would not justify a relative-density application.

Corollary 10.6. *For every fixed $m \geq 2$ and every fixed lower cutoff, there are infinitely many nonconstant arithmetic progressions*

$$a_j = A + jD \quad (0 \leq j < m), \qquad D > 0,$$

above that cutoff, consisting of primes in the class 43 (mod 186)*. The corresponding m cubic Pisot numbers β_{a_j} have prime integer parts in that exact progression and share both composite-index classes of Proposition 10.5.*

Proof. Deleting a finite initial segment preserves the positive density (119). Apply the relative Green–Tao theorem [24, Theorem 1.2], followed by Proposition 10.5. □

The *Pisot values themselves are not asserted to form an AP*. Indeed the root equation is

$$a = g(\beta_a) = \beta_a - \frac{4}{\beta_a} + \frac{4}{\beta_a^2}, \qquad 0 < \beta_a - a < \frac{4}{a}.$$

For $\beta > 8$, $g' > 0$ and $g'' = 8(3-\beta)/\beta^4 < 0$. Its inverse is strictly convex, so an equally spaced coefficient triple gives a non-equally-spaced root triple. The roots are, however, within $4/\min_j a_j$ of their coefficient AP; this error can be made arbitrarily small by raising the cutoff.

10.4.2 Bounded clusters from admissible linear forms

Corollary 10.7. *For each fixed $m \geq 1$, there exists a finite H_m such that infinitely many intervals of length $H_m + 1$ contain at least m distinct totally real cubic Pisot numbers from* (118) *with prime integer parts. Every selected number has the composite-power progressions in Proposition 10.5.*

Proof. Choose K sufficiently large for Maynard's positive-linear-forms conclusion to give at least m primes. Put

$$M = \prod_{\substack{\ell \leq K \text{ prime} \\ \ell \nmid 186}} \ell, \qquad L_i(n) = 186n + 43 + 186Mi \quad (0 \leq i < K).$$

This tuple is admissible. If $\ell \mid 186$, all its values are $43 \not\equiv 0 \pmod{\ell}$. If $\ell \leq K$ and $\ell \nmid 186$, then $\ell \mid M$, so just one residue of n modulo ℓ is forbidden. If $\ell > K$ and $\ell \nmid 186$, at most $K < \ell$ residues are forbidden. Thus no prime divides the product for every integer n.

The linear-forms statement in [13, p. 2] supplies infinitely many n with at least m prime values. Their coefficient span is at most $H_m = 186M(K-1)$. Since $\beta_a \in (a, a+1)$, the root span is less than $H_m + 1$. Distinct coefficients have distinct roots, and the fixed index certificates remain valid. □

We have not claimed simultaneous primality of all K forms, nor used a general assertion that arbitrary positive-density prime subsets inherit Maynard's theorem. The explicit admissibility argument is why this application is justified.

10.5 The shifted-prime polynomial theorem meets the trace recurrence

Proposition 10.8. *Fix $C \geq 1$ and $s \in \{1, -1\}$. For each odd $n \geq 1$, define*

$$P_n(m) = T_n(m - s, C, s) + s.$$

Then $P_n \in \mathbb{Z}[m]$ and $P_n(0) = 0$. For every finite set I of positive odd exponents, the shifted-prime theorem of Lê–Wolf gives infinitely many configurations in which all numbers

$$x + P_n(p+1) = x + T_n(p+1-s, C, s) + s, \qquad n \in I, \tag{120}$$

are prime, with p prime and x an integer. The same holds in a prime subset of positive relative upper density.

Proof. Integrality follows from (115). At $a = -s$ the polynomial factors as

$$f_{-s,C,s}(X) = (X+s)(X^2 - C).$$

The two square-root contributions cancel at odd powers, leaving $T_n(-s, C, s) = -s$. Hence $P_n(0) = 0$. These are precisely the polynomial hypotheses of [25, Theorem 4], including its $p+1$ version. □

This is an exact algebraic connection, not an analogy based on the word "prime." The theorem still includes a free translation:

sign	coefficient $a = p+1-s$	required x for actual traces
$s = -1$	$p+2$	1
$s = 1$	p	-1

Only on the slice $x = -s$ do the values in (120) equal the untranslated traces. The cited theorem does not prescribe this slice. For real Pisot parameters satisfying Lemma 10.2, actual odd traces are the desired floors. In a nonreal family a shadow-sign condition remains necessary even after the trace values are fixed.

The obstruction to fixing x is concrete. Since $P_1(m) = m$, the real slice $x = 1$ already asks for p and $p+2$ both prime when exponent one is included. Infinitely many such configurations would imply infinitely many twin primes. This is a warning about the proposed strengthening, not an equivalence between the Pisot problem and the twin-prime problem. For the $s = 1$ slice, the first trace is automatically p, but the third is

$$p^3 + 3C(p+1),$$

a prescribed cubic at prime arguments. A linear-pattern theorem does not supply its infinitely many prime values. Tao–Ziegler's polynomial progression theorem also retains an additive translation [26]; "polynomial" does not remove the fixed-slice issue.

10.5.1 Why the translation is not another choice of Pisot root

Lemma 10.9. *Let $\beta, \gamma > 1$ be Pisot numbers. If, on an unbounded sequence n_j of positive integers,*

$$\mathrm{Tr}(\gamma^{n_j}) = \mathrm{Tr}(\beta^{n_j}) + c$$

for one fixed constant c, then $\gamma = \beta$ and $c = 0$. Within a fixed (C, s) signed family, even a common translation of both T_1 and T_3 at two positive coefficients forces equality of the coefficients and zero translation.

Proof. Each full trace is its dominant power plus an exponentially decaying error. If $\gamma > \beta$, the difference of dominant powers tends to infinity; if $\gamma < \beta$, it tends to minus infinity. Thus the roots are equal, and their full traces are equal.

For the finite assertion, $T_1(a') - T_1(a) = c$ gives $c = a' - a$. But

$$T_3(a') - T_3(a) = (a'-a)(a'^2 + a'a + a^2 + 3C).$$

The factor in parentheses exceeds one, so equality with c forces $a' = a$ and $c = 0$. □

This does not forbid fitting a finite list by unrelated algebraic data. It forbids silently absorbing a constant shift into one fixed Pisot trace profile, or into the same signed family at its first two odd trace coordinates.

10.6 What linearization and prime witnesses would have to accomplish

The first trace polynomials are

$$T_1 = a, \qquad T_3 = a^3 + 3C(a+s), \qquad T_5 = a^5 + 5C(a+s)(a^2+C).$$

For fixed C, s, T_n is monic of degree n in a, by the recurrence. Choosing a prime or one above a prime does not linearize its higher traces. Fixing a instead makes the real third trace $a^3 + 3(a-1)C$ affine-linear in C, but the Pisot locus $a > 2C$ restricts C to a finite interval. Dirichlet's theorem for an unrestricted progression does not guarantee a prime in this particular interval. Its residue is locally admissible exactly when $3 \nmid a$, since $\gcd(a^3, 3(a-1)) = \gcd(a^3, 3)$; and the fifth trace is already quadratic in C. A successful isolated step would not prove a whole prime clock.

Similarly, abundance of primes cannot replace dependence on a fixed candidate. A criterion requiring a divisor $\ell \mid a$ with $\ell \equiv 2 \pmod 3$ cannot apply to $a = 7^j$, however rich its witness prime set may be. For $C = 4$ and $j \geq 2$, these coefficients still lie in the real Pisot locus. This is only a limitation of that particular criterion, not evidence that their scheduled floors are prime.

Here is an exact candidate-dependent alternative. Let β satisfy Lemma 10.2, let $\ell \nmid C$ be prime, and let M be its companion matrix modulo ℓ. The matrix is invertible, and $\mathrm{Tr}(M^n) \equiv T_n \pmod \ell$. Put

$$E = \mathrm{ord}(M) = 3^v U, \quad \gcd(U, 3) = 1, \qquad Z_\ell = \{e \bmod E : \mathrm{Tr}(M^e) = 0\}.$$

For a fixed positive integer r, the exponent sequence $r3^k - 1$ is periodic modulo E once $k \geq \max(v, 1)$. Consequently

$$Z_\ell \text{ intersects that eventual orbit} \iff \ell \mid T_{r3^k - 1} \text{ for infinitely many } k. \tag{121}$$

Indeed if $k_0 \geq \max(v, 1)$ is a hit and $3^h \equiv 1 \pmod U$, then $E \mid r3^{k_0}(3^{th} - 1)$ for every $t \geq 0$; conversely, infinitely many hits include an eventual one. For even r, these exponents are odd, so (114) turns the hits into composite floors once the trace exceeds ℓ. The same argument applies to the classical orbit 3^k.

Thus one genuine exclusion strategy is to find such a prime ℓ *for each candidate*. Prime patterns in an unrelated additive parameter do not themselves establish (121). A different genuine strategy would be to establish actual finite prime-floor prefixes with a bound independent of prefix length. The next section shows exactly why that latter input would suffice for our two clocks.

10.7 A boundary-safe equivalence for returning exponent clocks

Individual prime-floor constraints are not closed: numbers tending to 4 from below may have prime floor 3, but their limit has floor 4. A compactness argument must therefore address integer endpoints. The following return condition supplies the necessary arithmetic.

Theorem 10.10 (Divisibility-return compactness)**.** *Let $(C_k)_{k \geq 1}$ be a sequence of positive integers such that*

$$\text{for every } m \text{ there are } l > m \text{ and } q \in \mathbb{Z}_{\geq 2} \text{ with } C_l = qC_m. \tag{122}$$

Fix $1 < B < \infty$. The following are equivalent:

(i) For every N, there is $A_N \in (1, B]$ such that $\left\lfloor A_N^{C_k} \right\rfloor$ is prime for every $1 \le k \le N$.

(ii) There is $A \in (1, B]$ such that $\left\lfloor A^{C_k} \right\rfloor$ is prime for every $k \ge 1$.

The finite witnesses need not have a common prime prefix.

Proof. Only (i)⇒(ii) needs proof. The first constraint gives $A_N \ge 2^{1/C_1} > 1$. Pass to a convergent subsequence with indices tending to infinity and limit $A \in [2^{1/C_1}, B]$. Fix m. If A^{C_m} is not an integer, its floor is locally constant and hence prime. If it is a prime integer, the desired condition already holds.

Otherwise $A^{C_m} = t$ is a composite integer, with $t \ge 4$. For all sufficiently late terms, $A_N \ge A$ would force $\left\lfloor A_N^{C_m} \right\rfloor = t$, contrary to (i). Thus these terms approach A from below. Choose l, q from (122). We then have $A_N^{C_l} \to t^q$ from below, so eventually, with $N \ge l$,

$$\left\lfloor A_N^{C_l} \right\rfloor = t^q - 1 = (t-1)(1 + t + \cdots + t^{q-1}).$$

Both factors exceed one. This contradicts primality at coordinate l and excludes every forbidden integer boundary. □

Both clocks (113) satisfy (122). For 3^k, take $l = m + 1$, $q = 3$. For $C_m = 2 \cdot 3^m - 1$, we have $\gcd(C_m, 3) = 1$. If $L = \operatorname{ord}_{C_m}(3)$, then $C_{m+L} \equiv C_m \equiv 0 \pmod{C_m}$ and $C_{m+L} > C_m$. If one instead starts the negative clock at $k = 0$, its first exponent is 1, which divides the next exponent 5; the other returns are unchanged.

Corollary 10.11 (Finite-prefix minima)**.** *Let $W_N = \{A > 1 : \left\lfloor A^{C_k} \right\rfloor$ is prime for $1 \le k \le N\}$. Every nonempty W_N has a least element a_N. If all W_N are nonempty, the sequence (a_N) is nondecreasing. Under (122), it is bounded if and only if an all-prime base exists; when bounded, its limit is exactly the least all-prime base.*

Proof. Choose a witness $U \in W_N$. Below U only finitely many tuples $(p_1, \ldots, p_N)$ of prime floors occur. The cell of each tuple is

$$[2^{1/C_1}, U] \cap \bigcap_{k=1}^{N} [p_k^{1/C_k}, (p_k + 1)^{1/C_k}).$$

Every nonempty such interval contains its left endpoint. Their finite union therefore has a minimum, which is also the global minimum of W_N. The inclusion $W_{N+1} \subseteq W_N$ gives monotonicity. If the minima are bounded, apply the proof of Theorem 10.10 to their convergent sequence. Its limit is an all-prime base. Every all-prime base is at least every a_N, so this limit is the least one. Conversely any all-prime base bounds all a_N above. □

If some W_N is empty, no all-prime base exists. With the convention $\min \varnothing = +\infty$, the statement can equivalently be written as convergence of the extended finite-prefix minima to the extended least all-prime base.

For comparison, a fixed bound B also gives a purely finite argument if only cubic Pisot candidates are allowed: there are finitely many such numbers at most B. Their integer polynomial coefficients are bounded, for example by $B + 2$, $2B + 1$, and B in absolute value for the three elementary symmetric functions. Decreasing nonempty finite sets of surviving candidates have nonempty intersection. Theorem 10.10 is stronger: it treats all real bases and handles the nonclosed endpoints by an arithmetic return.

10.8 Conclusions and the remaining input

There are genuine bridges, but they prove different things. Proposition 10.1 realizes trace $p+1$ and prime first floor p. Proposition 10.3 realizes a nearby real cubic whose odd floors are all composite. Corollaries 10.6 and 10.7 organize many distinct Pisot numbers by prime coefficients without changing their recurring composite powers. Proposition 10.8 identifies the exact shifted-prime polynomial pattern and the missing translation $x = -s$.

For either returning clock, the correct positive existence target is *every full finite prime-floor prefix with one uniform bound on the base*. Theorem 10.10 then supplies one fixed infinite base; compatibility is not an additional assumption under its return hypothesis. The prime-pattern results used here supply neither those full power prefixes nor that uniform bound. Conversely, the modular criterion (121) requires a witness depending on each fixed candidate, not merely an abundant set of primes.

No assertion here closes all odd-coefficient cubic candidates, constructs an algebraic base with a full prime power clock, or proves a new transcendence statement for a Mills-type minimum. In particular the classical Mills case and the negative-shift minimum with exponents $2 \cdot 3^k - 1$ are not resolved by these bridges.

10.9 Portable exact checks

The accompanying `certificates/` directory contains standard-library Python checks and their JSON outputs. Run from the package directory:

```
python3 certificates/shifted_prime_trace_verify.py
python3 certificates/prime_pattern_bridge_verify.py
```

The first checks the rational isolation in (116), primality of 257, sample parity identities, and exact trace normalization. The second checks polynomial coefficients, the full residue periods used in the prime-31 proof, and finite instances of the unit and admissibility calculations. Neither performs a numerical test of an infinite prime-pattern theorem. The universal algebraic, admissibility, and endpoint arguments are the proofs in the paper, not extrapolations from the checks.

Portable file	Purpose
`shifted_prime_trace_verify.py`	Exact trace and root checks
`shifted_prime_trace_verify.json`	Recorded result
`prime_pattern_bridge_verify.py`	Coefficient and period checks
`prime_pattern_bridge_verify.json`	Recorded result

The package README specifies scope and execution. External papers are cited by links and are not redistributed. No third-party proof is claimed to be reproduced by the finite certificates.

Part III

Coefficient density and obstructions to fixed congruence covers

11 Nonunit cubic families and fixed modulus limitations

The next sections turn from a distinguished least constant to coefficient families of cubic Pisot numbers. This first step supplies pointwise congruence certificates and proves why any fixed finite collection of primes can miss infinitely many members.

11.1 The families and the main results

For integers $C > 1$ and $s \in \{1, -1\}$, put

$$f_{a,C,s}(X) = X^3 - aX^2 - CX - sC. \tag{123}$$

A Pisot number is a real algebraic integer $\beta > 1$ all of whose other conjugates have modulus below one. When (123) is its minimal polynomial, let T_n be the sum of the nth powers of its three conjugates. Then

$$\begin{aligned} T_0 = 3, \quad T_1 = a, \quad T_2 = a^2 + 2C, \\ T_n = aT_{n-1} + CT_{n-2} + sCT_{n-3} \qquad (n \geq 3). \end{aligned} \tag{124}$$

In the ranges considered below, $s = 1$ gives a nonreal conjugate pair, and $s = -1$ gives two small real conjugates. The norm is sC, so these numbers are nonunits.

Composite floors on the classical schedule 3^k relate to the Pisot problem in [9, Problem 1.1]. The negative schedule $2 \cdot 3^k - 1$ also arises in algebraic-minimum reductions for shifted prime-representing constants [4, 8]. Our theorems are stated directly for explicit polynomials and do not assume that their roots are least constants.

The author's earlier manuscript [6, theorem "Nonreal cubic conjugates"] already proves infinitely many composite classical floors for every non-totally-real cubic Pisot number. Thus the complex classical families below are explicit recurring-divisor refinements, not newly resolved open cases. The totally real nonunit families are restricted progress in the remaining direction; the complex negative-clock statement concerns a different schedule.

Theorem 11.1 (An adaptive discriminant-prime certificate)**.** *Let $C > 1$, let $a > 2C$ be odd, and let $s \in \{1, -1\}$. Suppose a prime $p > 3$ satisfies*

$$p \mid C + 27, \qquad 4a \equiv 45s \pmod{p}.$$

Then $f_{a,C,s}$ is irreducible cubic Pisot. Put $d = \operatorname{ord}_p(2)$. For every positive odd n,

$$p \mid T_n \quad \Longleftrightarrow \quad d \mid 3n + 1.$$

For $k \geq 1$ this yields the following exact trace-divisibility schedules.

(i) *If $3 \nmid d$, put $D = d/\gcd(d, 2)$ and $h_- = \operatorname{ord}_D(3)$. The negative clock has exactly the hits $k \equiv -1 \pmod{h_-}$; at every such index $\left\lfloor \beta^{2 \cdot 3^k - 1} \right\rfloor$ is composite.*

(ii) *Suppose $3 \nmid d$ and -1 belongs to the subgroup generated by 3 modulo d. Then $h_+ = \operatorname{ord}_d(3)$ is even, and the classical clock has exactly the hits $k \equiv h_+/2 - 1 \pmod{h_+}$. At each such index $\left\lfloor \beta^{3^k} \right\rfloor$ is composite.*

If $3 \mid d$, neither clock has a hit from this certificate. For the classical clock, failure of the subgroup condition likewise means there are no hits from this prime.

Corollary 11.2 (The prime-31 families)**.** *Let $C > 1$ satisfy $C \equiv 4 \pmod{31}$, and let $a > 2C$ be odd. Suppose either*

$$s = 1,\ a \equiv 19 \pmod{31}, \qquad \text{or} \qquad s = -1,\ a \equiv 12 \pmod{31}.$$

Then (123) is an irreducible cubic Pisot polynomial. Its dominant root β satisfies

$$\left\lfloor \beta^{3^k} \right\rfloor \text{ is composite whenever } k \geq 1,\ k \equiv 1 \pmod{4},$$
$$\left\lfloor \beta^{2 \cdot 3^k - 1} \right\rfloor \text{ is composite whenever } k \geq 1,\ k \equiv 3 \pmod{4}.$$

Both conclusions hold at every stated index, not merely eventually.

In particular, at $C = 4$ the classical theorem covers the entire positive progressions $a = 19 + 62t$ for $s = 1$ and $a = 43 + 62t$ for $s = -1$, $t \geq 0$. If one also imposes the negative-shift coefficient condition $a \equiv -s \pmod{3}$, the progressions become $143 + 186t$ and $43 + 186t$. For general norms these additional conditions can be combined with $C \equiv 4 \pmod{93}$; coprimality conditions, if desired for another application, remain separate restrictions.

Theorem 11.3 (No fixed modulus excludes the real shadow locus)**.** *Fix $C > 1$ with $C \equiv 1 \pmod{3}$, and an integer $m \geq 1$. There are infinitely many odd integers $a \equiv 1 \pmod{3}$ with $\gcd(a, C) = 1$ such that $f_{a,C,-1}$ is irreducible and totally real Pisot, with dominant root β, and*

$$\mathrm{N}(\beta) = -C, \qquad \mathrm{Tr}(\beta^{-1}) = 1, \qquad C < \beta^{3/20},$$
$$0 < \beta^n - T_n < 2^{-n}\beta^{-17n/40}, \qquad \lfloor \beta^n \rfloor = T_n \equiv 1 \pmod{m} \quad \text{for every positive odd } n.$$

The numbers β depend on m. No assertion is made that their traces are prime or that they are least prime-representing constants.

Thus a finite set of fixed primes can exclude useful infinite subfamilies, as in Corollary 11.2, but cannot exclude the entire unbounded real coefficient and shadow locus of Theorem 11.3. Sections 11.3–11.4 explain the nonunit discriminant structure, and Section 11.6 gives a separate limitation of a purely phase-based approach.

11.2 Root configuration and prime certificates

Lemma 11.4. *If $C \geq 2$ and $a > 2C$, both signs in (123) give an irreducible cubic Pisot polynomial with $\beta \in (a, a+1)$. For $s = -1$, write the other roots as $-w, v$, where $0 < v < w < 1$. Then $\lfloor \beta^n \rfloor = T_n$ for every positive odd n. For $s = 1$, the other roots are nonreal, of common modulus $\sqrt{C/\beta}$, and $|T_n - \beta^n| < 1$ for every $n \geq 3$. If a is odd, T_n is odd for every $n \geq 1$.*

Proof. For $s = -1$, the signs at $-1, 0, 1, a, a+1$ are respectively

$$2C - a - 1 < 0, \quad C > 0, \quad 1 - a < 0, \quad C(1-a) < 0, \quad (a+1)^2 - Ca > 0.$$

Thus the three roots lie in $(-1, 0), (0, 1), (a, a+1)$. They are all nonintegral, so the monic integral cubic is irreducible. Vieta gives $v - w = a - \beta < 0$, and therefore

$$-1 < v^n - w^n < 0 \quad (n \text{ positive and odd}).$$

The trace identity $T_n = \beta^n + v^n - w^n$ proves the floor assertion.

For $s = 1$, the discriminant is

$$\Delta = CH_{C,s}(a), \qquad H_{C,s}(a) = -4sa^3 + C(a^2 - 18sa - 27) + 4C^2. \tag{125}$$

For $s = 1$ and $a \geq C \geq 2$ this is negative: $-4a^3 + Ca^2 \leq -3a^3$ and $-3a^3 + 4C^2 < 0$. Also $f(a) = -C(a+1) < 0$ and $f(a+1) = (a+1)^2 - C(a+2) > 0$. There is one real root, in $(a, a+1)$, hence no rational root. The other-root product is $C/\beta < 1$, so the pair has modulus $\sqrt{C/\beta}$. Since $a > 2C$,

$$|T_n - \beta^n| \leq 2(C/\beta)^{n/2} < 2(1/2)^{3/2} < 1 \qquad (n \geq 3).$$

Finally, if a is odd, the first three traces are odd and the sum of the recurrence coefficients in (124) is odd modulo two. Induction gives oddness of every trace. □

Proof of Theorem 11.1. Lemma 11.4 gives the root configuration. In $\mathbb{F}_p[X]$,

$$f_{a,C,s}(X) = (X - 6s)^2(X + 3s/4). \tag{126}$$

The roots are nonzero and distinct apart from the displayed double root, since $p > 3$. Their ratio is $-1/8$, so at odd exponents

$$T_n \equiv (6s)^n(2 - 8^{-n}) \pmod{p}.$$

This vanishes exactly when $2^{3n+1} = 1$, or $d \mid 3n + 1$. For $n = 2 \cdot 3^k - 1$ this is $D \mid 3^{k+1} - 1$. It is impossible if $3 \mid d$; otherwise it is exactly $k + 1 \equiv 0 \pmod{h_-}$. For $n = 3^k$ it is $3^{k+1} \equiv -1 \pmod{d}$. Here $d > 2$, because orders one or two would force $p \leq 3$. Within the cyclic group generated by 3, its element of order two, when present, is $3^{h_+/2}$. Thus the stated subgroup condition and index class are necessary and sufficient. Merely knowing that h_+ is even would not suffice in a noncyclic ambient unit group.

At every certified $k \geq 1$, $n \geq 3$ and the floor is T_n or $T_n - 1$, by Lemma 11.4. Since a is odd, T_n is odd. Also

$$T_n > \beta^n - 1 > 8C^3 - 1 > C + 27 \geq p \qquad (C \geq 2).$$

Thus T_n is a composite multiple of p, and $T_n - 1$ is an even integer greater than two. This proves the floor assertions. □

The test requires factoring the comparatively small integer $C + 27$ and checking orders and a coefficient residue. It is not universal: $C = 37$ gives $C + 27 = 64$, with no eligible odd prime, while $C = 22$ gives 49 and $\mathrm{ord}_7(2) = 3$. Even when an eligible prime exists, its coefficient congruence need not hold. Failure of this test is not evidence that any floor is prime. The certificate uses repeated reduction of the polynomial, not a claim that p ramifies in the cubic number field: an index factor can absorb a polynomial-discriminant factor. No field-ramification assertion is needed in the argument.

Proof of Corollary 11.2. For $p = 31$ one has $d = \mathrm{ord}_{31}(2) = 5$ and $\mathrm{ord}_5(3) = 4$, with $3^2 \equiv -1 \pmod{5}$. Also $45s/4 \equiv 19s \pmod{31}$, giving residues 19 and 12. Theorem 11.1 therefore gives precisely the stated index classes. The direct factorizations are

$$\begin{aligned} f_{19,4,1}(X) &= (X - 6)^2(X - 7), \\ f_{12,4,-1}(X) &= (X + 6)^2(X + 7) \end{aligned} \qquad \text{over } \mathbb{F}_{31}. \tag{127}$$

Equivalently, $31 \mid T_n$ at odd exponents exactly when $n \equiv 3 \pmod{5}$. □

This direct factorization proof is important: when $C = 4$, the prime 31 divides $C + 27$, so the generic local criterion in Proposition 11.12 below does not automatically cover it.

Proposition 11.5 (An additional prime-53 family)**.** *For $C = 4$, every dominant root of*

$$f_{239+318t,4,1} \quad \text{or} \quad f_{79+318t,4,-1}, \qquad t \geq 0,$$

has a composite floor at exponent $2 \cdot 3^k - 1$ whenever $k \geq 1$ and $k \equiv 1 \pmod 3$.

Proof. Modulo 53, the complex coefficient is $a = 27$ and

$$f = (X - 19)^2(X - 42), \qquad 42/19 = -1/21.$$

The element 21 has order 52 and $21^5 = 27 = 1/2$. The exponent $2 \cdot 3^k - 1$ is respectively $1, 5, 17$ modulo 52 when $k \equiv 0, 1, 2 \pmod 3$, because $3^3 \equiv 27 \pmod{52}$ and $2 \cdot 3^3 \equiv 2 \pmod{52}$. Thus the trace vanishes modulo 53 exactly when $k \equiv 1 \pmod 3$. Reflection gives the real coefficient $a = 26$ modulo 53. The displayed progressions impose these residues, oddness, and $a \equiv -s \pmod 3$. Lemma 11.4 applies, and the same odd-trace/even-predecessor argument gives composite floors. All relevant traces already exceed 53 at the first exponent 5. □

For each fixed C, arbitrarily large members of the prime-31 families also satisfy $C \leq \beta^{3/20}$ and small-conjugate modulus below $\beta^{-17/40}$. In the complex case this follows from $\rho = \sqrt{C/\beta}$; in the real case it follows from the estimate in Section 11.5. These additional inequalities are not needed for the composite-floor proofs.

11.2.1 Extension to every admissible even multiplier

Corollary 11.6 (The prime-31 families on every even multiplier)**.** *Let $C > 1$ and $a > 2C$ be integers, with a odd and $C \equiv 4 \pmod{31}$. Let $s \in \{1, -1\}$ and suppose $a \equiv 19s \pmod{31}$. Let β be the dominant root of*

$$X^3 - aX^2 - CX - sC.$$

For every positive even integer r with $5 \nmid r$, there is exactly one residue class $k_r \pmod 4$ such that

$$3^{k_r+1} \equiv 2r^{-1} \pmod 5.$$

For every $k \geq 1$ with $k \equiv k_r \pmod 4$, the integer $\lfloor \beta^{r3^k-1} \rfloor$ is composite. In particular, each such β is excluded from the prime-representing set of every one of these clocks. No assertion about the minimum for any clock is needed.

Proof. The root-location argument for this polynomial gives a cubic Pisot number $\beta \in (a, a+1)$. If $s = -1$, its small roots are $-w, v$ with $0 < v < w < 1$, so at every positive odd exponent n its floor is exactly its integral power trace T_n. If $s = 1$, the small roots are nonreal of modulus $\sqrt{C/\beta} < 1/\sqrt{2}$, and hence $|T_n - \beta^n| < 1$ for $n \geq 3$. In either case the floor at such an odd exponent is T_n or $T_n - 1$. The Newton recurrence

$$T_0 = 3, \quad T_1 = a, \quad T_2 = a^2 + 2C, \qquad T_n = aT_{n-1} + CT_{n-2} + sCT_{n-3}$$

shows that every T_n is odd.

Modulo 31, the polynomial factors as

$$(X - 6s)^2(X + 3s/4).$$

Thus for odd n,

$$T_n \equiv (6s)^n(2 - 8^{-n}) \pmod{31}.$$

Since 2 has order 5 modulo 31, this trace vanishes exactly when $5 \mid 3n+1$. For $n = r3^k - 1$ this is $r3^{k+1} \equiv 2 \pmod 5$. The number 3 generates the nonzero residues modulo 5, giving the claimed unique class modulo 4. As r is even, every prescribed n is odd and at least 5. Furthermore

$$T_n > \beta^n - 1 > (2C)^3 - 1 > C + 27 \geq 31.$$

At every indicated index, T_n is therefore an odd composite multiple of 31, while $T_n - 1$ is an even integer greater than two. Either possible floor is composite. □

Remark 11.7. The condition $5 \nmid r$ is exact for this particular witness: if $5 \mid r$, the prime 31 has no zero trace on this orbit. This failure says nothing about other possible divisors. The corollary is a strengthened family exclusion, not a solution of ξ_2 or of the general even-multiplier minimum problem.

11.3 Nonunit discriminants and the rational-square locus

The factorization at $a = -s$ is

$$\begin{gathered} f_{-s,C,s}(X) = (X+s)(X^2 - C), \\ T_n \equiv -s \pmod{a+s} \quad (n \text{ odd}), \qquad \Delta \equiv 4C(C-1)^2 \pmod{a+s}. \end{gathered} \tag{128}$$

The trace congruence follows because the odd power sums of the two roots of $X^2 - C$ cancel. Equivalently, it is a polynomial identity in the Newton sums, so is valid also modulo nonprime integers.

Proposition 11.8. *For every integer $C > 1$, $H_{C,s}$ is irreducible over $\mathbb{Q}$, and*

$$\operatorname{disc}_a H_{C,s} = -16C^2(C+27)^3. \tag{129}$$

Its root field is generated by a root u of $u^3 + Cu + 2sC = 0$, and its quadratic resolvent is $\mathbb{Q}(\sqrt{-(C+27)})$.

Proof. Eliminating a from $f(u) = f'(u) = 0$ gives

$$u^3 + Cu + 2sC = 0, \qquad a = \frac{3u^2 - C}{2u}. \tag{130}$$

With $t = -su$, the cubic becomes $t^3 + Ct - 2C$. Its derivative is positive and its unique real root lies in $(1, 2)$, since its values at $1, 2$ are $1 - C < 0$ and $8 > 0$. A rational root of a monic integral cubic is integral, so this cubic is irreducible. The inverse rational formula is

$$u = -\frac{C(a+9s)}{2(a^2+3C)}. \tag{131}$$

Substitution verifies both maps. If the denominator in (131) vanished at a root of H, then reducing H under $a^2 = -3C$ would give $C(C - 27 - 6sa) = 0$, forcing $C = -27$; this is impossible here. Thus the two generated fields coincide and have degree three, proving irreducibility of H. Expansion of its cubic discriminant gives (129). Removing its rational square factor gives the stated resolvent. □

For comparison, at $C = 1$ the exceptional factorization is

$$H_{1,s}(a) = -s(a+s)(4a^2 - 5sa + 23).$$

This linear factor is responsible for the quadratic, Pell-type reduction used in the unit case [17].

Proposition 11.9 (No integral cancellation under the norm bound)**.** *Suppose a is odd, (123) is cubic Pisot, $C > 1$, and $C \le \beta^{3/20}$. Then neither $(a+s)/2$ nor $a+s$ divides Δ.*

Proof. The two small conjugates give $\beta < a + 2$, so

$$a - 1 > C^{20/3} - 3 \ge C^6 - 3 > 8C(C-1)^2.$$

The last inequality follows from $C^6 \ge 8C^3$ for $C \ge 2$, and $8C^3 - 8C(C-1)^2 = 8C(2C-1) > 3$. Therefore $0 < 4C(C-1)^2 < (a+s)/2$. The discriminant congruence in (128) excludes both divisibilities. □

This does not exclude rational squares whose denominators cancel square factors of $a+s$. The next theorem treats that exact condition.

Theorem 11.10 (The direct nonunit square relation)**.** *Fix $C > 1$ and $s \in \{1, -1\}$. For integer $a \ne -s$, the condition*

$$\frac{\Delta}{-2s(a+s)} \in \mathbb{Q}^2 \tag{132}$$

is equivalent to the existence of an integer Y satisfying

$$Y^2 = -2sC(a+s)H_{C,s}(a). \tag{133}$$

The smooth projective model of this quartic has genus one. Consequently only finitely many integer coefficients $a \ne -s$ satisfy (132) for each fixed C, s.

Proof. Multiplying (132) by $[-2s(a+s)]^2$ gives the right side of (133). A rational square which is an integer is an integer square; division by the same nonzero square proves the converse. Proposition 11.8 shows that H has three simple roots, and $H(-s) = 4(C-1)^2 \ne 0$. Thus the quartic has four distinct roots. Its double cover of the a-line has four simple branch points, so Riemann–Hurwitz gives $2g - 2 = -4 + 4 = 0$. The affine curve is smooth and of genus one. Siegel's integral-point theorem [5, 16] gives finitely many integral pairs (a, Y), hence finitely many such a. □

For the positive range $M = a+s > 0$, put $B = -2s\Delta$ and suppose $B > 0$. If $g = \gcd(M, B)$, the discriminant congruence gives $g \mid 8C(C-1)^2$. The ratio in (132) is $B/(4M)$; it is a rational square exactly when the coprime integers $M/g, B/g$ are both squares. Hence the condition is equivalent to

$$\begin{aligned} a = gu^2 - s, \qquad v^2 = F_{C,s,g}(u), \qquad \gcd(u,v) = 1, \\ F_{C,s,g}(u) = \frac{-2sC}{g} H_{C,s}(gu^2 - s), \qquad u, v > 0. \end{aligned} \tag{134}$$

For every divisor g of $8C(C-1)^2$, this F has integral coefficients: its constant term is $-8sC(C-1)^2/g$, and the nonconstant terms of the substituted H contain g. Its degree is six, its leading coefficient is $8Cg^2$, and it has six distinct roots because H is squarefree and $H(-s) \ne 0$. Thus each displayed hyperelliptic curve has genus two. The coprimality requirement ensures the stated converse, including the identification of g as the gcd.

At $C = 1$, the repeated factor $a + s$ in (133) leaves a quadratic after cancellation. For $C > 1$ this degeneration disappears. Theorem 11.10 does not provide an effective bound, a bound uniform in C, or a classification of all discriminant-character survivors. It excludes only an infinite fixed-C family obtained by imposing this particular direct rational-square relation.

11.4 A cubic-field norm for trace resultants

Set

$$w = su + 2, \qquad h_C(W) = (W-2)^3 + CW = W^3 - 6W^2 + (12+C)W - 8. \tag{135}$$

Then $\mathrm{N}(w) = 8$. At a root a of $H_{C,s}$ given by (130), the polynomial has roots u, u, v, where $v = sC/u^2$ and $v/u = -1/w$. For odd n it follows that

$$T_n(a) = u^n(2 - w^{-n}). \tag{136}$$

Proposition 11.11. *Use the ordered resultant convention for which* $\mathrm{Res}_a(a, H) = H(0)$. *For every positive odd* n,

$$\mathrm{Res}_a(T_n, H_{C,s}) = C^n \,\mathrm{N}_{\mathbb{Q}(w)/\mathbb{Q}}(1 - 2w^n) = C^n(1 - 2L_n + 4M_n - 8^{n+1}), \tag{137}$$

where $L_n = \sum_i w_i^n$ *and* $M_n = \sum_{i<j}(w_i w_j)^n$ *for the three roots of* h_C. *In particular this resultant is independent of* s *at odd exponents.*

Proof. As a polynomial in a, T_n is monic of degree n. The leading coefficient of H is $-4s$, and $\prod_i u_i = -2sC$. Since n is odd, switching the resultant order contributes $(-1)^{3n} = -1$. Thus (136) gives

$$\mathrm{Res}(T_n, H) = -(-4s)^n(-2sC)^n \prod_i (2 - w_i^{-n}) = -C^n \prod_i (2w_i^n - 1).$$

There are three factors, so this is $C^n \prod_i (1 - 2w_i^n)$. Expanding the product and using $\prod_i w_i = 8$ gives the last equality. □

The integers in (137) can be computed without algebraic approximations, using

$$\begin{aligned} &L_0 = 3, \quad L_1 = 6, \quad L_2 = 12 - 2C, \\ &L_n = 6L_{n-1} - (12+C)L_{n-2} + 8L_{n-3}, \\ &M_0 = 3, \quad M_1 = 12 + C, \quad M_2 = (12+C)^2 - 96, \\ &M_n = (12+C)M_{n-1} - 48M_{n-2} + 64M_{n-3}. \end{aligned}$$

Writing $E_C(n) = \mathrm{N}(1 - 2w^n)$, examples are

$$E_C(1) = 4C - 27,$$

$$E_C(5) = 4C^5 + 240C^4 + 4800C^3 + 35780C^2 + 77600C - 250047.$$

In particular $E_4(3) = 31 \cdot 127$ and $E_4(5) = 53 \cdot 18973$.

Proposition 11.12 (A local scalar criterion)**.** *Let* p *be odd with* $p \nmid C(C+27)$ *and* $H_{C,s}(a) \equiv 0 \pmod p$. *Define* u *by* (131) *modulo* p *and* $w = su + 2$. *These are defined and nonzero, and for every positive odd* n,

$$p \mid T_n(a) \quad \Longleftrightarrow \quad w^n = 1/2 \text{ in } \mathbb{F}_p.$$

Proof. The denominator calculation in Proposition 11.8 works modulo p under the excluded factors. The identities (130) and (135) imply $u, w \neq 0$, and factor f as $(X-u)^2(X-v)$ over $\mathbb{F}_p$. Equation (136) gives the equivalence. □

For the negative clock the scalar equation is $(w^2)^{3^k} = w/2$. If it has a hit at an index $k \geq v_3(\mathrm{ord}_p(w))$, that hit recurs infinitely often: write $\mathrm{ord}_p(w) = 3^e U$ with $3 \nmid U$, and choose $h > 0$ with $3^h \equiv 1 \pmod U$. Then $\mathrm{ord}_p(w)$ divides $2 \cdot 3^k(3^{th} - 1)$ for all $t \geq 0$. This is a recurrence implication, not an existence theorem for a suitable prime or a suitable initial hit. Similarly, divisibility of $E_C(n)$ is only a necessary condition for a common trace/discriminant prime outside C; a coefficient specialization is still required for a converse.

11.5 The fixed-modulus obstruction and its quantitative form

Proof of Theorem 11.3. Let $A_0 = 2^{27}C^7$ and $L = \operatorname{lcm}(6C, m)$. Choose any $a \geq A_0$ with $a \equiv 1 \pmod{L}$. Then a is odd, $a \equiv 1 \pmod 3$, $\gcd(a, C) = 1$, and $a > 2C$. Lemma 11.4 gives roots $-w, v, \beta$ with $0 < v < w < 1$ and $\beta \in (a, a+1)$. The negative-root equation gives

$$aw^2 = C(1+w) - w^3 < 2C.$$

Since $\beta < a + 1 < 2a$, this yields $w < 2\sqrt{C/\beta}$. Moreover

$$a^3 \geq A_0^3 = 2^{81}C^{21} > 2^{80}C^{20} = (4\sqrt{C})^{40},$$

so $\beta^{3/40} > 4\sqrt{C}$ and

$$w < \tfrac{1}{2}\beta^{-17/40}, \qquad C < \beta^{3/20}.$$

For positive odd n,

$$0 < \beta^n - T_n = w^n - v^n < w^n < 2^{-n}\beta^{-17n/40}.$$

In particular its floor is T_n; also $T_n > 0$ and $\beta^{-17n/40} < T_n^{-17/40}$, if the numerical shadow bound is expressed in terms of the integer trace instead.

At $a = 1$, the factorization $f = (X-1)(X^2-C)$ gives the polynomial identity $T_n(1, C, -1) = 1$ for every odd n. Because (124) consists of polynomials with integral coefficients, $a \equiv 1 \pmod m$ implies $T_n \equiv 1 \pmod m$ for all those exponents. This includes primes dividing C, the prime two, and arbitrary prime powers; no division in a residue ring is involved. The norm and inverse trace follow from the coefficients. Finally there are infinitely many such a, and their distinct intervals $(a, a+1)$ give distinct Pisot numbers. □

Corollary 11.13 (Necessary product size for a finite-prime cover)**.** *Fix $C > 1$ with $C \equiv 1 \pmod 3$, let $A_0 = 2^{27}C^7$, and let $H > A_0$. Suppose a finite set $\mathcal{P}$ of primes excludes every coefficient in the real locus of Theorem 11.3 with $a \leq H$ by divisibility of some odd scheduled trace. Put $Q = \prod_{p\in\mathcal{P}} p$. Then*

$$6CQ > H - A_0, \qquad \sum_{p\in\mathcal{P}} \log p > \log\left(\frac{H - A_0}{6C}\right).$$

For $C = 4$, this reads $24Q > H - 2^{41}$.

Proof. The first integer $a \geq A_0$ congruent to one modulo $6CQ$ lies strictly below $A_0 + 6CQ$. If $6CQ \leq H - A_0$, this coefficient lies below H and satisfies every conclusion of Theorem 11.3, while all its odd traces are one modulo every $p \in \mathcal{P}$. That contradicts the proposed cover. Taking logarithms proves the second inequality. □

This is not a lower bound for arbitrary algorithms. It does not exclude coefficient-dependent divisors, unbounded prime sets, additional discriminant restrictions, or arguments using minimality. Nor does it construct primes: the trace could have a prime divisor outside any specified finite set.

11.6 Irrational phase survivors near zero

For the complex branch, write its small conjugates as $\rho e^{\pm i\theta}$. Vieta gives

$$2\rho\cos\theta = -\frac{C}{\beta}\left(1 + \frac{1}{\beta}\right) < 0,$$

so $\theta = \pi/2 + \delta$ for $0 < \delta < \pi/2$. For $n_k = 2 \cdot 3^k - 1 \equiv 1 \pmod 4$,

$$T_{n_k} - \beta^{n_k} = -2\rho^{n_k} \sin(n_k \delta).$$

Once this shadow has magnitude below one, the floor equals T_{n_k} if and only if $\sin(n_k\delta) \geq 0$. Writing $t = \delta/\pi \in (0, 1/2)$ gives the exact phase condition

$$\{3^k t\} \in [t/2, (1+t)/2]. \tag{138}$$

Here $n_k\delta/(2\pi) = 3^k t - t/2$, and the translated sine half-circle does not wrap around the unit interval.

Theorem 11.14. *For every integer $m \geq 1$, there is a set of Hausdorff dimension $\log 2/(2\log 3)$ inside $[3^{-(m+1)}, \frac{1}{2}3^{-m}]$ such that every member satisfies (138) strictly for every $k \geq 0$. All but countably many members are irrational.*

Proof. Use the ternary expansions

$$t = 0.\underbrace{00\cdots0}_{m \text{ digits}} 1b_1, 1b_2, 1b_3 \cdots \quad \text{in base three,} \qquad b_j \in \{0, 1\}.$$

All digits are zero or one, and every zero run has length at most m. Every shifted tail therefore satisfies

$$3^{-(m+1)} \leq \{3^k t\} \leq \tfrac{1}{2}.$$

The initial prefix gives $3^{-(m+1)} \leq t \leq \frac{1}{2}3^{-m}$, and hence

$$t/2 \leq \tfrac{1}{4}3^{-m} < 3^{-(m+1)}, \qquad (1+t)/2 > \tfrac{1}{2}.$$

Thus all survivor inequalities are strict.

After the fixed prefix, the two allowed digit blocks are 10 and 11. They give two separated similarity maps of contraction $1/9$. At depth j there are 2^j disjoint cylinder sets of diameter proportional to 9^{-j}. These covers give dimension at most $\log 2/\log 9$. Assigning mass 2^{-j} to each cylinder gives the reverse inequality by the mass distribution principle, since an interval at scale 9^{-j} meets a bounded number of cylinders. A fixed finite prefix does not alter the dimension. Rational numbers have eventually periodic ternary expansions and are countable, whereas this separated binary construction is uncountable. Removing its rational members leaves the stated irrational survivors. □

Taking m arbitrarily large places these sets arbitrarily close to zero. No claim is made that their angles arise from conjugates of cubic algebraic integers. The theorem shows precisely that smallness of the angle, irrationality, and the ternary recurrence alone do not eliminate all phase survivors. A complete argument must use additional arithmetic or minimality; equidistribution of an algebraic-angle orbit must not be assumed without proof.

11.7 Scope

The explicit prime certificates prove composite floors for proper infinite nonunit subfamilies on both schedules. The fixed-modulus construction proves that a finite prime-divisor cover cannot settle the whole unbounded real coefficient/shadow locus. The genus-one finiteness theorem concerns one specified rational-square relation for each fixed norm, not all possible local or character conditions. The phase construction concerns real angles, not a realized family of algebraic Pisot conjugates.

None of these results proves transcendence of the least constant for $2 \cdot 3^k - 1$, transcendence of the classical Mills constant, or the full assertion that every cubic Pisot number has infinitely many composite floors along 3^k. Their roles are complementary: explicit excluded families, an exact discriminant-norm calculation, and limitations of several otherwise natural uniform methods.

11.8 Portable exact verification

The universal proofs are given above. The supplementary files in `certificates` check symbolic identities, conventional Sylvester-determinant signs, finite-field factorizations, exponent cycles, and exact sample root/shadow inequalities.

File	Role
`adaptive_ramification_verify.py`	Adaptive order and clock certificates
`adaptive_ramification_verify.json`	Adaptive certificate results
`audit_nonunit_discriminant_verify.py`	Independent norm and local checks
`audit_nonunit_discriminant_verify.json`	Independent terminal results
`nonunit_discriminant_verify.py`	Discriminant, norm and prime certificates
`nonunit_discriminant_verify.json`	Exact terminal results
`audit_fixed_nonunit_obstruction.py`	Real-locus identities and modulus checks
`audit_fixed_nonunit_obstruction.json`	Real-locus terminal results
`phase_survivors_verify.py`	Exact rational survivor samples
`phase_survivors_verify.json`	Phase sample results

The algebraic checking programs use SymPy for exact polynomial arithmetic. The phase program uses the Python standard library. Run without Python's optimization flag:

```
cd certificates
python3 adaptive_ramification_verify.py
python3 nonunit_discriminant_verify.py
python3 audit_nonunit_discriminant_verify.py
python3 audit_fixed_nonunit_obstruction.py
python3 phase_survivors_verify.py
```

The phase file checks 1008 periodic rational examples. Those checks are illustrations only; irrationality, Hausdorff dimension and the all-index conclusion are proved symbolically in Theorem 11.14. Likewise finite samples do not establish the universal modulus theorem or the genus statements. For sign-sensitive resultants the supplied code constructs the ordered Sylvester determinant directly, with $\mathrm{Res}(X, X^3 + 2) = 2$ as a convention check.

12 Density on fixed norm lines

Failure of a finite universal cover is compatible with a density-one theorem. Varying the auxiliary prime with the coefficient gives the following result on every fixed norm line, without deciding any prescribed exceptional coefficient.

12.1 The coefficient line and the main theorem

For $C \geq 1$, $s \in \{1, -1\}$ and an integer $a > 2C$, set

$$f_{a,C,s}(X) = X^3 - aX^2 - CX - sC. \tag{139}$$

Its dominant real root will be denoted by $\beta_{a,C,s}$. Lemma 12.2 shows that it is cubic Pisot: an algebraic integer larger than 1 whose other conjugates have modulus below 1. Its norm is sC, so $C > 1$ gives nonunits, while $C = 1$ is included too.

Let M be a fixed positive even integer and let $\mathcal{R}$ be a nonempty collection of odd residue classes modulo M. Write

$$\mathcal{A}(H) = \{a \in \mathbb{N} : 2C < a \le H,\ a \bmod M \in \mathcal{R}\}.$$

All density statements below concern coefficients in this explicitly specified one-dimensional set.

Theorem 12.1 (Simultaneous fixed-line density one)**.** *Fix $C \ge 1$, M and $\mathcal{R}$ as above. For relative natural density one of the coefficients $a \in \mathcal{A}(H)$, there is an integer $L = L(a) \ge 1$ such that, for both $s = 1$ and $s = -1$, all four integers*

$$\left\lfloor \beta_{a,C,s}^{3^k} \right\rfloor, \qquad \left\lfloor \beta_{a,C,s}^{2\cdot 3^k - 1} \right\rfloor$$

are composite whenever $k \ge 1$ and $L \mid k$. More precisely, if $\mathcal{G}(H)$ denotes the coefficients with this property, then

$$\frac{\#\mathcal{G}(H)}{\#\mathcal{A}(H)} \longrightarrow 1.$$

A single suitable prime divisor of a provides the common progression.

The progression has density $1/L$ in the index k, but L may depend on a. Neither a uniform lower bound on this index density nor uniformity in C is asserted.

Relation to the earlier corpus. The small-prime return mechanism for the classical clock already occurs in [6, lemma "A uniform small-prime return condition"]. The same manuscript's theorem "Density one at every fixed nonzero norm" varies both coefficients (a, b) in $X^3 - aX^2 + bX - c$ and gives $o(B^2)$ exceptional pairs. Its theorem "Density one within Saito's shrinking conjugate locus" still varies b over a growing interval and gives $o(B^{63/40})$ exceptions in a family of that order. Neither estimate controls the line $b = -C$, $c = sC$, containing only $O(B)$ points: the entire line could lie in either exceptional set. Theorem 12.1 supplies a separate line-specific argument and a common return for two clocks.

The nonreal cubic Pisot case of infinitely many composite classical floors was also established in [6, theorem "Nonreal cubic conjugates"]. We retain that antecedent. Our simultaneous certificate is not presented as a new resolution of that known case, and no literature-wide priority claim is made.

12.2 Root locations, traces, and actual floors

Let T_n be the sum of the nth powers of the three roots of (139). Newton's identities give integers

$$\begin{aligned} T_0 = 3, \qquad T_1 = a, \qquad T_2 = a^2 + 2C, \\ T_n = aT_{n-1} + CT_{n-2} + sCT_{n-3} \qquad (n \ge 3). \end{aligned} \tag{140}$$

Lemma 12.2. *For $C \ge 1$ and $a > 2C$, both polynomials in (139) are irreducible cubic Pisot, with $\beta \in (a, a+1)$. For $s = -1$, their other roots are $-\rho, \sigma$ with $0 < \sigma < \rho < 1$, and*

$$\lfloor \beta^n \rfloor = T_n \quad (n \ge 1 \text{ odd}).$$

For $s = 1$, the other roots are nonreal, of modulus $\sqrt{C/\beta}$, and $|T_n - \beta^n| < 1$ for every $n \ge 3$. If a is odd, every T_n is odd.

Proof. For $s = -1$, the signs at $-1, 0, 1, a, a+1$ are respectively

$$2C - a - 1 < 0, \quad C > 0, \quad 1 - a < 0, \quad C(1-a) < 0, \quad (a+1)^2 - Ca > 0.$$

Thus the roots lie in $(-1, 0), (0, 1), (a, a+1)$. None is an integer, so the monic cubic has no rational root and is irreducible. Vieta gives $\sigma - \rho = a - \beta < 0$. For positive odd n,

$$0 < \beta^n - T_n = \rho^n - \sigma^n < 1,$$

proving the floor identity.

In general, direct expansion gives

$$\Delta = CH_{C,s}(a), \qquad H_{C,s}(a) = -4sa^3 + C(a^2 - 18sa - 27) + 4C^2. \tag{141}$$

For $s = 1$ this is negative when $a > 2C$: $-4a^3 + Ca^2 + 4C^2 < -3a^3 + a^2 < 0$, and the remaining terms are negative. Also $f(a) = -C(a+1) < 0$ and $f(a+1) = (a+1)^2 - C(a+2) > 0$. Hence the sole real root lies in $(a, a+1)$, giving irreducibility. The other roots have product C/β and common modulus $\sqrt{C/\beta} < 1/\sqrt{2}$. Consequently

$$|T_n - \beta^n| \le 2(C/\beta)^{n/2} < 1 \qquad (n \ge 3).$$

Finally, if a is odd, the first three traces are odd and the sum of the recurrence coefficients is $1 + C + C = 1 \pmod 2$. Induction proves the parity assertion. □

Lemma 12.3 (The coefficient-divisor floor bridge)**.** *Let $a > 2C$ be odd and let a prime $p \mid a$. At any odd $n \ge 3$ for which $p \mid T_n$, the integer $\left\lfloor \beta_{a,C,s}^n \right\rfloor$ is composite.*

Proof. We have $p \le a$ and

$$T_n > \beta^n - 1 > a^3 - 1 > a \ge p.$$

For $s = -1$, the floor is this proper multiple of p. For $s = 1$, it is either the odd integer T_n or the even integer $T_n - 1 > 2$, by Lemma 12.2. Both are composite. No assumption on the sign of the complex conjugate sum is needed. □

12.3 A common modular return for both clocks

Put

$$h_{C,s}(X) = X^3 - CX - sC,$$

which is the reduction of $f_{a,C,s}$ whenever $p \mid a$.

Lemma 12.4. *Suppose $p \nmid C$ and there is an integer $N \ge 1$, coprime to 3, such that*

$$X^N = 1 \quad \text{in } \mathbb{F}_p[X]/(h_{C,s}).$$

If $p \mid a$ and $L = \operatorname{ord}_N(3)$, with $L = 1$ when $N = 1$, then

$$T_{3^k} \equiv T_{2 \cdot 3^k - 1} \equiv 0 \pmod p \qquad (L \mid k).$$

A common N for the two signs gives a common progression for all four floors.

Proof. At $L \mid k$, both exponents are $1 \pmod N$, so every root power returns to its first power. Their sum is $T_1 = a \equiv 0 \pmod p$. Equivalently, apply the trace functional in the quotient ring to $X^{3^k} = X^{2 \cdot 3^k - 1} = X$. □

For an elementary source of these identities, define

$$D_C = 4C - 27, \qquad \mathcal{P}_C = \{p > 3 : p \equiv 2 \pmod 3,\ p \nmid CD_C, \left(\frac{D_C}{p}\right) = 1\}. \tag{142}$$

Here and below p denotes a prime.

Proposition 12.5. *For every $p \in \mathcal{P}_C$ and both signs,*

$$X^{p^3-1} = 1 \quad \text{in } \mathbb{F}_p[X]/(h_{C,s}).$$

In particular $N = p^3 - 1$ is a common prime-to-3 order bound.

Proof. The discriminant of $h_{C,s}$ is $C^2 D_C$, a nonzero square modulo p. Its Frobenius permutation is therefore even. To see this directly, Frobenius acts on a Vandermonde square root of the discriminant by the sign of its permutation; that square root lies in $\mathbb{F}_p$ precisely when the discriminant is a square. A separable cubic with even Frobenius has splitting type $1+1+1$ or 3. All roots are nonzero and lie in $\mathbb{F}_{p^3}$, so their orders divide $p^3 - 1$. Separability gives the quotient-ring identity. Since $p \equiv -1 \pmod 3$, $3 \nmid p^3 - 1$. The argument is independent of s; alternatively the two root triples are negatives of one another. □

12.4 Divergent good-prime sets, including the exceptional norm

12.4.1 Explicit prime progressions when $C \neq 6$

Proposition 12.6. *For every positive integer $C \neq 6$,*

$$\sum_{p \in \mathcal{P}_C} \frac{1}{p} = \infty.$$

Proof. The odd integer $D = D_C$ is always $1 \pmod 4$ and never zero. Quadratic reciprocity, including its sign convention for negative D, gives $\left(\frac{D}{p}\right) = \left(\frac{p}{|D|}\right)$; on the right is the Jacobi symbol, with the denominator-one value defined as 1. The following reduced residue classes supply usable primes, apart from the finitely many divisors of $2CD$:

(i) If $C \geq 7$, then $D > 0$ and take $p \equiv -1 \pmod{3D}$. Here $p \equiv 2 \pmod 3$ and $\left(\frac{p}{D}\right) = \left(\frac{-1}{D}\right) = 1$.

(ii) If $C \in \{1, 2, 4, 5\}$, impose $p \equiv 1 \pmod{|D|}$ and $p \equiv 2 \pmod 3$. The moduli are coprime, and CRT gives a reduced class with Jacobi symbol 1.

(iii) If $C = 3$, take $p \equiv 2 \pmod{15}$. Then $D = -15$ and $\left(\frac{p}{15}\right) = \left(\frac{2}{3}\right)\left(\frac{2}{5}\right) = 1$.

The reciprocal sum of primes in a reduced arithmetic progression diverges, by the harmonic form of Dirichlet's theorem [21, Theorem 4.11 and its proof]; see also [22, Equation 27.11.9]. Removing finitely many primes does not change divergence. □

At $C = 6$, $D_C = -3$ and every $p \equiv 2 \pmod 3$ has $\left(\frac{D_C}{p}\right) = -1$. Thus this particular prime selection is empty. The next argument supplies orders directly rather than inferring them from a splitting condition.

12.4.2 The case $C = 6$: prescribed reduction orders

We use the following precise specialization of a published theorem. For a point R on an algebraic group, write G_R for the Zariski closure of the cyclic subgroup it generates.

Theorem 12.7 (Perucca, Theorem 3.1 [20])**.** *Let G be a torus over a number field K, let F/K be finite, and let $R \in G(F)$ with G_R connected. For any nonzero integer m, there is a positive-Dirichlet-density set of primes $\mathfrak{p}$ of K such that some prime $\mathfrak{q}$ of F above $\mathfrak{p}$ gives a reduction of R whose order is coprime to m.*

Proposition 12.8. *There is a set $\mathcal{P}_6$ of rational primes with divergent reciprocal sum such that, for each $p \in \mathcal{P}_6$, a common integer N_p coprime to 3 satisfies*

$$X^{N_p} = 1 \quad \text{in } \mathbb{F}_p[X]/(X^3 - 6X - 6s) \quad (s = 1, -1).$$

Proof. Let $\alpha_1, \alpha_2, \alpha_3$ be the roots of $X^3 - 6X - 6$, and let F be its splitting field. The polynomial is Eisenstein at 3 and has discriminant -108. Its root absolute values are b, ρ, ρ, where $b > 0$ is the real root and $\rho^2 = 6/b$. Since $b^3 = 6b + 6 > 6$, we have $b > \rho$.

We verify multiplicative independence, which is the essential hypothesis. Suppose $\prod_i \alpha_i^{e_i} = 1$ with $e_i \in \mathbb{Z}$, and put $S = e_1 + e_2 + e_3$. Transitivity of the Galois action lets us send each coordinate in turn to the real root. Taking absolute values gives

$$e_i \log b + (S - e_i) \log \rho = 0 \qquad (i = 1, 2, 3).$$

Subtracting shows $e_1 = e_2 = e_3 = e$, because $b \neq \rho$. The product of the root absolute values is 6, so $6^e = 1$ and $e = 0$. Thus there is no nontrivial multiplicative relation. Every proper algebraic subgroup of the split torus is annihilated by a nontrivial integer character, possibly after multiplying the character to remove finite components. Hence the point $R = (\alpha_1, \alpha_2, \alpha_3)$ has $G_R = \mathbb{G}_m^3$, which is connected.

Apply Theorem 12.7 with

$$K = \mathbb{Q}, \qquad G = \mathbb{G}_m^3, \qquad m = 3.$$

It supplies rational primes directly, although R is only F-rational. Discard finitely many bad reductions, including 2 and 3. At a prime $\mathfrak{q}$ above each remaining p, the reduced roots are distinct nonzero elements of the common finite field $\kappa(\mathfrak{q})$. The order of their tuple is the least common multiple of their orders, and is coprime to 3. Consequently their separable polynomial divides $X^N - 1$ over that field, and hence over $\mathbb{F}_p$. Negating the roots changes orders only by a possible factor 2; replacing N by $\operatorname{lcm}(N, 2)$ gives a common bound for both signs.

Positive Dirichlet density implies reciprocal divergence: otherwise $\sum p^{-z}$ would remain bounded as real z decreases to 1, contradicting the definition of positive Dirichlet density. □

Theorem 12.7 is the only additional input needed for $C = 6$. No residue-degree-one restriction or descent of prime-ideal density is used. In particular, complete splitting alone would not justify the claimed prime-to-3 orders.

12.5 From good primes to density-one coefficients

Proof of Theorem 12.1. For each fixed C, Propositions 12.6 and 12.8 give a set of good primes with divergent reciprocal sum. Delete the finitely many primes dividing M. For a fixed finite set $\mathcal{F}$ of remaining good primes, the Chinese remainder theorem gives

$$\lim_{H \to \infty} \frac{\#\{a \in \mathcal{A}(H) : p \nmid a \text{ for all } p \in \mathcal{F}\}}{\#\mathcal{A}(H)} = \prod_{p \in \mathcal{F}} \left(1 - \frac{1}{p}\right). \tag{143}$$

Indeed each allowed residue modulo M has $\prod_{p\in\mathcal{F}} p$ extensions, of which exactly $\prod_{p\in\mathcal{F}}(p-1)$ avoid all those divisibilities.

The coefficients with no good prime divisor avoid every such finite set. First take their relative upper density as $H \to \infty$, and then increase $\mathcal{F}$. Since $\sum 1/p = \infty$, the products in (143) tend to zero. Thus relative density one have a good prime divisor $p \mid a$. For each such coefficient, Lemma 12.4 supplies a common progression $L \mid k$ of trace divisors for both signs and clocks. Lemma 12.3 turns every positive index in that progression into four composite floors, proving the theorem. □

Corollary 12.9 (The earlier residual congruence line)**.** *Fix $C > 1$ with $C \equiv 1 \pmod 3$, and fix $s \in \{1, -1\}$. Theorem 12.1 holds relative to the denominator*

$$\mathcal{A}_{C,s}(H) = \{2C < a \le H : a \text{ odd},\ a \equiv -s \pmod 3, \gcd(a, C) = 1\}.$$

Its cardinality is $\delta_C H + O_C(1)$, where

$$\delta_C = \frac{\varphi(6C)}{12C} = \frac{1}{6}\prod_{\substack{q\mid C\\ q>3}}\left(1 - \frac{1}{q}\right) > 0.$$

Proof. These conditions form a nonempty union of odd classes modulo $6C$. There are $\varphi(6C)/2$ such classes, since exactly half of the units modulo $6C$ have the chosen nonzero residue modulo 3. Apply Theorem 12.1. The factor at 2 is already included by oddness; it must not be counted again. □

Because $a < \beta_{a,C,s} < a + 1$, ordering either sign's roots by $\beta < B$ rather than $a \le B$ changes the cutoff by at most one coefficient. The same density conclusion follows. These statements do not permit conditioning on an arbitrary thin nonlinear subset, such as a selected Pell family or a phase-survivor set.

12.6 Every odd norm seed supplies a linear residue specialization

The main proof uses unramified coefficient divisors and is distinct from the preceding discriminant-prime method [23]. We next give a positive result about that other method, with its precise existential quantifiers.

For $C > 1$ define

$$g_C(W) = (W-2)^3 + CW, \qquad E_C(n) = \mathrm{N}_{\mathbb{Q}(w)/\mathbb{Q}}(1 - 2w^n), \qquad g_C(w) = 0. \tag{144}$$

The derivative $3(W-2)^2 + C$ is positive, and the unique real root lies in $(0, 1)$ since $g_C(0) = -8$ and $g_C(1) = C - 1 > 0$. Thus the monic cubic has no integer root and is irreducible; the norm in (144) is well defined.

Theorem 12.10 (A genuine degree-one prime at every odd seed)**.** *Fix $C > 1$ with $C \equiv 1 \pmod 3$, and let $j \ge 1$ be odd. Put $n = 2 \cdot 3^j - 1$. There are a prime $p \equiv 2 \pmod 3$, $p \nmid C$, and $w_0 \in \mathbb{F}_p$ such that*

$$g_C(w_0) = 0, \qquad 2w_0^n = 1.$$

For either sign, the formulas

$$u = s(w_0 - 2), \qquad a_0 = \frac{3u^2 - C}{2u} \quad \text{in } \mathbb{F}_p \tag{145}$$

give a repeated root of $f_{a_0,C,s}$ and a zero trace at exponent n. The trace zero recurs on an arithmetic progression of negative-clock indices containing j.

Proof. Here $C \geq 4$. The real root r of g_C lies in $(0, 2/3)$, since $g_C(2/3) = -64/27 + 2C/3 > 0$. If $z, \overline{z}$ are the other roots, their product has modulus $|z|^2 = 8/r > 12$. As $n \geq 5$,

$$E_C(n) = (1 - 2r^n)|1 - 2z^n|^2 > 0.$$

This integer is odd: the multiplication matrix of $1 - 2w^n$ is the identity modulo 2.

Modulo 3,

$$g_C(W) = (W - 1)(W^2 + W + 2).$$

For a root ζ of the irreducible quadratic in $\mathbb{F}_9$, $\zeta^4 = -1$ and $\mathrm{N}_{\mathbb{F}_9/\mathbb{F}_3}(1 - \zeta) = 1$. Since j is odd, $n \equiv 5 \pmod 8$, and reduction of the norm gives

$$E_C(n) \equiv (1 - 2)\,\mathrm{N}_{\mathbb{F}_9/\mathbb{F}_3}(1 + 2\zeta) = -1 \pmod 3.$$

Consequently some prime $p \equiv 2 \pmod 3$ divides $E_C(n)$ to an odd valuation. It is not 2, because $E_C(n)$ is odd, and hence $p > 3$.

If $p \mid C$, then $g_C(W) = (W - 2)^3 \pmod p$, so norm zero would give $2^{n+1} = 1 \pmod p$. The order of 2 would divide both $n + 1 = 2 \cdot 3^j$ and $p - 1$, forcing it to divide 2. This would imply $p \mid 3$, a contradiction. Thus $p \nmid C$.

If $p \mid C + 27$, then

$$g_C(W) = (W - 8)(W + 1)^2, \qquad E_C(n) \equiv 9(1 - 2 \cdot 8^n) \pmod p.$$

Therefore $w_0 = 8$ supplies the asserted root. Otherwise

$$\mathrm{disc}(g_C) = -4C^2(C + 27)$$

is nonzero modulo p. The order $\mathbb{Z}[w]$ is maximal at p, and the norm valuation is the sum of local valuations weighted by residue degrees. A degree-two factor contributes an even integer to that valuation. A degree-three factor cannot contribute: a root $\xi \in \mathbb{F}_{p^3}$ with $\xi^n = 1/2$ would have field norm 8, yielding

$$8^n = 2^{-3}, \qquad 2^{3(n+1)} = 1 \pmod p.$$

Again the order of 2 would divide 2, since $3 \nmid p - 1$, which is impossible. The total norm valuation is odd, so a degree-one factor must contribute. Its root $w_0 \in \mathbb{F}_p$ satisfies both equations.

The values $w_0 = 0, 2$ are impossible, since $g_C(0) = -8$ and $g_C(2) = 2C$. Thus (145) is defined, and substitution gives

$$u^3 + Cu + 2sC = 0, \qquad f(u) = f'(u) = 0.$$

The remaining root v satisfies $v/u = -1/w_0$. At odd exponents m, the trace is consequently

$$T_m = u^m(2 - w_0^{-m}) \pmod p. \tag{146}$$

Let $d = \mathrm{ord}_p(w_0)$ and $L = \mathrm{ord}_d(3)$. Because $d \mid p - 1$, $3 \nmid d$; at every $k \equiv j \pmod L$, $2 \cdot 3^k - 1 \equiv n \pmod d$, so (146) vanishes. □

For the real sign $s = -1$, CRT combines $a \equiv a_0 \pmod p$ with $a \equiv 1 \pmod{6C}$, because $p \nmid 6C$. This gives arbitrarily large admissible real Pisot coefficients. Every odd floor equals its trace, so the recurring divisible floors are composite once they exceed p. Unlike Theorem 12.1, this construction need not have $p \mid a$, and an eventual size qualification is appropriate.

The $C = 37$ example has $C + 27 = 64$, so it lies beyond the special method using only odd primes dividing $C + 27$. The theorem does not place an arbitrary prescribed coefficient in the constructed residue class. It also does not assert that primes obtained from different seeds are distinct or have a divergent reciprocal sum.

C	p	$w_0 \pmod p$	$a_0 \pmod p$ for $s=1$
4	53	21	27
7	17	8	7
22	557	295	45
37	1021912841	691653895	197820572

Table 14: Exact examples at the seed $j=1$, $n=5$. The real coefficient residue is the negative of the displayed one.

12.7 The entire discriminant-prime method is not complete

By a discriminant-prime trace certificate we mean a prime divisor of Δ that divides the trace at a relevant odd exponent. The next example excludes all such primes, not just a selected finite list.

Theorem 12.11 (An exact method-completeness counterexample)**.** *Let β be the largest root of*

$$f(X) = X^3 - 1000003X^2 - 4X + 4.$$

It is an irreducible totally real cubic Pisot number, and its floors at all positive odd exponents equal its traces. No prime dividing the polynomial discriminant divides any of these odd traces. Nevertheless

$$\left\lfloor \beta^{2\cdot 3^k - 1} \right\rfloor \text{ is composite} \quad \text{whenever } k \geq 1, \quad k \equiv 10 \pmod{30}.$$

Thus the example disproves completeness of the discriminant-prime-only certificate method; it is not an example of prime floors on the full clock.

Proof. Here $C=4$, $a=1000003$ and $s=-1$. Lemma 12.2 gives the asserted root and floor properties. The complete discriminant factorization is

$$\Delta = 4H, \qquad H = 4000040000204000316 = 2^2 \cdot 3 \cdot 3373 \cdot 98824982710841. \tag{147}$$

Both factors above 3 are prime; the accompanying exact recursive Lucas certificates prove their primality and are described in Section 12.9.

At the two large primes the reductions are $(X-u)^2(X-v)$, with

p	u	v	$w = 2-u \pmod p$
3373	2495	3351	880
98824982710841	70198375859310	57253214703065	28626606851533

All displayed roots are nonzero, and direct multiplication verifies the two factorizations. Formula (146) applies. The exact quadratic-character data are

p	$\left(\frac{w}{p}\right)$	$\left(\frac{2}{p}\right)$
3373	1	-1
98824982710841	-1	1

For odd n, the character of w^n equals that of w, whereas $1/2$ has the character of 2. These mismatches rule out $w^n = 1/2$ and hence $p \mid T_n$ for every odd n. Parity rules out $p=2$. Modulo 3 the polynomial is $(X-1)^2(X+1)$, with odd traces $2-1=1$. Equation (147) now excludes every discriminant prime.

A separate unramified prime gives actual composite floors. In $\mathbb{F}_5[X]/(f)$ put $U_k = X^{2\cdot 3^k-1}$. The recurrence is $U_{k+1} = X^2 U_k^3$; exact multiplication gives

$$U_0 = X, \qquad U_{10} = 4 + 2X + X^2, \qquad U_{30} = X.$$

The trace of U_{10} is 0 (mod 5), so $5 \mid T_{2\cdot 3^k-1}$ for every $k \equiv 10 \pmod{30}$. These real floors equal their traces and exceed 5. Also $5 \nmid \Delta$, as required. □

This example satisfies the usual strict real conjugate bounds as well. Writing its small roots as $-\rho, \sigma$, the exact checks

$$f(-1/400) = -\frac{143361201}{64000000} < 0, \quad 400^{40} > (1000004)^{17}, \quad 4^{20} < (1000003)^3$$

give

$$\rho < \beta^{-17/40}, \qquad 4 < \beta^{3/20}.$$

Since $\rho\sigma = 4/\beta$, they also imply $\rho < \sigma^{17/23}$. Thus the counterexample to that method does not disappear by imposing these strict shrinking-conjugate inequalities.

12.8 What the density theorem does and does not cover

For every fixed C, Theorem 12.1 gives density one on each fixed odd coefficient progression. It does not prove that the exceptional set is empty, that a specified coefficient is certified, or that the coefficient of a hypothetical algebraic least constant is typical. Neither a uniform error term in C nor a uniform positive lower bound on the composite-index density is supplied.

Theorem 12.10 concerns existence of a prime and a coefficient residue at each seed, not coverage of every coefficient. If $\mathcal{B}_{C,p}$ denotes the coefficient residues certified by a recurrent discriminant prime p, a sufficient condition for the same CRT strategy to yield density one for that restricted method would be

$$\sum_p \frac{\#\mathcal{B}_{C,p}}{p} = \infty.$$

No such divergence is proved here. Infinitely many norm seeds, or even infinitely many distinct prime divisors, would not establish it. Theorem 12.11 already shows that completeness is false.

Finite unions of fixed residue classes are covered; arbitrary thin nonlinear conditioning is not. In particular, one cannot transfer this density statement to a selected Pell ray or a complex phase-survivor set without a separate distribution theorem. These distinctions are essential when using coefficient statistics in questions about individual Pisot numbers or global prime-representing minima.

12.9 Portable exact certificates

The accompanying `certificates/` directory contains the mathematical verifiers and their outputs. Run the following commands from the corresponding extracted source archive's directory:

```
python3 certificates/fixed_norm_adaptive_verify.py
python3 certificates/C6_kummer_verify.py
python3 certificates/audit_fixed_line_density_verify.py
python3 certificates/audit_norm_seed_verify.py
python3 certificates/audit_discriminant_counterexample_verify.py
(cd certificates && sha256sum -c SHA256SUMS.txt)
```

The first and fourth programs require SymPy. The other three use only the Python standard library. The `audit_` prefix distinguishes separate implementations of the arithmetic, not an additional mathematical hypothesis. Each program checks exact assertions and prints a PASS result; JSON records contain the finite data.

The principal verifier checks 984 local returns, 932 explicit prime-progression selections, 10 complete-period CRT counts, 99 norm sign/residue checks and 46 degree-one norm specializations. The independent line verifier checks 1188 local cases, 11 progressions, 24 CRT periods and 990 root/parity cases. The independent norm verifier computes the determinant of $I - 2W^n$, where

$$W = \begin{pmatrix} 0 & 0 & 8 \\ 1 & 0 & -C-12 \\ 0 & 1 & 6 \end{pmatrix},$$

and checks 99 norms and 43 linear-root specializations. These finite ranges supplement the proofs; none is used as evidence for prime reciprocal divergence.

For $C = 6$, the exact checker uses the fact that every factor of a cubic has degree 1, 2 or 3, so all roots lie in $\mathbb{F}_{p^6}$. Removing the full 3-part from $p^6 - 1$ and testing $X^N = 1$ in the quotient ring is an exact good-prime test. The sample $p = 11$ gives $X^{40} = 1$ for both signs and $\mathrm{ord}_{40}(3) = 4$. The prime-density conclusion comes from Perucca's theorem, not the search.

12.9.1 Primality and the counterexample

The file `audit_discriminant_counterexample_certificates.json` contains complete recursive Lucas certificates for the two prime factors in (147). The standalone verifier checks 15 primes in the recursion using only integer products, modular powers and gcds. For every non-base entry n, it verifies a complete factorization of $n - 1$ into recursively certified primes and a base b satisfying

$$b^{n-1} \equiv 1 \pmod{n}, \qquad \gcd(b^{(n-1)/q} - 1, n) = 1 \quad (q \mid n-1).$$

These conditions prove primality: for every prime divisor ℓ of n, the order of $b \pmod{\ell}$ contains the full power of every $q \mid n - 1$, so $n - 1 \mid \ell - 1$, forcing $\ell = n$. Thus the factorization is not accepted on a probable-prime label. The same program independently verifies both local factorizations, Euler-criterion mismatches, the strict endpoint inequalities, and the unramified prime-5 cycle.

13 Prime coefficients and simultaneous composite floors

The coefficient itself can be restricted to primes if the chosen local residue conditions are compatible with prime distribution. The Frobenius construction below verifies that compatibility rather than assuming it from a splitting condition.

13.1 Prime-relative density and the main theorem

Fix $C \geq 1$ with $C \equiv 1 \pmod 3$. For $s \in \{1, -1\}$ and an integer $a > 2C$, put

$$f_{a,C,s}(X) = X^3 - aX^2 - CX - sC. \tag{148}$$

Its dominant real root is denoted by $\beta_{a,C,s}$. Section 13.6 proves that these are irreducible cubic Pisot polynomials, with $\beta_{a,C,s} \in (a, a+1)$. The sign $s = -1$ gives the totally real branch; $s = 1$ gives two nonreal small conjugates.

Fix an even integer $M \geq 2$ and a reduced residue class $b \pmod M$. Write

$$\pi(H; M, b) = \#\{a \leq H : a \text{ prime}, \ a \equiv b \pmod M\}.$$

The reduced class is odd because M is even.

Theorem 13.1 (Density one among prime coefficients)**.** *Relative density one of the prime coefficients $a \equiv b \pmod M$, $a > 2C$, have an integer $L = L(a) \geq 1$ such that, for both signs,*

$$\left\lfloor \beta_{a,C,s}^{3^k} \right\rfloor, \qquad \left\lfloor \beta_{a,C,s}^{2 \cdot 3^k - 1} \right\rfloor$$

are composite at every index

$$k = 2 + tL, \qquad t \geq 1.$$

Equivalently, the exceptional prime coefficients up to H are $o(\pi(H; M, b))$ as $H \to \infty$. The same conclusion holds in any fixed nonempty finite union of reduced odd residue classes.

This density is relative to *primes*, not to all integers. The preceding fixed-line theorem [28] proved integer-relative density one using witness primes dividing a. It cannot simply be restricted to prime coefficients, which form a zero-density subset of the integers. Here the witness primes supply *nonzero* trace-zero residues.

The common seed index is $k = 2$, with exponents 9 and 17. We do not assert compositeness at the seed: a seed trace may equal its witness prime. The theorem concerns all subsequent returns. No uniform bound on L, uniformity in C or M, or exceptional-set rate is asserted.

The nonreal cubic Pisot result for infinitely many composite classical floors was already established in [6, theorem *Nonreal cubic conjugates*]. Its inclusion here serves the simultaneous certificate, not a claim to resolve that known case anew.

13.2 Two trace polynomials and exact residue identities

Write $T_{n,C,s}(A)$ for the Newton trace polynomial of $X^3 - AX^2 - CX - sC$, and set $P_n(A) = T_{n,C,-1}(A)$. Its recurrence is

$$\begin{aligned} P_0 = 3, \qquad P_1 = A, \qquad P_2 = A^2 + 2C, \\ P_n = AP_{n-1} + CP_{n-2} - CP_{n-3} \qquad (n \geq 3). \end{aligned} \tag{149}$$

Induction on the leading term shows that every P_n, $n \geq 1$, is monic of degree n in $\mathbb{Z}[A]$.

Lemma 13.2. *For every positive odd n,*

$$P_n(1) = 1. \tag{150}$$

Moreover

$$P_9(0) = -3C^3(3C + 1), \qquad P_{17}(0) = -17C^6(C^2 + 5C + 1). \tag{151}$$

If $C \equiv 1 \pmod 3$ and $A \equiv 5 \pmod 9$, then

$$P_9(A) \equiv 2, \qquad P_{17}(A) \equiv 4 \pmod 9. \tag{152}$$

Proof. The factorization $f_{1,C,-1}(X) = (X-1)(X^2 - C)$ proves (150), since the latter two roots have cancelling odd powers. Substitution of $A = 0$ into (149) gives (151).

The recurrence modulo 9 gives the following complete table. Each row lists values at $A = 0, 1, \ldots, 8$:

$C \bmod 9$	$P_9(A) \bmod 9$	$P_{17}(A) \bmod 9$
1	$(6, 1, 2, 6, 1, 2, 6, 1, 2)$	$(7, 1, 7, 4, 1, 4, 1, 1, 1)$
4	$(6, 1, 2, 6, 1, 2, 6, 1, 2)$	$(1, 1, 7, 7, 1, 4, 4, 1, 1)$
7	$(6, 1, 2, 6, 1, 2, 6, 1, 2)$	$(4, 1, 7, 1, 1, 4, 7, 1, 1)$

These are all possible $C \pmod 9$ under the hypothesis. The entry at $A = 5$ proves (152). Section 13.9 supplies exact implementations of these finite identities. □

Both constant terms in (151) are nonzero for $C > 0$. This will keep the trace-zero residues accessible to prime coefficients.

13.3 An integer anchor and a realized Frobenius class

We use Chebotarev in the following standard form: if $E/\mathbb{Q}$ is finite Galois and $\mathcal{D}$ is a conjugacy class in $\mathrm{Gal}(E/\mathbb{Q})$, the unramified rational primes with that Frobenius class have natural prime density $|\mathcal{D}|/[E : \mathbb{Q}]$. Away from the discriminant of a monic squarefree integral polynomial, its factor degrees modulo a prime are the cycle lengths of Frobenius on its roots [27, Fact 2.1 and Theorem 3.1]. The cyclotomic specialization also gives equidistribution of primes among fixed reduced residue classes.

Proposition 13.3. *For each $n \in \{9, 17\}$ there is a set $\mathcal{S}_n$ of rational primes of positive natural prime density such that every $p \in \mathcal{S}_n$ satisfies*

$$p \nmid 6C, \qquad p \bmod 9 \in \{2, 4, 5, 7\},$$

and P_n has a nonzero root modulo p. In particular, $\sum_{p \in \mathcal{S}_n} 1/p = \infty$.

Proof. Fix n and write $P = P_n$. Factor it over $\mathbb{Q}$ as

$$P = \prod_i G_i^{e_i}, \qquad F = \prod_i G_i,$$

where the distinct G_i are monic irreducible integer polynomials, by Gauss's lemma. Thus $F \in \mathbb{Z}[A]$ is monic and squarefree, with $\mathrm{disc}(F) \neq 0$ and $F(0) \neq 0$. For *every* prime p and residue r,

$$P(r) = 0 \pmod p \quad \Longleftrightarrow \quad F(r) = 0 \pmod p. \tag{153}$$

This follows from the product of integral factors, not an assumption that taking squarefree parts commutes with reduction.

Let $\mathcal{B}$ be the finite set of prime divisors of $6C \,\mathrm{disc}(F)\, F(0)$. Choose an arbitrarily large positive *integer* A_0 satisfying

$$A_0 \equiv 5 \pmod 9, \qquad A_0 \equiv 1 \pmod q \quad (q \in \mathcal{B},\ q \neq 3). \tag{154}$$

CRT makes these compatible and gives an odd A_0. Since P is monic, we may also require $P(A_0) > 0$. For every $q \in \mathcal{B} \setminus \{3\}$, equation (150) gives $P(A_0) \equiv 1 \pmod q$; equation (152) excludes 3. Thus $P(A_0)$ is coprime to every prime in $\mathcal{B}$.

The positive integer $P(A_0)$ is 2 or $4 \pmod 9$. If all its prime divisors were 1 or $-1 \pmod 9$, their product, with multiplicities, would also be 1 or $-1 \pmod 9$. Hence some prime divisor p_0 satisfies

$$p_0 \bmod 9 \in \{2, 4, 5, 7\}. \tag{155}$$

It lies outside $\mathcal{B}$. By (153), $A_0 \pmod{p_0}$ is a root of F. It is nonzero since $p_0 \nmid F(0)$, and simple since $p_0 \nmid \operatorname{disc}(F)$. The corresponding root of P need not be simple.

Let K be the splitting field of F, and put $E = K\mathbb{Q}(\zeta_9)$. The prime p_0 is unramified in both fields and hence in E. Take its *actual* Frobenius conjugacy class $\mathcal{D} \subset \mathrm{Gal}(E/\mathbb{Q})$. Its restriction to K fixes a root of F, and its cyclotomic restriction sends ζ_9 to $\zeta_9^{p_0}$. The fixed-root property is invariant under conjugation; the cyclotomic restriction is constant on the class because the cyclotomic group is abelian.

Chebotarev supplies a prime set of density $|\mathcal{D}|/[E:\mathbb{Q}] > 0$ with both properties. Delete the finitely many primes in $\mathcal{B}$. Each remaining prime has a nonzero root of F, hence of P, and satisfies (155). This is $\mathcal{S}_n$. Positive prime density implies reciprocal divergence, for example by partial summation and the prime number theorem. □

The anchor is not a final candidate coefficient: it need not satisfy $a \equiv b \pmod{M}$ or any other residual condition. It realizes a compatible Frobenius class. No disjointness between K and $\mathbb{Q}(\zeta_9)$ is asserted. One could choose the anchor prime by Dirichlet's theorem, since (154) is reduced, but an integer anchor already suffices.

13.4 A universal exponent for cubic quotient rings

The cubic at a witness prime may have repeated factors. The next bound handles this without another separability requirement.

Lemma 13.4. *Let $p \geq 5$, $p \nmid C$, $a \in \mathbb{F}_p$, and $s \in \{1, -1\}$. In*

$$R = \mathbb{F}_p[X]/(X^3 - aX^2 - CX - sC),$$

the class x of X satisfies

$$x^{E_p} = 1, \qquad E_p = p(p^2-1)(p^2+p+1). \tag{156}$$

For $p \bmod 9 \in \{2,4,5,7\}$, write $E_p = 3^e m_p$, $3 \nmid m_p$. Then

$$e = \begin{cases} 1, & p \equiv 2,5 \pmod 9, \\ 2, & p \equiv 4,7 \pmod 9. \end{cases} \tag{157}$$

With $L_p = \mathrm{ord}_{m_p}(3)$, taking $L_p = 1$ if $m_p = 1$, every $k \geq 2$ with $k \equiv 2 \pmod{L_p}$ satisfies

$$3^k \equiv 9, \qquad 2 \cdot 3^k - 1 \equiv 17 \pmod{E_p}. \tag{158}$$

Proof. The constant term is nonzero, so x is a unit. Modulo the nilradical, each field factor has degree $d \in \{1,2,3\}$, and x has order dividing a common multiple of $p^d - 1$. The integer $B_p = (p^2-1)(p^2+p+1)$ is such a multiple. Therefore $x^{B_p} = 1 + z$ with z nilpotent. Every irreducible-factor multiplicity is at most 3, so $z^3 = 0$. In characteristic $p \geq 5$, $(1+z)^p = 1 + z^p = 1$, proving (156) and explaining its extra factor p.

The four residue classes modulo 9 give (157) by direct valuation of the two factors of B_p. Modulo m_p the order definition gives $3^k \equiv 3^2$. Modulo 3^e both 3^k and 9 vanish, since $k \geq 2$ and $e \leq 2$. CRT gives the first congruence in (158), and the second follows. □

The multiplication-map trace of x^n equals $T_{n,C,s}(a)$ also in a nonreduced algebra: use the companion matrix and Newton's identities. Consequently

$$\begin{aligned} T_{3^k,C,s}(a) &\equiv T_{9,C,s}(a) \pmod p, \\ T_{2\cdot 3^k-1,C,s}(a) &\equiv T_{17,C,s}(a) \pmod p \end{aligned} \qquad (k \geq 2,\ k \equiv 2 \pmod{L_p}). \tag{159}$$

13.5 The prime-coefficient sieve and its denominator

For $p \in \mathcal{S}_n$ put

$$\mathcal{R}_{n,p} = \{r \in \mathbb{F}_p^\times : P_n(r) = 0\}, \qquad r_{n,p} = \#\mathcal{R}_{n,p} \geq 1.$$

Delete the finitely many witness primes dividing M.

Lemma 13.5 (Prime-relative CRT)**.** *For each fixed finite set $\mathcal{Q} \subset \mathcal{S}_n$ of remaining primes,*

$$\lim_{H\to\infty} \frac{\#\{a \leq H : a \text{ prime},\ a \equiv b \pmod M,\ a \bmod p \notin \mathcal{R}_{n,p}\ (p \in \mathcal{Q})\}}{\pi(H; M, b)} = \prod_{p\in\mathcal{Q}} \left(1 - \frac{r_{n,p}}{p-1}\right). \tag{160}$$

Proof. Apart from the finitely many coefficients equal to a prime in $\mathcal{Q}$, prime coefficients occupy only nonzero residues modulo each p. There are $p-1$ reduced choices, of which $r_{n,p}$ are forbidden. CRT combines these with the fixed reduced class modulo M. All resulting reduced classes modulo $M\prod_{p\in\mathcal{Q}} p$ have the same limiting prime density, by cyclotomic Chebotarev. Counting them gives (160). □

The denominator is $p-1$, *not* p. Since $r_{n,p} \geq 1$ and $\sum_{p\in\mathcal{S}_n} 1/p = \infty$, the products tend to zero as $\mathcal{Q}$ increases. Coefficients missing every trace-zero residue avoid each fixed $\mathcal{Q}$. Take their relative upper density as $H \to \infty$ *before* enlarging $\mathcal{Q}$. Thus relative density one of the prime coefficients have a witness prime for this seed and the real sign.

For the complex sign, reflection of the cubic gives, for odd n,

$$T_{n,C,1}(a) = -P_n(-a). \tag{161}$$

The same proof therefore applies with root sets $-\mathcal{R}_{n,p}$. Apply it to the two seeds and the two signs. The intersection of the resulting four sets still has relative density one: their four exceptional sets have a union of relative density zero. No independence among these four conditions is assumed.

For a coefficient in this intersection choose one witness prime for each seed/sign pair, and let $L(a)$ be the least common multiple of their four periods L_p. Equation (159) makes all four traces divisible by their witnesses whenever $k \geq 2$ and $k \equiv 2 \pmod{L(a)}$. The common offset 2 matters: four arbitrary progressions need not intersect, whereas these ones do.

13.6 Returned traces and actual composite floors

Lemma 13.6. *For odd $a > 2C$, both signs give irreducible cubic Pisot numbers $\beta \in (a, a+1)$, and all their traces are odd. For $s = -1$, $\lfloor \beta^n \rfloor = T_{n,C,-1}(a)$ at every positive odd n. For $s = 1$ and $n \geq 3$, the floor is either $T_{n,C,1}(a)$ or $T_{n,C,1}(a) - 1$. If a prime divides the seed trace at exponent $n_0 = 9$ or 17, every later return on the corresponding clock gives a composite floor.*

Proof. For $s = -1$ the signs at $-1, 0, 1, a, a+1$ are

$$2C - a - 1 < 0, \quad C > 0, \quad 1 - a < 0, \quad C(1-a) < 0, \quad (a+1)^2 - Ca > 0.$$

The roots lie in $(-1, 0), (0, 1), (a, a+1)$. Writing the small roots as $-\rho, \sigma$, their sum $\sigma - \rho = a - \beta < 0$ gives $\rho > \sigma$. For odd n,

$$0 < \beta^n - T_n = \rho^n - \sigma^n < 1.$$

For $s = 1$ there is no nonpositive root: at $X = -y$ the value is $-y^3 - ay^2 + Cy - C < 0$ for $y \geq 0$. For $y \leq 1$ use $C(y-1) \leq 0$; for $y \geq 1$ use $a > 2C$. Descartes's rule gives exactly

one positive root, and $f(a) < 0 < f(a+1)$ places it in $(a, a+1)$. The other roots are nonreal, with common modulus $\sqrt{C/\beta} < 1/\sqrt{2}$. Thus

$$|T_n - \beta^n| \le 2(C/\beta)^{n/2} < 1 \qquad (n \ge 3).$$

In either sign no root is an integer: the large root lies between consecutive integers, while any other real root is nonzero with modulus below 1. The monic cubics are irreducible.

The initial traces $3, a, a^2 + 2C$ are odd, and their recurrence preserves parity because its coefficient sum is $1 + C + C = 1 \pmod 2$. This proves the floor alternatives, including the even predecessor in the complex case.

Let p be a seed witness. Its seed trace T_{n_0} is positive and divisible by p, hence $T_{n_0} \ge p$. At a later index $k \ge 3$, the clock exponent n exceeds n_0. Since $\beta > a \ge 3$,

$$T_n > \beta^n - 1 > \beta^{n_0} + 1 > T_{n_0} \ge p.$$

At a return, T_n is a proper multiple of p. The real floor is this composite trace. The complex floor is either that trace or its even predecessor, which exceeds 2; both are composite. □

Completion of Theorem 13.1. Section 13.5 supplies a relative-density-one set of prime coefficients and a common period $L(a)$ for their four trace witnesses. Every $k = 2 + tL(a)$, $t \ge 1$, is strictly later than the seed. Lemma 13.6 proves all four floors composite.

For a finite union of reduced classes, sum the finitely many exceptional estimates. Different fixed moduli may first be replaced by a common modulus; nonreduced lifts contain only finitely many primes and can be discarded. □

One may in particular impose either residual condition $a \equiv -s \pmod 3$ along with oddness. For prime coefficients, $\gcd(a, C) = 1$ deletes only finitely many values. In the real branch, writing the small roots as $-\rho, \sigma$, the equation at $-\rho$ gives $a\rho^2 = C(1+\rho) - \rho^3 < 2C$. Since $\beta \in (a, a+1)$, for fixed C this yields $\rho < \beta^{-17/40}$ and $C < \beta^{3/20}$ for all sufficiently large a. The norm relation $\rho\sigma = C/\beta$ also gives $\rho < \sigma^{17/23}$ in that range. These cofinite real restrictions preserve the prime-relative conclusion. No such transfer to an arbitrary thin phase-defined subset is asserted.

13.7 Arithmetic progressions of certified prime coefficients

We use Green–Tao's relative-prime theorem: every subset of the primes of positive relative upper density contains arbitrarily long nonconstant arithmetic progressions [24, Theorem 1.2]. This is an imported theorem, not a new transference principle.

Corollary 13.7. *For every length $\ell \ge 2$ and bound B, there is an arithmetic progression of prime coefficients*

$$a_j = u + jd > B, \qquad j = 0, \dots, \ell - 1, \qquad d > 0,$$

all congruent to $b \pmod M$, and an integer $L^* \ge 1$ such that at every $k = 2 + tL^*$, $t \ge 1$, all 4ℓ associated floors, from both signs and both clocks, are composite.

Proof. Let $\mathcal{G}$ be the prime set certified in Theorem 13.1. Since $\pi(H; M, b)/\pi(H) \to 1/\varphi(M)$, its density relative to all primes is $1/\varphi(M) > 0$. Remove the finite segment up to $\max(B, 2C)$ and apply Green–Tao.

For the resulting finite coefficient progression put

$$L^* = \operatorname{lcm}_{0 \le j < \ell} L(a_j).$$

Every $k = 2 + tL^*$, $t \geq 1$, is a later common return for every coefficient, proving the assertion. □

The exact arithmetic progression is in the coefficients, equivalently in the integer parts of the large roots, not necessarily in the algebraic roots themselves. No bounded-gap or bounded-cluster conclusion is drawn. Prime-relative density is not substituted for the distribution input of a Maynard-type argument.

13.8 Scope and relation to earlier work

The preceding integer-density proof used primes dividing a. That mechanism does not establish density among prime coefficients. Here nonzero seed-root classes and reduced-residue equidistribution supply a separate result.

The local corpus [6] already treats projective returns and gives a counterexample showing that an arbitrary observed prime trace need not recur after a 3-primary transient. We do not assume automatic recurrence of such a seed. The chosen residues modulo 9 force the relevant transient to have ended by index 2, as quantified in (157). The inspected antecedent passages do not provide the prime-coefficient density argument used here; no literature-wide novelty claim is made.

The anchor in (154) certifies compatibility through an actual Frobenius class. There is no hidden splitting-field disjointness, primitive-root hypothesis, or separability assumption on the later cubic quotient algebra.

The result concerns each fixed $C \equiv 1 \pmod 3$ and fixed reduced coefficient progression. It does not exclude every coefficient, give an effective exceptional-set rate, or identify a coefficient attached to a hypothetical least prime-representing constant. An arbitrary further thin conditioning requires another theorem. Neither the density result nor Corollary 13.7 gives a universal cubic Pisot or Mills-transcendence closure.

13.9 Portable exact checks

The original source archive includes two separately implemented mathematical verifiers with JSON/log outputs. Both require Python 3 and SymPy. Run from the corresponding extracted source archive's directory:

```
python3 certificates/prime_coefficient_mod9_verify.py
python3 certificates/audit_prime_coefficient_density_verify.py
(cd certificates && sha256sum -c SHA256SUMS.txt)
```

The second filename identifies the independent implementation.

The first program checks:

- monicity and degrees through $n = 17$, odd boundary and reflection identities, and the two symbolic constant terms;
- all 81 pairs $(A, C) \pmod 9$ for the integer-valued congruence $P_9(A) \equiv P_3(A) \pmod 9$, and the complete seed table;
- 42400 residue-root equivalence checks for sampled squarefree parts, including primes of bad specialization;
- 5152 cubic unit-exponent checks, including 210 nonreduced cases;
- 68 exponent returns, 282 seed-trace returns, and 24 exact reduced-CRT counts.

The congruence $P_9(A) \equiv P_3(A) \pmod 9$ is integer-valued, not a coefficientwise assertion in $\mathbb{Z}[A, C]$.

The independent program checks the symbolic identities, 10 exact squarefree/CRT anchors, 12044 quotient algebras (including 370 nonreduced cases), and 24 reduced-CRT counts. All acceptance arithmetic is exact.

These computations validate finite algebraic inputs and sample specializations. They do not prove Chebotarev, prime equidistribution, or Green–Tao, and do not compute a numerical prime density or an exceptional-set rate. The analytic conclusions use the general arguments and the precisely identified imported theorems.

14 Signature controlled density and quantitative exceptional sets

The signature argument generalizes the fixed-norm construction to fixed pairs of coefficients and quantifies the exceptional set. The projective synthesis incorporated here improves its original logarithmic exponent from 1/6 to 2/3 and retains the full sieve proof.

The full-state return in the fixed-line density argument can be replaced by a scalar return. This admits four, rather than one, reduced residue classes modulo 9. The resulting improvement is a consequence of the signature and projective mechanisms developed in this article, not an assumption on primitive roots or on the distribution of a distinguished coefficient.

Fix positive integers B, q and consider

$$f_{a,B,q,s}(X) = X^3 - aX^2 - BX - sq, \qquad s \in \{-1, 1\}.$$

Write $\beta_{a,B,q,s}$ for its dominant root and put

$$A_- = B + q + 1, \qquad A_+ = \max\{B + q + 1, B^2/(4q), 2q\}.$$

Fix an even modulus M and a reduced residue class $b \pmod M$. Let $E_{\mathbb{Z}}(X)$ count the integer coefficients $a \equiv b \pmod M$, $A_- < a \le X$, for which there is no positive integer L such that both real-branch floors

$$\left\lfloor \beta_{a,B,q,-1}^{3^k} \right\rfloor, \qquad \left\lfloor \beta_{a,B,q,-1}^{2\cdot 3^k - 1} \right\rfloor$$

are composite for every $k = 1 + tL$, $t \ge 1$. Let $E_{\mathbb{P}}(X)$ count the prime coefficients among these exceptions. For the simultaneous four-floor conclusion use $a > A_+$ and include both signs; this version will require $B \equiv q \pmod 2$.

Theorem 14.1 (Projective improvement of the fixed-line rates)**.** *For each fixed B, q, M as above,*

$$E_{\mathbb{Z}}(X) \ll_{B,q,M} \frac{X}{(\log X)^{2/3}}, \qquad E_{\mathbb{P}}(X) \ll_{B,q,M} \frac{X}{(\log X)^{5/3}}.$$

The same estimates hold for exceptions to the simultaneous four-floor conclusion when $B \equiv q \pmod 2$. Thus its exceptional proportion among prime coefficients is $O_{B,q,M}((\log X)^{-2/3})$. Finite unions of fixed reduced odd classes are allowed. On the diagonal $B = q = C$, the sufficient root threshold may be replaced by $a > 2C$, for every fixed positive integer C. No bound on L, uniformity in the fixed parameters, or exclusion of an individual coefficient is asserted.

14.0.1 A universal projective exponent

Lemma 14.2 (Scalar rather than full-state returns)**.** *Let $p \geq 5$ be prime, let $f \in \mathbb{F}_p[X]$ be a monic cubic, and let u be any unit in $R = \mathbb{F}_p[X]/(f)$. Set*

$$D_p = p \operatorname{lcm}(p^2 - 1, p^2 + p + 1) = \frac{p(p^2 - 1)(p^2 + p + 1)}{\gcd(p - 1, 3)}.$$

Then $u^{D_p} \in \mathbb{F}_p^{\times}$, viewed as the scalar units of R. Moreover

$$v_3(D_p) = \begin{cases} v_3(p + 1), & p \equiv 2 \pmod 3, \\ v_3(p - 1), & p \equiv 1 \pmod 3. \end{cases}$$

In particular $v_3(D_p) = 1$ for $p \pmod 9 \in \{2, 4, 5, 7\}$.

Proof. If f is irreducible, R is the field $\mathbb{F}_{p^3}$ and u^{p^2+p+1} is its nonzero field norm, hence is scalar. Since $p^2 + p + 1$ divides D_p, the assertion follows. If f is reducible, every field factor of the reduced algebra has degree at most two. Thus $u^{p^2-1} = 1 + z$ with z in the nilradical. The nilradical of a cubic quotient has cube zero. Consequently $(1 + z)^p = 1 + z^p = 1$, and $u^{p(p^2-1)} = 1$. This exponent divides D_p, proving the result even at repeated reductions. Finally $\gcd(p^2 - 1, p^2 + p + 1) = \gcd(p - 1, 3)$. If $p \equiv 1 \pmod 3$, then $v_3(p^2 + p + 1) = 1$; if $p \equiv 2 \pmod 3$, that factor is prime to 3. These facts give the displayed valuations. □

For the cubic $f_{a,B,q,s}$, exclude the primes dividing q so that $x = X \bmod f$ is a unit. At any remaining prime in the four indicated classes put

$$m_p = D_p/3, \qquad L_p = \operatorname{ord}_{m_p}(3),$$

with order one if the modulus is one. For $k \geq 1$ and $k \equiv 1 \pmod{L_p}$, the differences

$$3^k - 3, \qquad (2 \cdot 3^k - 1) - 5 = 2(3^k - 3)$$

are multiples of D_p. They are divisible by m_p by definition of the order and by 3 directly. Therefore each returned ring state is a nonzero scalar multiple of its seed state. The trace of multiplication is linear over $\mathbb{F}_p$, so

$$\begin{aligned} T_{3,s}(a) \equiv 0 \pmod p &\implies T_{3^k,s}(a) \equiv 0 \pmod p, \\ T_{5,s}(a) \equiv 0 \pmod p &\implies T_{2\cdot 3^k-1,s}(a) \equiv 0 \pmod p. \end{aligned} \tag{162}$$

The equalities of the trace values themselves are not required and are not asserted. The multiplication trace agrees with the Newton trace even in a nonreduced cubic algebra, by the companion matrix.

14.0.2 Four compatible Frobenius fibers

For the real branch the two seed polynomials are

$$P_3(A) = A^3 + 3BA - 3q, \qquad P_5(A) = A^5 + 5BA^3 - 5qA^2 + 5B^2A - 5Bq.$$

Each has exactly one real root, which is positive and simple. For P_3 this follows from its positive derivative and negative value at zero. For P_5, write

$$P_5(A) = 5(A^2 + B)(Q_B(A) - q), \qquad Q_B(A) = \frac{A(A^4 + 5BA^2 + 5B^2)}{5(A^2 + B)}.$$

The derivative

$$Q_B'(A) = \frac{3A^6 + 10BA^4 + 10B^2A^2 + 5B^3}{5(A^2 + B)^2} > 0$$

and $Q_B(A) \sim A^3/5$ at both ends prove the assertion. Choose the monic irreducible factor $h \in \mathbb{Z}[A]$ containing this unique real root α. Its degree d is 1, 3, or 5, its constant term is nonzero, and $K = \mathbb{Q}(\alpha)$ has exactly one real embedding.

Let $Z = \mathbb{Q}(\zeta_9)$. Then $K \cap Z = \mathbb{Q}$. Indeed, degrees one and five are immediate from the degrees; in degree three a nontrivial intersection would identify K with the unique cubic subfield of Z, which is totally real, contradicting its signature. Let E be the compositum of the splitting field of h with Z, and write

$$G = \mathrm{Gal}(E/\mathbb{Q}), \qquad \pi : G \longrightarrow \mathrm{Gal}(Z/\mathbb{Q}).$$

The stabilizer H of α surjects onto this cyclic quotient of order six: the fixed field in Z of $\pi(H)$ is $Z \cap E^H = Z \cap K = \mathbb{Q}$. Every other root stabilizer also surjects, by conjugacy. This establishes compatibility, without a linear-disjointness assumption on the splitting field.

For each $r \in \{2, 4, 5, 7\}$ let $\tau_r(\zeta_9) = \zeta_9^r$. Since each root stabilizer has order $|G|/d$, its intersection with each fiber $\pi^{-1}(\tau_r)$ has size $|G|/(6d)$. Counting pairs consisting of an element and one of its fixed roots therefore gives

$$\frac{1}{|G|} \sum_{g \in \pi^{-1}(\{\tau_2, \tau_4, \tau_5, \tau_7\})} |\operatorname{Fix}(g)| = \frac{4}{6} = \frac{2}{3}. \tag{163}$$

Delete the finitely many primes at most seven, the primes dividing $qMh(0)\operatorname{disc}(h)$, and any additional ramified primes of E. Let $r(p)$ be the number of roots of $h \bmod p$ when $p \bmod 9 \in \{2, 4, 5, 7\}$ is not deleted, and zero otherwise. All these roots are nonzero coefficient residues; $0 \le r(p) \le 5$. Factorization–Frobenius compatibility and Chebotarev [27, Fact 2.1 and Theorem 3.1], together with the Chebotarev Mertens theorem [29, Theorem A], imply

$$\sum_{p \le y} \frac{r(p)}{p} = \frac{2}{3} \log\log y + O_{B,q,M}(1). \tag{164}$$

For clarity, the Mertens input is applied separately to the finitely many conjugacy classes and weighted by their fixed-root counts; (163) is exactly the resulting coefficient. Thus the value $2/3$ does not come from an unproved independence heuristic.

A root $a \bmod p$ is a seed-trace zero because h divides P_n. For the partner sign, the identity $T_{n,1}(a) = -P_n(-a)$ for odd n gives the reflected nonzero root classes, with the same cardinalities and the same Mertens estimate.

14.0.3 Quantitative sieve and simultaneous floor returns

Here is a self-contained upper-sieve calculation recording the logarithmic exponent and its error term. Fix one seed/sign pair. For integer coefficients forbid its $r(p)$ root classes, writing $\rho(p) = r(p)$. For prime coefficients additionally forbid the zero class at every undeleted prime, writing $\rho(p) = 1 + r(p)$ there. At deleted primes impose no condition. Omitting primes with zero local count, we have $0 < \rho(p) \le 6 < p$ and

$$\sum_{p \le y} \frac{\rho(p)}{p} = \kappa \log\log y + O(1), \qquad \kappa = \frac{2}{3} \text{ or } \frac{5}{3},$$

respectively. Nonzero seed roots are essential for the extra class to be distinct in the prime case.

Let N be the number of integer coefficients in the fixed progression and relevant interval up to X. On squarefree integers supported on the sieve primes extend ρ multiplicatively. CRT gives $N\rho(v)/v + R_v$ coefficients forbidden at every prime dividing v, where $|R_v| \le \rho(v)$. Put

$$u(p) = \rho(p)/p, \qquad g(p) = u(p)/(1-u(p)), \qquad G(z) = \sum_{\substack{d \le z \\ d \text{ squarefree, supported}}} g(d).$$

The Selberg quadratic weights, supported on $d \le z$, are

$$\lambda_d = \mu(d) \prod_{p \mid d} (1-u(p))^{-1} \frac{G_d(z/d)}{G(z)}, \qquad G_d(v) = \sum_{\substack{m \le v \\ (m,d)=1}} g(m).$$

All sums here use squarefree supported integers. They satisfy $\lambda_1 = 1$, $|\lambda_d| \le 1$, and

$$\sum_{d,e \le z} \lambda_d \lambda_e u([d,e]) = G(z)^{-1}.$$

For verification, set $Y_r = \sum_{r \mid d} \lambda_d u(d)$. Then the quadratic form is $\sum_r Y_r^2/g(r)$ and $\sum_r \mu(r) Y_r = 1$; its minimizer is $Y_r = \mu(r)g(r)/G(z)$. Divisor inversion gives the displayed weights. Their absolute bound follows by including in $G(z)$ all distinct terms ℓm with $\ell \mid d$, $m \le z/d$, and $(m,d) = 1$. Squaring the divisor sum majorizes the indicator of survival, so

$$S(X,z) \le \frac{N}{G(z)} + \sum_{d,e \le z} \rho([d,e]) \le \frac{N}{G(z)} + z^2 (1+\log z)^{10}.$$

For the last inequality use $\rho([d,e]) \le \rho(d)\rho(e)$, $\rho(d) \le 6^{\omega(d)} \le \tau_6(d)$, and $\sum_{d \le z} \tau_6(d) \le z(1+\log z)^5$.

Finally $G(z) \gg (\log z)^\kappa$. Indeed set $y = z^{1/24}$ and give a squarefree product D of primes at most y probability proportional to $g(D)$. Its expected logarithm is

$$\sum_{p \le y} \frac{\rho(p) \log p}{p} \le 6 \log y + O(1) = \tfrac{1}{4} \log z + O(1).$$

Markov's inequality puts at least half the product weight on $D \le z$ for all sufficiently large z. This weight is included in $G(z)$, while

$$\prod_{p \le y} (1+g(p)) = \prod_{p \le y} (1-\rho(p)/p)^{-1} \asymp (\log y)^\kappa$$

follows by taking logarithms and using the convergence of $\sum_p \rho(p)^2/p^2$. Choose $z = X^{1/4}$. The result is

$$S(X,z) \ll \frac{X}{(\log X)^\kappa}.$$

In the prime case the coefficients at most z contribute at most z, and every larger prime avoids the zero classes at sieve primes. Thus the same estimate applies to prime coefficients lacking this seed/sign witness. This does not require prime equidistribution in a growing modulus.

It remains to check that avoiding these exceptional sets supplies actual composite floors at common indices. For $a > A_-$, the real cubic has roots $-\rho, \sigma, \beta$, with $0 < \sigma < \rho < 1$ and $a < \beta < a+1$. This follows from its signs at $-1, 0, 1, a, a+1$, and the ordering of the small roots from their sum $a - \beta < 0$. At every positive odd exponent,

$$0 < \beta^n - T_{n,-1}(a) = \rho^n - \sigma^n < 1,$$

so its floor is its trace. For $a > A_+$ the partner has no nonpositive real root, since for $y \geq 0$

$$f_{a,B,q,1}(-y) = -y^3 - ay^2 + By - q \leq -y^3 + B^2/(4a) - q < 0.$$

Descartes's rule and the signs at $a, a+1$ give its sole real root in $(a, a+1)$. Its other two roots have modulus $\sqrt{q/\beta} < 1/\sqrt{2}$, so $|T_{n,1}(a) - \beta^n| < 1$ for $n \geq 3$. These cubics are irreducible since they have no integer root. When a is odd and $B \equiv q \pmod 2$, Newton's recurrence preserves oddness of every trace. Thus the partner floor is either its odd trace or its even predecessor.

Outside the union of the two one-pair exceptional sets, choose a witness prime for each real seed; for four floors choose witnesses for both seeds and signs. Take L to be the least common multiple of their L_p. All seed trace zeros recur at every $k = 1 + tL$, $t \geq 1$, by (162). The common offset 1 is crucial; arbitrary progressions need not intersect. If p divides a positive seed trace $T_{n_0,s}(a)$, then $p \leq T_{n_0,s}(a)$. At a later clock index, since $\beta > a \geq 3$,

$$T_{n,s}(a) > \beta^n - 1 > \beta^{n_0} + 1 > T_{n_0,s}(a) \geq p.$$

The returned trace is a proper multiple of p. The real floor is that trace. In the partner case the alternative is an even integer larger than two, also composite. Taking a finite union of the exceptional estimates proves the theorem. Division by $\pi(X; M, b) \sim X/(\varphi(M) \log X)$ gives its relative prime form. The sharper diagonal threshold follows from the same root signs and the bound $C/\beta < 1/2$; it changes only a finite set of coefficients.

Corollary 14.3 (A finite catalogue with one index progression)**.** *Fix finitely many positive pairs (B_i, q_i) and one fixed reduced odd coefficient progression. For all but $O(X/(\log X)^{2/3})$ integer coefficients up to X, and all but $O(X/(\log X)^{5/3})$ prime coefficients up to X, there is one progression $k = 1 + tL(a)$, $t \geq 1$, on which the two real floors are composite simultaneously for every pair in the catalogue. The partner floors may simultaneously be included for every pair with $B_i \equiv q_i \pmod 2$. All coefficients are taken above the largest applicable fixed threshold.*

Proof. Take a finite union of the preceding exceptional sets and the least common multiple of the finitely many resulting periods. The common offset remains one. Constants may depend on the entire fixed catalogue; no assertion is made for a catalogue growing with X. □

Relation to the earlier rate. The original full-state exponent used only the residue class 2 (mod 9) and gave logarithmic exponents 1/6 and 7/6. The scalar-return argument above replaces that proof and improves those exponents to 2/3 and 5/3 with the same fixed-parameter scope.

14.1 Prime coefficient progressions and scope

Corollary 14.4 (Finite coefficient progressions with common returns)**.** *For each fixed B, q, M, b, every length $\ell \geq 2$ and every lower bound admit a nonconstant arithmetic progression of certified prime coefficients above that bound. All its coefficients have a common progression $k = 1 + tL_*$, $t \geq 1$, of composite floors for both real clocks. If $B \equiv q \pmod 2$, all four sign and clock combinations can be included simultaneously.*

Proof. The certified prime set has relative density $1/\varphi(M) > 0$ among all primes. Removing an initial segment preserves that density. The relative-prime form of the Green–Tao theorem [24, Theorem 1.2] gives an arithmetic progression of any prescribed finite length. For its finitely many coefficients choose their periods $L(a)$ and take $L_* = \operatorname{lcm}_a L(a)$. Their common offset is 1, so this least common multiple synchronizes every asserted return. □

The arithmetic progression is in the coefficients, equivalently in the integer parts of the dominant roots, not necessarily in the roots themselves. No bounded-gap or Maynard bounded-cluster conclusion is inferred.

The mechanism differs from the earlier integer-line result [28] and from the congruence-restricted prime-line argument [32]. Here the real signature of a factor of P_3 or P_5 replaces the earlier integer CRT anchor. It proves precisely the needed compatibility even if the splitting field has nontrivial cyclotomic intersection. It also avoids assuming that every quintic is irreducible.

A close methodological precedent is Brüdern–Soundararajan [30, Section 2, Lemma 3]. They characterize containment of the mth roots of unity in an irreducible polynomial's root field by the condition that all but finitely many prime divisors of its values lie in 1 (mod m), using Chebotarev. The same lemma is numbered 2 in their arXiv version 1. This relates polynomial prime divisors to root-field cyclotomic information, as does our construction. The restriction-image argument in the root-stabilizer argument is standard Galois correspondence, not a new general Galois principle.

For a neighboring Pisot-floor construction, Hahn–Ismailescu–Kim–Kim [31, Section 4, Theorem 4] prove compositeness at every positive power for the dominant root of $X^3 - aX^2 + cX - 1$ when $c \geq 0$, $a \geq 2c + 5$, and $a \equiv c \equiv 3 \pmod 6$. Those conditions force $a > 3$ to be divisible by 3, so the coefficient a is composite. Their all-power construction concerns a different linear-coefficient sign; it is not the fixed negative-linear, prime-coefficient density assertion proved here.

The existing corpus [6] already discusses modular and projective returns and warns that a completely split prime seed need not recur along a prescribed clock. The original choice $p \equiv 2 \pmod 9$ controlled the 3-primary part of a universal cubic exponent, which makes recurrence at the first seed valid. No literature-wide priority claim is made.

These statements concern each fixed positive pair (B, q). They do not cover arbitrary coefficient signs or positive-norm totally real cubics. With $B \not\equiv q \pmod 2$, the partner trace congruences still hold, but this proof gives no corresponding partner-floor conclusion. The estimates are not uniform in the parameters, the periods, or an index-density parameter. They supply neither an explicit onset nor emptiness of the exceptional set. Powering a Pisot number generally changes B, q, so the fixed-parameter estimates do not control such a nonlinear orbit. An arbitrary thin phase condition, a specified individual survivor, or a candidate for a least prime-representing constant needs a separate argument. In particular no universal Mills or Pisot-transcendence closure follows.

14.2 Portable exact certificates

The accompanying programs use exact integer, rational, and symbolic arithmetic. In the original source archive, run from its extracted directory:

```
python3 certificates/signature_prime_density_verify.py
python3 certificates/fixed_B_q_signature_verify.py
python3 certificates/quantitative_sieve_verify.py
(cd certificates && sha256sum -c SHA256SUMS.txt)
```

The first two programs require Python 3 and SymPy. The third uses only the Python standard library. Their JSON outputs and concise logs are included.

The diagonal checker verifies the symbolic trace and derivative identities, 252 exact Sturm and factor samples, the order-six group modulo 9, 6168 cubic quotient algebras including 162 nonreduced cases, 44 exponent returns, 40 usable sample prime witnesses, 260 reflected trace returns, and 48 exact reduced-CRT counts.

The general (B, q) checker verifies the displayed derivative identity, 362 real-root and factor samples including three linear-factor cases, 63 two-sign root-location and parity samples, 8916 cubic quotient algebras including 418 repeated reductions, and 17832 seed-trace equalities.

The original quantitative checker verifies three exact finite group models of the single-fiber identity underlying (163), including a non-disjoint subdirect product. It uses the original restriction $p \equiv 2 \pmod 9$ and weighted density $1/6$, not the new four-fiber density $2/3$. It also verifies four rational Selberg quadratic-form and residue-sieve calculations. It checks the minimizing weights, their absolute bound by 1, the exact quadratic minimum, the squared-sum majorant, and the remainder estimate.

These finite tests supplement the general proofs. They do not prove the imported Chebotarev, Mertens, or Green–Tao theorems; do not turn the asymptotic constants into numerical ones; and do not establish exclusion of every coefficient.

Part IV

Thin families and adaptive projective witnesses

15 The thin prime power family

Density statements need not resolve a thin sequence such as $a = 7^m$. We therefore keep this family separate and prove explicit residue-class returns, together with an obstruction to covering it by finitely many fixed primes.

15.1 The thin family and the two conclusions

For $m \geq 2$ put $a = 7^m$ and

$$f_m(X) = X^3 - aX^2 - 4X + 4. \tag{165}$$

We will show that its dominant root β_m is totally real cubic Pisot. Recall that a Pisot number is a real algebraic integer greater than one whose other conjugates have modulus less than one. Let $T_n(a)$ be the sum of the nth powers of the three roots, and retain the original clock indices

$$E_k^+ = 3^k, \qquad E_k^- = 2 \cdot 3^k - 1, \qquad k \geq 1. \tag{166}$$

Here $\lfloor \beta_m \rfloor = 7^m$ is composite: the family is not already a classical Mills power ξ^{3^j} at a required prime index, even though it satisfies the root, norm, and small-conjugate inequalities proved below. Our clocks start at $k = 1$, not at exponent one.

This exponential coefficient family is thin among integer coefficients. A density-one result in the ambient coefficient line does not imply that any particular 7^m belongs to its successful set. The arguments below work directly on this family.

Theorem 15.1. *For every $m \geq 2$ with $m \not\equiv 0, 4 \pmod{12}$, each sequence*

$$\left\lfloor \beta_m^{E_k^+} \right\rfloor, \qquad \left\lfloor \beta_m^{E_k^-} \right\rfloor$$

has an explicit infinite arithmetic progression of composite terms in its original index k. The negative clock is certified for every $12 \nmid m$. The classical clock is certified whenever $4 \nmid m$ or $m \equiv 2 \pmod{3}$.

Every indicated term is composite, including the first index in its certificate. The two clocks may use different *progressions; no common simultaneous progression is asserted.*

Theorem 15.2. *For every finite set P of primes, there are infinitely many $m \geq 2$ such that, for every $p \in P$ and every $k \geq 1$,*

$$p \nmid \left\lfloor \beta_m^{E_k^+} \right\rfloor, \qquad p \nmid \left\lfloor \beta_m^{E_k^-} \right\rfloor.$$

These m can be arbitrarily large and their Pisot numbers satisfy

$$4 < \beta_m^{3/20}, \qquad |\gamma| < \tfrac{1}{2}\beta_m^{-17/40} \quad \text{for both other conjugates } \gamma.$$

Theorem 15.2 does not say that the floors are prime. Its quantifiers are $\forall P$ finite, $\exists$ infinitely many m, not the reverse. It excludes a fixed finite-prime cover of the whole family, but does not exclude a witness prime depending on m. All results needed here are proved below; no prime-distribution theorem or ambient density assertion is used.

15.2 Actual Pisot roots and exact odd floors

Lemma 15.3. *For every integer $a > 8$, the polynomial $X^3 - aX^2 - 4X + 4$ is irreducible and has roots $-w, v, \beta$ with*

$$0 < v < w < 1, \qquad a < \beta < a + 1.$$

For every positive odd n,

$$0 < \beta^n - T_n(a) = w^n - v^n < 1, \qquad \lfloor \beta^n \rfloor = T_n(a). \tag{167}$$

If $a = 7^m$ and $m \geq 15$, both small conjugates and the norm satisfy the bounds in Theorem 15.2.

Proof. The polynomial values at $-1, 0, 1$ have signs $-, +, -$, since $f(-1) = 7-a$ and $f(1) = 1-a$. Also $f(a) = 4(1-a) < 0$ and $f(a+1) = (a-1)^2 > 0$. There is one root in each of $(-1, 0), (0, 1), (a, a+1)$. These intervals contain no integer, so the monic integral cubic has no rational root and is irreducible. Vieta gives $v - w = a - \beta < 0$, which proves (167).

The equation at $-w$ gives $aw^2 = 4 + 4w - w^3 < 8$. For $m \geq 15$ we have $a > 2^{41}$ and $\beta < a + 1 < 2a$, hence

$$(2w)^{40}\beta^{17} < (32/a)^{20}(2a)^{17} = \frac{2^{117}}{a^3} < 1.$$

This gives the asserted strict small-conjugate bound, also for $v < w$. The norm has absolute value four and $4^{20} < \beta^3$. □

Newton's identities give the integral trace recurrence

$$\begin{aligned} &T_0(a) = 3, \quad T_1(a) = a, \quad T_2(a) = a^2 + 8, \\ &T_n(a) = aT_{n-1}(a) + 4T_{n-2}(a) - 4T_{n-3}(a), \qquad n \geq 3. \end{aligned} \tag{168}$$

It also defines trace polynomials at arbitrary integer parameters, including reducible specializations. When $a = 7^m$ all positive traces are odd. At every clock index $k \geq 1$, the exponent is odd and at least three, so

$$T_{E_k^{\pm}}(a) > \beta^{E_k^{\pm}} - 1 > a^3 - 1 \geq 49^3 - 1 > 19. \tag{169}$$

Thus a divisor in $\{5, 11, 13, 19\}$ proves actual floor compositeness at every certified index, not merely eventually.

15.3 Exact returns in a cubic finite algebra

Fix a prime p and a coefficient residue $a \in \mathbb{F}_p$. Work in

$$R_{p,a} = \mathbb{F}_p[X]/(X^3 - aX^2 - 4X + 4), \qquad x = X \bmod f.$$

The trace of multiplication by an element of this three-dimensional algebra is the linear functional

$$\tau(c_0 + c_1 x + c_2 x^2) = 3c_0 + ac_1 + (a^2 + 8)c_2. \tag{170}$$

In particular $\tau(x^n) = T_n(a) \pmod p$. This follows from (168), or directly from the multiplication matrix

$$M_{p,a} = \begin{pmatrix} 0 & 0 & -4 \\ 1 & 0 & 4 \\ 0 & 1 & a \end{pmatrix}. \tag{171}$$

These identities do not require a separable reduction.

Starting from $u_0 = x$, define

$$F_+(u) = u^3, \qquad F_-(u) = x^2u^3.$$

Their respective iterates are $u_k = x^{E_k^+}$ and $u_k = x^{E_k^-}$, now allowing $k = 0$ just for initialization.

Lemma 15.4 (Exact return certificate)**.** *For either map, suppose k_0, h are positive integers and*

$$u_{k_0+h} = u_{k_0}, \qquad \tau(u_{k_0}) = 0.$$

Then $p \mid T_{E_k^\pm}(a)$ at every original index $k = k_0 + th$, $t \geq 0$.

Proof. Equality of two states under a deterministic map makes their later iterates equal. Repeating the return gives $u_{k_0+th} = u_{k_0}$, and taking (170) proves the claim. No purely periodic behavior from $k = 0$ is assumed. □

15.3.1 The finite certificate table

The orders of 7 modulo $5, 11, 13, 19$ are $4, 10, 12, 3$, respectively. Table 15 gives exact return certificates, with the coefficient residue forced by its condition on m. The state column lists (c_0, c_1, c_2) for u_{k_0}; in every row, $u_{k_0+h} = u_{k_0}$ and (170) is zero. The displayed h need not be the minimal period.

Label	Clock	p	m class	$a \bmod p$	k_0	h	u_{k_0}
C1	+	5	1 mod 4	2	1	2	$(1, 4, 2)$
C2	+	5	2 mod 4	4	1	30	$(1, 4, 4)$
C3	+	5	3 mod 4	3	6	30	$(0, 1, 1)$
N2	−	5	2 mod 4	4	5	30	$(4, 3, 4)$
N3	−	5	3 mod 4	3	10	30	$(4, 2, 1)$
N11a	−	11	1 mod 10	7	2	4	$(7, 8, 0)$
N11b	−	11	5 mod 10	10	13	36	$(6, 9, 10)$
N11c	−	11	7 mod 10	6	2	4	$(2, 10, 8)$
N11d	−	11	9 mod 10	8	8	36	$(10, 7, 4)$
N19a	−	19	1 mod 3	7	5	4	$(11, 17, 5)$
N19b	−	19	2 mod 3	11	40	126	$(13, 2, 1)$
N13a	−	13	9 mod 12	8	2	6	$(3, 11, 1)$
N13b	−	13	4 mod 12	9	9	10	$(9, 4, 12)$
N13c	−	13	8 mod 12	3	4	10	$(5, 5, 12)$
C19	+	19	2 mod 3	11	19	126	$(5, 1, 16)$

Table 15: Each row certifies all $k = k_0 + th$, $t \geq 0$, on its stated clock.

For a transparent reproduction of the finite identities, multiply two coefficient triples by convolution to obtain $z_0, \ldots, z_4$. For $j = 4, 3$, replace $z_j x^j$ using $x^3 = ax^2 + 4x - 4$:

$$z_{j-1} \leftarrow z_{j-1} + az_j, \qquad z_{j-2} \leftarrow z_{j-2} + 4z_j, \qquad z_{j-3} \leftarrow z_{j-3} - 4z_j. \tag{172}$$

Reduce the first three coefficients modulo p, iterate the indicated map from x, and compare its states at k_0 and $k_0 + h$. This requires at most 166 cubing steps for any row. The appendix provides this exact algorithm and a separate matrix-exponentiation checker for the same identities. Thus the infinite conclusions rest on finite algebraic equalities and Lemma 15.4, not on extrapolating a sample of clock terms.

The polynomial discriminant is

$$\Delta(a) = 16(a^3 + a^2 + 18a - 11). \tag{173}$$

It is nonzero modulo p at every table row, and $p \nmid a$. These primes are therefore unramified in the cubic number field: the field discriminant divides the polynomial discriminant. In particular the table does not use coefficient-divisor witnesses or primes of repeated reduction.

15.3.2 Why the prime five rows have a simple explanation

For $a = 2 \pmod 5$,

$$f = (X + 1)(X^2 + 2X + 4).$$

The quadratic roots are $2\zeta, 2\zeta^2$, with $\zeta^3 = 1$ and $\zeta \neq 1$. For $n = 3^k$, $k \geq 1$, the trace is $-1 + 2^{n+1}$ modulo five, which vanishes when k is odd. This is C1.

For $a = 3, 4 \pmod 5$, the cubics have no root in $\mathbb{F}_5$ and are irreducible. Their roots lie in $\mathbb{F}_{125}$ and have norm $-4 = 1$, so belong to the cyclic norm-one group of order 31. Each root is nonidentity, hence has order 31. This group contains a trace-zero element: any root of the irreducible polynomial $X^3 + 2X + 4$ supplies one. Since 3 generates $(\mathbb{Z}/31\mathbb{Z})^*$, the classical exponent orbit reaches every nonzero exponent residue. The negative exponent orbit reaches every residue except -1. The traces at exponents 0 and -1 are 3 and 1, respectively, so neither can be the trace-zero exponent. Both clocks therefore reach a trace-zero group element. The table records explicit indices and returns for this mechanism.

15.3.3 The coefficient classes are covered pointwise

Proof of Theorem 15.1. C1–C3 cover every $4 \nmid m$ on the classical clock, and C19 adds $m \equiv 2 \pmod 3$. Their union is precisely the classes $m \not\equiv 0, 4 \pmod{12}$.

For the negative clock, N2 and N3 cover $m \equiv 2, 3 \pmod 4$. Suppose $m \equiv 1 \pmod 4$. If $m \bmod 10$ is in $\{1, 5, 7, 9\}$, one of N11a–N11d applies. Otherwise $m \equiv 3 \pmod{10}$, so $m \equiv 13 \pmod{20}$. If $3 \nmid m$, use N19a or N19b. If $3 \mid m$, then $m \equiv 33 \pmod{60}$ and hence $m \equiv 9 \pmod{12}$, covered by N13a. This covers every odd m. Finally, $4 \mid m$ and $12 \nmid m$ give $m \equiv 4, 8 \pmod{12}$, covered by N13b or N13c.

Lemma 15.4 supplies the index progression for each row. Equations (167) and (169) turn its divisor into actual compositeness at every indicated index. No matching of the two clocks' offsets is needed or asserted. □

This is a complete argument on the stated coefficient-exponent classes, not a density transfer. Coefficients outside the table's guarantee may still have infinitely many composite floors.

15.4 Every fixed finite set of primes misses infinitely many members

The prime 7 divides the coefficient, but never a scheduled trace. At $a = 0 \pmod 7$, the two finite-algebra orbits have the following exact states:

clock	u_1	u_2	$(\tau(u_1), \tau(u_2))$
$+$	$(3, 4, 0)$	$(2, 0, 4)$	$(2, 3)$
$-$	$(5, 2, 3)$	$(3, 3, 6)$	$(4, 1)$

For each clock, its map sends u_2 back to u_1. Thus the indicated nonzero trace residues alternate for all $k \geq 1$. This two-cycle is checked directly by (172).

Proof of Theorem 15.2. Given finite P, set

$$L = \operatorname{lcm}\{\operatorname{ord}_p(7) : p \in P,\ p \neq 7\}, \tag{174}$$

with the empty least common multiple equal to one. For every positive multiple m of L and every $p \in P \setminus \{7\}$, we have $7^m \equiv 1 \pmod{p}$. The integral factorization

$$X^3 - X^2 - 4X + 4 = (X - 1)(X^2 - 4)$$

gives, at every positive odd exponent,

$$T_n(1) = 1 + 2^n + (-2)^n = 1. \tag{175}$$

The trace polynomial therefore specializes to $1 \pmod{p}$. This also holds at primes where the factors meet; it is an integral identity and does not assume separability. The separate two-cycle excludes $p = 7$. Equation (167) identifies all scheduled floors with these traces. Taking arbitrarily large multiples m of L, in particular $m \geq 15$, proves the conclusion and the strict bounds through Lemma 15.3. □

Corollary 15.5 (A quantitative fixed-prime limitation)**.** *Suppose a finite prime set P supplies a scheduled trace divisor for every integer $m \in [M_0, M]$, where $M \geq M_0 \geq 2$. Then the L in (174) satisfies*

$$L > M - M_0 + 1, \qquad \prod_{\substack{p \in P \\ p \neq 7}} (p - 1) > M - M_0 + 1.$$

Proof. If $L \leq M - M_0 + 1$, the first multiple of L at least M_0 is at most $M_0 + L - 1 \leq M$. The proof of Theorem 15.2 shows that this member has no such divisor. Finally $L \leq \prod_{p \in P,\, p \neq 7}(p - 1)$. □

The limitation concerns fixed-prime covers, not every possible algorithm or Diophantine proof. In particular it is compatible with a prime witness that changes with m.

15.5 Exact classical three adic persistence

Let v_3 denote the usual 3-adic valuation of a nonzero rational number, with $v_3(0) = +\infty$ when needed for coefficient bounds.

Theorem 15.6. *For every $m \geq 1$ and $k \geq 1$,*

$$v_3\big(T_{3^k}(7^m) - 1\big) = k + 1 + v_3(m). \tag{176}$$

More generally, for every integer $a \equiv 1 \pmod{3}$, $a \neq 1$,

$$v_3\big(T_{3^k}(a) - 1\big) = k + v_3(a - 1).$$

Proof. Put $a = 1 + h$ and $D(z) = (1 - z)(1 - 4z^2)$. The formal trace identity

$$\sum_{n \geq 1} \frac{T_n(a)}{n} z^n = -\log(1 - az - 4z^2 + 4z^3)$$

follows by factoring into the three root factors and expanding the formal logarithms. Subtracting the identity at $a = 1$ gives

$$\sum_{n \geq 1} \frac{T_n(1 + h) - T_n(1)}{n} z^n = \sum_{j \geq 1} \frac{h^j z^j}{j D(z)^j}.$$

Consequently the coefficient $A_{n,j}$ of h^j is exactly

$$A_{n,j} = \frac{n}{j} [z^{n-j}] D(z)^{-j}. \tag{177}$$

The bracket is an integer. When $n = 3^k$, we have $v_3(A_{n,j}) \geq k - v_3(j)$. For $j = 1$,

$$A_{n,1} = n \sum_{r=0}^{(n-1)/2} 4^r = n\frac{4^{(n+1)/2} - 1}{3}.$$

The geometric sum is a 3-adic unit: modulo three it is $(n+1)/2 \not\equiv 0$. Thus $v_3(A_{n,1}) = k$.

Writing $b = v_3(h) \geq 1$, every term with $j \geq 2$ satisfies

$$v_3(A_{n,j}h^j) \geq k - v_3(j) + jb > k + b,$$

since $v_3(j) < j - 1$. The linear term is therefore the unique term of least valuation and cannot cancel. Equation (175) gives $T_n(1) = 1$, proving the general assertion.

Finally $v_3(7^m - 1) = 1 + v_3(m)$. For completeness, if $3 \nmid r$, the binomial expansion gives $v_3((1+6)^r - 1) = 1$; each cubing of a number $u \equiv 1 \pmod 3$ adds exactly one, since $u^2 + u + 1 = 3 + 3(u-1) + (u-1)^2$ has valuation one. Writing $m = 3^t r$ proves the formula and hence (176). □

This describes congruence persistence near trace one, not compositeness. Its negative-clock analogue is false, even though the first derivative alone has the suggestive valuation. Indeed

$$T_5(1+h) - 1 = 105h + 50h^2 + 30h^3 + 5h^4 + h^5.$$

At $m = 4$, $a = 2401$, one obtains exactly

$$T_5(2401) - 1 = 79792543008252000 \equiv 54 \pmod{81}.$$

The valuation is three, whereas $k + 1 + v_3(m) = 2$ for $k = 1$. Higher Taylor terms can cancel the leading term. No general negative-clock valuation formula is claimed.

15.6 The precise remaining adaptive problem

A coefficient-divisor criterion requiring a prime $p \equiv 2 \pmod 3$ with $p \mid a$ cannot apply to $a = 7^m$, whose only prime divisor is $7 \equiv 1 \pmod 3$. More generally, constructing useful coefficient residue classes does not automatically place a power of seven in them. Two concrete specializations illustrate this obstruction:

$$\operatorname{ord}_{31}(7) = 15, \quad 12^{15} = -1 \pmod{31}; \qquad \operatorname{ord}_{53}(7) = 26, \quad 26^{26} = -1 \pmod{53}.$$

Thus $a \equiv 12 \pmod{31}$ and $a \equiv 26 \pmod{53}$ are both disjoint from powers of seven. This only describes those residue classes; it does not say those primes are useless at all other coefficients.

For a fixed candidate and an odd prime p, the matrix in (171) is invertible. If its order is E, a recurring divisor certificate asks whether

$$\{e \bmod E : \operatorname{Tr}(M_{p,a}^e) = 0\}$$

intersects the eventual orbit of the desired exponent clock modulo E. The finite table proves particular intersections. To cover all remaining m by this strategy would require a suitable prime *for each specified* m; no such universal statement is proved. Infinitely many composite terms need not logically come from one prime recurring, so this is a sufficient strategy, not an equivalent reformulation of the entire compositeness problem.

If a splitting or Chebotarev argument instead supplies a useful nonzero residue $r \pmod p$, transferring it to the thin family first requires

$$r \in \langle 7 \rangle \subseteq \mathbb{F}_p^*.$$

With $d = \mathrm{ord}_p(7)$, this is exactly $r^d = 1 \pmod p$. When true, a discrete logarithm supplies one exponent class $m \equiv m_0 \pmod d$, not coverage of every preassigned m. A splitting or quadratic-character condition alone does not imply this membership. An unproved primitive-root assertion for seven would be an additional assumption; even full subgroup membership would not force the residue to equal 7^m for a given m.

There is also a direct obstruction to simply taking many prime factors of $7^m - 1$: (175) makes every odd trace congruent to one modulo each such prime. Theorem 15.2 similarly explains why enlarging a fixed finite table cannot close the entire family. The classes not covered by this initial table include $12 \mid m$. Adaptive certificates in Sections 17–19 handle some of them, including $m = 12, 60$, but not all. The results here make no claim of all-m closure, of a distinguished least prime-representing base, or of a new Mills-type transcendence theorem.

15.7 Portable verification

Two standard-library Python programs accompany the paper:

```
python3 certificates/prime_power_coefficient_verify.py
python3 certificates/audit_prime_power_coefficient_verify.py
```

The first uses the quotient-algebra arithmetic (172). The second uses binary exponentiation of the 3×3 matrices (171), with no import of the first program's arithmetic. Both verify all fifteen returns, the coefficient classes, nonzero discriminants, and the nonzero prime-seven cycles. The complete coefficient-class comparison is performed on $\mathbb{Z}/60\mathbb{Z}$, since every table modulus divides sixty. It is not a sample of large m.

The matrix program also verifies exact matrix orders. In table order they are

$$12, 31, 31, 31, 31, 10, 1330, 120, 1330,$$
$$90, 6858, 168, 549, 549, 6858.$$

The starting indices satisfy $k_0 \geq v_3(E)$ in every row, and the two original exponents differ by a multiple of E. These checks confirm the handling of possible initial transients. Supplemental finite valuation tests support the examples; the universal valuation statement is established by Theorem 15.6.

Each program has a corresponding `.json` result file. The README specifies execution and scope. The finite identities and the mathematical proofs together establish the stated infinite conclusions.

16 A prime seed outlier and certified returns

The following example shows that prime leading coefficients and several prime seeds can coexist with later recurring composites. It supplies exact prime certificates rather than treating probable-prime output as proof.

16.1 One coefficient and the two exponent clocks

For an integer $a > 8$, put

$$f_a(X) = X^3 - aX^2 - 4X + 4$$

and let β_a be its largest real root. The two clocks have exponents 3^k and $2 \cdot 3^k - 1$, with $k \geq 1$; their first exponents are 3 and 5. We call $a = \lfloor \beta_a \rfloor$ the leading coefficient or first floor.

Theorem 16.1. *Let $a = 100981$ and $\beta = \beta_a$. The polynomial f_a is irreducible and totally real, with roots $-\rho, \sigma, \beta$ satisfying*

$$0 < \sigma < \rho < \beta^{-17/40} < 1, \qquad a < \beta < a+1, \qquad 4 < \beta^{3/20}, \qquad \rho < \sigma^{17/23}. \tag{178}$$

Moreover, $\mathrm{Norm}(\beta) = -4$ *and* $\mathrm{Tr}(\beta^{-1}) = 1$. *The three integers*

$$\begin{aligned} \lfloor \beta \rfloor &= 100981, \\ \left\lfloor \beta^3 \right\rfloor &= 1029719653587901, \\ \left\lfloor \beta^5 \right\rfloor &= 1050021850218617833198290 1 \end{aligned} \tag{179}$$

are primes congruent to 1 *modulo* 9. *Both* $\lfloor \beta^9 \rfloor$ *and* $\lfloor \beta^{17} \rfloor$ *are composite. There are also the separate infinite divisibility certificates*

$$\begin{array}{lll} 13 \mid \left\lfloor \beta^{3^k} \right\rfloor, & k = 6t, & t \geq 1, \\ 41 \mid \left\lfloor \beta^{2 \cdot 3^k - 1} \right\rfloor, & k = 23 + 3444t, & t \geq 0. \end{array} \tag{180}$$

Every multiple in (180) *is proper, so every indicated floor is composite.*

The fixed-seed criterion under discussion requires a prime $p \equiv 2 \pmod 9$ dividing the trace at exponent 3 or 5, possibly with further finite prime exclusions. Neither seed in (179) has such a divisor. This is a point outside that sufficient criterion, not a counterexample to a density-one conclusion. In particular, the obstruction is not merely that the first floor was composite: all three displayed initial values are proved prime.

16.2 Roots and exact floors

For every integer $a > 8$, direct substitution gives

$$\begin{gathered} f_a(-1) = 7 - a < 0, \qquad f_a(0) = 4 > 0, \qquad f_a(1) = 1 - a < 0, \\ f_a(a) = 4(1-a) < 0, \qquad f_a(a+1) = (a+1)^2 - 4a > 0. \end{gathered}$$

There is one root in each of $(-1, 0)$, $(0, 1)$ and $(a, a+1)$, and these exhaust the three roots. Write them as $-\rho, \sigma, \beta_a$. Their sum gives $\sigma - \rho = a - \beta_a < 0$, hence $0 < \sigma < \rho < 1$. None is an integer. A reducible monic cubic over $\mathbb{Q}$ would have an integer root, so f_a is irreducible. Thus β_a is a totally real cubic Pisot number. The elementary symmetric functions give

$$\mathrm{Norm}(\beta_a) = -4, \qquad \mathrm{Tr}(\beta_a^{-1}) = \frac{-4}{-4} = 1, \qquad \rho\sigma = \frac{4}{\beta_a}. \tag{181}$$

Define the integral trace polynomials by

$$\begin{gathered} T_0(a) = 3, \qquad T_1(a) = a, \qquad T_2(a) = a^2 + 8, \\ T_n(a) = aT_{n-1}(a) + 4T_{n-2}(a) - 4T_{n-3}(a) \quad (n \geq 3). \end{gathered} \tag{182}$$

Newton's identities and the equation for each root show that $T_n(a) = \beta_a^n + (-\rho)^n + \sigma^n$. For every positive odd n,

$$0 < \beta_a^n - T_n(a) = \rho^n - \sigma^n < 1, \qquad \lfloor \beta_a^n \rfloor = T_n(a). \tag{183}$$

This identity holds from the first odd exponent, not just eventually.

For $a = 100981$, exact rational evaluation gives

$$f_a(-1/150) = -\frac{1557151}{3375000} < 0.$$

The unique negative root therefore lies in $(-1/150, 0)$, so $\rho < 1/150$. The exact integer inequalities

$$150^{40} > 100982^{17}, \qquad 4^{20} < 100981^3 \tag{184}$$

imply $\rho < \beta^{-17/40}$ and $4 < \beta^{3/20}$. The same small-conjugate bound holds for $\sigma < \rho$. Finally, (181) gives

$$\rho^{40} < \beta^{-17} = (\rho\sigma/4)^{17}, \qquad \rho^{23} < \sigma^{17}/4^{17} < \sigma^{17},$$

which proves the last inequality in (178). The coefficient is odd, is 1 modulo 3, and is coprime to 4.

16.3 Certified primality of the first seeds

We use the following complete-factor form of the Lucas criterion. Its proof is included to specify exactly what the certificates establish.

Lemma 16.2. *Let $N > 2$ and suppose the complete prime factorization of $N - 1$ is known. If an integer b satisfies*

$$b^{N-1} \equiv 1 \pmod{N}, \qquad \gcd(b^{(N-1)/r} - 1, N) = 1 \quad \text{for every prime } r \mid N-1, \tag{185}$$

then N is prime.

Proof. Let ℓ be any prime divisor of N. The first congruence makes b invertible modulo ℓ and gives $d = \mathrm{ord}_\ell(b) \mid N - 1$. If r^e is the exact power of r dividing $N - 1$, the gcd condition implies $b^{(N-1)/r} \not\equiv 1 \pmod{\ell}$. Consequently $r^e \mid d$. This holds for every r, so $d = N - 1$. As $d \mid \ell - 1$, we have $\ell \geq N$. Since $\ell \mid N$, it follows that $\ell = N$ and N is prime. □

Write N_1, N_3, N_5 for the three integers in (179). Their complete factorizations and certificate bases are

$$\begin{aligned} N_1 - 1 &= 2^2 \cdot 3^3 \cdot 5 \cdot 11 \cdot 17, && b_1 = 7, \\ N_3 - 1 &= 2^2 \cdot 3^4 \cdot 5^2 \cdot 11 \cdot 17 \cdot 131 \cdot 227 \cdot 22861, && b_3 = 6, \\ N_5 - 1 &= 2^2 \cdot 3^4 \cdot 5^2 \cdot 11 \cdot 17 \cdot 809 \cdot 2154553 \cdot 3977095391, && b_5 = 6. \end{aligned} \tag{186}$$

For each row, exact modular powers and gcds satisfy (185). All prime factors on the right are themselves recursively certified by the same criterion, terminating at the elementary prime 2. The complete certificate graph has 36 nodes and also certifies 13, 41 and 256603. The accompanying data and dependency-free acceptance programs are described in the appendix; no probable-prime test is an acceptance step.

At $a = 100981$, evaluation of (182) proves that $T_1(a) = N_1$, $T_3(a) = N_3$ and $T_5(a) = N_5$. Each is 1 modulo 9. Since $T_3(a)$ and $T_5(a)$ are themselves prime, neither has a prime divisor congruent to 2 modulo 9. Adding further exclusions to that witness criterion cannot detect this point.

16.4 The next floor on each clock

In this section $a = 100981$, and we abbreviate $T_n = T_n(a)$. The recurrence gives

$$T_9 = 109183498430798123598904347650717749806685962 1. \tag{187}$$

Its exact base-2 Fermat remainder is

$$2^{T_9-1} \bmod T_9 = 247023223208684492599063887120591198040833796 \neq 1. \tag{188}$$

As $T_9 > 1$ is odd, Fermat's necessary condition for primality proves it composite, without requiring its factorization.

For longer integers, a bracket $[\cdots]_{10}$ below means the single decimal integer obtained by concatenating its digit rows from top to bottom; no digit is omitted. Exact recurrence evaluation gives

$$T_{17} = \left[\begin{array}{c} 11805227047575212662824663918949265371 72 \\ 56949733735984684352155248793595555088042987461 \end{array}\right]_{10}. \tag{189}$$

There are the proper factorizations

$$T_{17} = 229247 \cdot \left[\begin{array}{c} 51495666453978515151014686861547873567 48 \\ 70117095255269139191157349032247099763 763 \end{array}\right]_{10} \tag{190}$$

and

$$T_{17} = 256603 \cdot \left[\begin{array}{c} 46005802923485745150386643643875034086 60 \\ 73076829717441668071516111633907440210 87 \end{array}\right]_{10}. \tag{191}$$

Each divisor and quotient exceeds 1. The first factorization alone proves compositeness; no primality assertion about 229247 is needed. The second uses the certified prime $256603 \equiv 4 \pmod 9$, which will also give a later-seed return. By (183), these computations prove compositeness of both clock floors at the common index $k = 2$.

16.5 Exact modular returns

For a prime $p \nmid 4$, let

$$R_p = \mathbb{F}_p[X]/(f_a), \qquad x = X \bmod f_a.$$

The element x is a unit. Multiplication by x in the basis $1, x, x^2$ has matrix

$$M_a = \begin{pmatrix} 0 & 0 & -4 \\ 1 & 0 & 4 \\ 0 & 1 & a \end{pmatrix}.$$

Consequently the multiplication trace of x^n is $T_n(a)$ modulo p, even when the quotient is not reduced. More generally,

$$\mathrm{Tr}_{R_p/\mathbb{F}_p}(v_0 + v_1 x + v_2 x^2) = 3v_0 + av_1 + (a^2 + 8)v_2. \tag{192}$$

16.5.1 The classical clock modulo 13

Here $a \equiv 10 \pmod{13}$ and

$$f_a(X) = (X + 5)(X^2 - 2X + 6) \quad \text{in } \mathbb{F}_{13}[X].$$

Exact reduction by $x^3 = ax^2 + 4x - 4$ gives

$$x^{168} = 1, \qquad \operatorname{ord}_{56}(3) = 6, \qquad x^{3^6} = 3 + 8x + 7x^2. \tag{193}$$

The last state's trace is $9 + 80 + 4 \cdot 7 = 117 \equiv 0 \pmod{13}$, since $a^2 + 8 \equiv 4 \pmod{13}$. For $k = 6t$, $t \geq 1$, the exponents 3^k and 3^6 agree modulo 56 and are both 0 modulo 3. They therefore agree modulo 168, proving $13 \mid T_{3^k}(a)$ for every such k. The range starts at $k = 6$, not at $k = 0$.

16.5.2 The negative clock modulo 41

Here $a \equiv 39 \pmod{41}$ and $f_a = X^3 + 2X^2 - 4X + 4$ modulo 41. Evaluation at all 41 residues gives no root, so this cubic is irreducible and $R_{41} = \mathbb{F}_{41^3}$. In particular,

$$x^{68920} = 1, \qquad \operatorname{ord}_{68920}(3) = 3444. \tag{194}$$

Starting at $u_0 = x$, the exact recurrence

$$u_{k+1} = x^2 u_k^3$$

represents $u_k = x^{2 \cdot 3^k - 1}$. Reduction in R_{41} gives

$$u_{23} = 18x + 3x^2.$$

As $a^2 + 8 \equiv 12 \pmod{41}$, its trace is $39 \cdot 18 + 12 \cdot 3 = 738 = 18 \cdot 41$. Equation (194) returns this state for every $k = 23 + 3444t$, $t \geq 0$. Thus $41 \mid T_{2 \cdot 3^k - 1}(a)$ on the asserted progression. The additional identity $x^{8615} = 1$ yields the same clock period, but is not needed for this argument.

On both displayed progressions, the odd exponent n is at least 3, and

$$T_n(a) > \beta_a^n - 1 > a^3 - 1 > 41. \tag{195}$$

Equations (183) and (195) turn the trace divisibilities into actual composite floors, completing the proof of Theorem 16.1.

The classical progression is 0 modulo 6. Since $3444 \equiv 0 \pmod{6}$, the displayed negative progression is $23 \equiv 5 \pmod{6}$. These two *displayed progressions* are disjoint; the certificates do not supply a common-index progression. The full sets of composite indices are not disjoint: The earlier direct floor calculation already proves that both contain $k = 2$.

16.5.3 A complementary later negative seed

For completeness, the prime $p = 256603$ in (191) gives another negative-clock certificate. We justify the universal exponent used for this return. If $p > 2$, g is a monic cubic with nonzero constant term over $\mathbb{F}_p$, and $x = X \bmod g$, then

$$x^{E_p} = 1, \qquad E_p = p(p^2 - 1)(p^2 + p + 1). \tag{196}$$

Indeed, factor $g = \prod h_i^{e_i}$ into distinct monic irreducibles. In the reduced quotient the factors are fields of degrees at most 3. Each possible multiplicative order divides $H = (p^2 - 1)(p^2 + p + 1)$, since $p^d - 1 \mid H$ for $d = 1, 2, 3$. Hence $x^H = 1 + z$ with $z^3 = 0$ in the original quotient. In characteristic $p \geq 3$, $(1 + z)^p = 1$, proving (196) also for nonreduced quotients.

For our prime $p \equiv 4 \pmod{9}$, $v_3(p^2 - 1) = v_3(p^2 + p + 1) = 1$, so $v_3(E_p) = 2$. Set $m = E_p/9$ and $L = \operatorname{ord}_m(3)$. If $k \geq 2$ and $k \equiv 2 \pmod{L}$, then $3^k \equiv 9 \pmod{m}$, while both sides are 0

modulo 9. Thus $2 \cdot 3^k - 1 \equiv 17 \pmod{E_p}$. The trace at every such exponent is divisible by 256603; (191) proves properness at $k = 2$, and larger exponents give still larger floors.

This is the offset-two, exponent-17 negative-seed mechanism. It complements the fixed exponent-3/5 test at this point. It makes no assertion about membership in a complete simultaneous four-way certificate set involving other signs or clocks. The much smaller prime 41 supplies the displayed negative progression in Theorem 16.1; no period optimization is needed for the 256603 certificate.

16.6 Coefficient families and finite prime filters

Proposition 16.3. *Every coefficient*

$$a = 1843 + 3198t, \qquad t \geq 0, \tag{197}$$

has the exact all-odd floor identity (183) *and both separate composite-floor progressions in* (180)*, with β replaced by β_a. There are infinitely many prime coefficients in this progression.*

Proof. Here $3198 = 6{\cdot}13{\cdot}41 = \operatorname{lcm}(6, 13, 41)$. Every coefficient in (197) is 1 modulo 6, 10 modulo 13, 39 modulo 41, and larger than 8. Thus the root proof, both quotient-ring calculations and the properness estimate remain valid. As $\gcd(1843, 3198) = 1$, the prime-progression theorem supplies infinitely many primes in this class; equivalently, use the cyclotomic case of Chebotarev [27, Theorem 3.1 and p. 4]. □

Our explicit point is the member $t = 31$. Primality of both first clock seeds is proved here only for $a = 100981$, not throughout (197). All sufficiently large members satisfy the same strict inequalities, as follows from the estimate below.

Lemma 16.4 (Prime-leading escape from each finite filter)**.** *For every fixed finite set P of primes, there are infinitely many prime integers $a > 8$ such that β_a satisfies* (178) *and*

$$\lfloor \beta_a^n \rfloor \equiv 1 \pmod{p} \quad \text{for every } p \in P \text{ and every positive odd } n. \tag{198}$$

These coefficients can be chosen arbitrarily large.

Proof. Put $M = \operatorname{lcm}(24, \prod_{p \in P} p)$, with empty product 1. Take primes $a \equiv 1 \pmod{M}$ above the threshold specified below. There are infinitely many by the same reduced-residue-class theorem. At the boundary coefficient 1,

$$f_1(X) = (X - 1)(X^2 - 4), \qquad T_n(1) = 1 + 2^n + (-2)^n = 1 \quad (n \text{ odd}).$$

Each $T_n(a)$ is an integer polynomial in a, by (182). Therefore $T_n(a) \equiv T_n(1) = 1 \pmod{p}$ for every $p \in P$, proving (198) by (183). This polynomial congruence does not require separability modulo p.

For an explicit sufficient cutoff, the negative root equation gives

$$a\rho^2 = 4(\rho + 1) - \rho^3 < 8, \qquad \rho < \sqrt{8/a}, \qquad \beta_a < a + 1 < 2a.$$

Consequently, if $a \geq 2^{27}$,

$$\rho \beta_a^{17/40} < 2^{77/40} a^{-3/40} \leq 2^{-1/10} < 1, \qquad 4^{20} < a^3 < \beta_a^3.$$

These prove the strict bounds for ρ and the norm magnitude; $\sigma < \rho$ and the argument following (184) give the remaining inequalities in (178). The congruence $a \equiv 1 \pmod{24}$ also gives oddness and $a \equiv 1 \pmod{3}$. □

Corollary 16.5. *Fix a finite prime set P, a length $\ell \geq 2$, and a lower bound A. There is a nonconstant arithmetic progression of ℓ prime coefficients $a > A$, each satisfying the conclusion of Lemma 16.4 for this same P.*

Proof. The prime coefficients in $1 \pmod M$ above $\max\{A, 2^{27}\}$ have relative density $1/\varphi(M) > 0$ among all primes, by the cyclotomic prime-progression theorem [27]. Apply the relative-prime Green–Tao theorem [24, Theorem 1.2] to this set. □

These finite arithmetic progressions are in the integer coefficients, not necessarily in their algebraic roots. The quantifiers are

$$\text{for every finite } P, \quad \text{there exist infinitely many prime coefficients escaping } P.$$

They do not produce one coefficient escaping every prime. Nor is simultaneous primality of $T_3(a)$ and $T_5(a)$ asserted for these families.

16.6.1 A common progression for the prime-seed outlier

The coefficient with prime leading floor and prime first seeds on both clocks also admits a small-prime simultaneous certificate. The following result does not claim that the earlier displayed separate progressions intersect; it supplies a different negative-clock witness.

Theorem 16.6 (Synchronization of the outlier)**.** *For every integer $a = 504 + 767u$, $u \geq 0$, let β_a be the largest root of $X^3 - aX^2 - 4X + 4$. Both floors*

$$\left\lfloor \beta_a^{3^k} \right\rfloor, \qquad \left\lfloor \beta_a^{2\cdot 3^k - 1} \right\rfloor$$

are composite at every $k = 30 + 1770t$, $t \geq 0$. The first is divisible by 13, and the second by 59. In particular this holds for the prime-seed outlier $a = 100981$.

Proof. Every such a satisfies $a \equiv 10 \pmod{13}$ and $a \equiv 32 \pmod{59}$; also $a \geq 504 > 59$. In each cubic quotient write x for the class of X and use

$$\tau(c_0 + c_1 x + c_2 x^2) = 3c_0 + ac_1 + (a^2 + 8)c_2.$$

This multiplication trace equals the reduced Newton trace at every power. The relation is $x^3 = ax^2 + 4x - 4$. Exact binary powering supplies the finite identities

modulus	scalar identity	zero-trace seed
13	$x^{168} = 1$	$x^{3^6} = 3 + 8x + 7x^2$
59	$x^{3541} = 55$	$x^{2\cdot 3^{30}-1} = 41 + 45x + 58x^2$.

Their seed traces are respectively

$$3 \cdot 3 + 10 \cdot 8 + 4 \cdot 7 = 117 = 9 \cdot 13, \qquad 3 \cdot 41 + 32 \cdot 45 + 1032 \cdot 58 = 61419 = 1041 \cdot 59.$$

The elementary congruences $3^6 \equiv 1 \pmod{56}$ and $3^{1770} \equiv 1 \pmod{3541}$ complete the certificates. If $6 \mid k$ and $k \geq 6$, then $3^k \equiv 3^6 \pmod{168}$: both are zero modulo 3 and one modulo 56. Thus 13 divides every indicated classical trace. Replacing $k = 30$ by $30 + 1770t$ changes the negative-clock exponent by a multiple of 3541. The second scalar identity then preserves zero trace modulo 59. Since $6 \mid 30$ and $6 \mid 1770$, both certificates hold on the stated common progression.

The real cubic root calculation gives $a < \beta_a < a + 1$ and $\lfloor \beta_a^n \rfloor = T_n(a)$ at every positive odd n. All indicated exponents are odd and at least three, so these floors exceed $a^3 - 1 > 59$. Both divisors are therefore proper, including at the first displayed index. Finally $100981 = 504 + 767 \cdot 131$, proving the specialization. □

The coefficient progression is reduced, since $\gcd(504, 767) = 1$, so it contains infinitely many prime coefficients. This assertion concerns the coefficient, not simultaneous primality of the initial seed traces throughout the progression. For odd coefficients it can equivalently be written $a = 1271 + 1534v$, $v \geq 0$.

16.7 Scope of the example

The explicit point separates a sufficient fixed-seed test from a universal pointwise test while retaining prime first floor and prime first clock seeds. Its later composite floors exclude it from an all-prime floor sequence. The separate return certificates also persist on a reduced coefficient progression, but their displayed index offsets do not synchronize. The new witness in Section 16.6.1 instead yields a common progression for the same outlier and an entire coefficient class. The fixed-filter lemma shows that requiring a prime leading coefficient does not make any fixed finite divisibility filter universal. None of these conclusions excludes every remaining candidate, establishes a global minimum, or resolves a universal Mills-type or Pisot prime-floor problem.

16.8 Portable exact certificates

The directory `certificates/` contains `prime_seed_lucas_certificates.json`, the full 36-node recursive primality graph, and two independent acceptance programs. Run from the paper directory, using Python 3 with no third-party dependency:

```
python3 certificates/prime_seed_outlier_verify.py
python3 certificates/audit_prime_seed_outlier_verify.py
(cd certificates && sha256sum -c SHA256SUMS.txt)
```

Each program has a corresponding `.json` report and `.log` file. The primality checks multiply each complete factorization to $N-1$, verify every recursive dependency, and check the modular powers and gcd conditions of Lemma 16.2. Every dependency is smaller than its parent, so induction from 2 proves all 36 asserted prime nodes.

The first verifier recomputes the traces, rational endpoint, strict integer inequalities, Fermat remainder, proper factorizations, quotient-ring states, multiplicative orders and coefficient congruences. The second uses direct 3-by-3 companion-matrix powers for the modular return checks, rather than importing the first program's quotient-ring implementation. It also checks the exact matrix orders 168 modulo 13 and 8615 modulo 41. The data at one return state, together with the proved exponent congruences, establish infinitely many indices; no finite sampling is extrapolated to an infinite claim. These exact checks do not replace the imported prime-density and Green–Tao theorems.

17 Adaptive returns and ancestral minimality

Adaptive certificates extend the thin-family coverage, while the second half of this section analyzes the actual minimality condition. Skipping several levels enlarges a raw interval but does not preserve every ancestral prime cell automatically.

17.1 Scope and the two exponent clocks

For a real number $B > 1$, the classical and negative clocks in this paper are respectively

$$\left\lfloor B^{3^k} \right\rfloor \quad \text{and} \quad \left\lfloor B^{2\cdot 3^k - 1} \right\rfloor, \qquad k \geq 1.$$

The word "negative" only names the second exponent sequence; both sequences have positive odd exponents. Their first exponents are 3 and 5.

The arithmetic part studies

$$f_a(X) = X^3 - aX^2 - 4X + 4, \qquad a = 7^m, \quad m \geq 2, \tag{199}$$

and its dominant real root β_a. We prove a polynomial factorization obstruction on the entire classical clock, a consecutive-coprimality identity, and separate numerical return certificates for two coefficients. In particular, polynomial irreducibility is not being confused with primality of polynomial values. The finite-ring certificates concern actual values and do not supply a divisor-existence theorem for all m.

The minimum problem has a different quantifier. For a set S of positive integers, write

$$\mathcal{W}(S) = \{B > 1 : \left\lfloor B^{3^k} \right\rfloor \in S \text{ for every } k \geq 1\}.$$

For the full set $\mathcal{P}$ of primes, its least element is the classical Mills constant, denoted here by ξ. We use Saito's conditional Pisot reduction for this least element, explicitly as an external input [4, Theorem 1.7]. We then give the necessary elementary proofs of the strict gap exponent and the eventual least-prime property before analyzing skipped levels.

The conclusions about altered sets are deliberately narrower than a conclusion about $\mathcal{W}(\mathcal{P})$. Our thinning theorem changes the allowed set S; its prime specialization is conditional on an admissible prime tower with a sufficiently small gap exponent. The unconditional example uses integers coprime to 6, not primes. No transfer of an ambient coefficient-density statement to the thin sequence 7^m is used, and no all-m exclusion or transcendence conclusion is obtained.

17.2 Exact floors and universal trace identities

Let $P_n(A)$ be the sum of the nth powers of the three formal roots of $X^3 - AX^2 - 4X + 4$. Newton's identities give

$$\begin{aligned} P_0 = 3, \qquad P_1 = A, \qquad P_2 = A^2 + 8, \\ P_n = AP_{n-1} + 4P_{n-2} - 4P_{n-3} \qquad (n \geq 3). \end{aligned} \tag{200}$$

Thus $P_n \in \mathbb{Z}[A]$, and P_n is monic of degree n for every $n \geq 1$: the first two cases are explicit, and the first summand in the recurrence supplies the unique term of degree n.

Lemma 17.1 (Roots and odd floors)**.** *For every integer $a > 8$, the polynomial f_a is irreducible over $\mathbb{Q}$ and has roots $-\rho, \sigma, \beta_a$ satisfying*

$$0 < \sigma < \rho < 1, \qquad a < \beta_a < a + 1.$$

In particular, β_a is a totally real cubic Pisot number, and for every positive odd integer n,

$$\lfloor \beta_a^n \rfloor = P_n(a), \qquad 0 < \beta_a^n - P_n(a) = \rho^n - \sigma^n < 1. \tag{201}$$

The floors in (201) strictly increase along the positive odd exponents.

Proof. Direct substitution gives

$$\begin{aligned} f_a(-1) = 7 - a < 0, \qquad f_a(0) = 4 > 0, \qquad f_a(1) = 1 - a < 0, \\ f_a(a) = 4(1 - a) < 0, \qquad f_a(a + 1) = (a + 1)^2 - 4a = (a - 1)^2 > 0. \end{aligned}$$

There is a root in each of $(-1, 0)$, $(0, 1)$, and $(a, a + 1)$. These exhaust the roots of the cubic. None is an integer, so the rational-root test proves irreducibility. Their sum is a, whence $\sigma - \rho = a - \beta_a < 0$. For odd n their power sum is $P_n(a) = \beta_a^n - \rho^n + \sigma^n$. The displayed bounds on ρ and σ prove (201). Finally, $\beta_a^{n+2} - \beta_a^n > 1$ because $\beta_a > a > 8$; taking floors proves strict increase. □

The next identity will be used for both irreducibility and coprimality. If the three formal roots are $\alpha_1, \alpha_2, \alpha_3$, put

$$Q_n(A) = \sum_{i<j} (\alpha_i \alpha_j)^n.$$

This symmetric expression lies in $\mathbb{Z}[A]$, by the fundamental theorem on symmetric polynomials. Applying the cubic power-sum identity to $\alpha_1^n, \alpha_2^n, \alpha_3^n$ gives

$$P_{3n} = P_n^3 - 3P_n Q_n + 3(-4)^n, \tag{202}$$

because $\alpha_1\alpha_2\alpha_3 = -4$.

Theorem 17.2 (All monomial substitutions on the classical clock)**.** *For every $k, r \geq 1$, the polynomial $P_{3^k}(X^r)$ is Eisenstein at 3, hence irreducible over $\mathbb{Q}$. Also $P_5(X^r)$ is Eisenstein at 5 for every $r \geq 1$.*

Proof. Reducing (202) modulo 3 and starting from $P_1(A) = A$ shows that

$$P_{3^k}(A) \equiv A^{3^k} \pmod 3 \qquad (k \geq 0). \tag{203}$$

Let $c_k = P_{3^k}(0)$. Then $c_0 = 0$, and $3 \mid c_k$ for every k. Specializing (202) at $A = 0$, with $n = 3^k$, yields

$$c_{k+1} = c_k^3 - 3c_k Q_{3^k}(0) + 3(-4)^{3^k} \equiv 6 \pmod 9.$$

Indeed, the first term is divisible by 27, the second by 9, and $(-4)^{3^k} \equiv -1 \pmod 3$. Thus every nonleading coefficient of P_{3^k} is divisible by 3, whereas its constant coefficient is not divisible by 9. Substitution $A = X^r$ preserves the leading and constant coefficients and only inserts zero coefficients. Eisenstein's criterion applies.

For completeness, the irreducibility implication can be seen without an additional factorization theorem. By Gauss's lemma, a nontrivial rational factorization of this monic integer polynomial would give a product of two positive-degree monic integer polynomials. Modulo 3 their product is a power of X, so both reductions are positive-degree powers of X. Both constant coefficients would be divisible by 3, making their product divisible by 9, a contradiction.

The recurrence gives

$$P_5(A) = A^5 + 20A^3 - 20A^2 + 80A - 80.$$

Every nonleading coefficient is divisible by 5 and the constant is not divisible by 25. The same substitution and Eisenstein argument, at 5, proves the last assertion. □

Remark 17.3 (Exact scope of the factorization obstruction). If $m = rt$, writing $7^m = (7^t)^r$ cannot lead to a nonconstant factorization of $P_{3^k}(X^r)$ in $\mathbb{Q}[X]$. This holds for every positive monomial substitution, not merely for the first classical trace. It says nothing about whether a particular integer value is prime, and it does not exclude congruence arguments, nonpolynomial factorizations, or other Diophantine mechanisms. The same proof works for the power sums of $X^3 - AX^2 - bX + c$, with fixed integers b, c and $3 \nmid c$: the constant coefficient at a classical exponent is $-3c \pmod 9$.

Theorem 17.4 (Consecutive classical coprimality)**.** *For every $m \geq 1$ and $k \geq 0$,*

$$\gcd\bigl(P_{3^k}(7^m), P_{3^{k+1}}(7^m)\bigr) = 1.$$

Proof. At an odd integer a, (200) modulo 2 shows that every positive trace is odd. At $a = 7^m$, equation (203) gives $P_{3^k}(a) \equiv 1 \pmod 3$. For $n = 3^k$, the identity (202) implies

$$\gcd(P_n(a), P_{3n}(a)) \mid 3 \cdot 4^n.$$

The gcd is neither even nor divisible by 3, and is therefore 1. □

This theorem concerns consecutive scheduled traces, not all pairs of traces. It prevents automatic propagation of a divisor to the very next cubing step. The nonconsecutive returns below are fully consistent with it.

17.3 Exact adaptive returns for two coefficients

17.3.1 The finite-ring mechanism

Let $d > 1$ be an integer and set

$$R_d = (\mathbb{Z}/d\mathbb{Z})[X]/(f_a), \qquad x = X \pmod{f_a}.$$

A triple (c_0, c_1, c_2) denotes $c_0 + c_1x + c_2x^2$ in this ring. The relation is $x^3 = ax^2 + 4x - 4$. Define

$$\tau(c_0 + c_1x + c_2x^2) = 3c_0 + ac_1 + (a^2 + 8)c_2 \pmod d. \tag{204}$$

The sequences $\tau(x^n)$ and $P_n(a) \pmod d$ obey the same recurrence and have the same first three values. Hence

$$\tau(x^n) = P_n(a) \pmod d \qquad (n \geq 0). \tag{205}$$

Equivalently, multiplication by x has the companion matrix

$$M = \begin{pmatrix} 0 & 0 & -4 \\ 1 & 0 & 4 \\ 0 & 1 & a \end{pmatrix} \pmod d, \tag{206}$$

and τ is the matrix trace of multiplication by a ring element.

Lemma 17.5 (Two return criteria)**.** *Suppose E, h are positive integers with $x^E = 1$ in R_d and $3^h \equiv 1 \pmod E$. If $d \mid P_{3^{k_0}}(a)$, then*

$$d \mid P_{3^{k_0+ht}}(a) \qquad (t \geq 0).$$

For the negative clock, put $u_0 = x$ and $u_{k+1} = x^2u_k^3$. If $u_{k_0+h} = u_{k_0}$ and $\tau(u_{k_0}) = 0$, then

$$d \mid P_{2\cdot 3^{k_0+ht}-1}(a) \qquad (t \geq 0).$$

Proof. The congruence $3^{k_0+ht} \equiv 3^{k_0} \pmod E$ implies equality of the corresponding powers of x, because $x^E = 1$. Apply (205). For the second assertion, induction gives $u_k = x^{2\cdot 3^k-1}$. The deterministic map $u \mapsto x^2u^3$ repeats every subsequent state after a return, so $u_{k_0+ht} = u_{k_0}$ for every $t \geq 0$. Apply (205) again. □

No primality of d is needed in this lemma, and h need not be a least period. If $a > 8$ and the seed trace is larger than d, Lemma 17.1 makes every displayed divisible floor a proper multiple, including the seed itself.

17.3.2 A common progression for the coefficient with exponent twelve

Set

$$a = 7^{12} = 13\,841\,287\,201.$$

The exact first classical trace factors as

$$\begin{aligned} P_3(a) &= a^3 + 12a - 12 \\ &= 2\,651\,730\,845\,859\,653\,471\,945\,118\,828\,001 \\ &= 3\,037\,679\,595\,827 \cdot 872\,946\,195\,346\,756\,763. \end{aligned} \tag{207}$$

Define

$$\begin{aligned} p &= 3\,037\,679\,595\,827, \\ E &= p^3 - 1 = 28\,030\,180\,350\,483\,512\,012\,134\,943\,524\,776\,874\,282, \\ h &= 91\,332\,334\,257\,806\,616\,700\,607\,778\,662\,532. \end{aligned} \tag{208}$$

The packaged exact verifiers establish

$$x^E = 1 \text{ in } R_p, \qquad 3^h \equiv 1 \pmod{E}, \qquad x^3 = (-4, 4, a), \qquad \tau(x^3) = 0 \pmod{p}. \tag{209}$$

These are finite integer identities, checked by binary powering; the verification method and filenames are specified in Section 17.8.

For the negative clock, reduce the same coefficient modulo 11. Then $a \equiv 5 \pmod{11}$, and the exact states are

$$u_1 = (5, 8, 7) = u_{37}, \qquad \tau(u_1) = 0 \pmod{11}. \tag{210}$$

The integer h in (208) is divisible by 36.

Theorem 17.6 (Simultaneous composite floors for one coefficient)**.** *For every $t \geq 0$, both floors*

$$\left\lfloor \beta_{7^{12}}^{3^k} \right\rfloor \quad \text{and} \quad \left\lfloor \beta_{7^{12}}^{2 \cdot 3^k - 1} \right\rfloor \quad \text{are composite at } k = 1 + ht,$$

where h is given in (208). The first is divisible by p and the second by 11.

Proof. Equations (209) and Lemma 17.5 give the classical divisibility at $k = 1 + ht$. The negative return (210) gives divisibility by 11 at $k = 1 + 36t$. Since $36 \mid h$, the same indices $1 + ht$ satisfy both certificates. Their offsets are both 1; no assertion about intersecting incompatible progressions is being made. The factorization (207) proves that p is a proper divisor at the first classical seed. The negative seed has exponent 5 and exceeds $a^3 - 1 > 11$. Strict increase from Lemma 17.1 makes all later multiples proper. □

The verifiers also prove that p is prime by exhaustive trial division through $\lfloor \sqrt{p} \rfloor = 1\,742\,894$. This is not a logical prerequisite for the return proof: a proper integer divisor and (209) already suffice. No iteration through all h states, unverified order computation, or prime-distribution hypothesis is used.

Remark 17.7 (Field interpretation of this certificate)*.* Here $p \equiv 2 \pmod{3}$. The independent matrix computation also gives

$$\det(M^p - M) \equiv 2\,129\,913\,761\,719 \not\equiv 0 \pmod{p}.$$

Thus $x^p - x$ is a unit in the cubic quotient. If f_a had a root in $\mathbb{F}_p$, evaluation at that root would send this unit to zero, which is impossible. The cubic is therefore irreducible modulo p, and $R_p \cong \mathbb{F}_{p^3}$. Since $\gcd(3, p^3 - 1) = 1$, this explains the absence of a 3-primary transient in the discovered return. The proof of Theorem 17.6 uses only the displayed ring identities, not this interpretation.

17.3.3 A prime first trace and later returns at exponent four

For $a = 7^4 = 2401$,

$$P_3(a) = 13\,841\,316\,001 \tag{211}$$

is prime. Both the quotient-ring verifier and the independent matrix verifier prove this by exhaustive trial division through $\left\lfloor \sqrt{P_3(a)} \right\rfloor = 117\,649$. In particular, universal compositeness at the first classical scheduled term is false even within the subfamily where $4 \mid m$.

For the later classical return, the exact data are

$$\begin{aligned} &d = 59, \qquad a \equiv 41 \pmod{59}, \qquad E = 59^3 - 1 = 205\,378, \qquad h_4 = 24\,780, \\ &x^E = 1, \qquad 3^{h_4} \equiv 1 \pmod{E}, \qquad x^9 = (36, 16, 24), \qquad \tau(x^9) = 0 \pmod{59}. \end{aligned} \tag{212}$$

For the negative clock, an independent matrix calculation modulo 13 establishes the state return

$$u_9 = u_{19} = (9, 4, 12), \qquad \tau(u_9) = 0 \pmod{13}. \tag{213}$$

Here the negative-clock state can equivalently be read as the first column of $M^{2\cdot 3^k - 1}$.

Corollary 17.8. *For every $t \geq 0$, the classical floor for $a = 7^4$ is composite at*

$$k = 2 + 24\,780t,$$

and the negative-clock floor is composite at

$$k = 9 + 10t.$$

Proof. Apply Lemma 17.5 to (212) and (213). The first corresponding exponents are 9 and $2 \cdot 3^9 - 1$. At every odd exponent $n \geq 3$, Lemma 17.1 gives $P_n(a) > \beta_a^n - 1 > a^3 - 1 > 59 > 13$. Thus every displayed divisor is proper. □

The two progressions in Corollary 17.8 are separate; no common-index assertion is needed or claimed for them. The certificates for $m = 4$ and $m = 12$ do not show that the same adaptive procedure must succeed for each m. In both examples the initial floor $\lfloor \beta_{7^m} \rfloor = 7^m$ is itself composite. Neither is an all-prime tower or a candidate for a prime leading floor.

17.4 The conditional classical Pisot setting

17.4.1 External prime and Pisot inputs

The prime-distribution input is the following every-interval lower bound. There exist constants $c > 0$ and x_0 such that, with $\theta_0 = 21/40$,

$$\#(\mathcal{P} \cap [x, x + x^{\theta_0}]) \geq c \frac{x^{\theta_0}}{\log x} \qquad (x \geq x_0). \tag{214}$$

This is the Baker–Harman–Pintz input [1], in the precise cardinality formulation stated by Saito [4, Theorem 1.6]. No numerical value for x_0 is assumed.

Saito's reduction [4, Theorem 1.7] says that, if the classical minimum ξ is algebraic, some power $\beta = \xi^{3^m}$ is a totally real cubic Pisot number. Its conjugates can be written as

$$\beta, \ -\rho, \ \gamma, \qquad 0 < \sigma = |\gamma| < \rho < 1, \qquad \rho \leq \min(\sigma^{17/23}, \beta^{-17/40}). \tag{215}$$

This is an inherited reduction, not a result established in this paper. It does not assert that any particular cubic with these root bounds has a prime tower. The subsequent elementary lemmas make explicit the further consequences needed here.

Put

$$M = 3^r, \qquad p_r = \mathrm{Tr}(\beta^{3^r}), \qquad e_r = \beta^{3^r} - p_r = \rho^{3^r} - \gamma^{3^r}, \\ u = -\frac{\log \rho}{\log \beta}, \qquad \kappa = \frac{2-u}{3}. \tag{216}$$

Unlike the family in (199), the sign of γ in this conditional setting is not prescribed. Since $|\gamma| < \rho$, however, $e_r > 0$, $e_r \sim \rho^{3^r}$, and $e_r < 1$ for all sufficiently large r. Therefore on a sufficiently late tail

$$p_r = \left\lfloor \beta^{3^r} \right\rfloor = \left\lfloor \xi^{3^{m+r}} \right\rfloor \in \mathcal{P}. \tag{217}$$

Lemma 17.9 (The strict gap exponent)**.** *Under* (215)*, the number u is irrational and*

$$\frac{17}{40} < u < \frac{1}{2}, \qquad \frac{1}{2} < \kappa < \frac{21}{40}. \tag{218}$$

Furthermore,

$$G_r := p_{r+1} - p_r^3 \sim 3(\beta^2\rho)^{3^r} \sim 3(p_r^3)^{\kappa}. \tag{219}$$

In particular, eventually $p_r^3 < p_{r+1} < (p_r+1)^3$.

Proof. The absolute norm $\beta\rho\sigma$ is a nonzero positive integer. Since $\sigma < \rho$, we have $\beta\rho^2 > 1$, so $u < 1/2$. The bound $\rho \le \beta^{-17/40}$ gives $u \ge 17/40$.

We prove irrationality to exclude equality at this endpoint. If $u = s/t$ with positive integers s, t, then $\beta^s\rho^t = 1$, or

$$\beta^s(-\rho)^t = (-1)^t.$$

The Galois group of the splitting field of an irreducible cubic is either S_3 or the cyclic group of order 3. In the first case, apply the transposition interchanging β and $-\rho$, and divide the two relations in absolute value. This gives $(\beta/\rho)^{s-t} = 1$, so $s = t$. It follows that $\beta\rho = 1$, contradicting $\beta\rho\sigma = \sigma < 1$ and the integral nonzero norm. In the cyclic case, let ℓ_i be the logarithms of the absolute values of the three conjugates in cyclic order. The relation and its two conjugates give

$$s\ell_1 + t\ell_2 = 0, \qquad s\ell_2 + t\ell_3 = 0, \qquad s\ell_3 + t\ell_1 = 0.$$

The coefficient matrix has determinant $s^3 + t^3 > 0$, forcing all $\ell_i = 0$, in contradiction with $\log\beta > 0$. This proves irrationality, and therefore (218).

With $M = 3^r$, expand exactly:

$$\begin{aligned} G_r &= \beta^{3M} - e_{r+1} - (\beta^M - e_r)^3 \\ &= 3\beta^{2M}e_r - 3\beta^M e_r^2 + e_r^3 - e_{r+1}. \end{aligned}$$

We have $e_r \sim \rho^M$ and $e_{r+1} \sim \rho^{3M}$. Dividing the last three terms by $\beta^{2M}\rho^M$ shows that each tends to zero, since $\rho/\beta < 1$. Thus $G_r \sim 3(\beta^2\rho)^M$. Since $p_r \sim \beta^M$ and $\rho = \beta^{-u}$, the second equivalence in (219) follows. The gap is eventually positive. On the tail where $e_r \in (0,1)$, $p_{r+1} < \beta^{3M} < (p_r+1)^3$, proving the upper nesting bound. □

17.4.2 Integer cells and the eventual least child

Lemma 17.10 (Exact ancestor of an integer cell)**.** *Let $q \geq 1$ and $D \geq 1$ be integers. Throughout the half-open interval*

$$[q^{1/D}, (q+1)^{1/D}),$$

the integer part is constant and equals $\left\lfloor q^{1/D} \right\rfloor$. Consequently a terminal label $q = \left\lfloor B^{LD} \right\rfloor$, where $L > 0$, determines its earlier label exactly by

$$\left\lfloor B^{L} \right\rfloor = \left\lfloor q^{1/D} \right\rfloor.$$

Proof. Put $j = \left\lfloor q^{1/D} \right\rfloor$. Then $j^D \leq q < (j+1)^D$. Both q and $(j+1)^D$ are integers, so $q + 1 \leq (j+1)^D$. Every point of the half-open interval is at least j and strictly less than $j + 1$. This proves both assertions, including the case where its excluded right endpoint is an integer. □

Lemma 17.11 (Continuation and least-prime choice)**.** *The estimate (214) implies that every sufficiently large prime has an infinite strict cubic prime continuation. If (217) comes from the actual global minimum ξ, then, for every sufficiently large r, p_{r+1} is the least prime strictly greater than p_r^3.*

Proof. At $x = q^3$, estimate (214) supplies a prime q' with

$$q^3 < q' \leq q^3 + q^{63/40} < (q+1)^3 - 1$$

for all sufficiently large primes q. The left endpoint is a composite cube. The last inequality holds eventually because $63/40 < 2$. Repeating this choice produces primes q_j with $q_j^3 < q_{j+1}$ and $q_{j+1} + 1 < (q_j + 1)^3$. The corresponding closed inverse-power cells are nested, with strict containment at both endpoints at each successive step. Their lengths tend to zero. Their unique common point lies below every right endpoint, since the next cell already does, and hence realizes all the prescribed prime floors. This proves continuation.

Now suppose, at a sufficiently late index, that a prime q obeys $p_r^3 < q < p_{r+1}$. In the original ξ clock, replace the label at level $m+r+1$ by q and continue from it as just constructed. The cell of this replacement label lies within the cell of p_r: its lower endpoint is above the p_r lower endpoint, while $q + 1 \leq p_{r+1} < (p_r + 1)^3$. Lemma 17.10 then preserves every earlier label as well. The resulting base B is strictly smaller than ξ, because

$$B \leq (q+1)^{1/3^{m+r+1}} \leq p_{r+1}^{1/3^{m+r+1}} < \xi.$$

It has prime floors at all original levels, contradicting global minimality. Thus no such prime q exists. Since p_{r+1} is itself prime and exceeds p_r^3, it is the least prime above that cube. □

The strict endpoint and the least-prime property are preparatory consequences of the cited reduction, the prime input, and actual global minimality. They are not consequences of a long skipped-level interval.

17.5 Ancestral conditions remove the apparent exponent gain

Work on the sufficiently late tail of the conditional minimum setting above. Fix a depth $d \geq 2$, put $M = 3^r$, and consider terminal labels at level $r + d$. A label q has the half-open cell

$$\left[q^{1/(M3^d)}, (q+1)^{1/(M3^d)}\right)$$

in the β coordinate. With $D = 3^{d-1}$, Lemma 17.10 makes its first descendant of level r exactly

$$\left\lfloor q^{1/D} \right\rfloor. \tag{220}$$

Proposition 17.12 (The terminal search corridor)**.** *Every prime terminal label in*

$$(p_r^{3^d},\ p_{r+1}^{3^{d-1}}) \tag{221}$$

has a composite ancestor at level $r+1$. More generally, for the search among earlier terminal labels $q < p_{r+d}$, preserving the first j descendants leaves the geometric corridor

$$I_{r,d}^{(j)} = (p_{r+j}^{3^{d-j}},\ p_{r+d}), \qquad 0 \le j < d. \tag{222}$$

For $j < d-1$, requiring the next ancestor to be prime reduces this corridor to the one with j replaced by $j+1$. After all intermediate conditions are imposed, the remaining corridor is

$$I_{r,d}^{(d-1)} = (p_{r+d-1}^3,\ p_{r+d}). \tag{223}$$

Here a corridor is a real search interval, not an assertion that it contains a prime.

Proof. For a prime q in (221), formula (220) gives an integer between p_r^3 and $p_{r+1} - 1$. The first is a composite cube, and every larger integer in this range is composite by Lemma 17.11. Thus the ancestor is not prime.

For the general step, suppose the label p_{r+j} has been preserved. Its cell gives the lower terminal bound $p_{r+j}^{3^{d-j}}$. Its upper terminal bound causes no additional restriction on $q < p_{r+d}$, because the candidate itself satisfies

$$p_{r+d} < \beta^{M3^d} < (p_{r+j}+1)^{3^{d-j}}.$$

For prime terminal labels the lower endpoint, a proper integer power, is already excluded. This gives the corridor (222).

If $j < d-1$, every terminal prime below $p_{r+j+1}^{3^{d-j-1}}$ has its next ancestor in $[p_{r+j}^3, p_{r+j+1})$. That ancestor is composite by the same least-prime argument. Conversely, within the remaining corridor, the next ancestor must equal p_{r+j+1}: its lower bound is that integer, and its upper bound follows from $p_{r+d} < (p_{r+j+1}+1)^{3^{d-j-1}}$. Thus imposing that next condition removes exactly the earlier forbidden portion of the geometric search and gives the next corridor. Iterate to obtain (223). □

A continuation after a prime terminal label cannot repair a composite earlier label. Also, this proposition concerns a first change by the chosen terminal depth. A smaller real base can still have terminal label $q = p_{r+d}$ and first differ at a deeper level; such a base is not being ruled out merely by the terminal-label search at depth d.

Theorem 17.13 (Fixed-depth width loss)**.** *Fix $d \ge 2$. As $r \to \infty$, put $X = \beta^{M3^d}$ and*

$$\eta_s = 1 - \frac{1+u}{3^s} \qquad (s \ge 1).$$

For each $0 \le j < d$,

$$|I_{r,d}^{(j)}| \sim 3^{d-j}\beta^{(3^d-3^j)M}\rho^{3^jM} = 3^{d-j}X^{\eta_{d-j}}. \tag{224}$$

In particular,

$$\eta_1 = \kappa \in \left(\frac{1}{2}, \frac{21}{40}\right), \qquad \eta_2 \in \left(\frac{5}{6}, \frac{101}{120}\right), \qquad \eta_s \longrightarrow 1. \tag{225}$$

The ratio of the final to the raw corridor width is

$$\frac{|I_{r,d}^{(d-1)}|}{|I_{r,d}^{(0)}|} \sim 3^{1-d}\left(\frac{\rho}{\beta}\right)^{(3^{d-1}-1)M} \longrightarrow 0. \tag{226}$$

Proof. Write $L = 3^{d-j}$ and $Y = \beta^{M3^j}$. Since $p_{r+j} = Y - e_{r+j}$ and $p_{r+d} = Y^L - e_{r+d}$,

$$|I_{r,d}^{(j)}| = Y^L - e_{r+d} - (Y - e_{r+j})^L.$$

For fixed d and j, the binomial expansion gives the leading term $LY^{L-1}e_{r+j}$. The next binomial term relative to this term is $O(e_{r+j}/Y)$, which tends to zero like $O((\rho/\beta)^{M3^j})$. Also

$$\frac{e_{r+d}}{Y^{L-1}e_{r+j}} \sim \left(\frac{\rho}{\beta}\right)^{M(3^d-3^j)} \longrightarrow 0.$$

Because $e_{r+j} \sim \rho^{M3^j}$, this proves the first asymptotic in (224). Substituting $\rho = \beta^{-u}$ proves the power of X. The bounds in (225) follow from (218). Dividing the expressions in (224) for $j = d-1$ and $j = 0$ yields (226). □

A raw two-level corridor therefore has an exponent greater than $5/6$, well above the BHP exponent. Its part compatible with every ancestor has exponent $\kappa < 21/40$, exactly the original one-step exponent at a later index. The gain is lost through the ancestral conditions, not through an imprecise estimate of the terminal interval. All these asymptotics have fixed depth; no uniform assertion for depth increasing with r is made.

17.6 Thinning while preserving short-interval richness

The following theorem isolates the information retained by a short-interval lower bound. It does not assume that S consists of primes, that the specified base is algebraic, or that it was already a minimum for S.

Theorem 17.14 (Altered-set minimum)**.** *Let $0 < \kappa < \theta < 1$, and suppose $S \subseteq \mathbb{N}$ satisfies*

$$\#(S \cap [x, x + x^\theta]) \geq c\frac{x^\theta}{\log x} \qquad (x \geq x_0) \tag{227}$$

for some $c > 0$ and $x_0 > 1$. Let $A > 1$, and suppose that for every $k \geq 1$,

$$P_k = \left\lfloor A^{3^k} \right\rfloor \in S, \qquad P_1 \geq 2, \qquad P_k^3 < P_{k+1} < (P_k + 1)^3. \tag{228}$$

Assume also that, for all sufficiently large k,

$$P_{k+1} - P_k^3 = O((P_k^3)^\kappa). \tag{229}$$

Define

$$S_A = (S \cap [P_1, \infty)) \setminus \bigcup_{k \geq 1} [P_k^3, P_{k+1}). \tag{230}$$

Then every P_k belongs to S_A, the number A is the attained global minimum of $\mathcal{W}(S_A)$, and for all sufficiently large x,

$$\#(S_A \cap [x, x + x^\theta]) \geq \frac{c}{2}\frac{x^\theta}{\log x}. \tag{231}$$

More precisely, the number of integers removed from such a window is $O(x^\kappa)$.

Proof. Because $P_1 \geq 2$ and $P_{k+1} > P_k^3$, the labels strictly increase and $P_k < P_k^3$ for every k. The deletion intervals are disjoint: the right endpoint of the kth is P_{k+1}, smaller than the left endpoint P_{k+1}^3 of the next. Their right endpoints are excluded. Thus no P_k is deleted, and all labels survive the initial cutoff. Hence $A \in \mathcal{W}(S_A)$.

Suppose $1 < B < A$ also belonged to $\mathcal{W}(S_A)$, and write $Q_k = \left\lfloor B^{3^k} \right\rfloor$. Monotonicity gives $Q_k \le P_k$. The cutoff gives $Q_1 \ge P_1$, so $Q_1 = P_1$. The two label sequences cannot agree forever. If they did, both A and B would belong to the same inverse-power cell at every k, so

$$0 < A - B \le (P_k + 1)^{1/3^k} - P_k^{1/3^k} \le 3^{-k}.$$

The last inequality is the mean value theorem, since the derivative of $y^{1/3^k}$ is at most 3^{-k} for $y \ge 1$. This is impossible as $k \to \infty$.

At the first differing index $k+1$, the preceding label is $Q_k = P_k$, whereas $Q_{k+1} < P_{k+1}$. The preceding equality implies $B^{3^k} \ge P_k$, and therefore

$$P_k^3 \le Q_{k+1} < P_{k+1}.$$

This is precisely a deleted interval. It contradicts $Q_{k+1} \in S_A$, proving global leastness.

For the quantitative statement put $X_k = P_k^3$ and $\Delta_k = P_{k+1} - X_k$. Since $\kappa < 1$, (229) implies $P_{k+1}/X_k \to 1$, and hence

$$X_{k+1} = P_{k+1}^3 \sim X_k^3. \tag{232}$$

The fixed cutoff and the finitely many early deletion intervals are irrelevant for sufficiently large x. If the window $[x, x + x^\theta]$ meets a late interval $[X_k, P_{k+1})$, then $X_k \le x + x^\theta$ and $x < P_{k+1} = X_k + O(X_k^\kappa)$. Since both θ and κ are below 1, these inequalities give $X_k \sim x$. Two late deletion intervals cannot both meet such a window: their starting points would have ratio tending to 1, whereas (232) makes the ratios of successive starting points tend to infinity. Thus at most one late deletion interval is met. It contains $O(x^\kappa)$ integers. Subtracting this loss from (227) gives

$$\#(S_A \cap [x, x + x^\theta]) \ge c\frac{x^\theta}{\log x} - O(x^\kappa).$$

Finally $x^\kappa = o(x^\theta/\log x)$ because $\kappa < \theta$. This proves (231) for every sufficiently large real x, not merely for an averaged collection of windows. □

Remark 17.15 (Why the first-label hypothesis matters). The hypothesis $P_1 \ge 2$ is essential to the stated general construction. If $P_1 = 1 < P_2$, its first deletion interval $[P_1^3, P_2) = [1, P_2)$ removes P_1 itself. Merely assuming $A > 1$ would not prevent this defect. In the prime applications and the explicit example below, the stronger hypothesis holds.

17.6.1 Conditional specialization to the actual primes

Take $S = \mathcal{P}$ and $\theta = 21/40$. If an all-prime classical candidate A has the gap bound (229) with $\kappa < 21/40$, its labels automatically have $P_1 \ge 2$ and strict nesting: a prime cannot equal the cube of an earlier prime, and the upper nesting bound follows from its floor cell. Theorem 17.14 then gives a subset $\mathcal{P}_A \subseteq \mathcal{P}$ which retains the entire candidate tower, still satisfies the BHP lower-bound scale, and has A as its exact global minimum. No composite has been added to the allowed outputs.

This is a conditional stability statement, not an unconditional construction of an algebraic prime tower. The existence of the required algebraic all-prime candidate has not been established. Moreover, even if such a candidate were supplied, the theorem changes the allowed set from all primes to a subset; it is not a counterexample about the full prime set. The conclusion is that the particular short-interval information (227) survives deletion of the competing-child intervals when their exponent is strictly smaller.

If only an eventually all-prime Pisot tail is given, its indexing must first be normalized. Choose R so that $\left\lfloor \beta^{3^{R+j}} \right\rfloor$ is prime for every $j \geq 0$, and put

$$\beta' = \beta^{3^R}, \qquad A_{\mathrm{tail}} = (\beta')^{1/3}, \qquad P_k = \left\lfloor (\beta')^{3^{k-1}} \right\rfloor \quad (k \geq 1). \tag{233}$$

Choose R late enough also for strict nesting. Then $P_k = \left\lfloor A_{\mathrm{tail}}^{3^k} \right\rfloor$ is a complete clock in the convention of Theorem 17.14. In the Pisot setting, (219) supplies its small-gap bound. The altered set has minimum A_{tail}, which need not be the original Mills minimum. An already complete classical candidate requires no such normalization.

The definition of primes and further arithmetic information about the full prime set can distinguish it from a thinned subset. Thus this theorem does not prohibit an argument using BHP together with additional arithmetic. It identifies the missing information in an argument that uses only the displayed richness bound, primality of the retained elements, and existence of an admissible tower.

17.7 An explicit algebraic minimum for integers coprime to six

The conditional prime specialization has an unconditional analogue for a larger integer set. It illustrates the scope of the thinning lemma without asserting that the outputs are primes.

Theorem 17.16 (An explicit nonprime-output witness)**.** *Let*

$$a = 2^{40} + 1, \qquad b = 2^{22}, \qquad f(X) = X^3 - aX^2 - bX + 2, \tag{234}$$

and let β be its largest real root. There is a set $S_* \subseteq \{n \geq 1 : \gcd(n, 6) = 1\}$ with the following properties:

(i) *$\beta^{1/3}$ is the attained global minimum of $\mathcal{W}(S_*)$.*

(ii) *For every sufficiently large real x,*

$$\#(S_* \cap [x, x + x^{21/40}]) = \left(\frac{1}{3} + o(1)\right) x^{21/40}.$$

(iii) *The roots are $\beta, -\rho, \gamma$ with $0 < \gamma < \rho < 1$ and*

$$\rho < \min(\gamma^{17/23}, \beta^{-17/40}).$$

Their trace gaps have an exponent $\kappa \in (1/2, 21/40)$.

The minimizing base is algebraic, but its first classical label is $2^{40} + 1$, which is composite.

Proof. The only possible rational roots of the monic cubic are $\pm 1, \pm 2$. Their values are

$$\begin{aligned} f(1) &= 3 - a - b, & f(-1) &= 1 - a + b, \\ f(2) &= 10 - 4a - 2b, & f(-2) &= -6 - 4a + 2b. \end{aligned}$$

All are negative and nonzero, so the cubic is irreducible. Exact rational substitutions give the signs

z	$-5 \cdot 10^{-6}$	$-4 \cdot 10^{-6}$	$3 \cdot 10^{-7}$	10^{-6}	a	$a+1$
$\operatorname{sgn} f(z)$	$-$	$+$	$+$	$-$	$-$	$+$

For the last two, $f(a) = 2 - ba < 0$ and $f(a+1) = (a+1)(a+1-b) + 2 > 0$. The other four are direct rational calculations, also checked exactly by the packaged portable verifier. Hence the three disjoint root brackets are

$$\begin{gathered} a < \beta < a + 1, \\ 4 \cdot 10^{-6} < \rho < 5 \cdot 10^{-6}, \qquad 3 \cdot 10^{-7} < \gamma < 10^{-6}. \end{gathered} \tag{235}$$

They exhaust the roots and give $0 < \gamma < \rho < 1$.

We record elementary bounds proving the required root-size inequalities. Set $t = 5 \cdot 10^{-6}$. Since $25 \cdot 2^{35} < 10^{12}$, we have $t^2 < 2^{-35}$. Also $a + 1 = 2^{40} + 2 < 2^{41}$. Therefore

$$t^{40}(a+1)^{17} < 2^{-700}2^{697} < 1,$$

which, with (235), gives $\rho < \beta^{-17/40}$. For the other inequality, use $t < 2^{-17}$ and $3 \cdot 10^{-7} > 2^{-22}$ to get

$$t^{23} < 2^{-391} < 2^{-374} < (3 \cdot 10^{-7})^{17}.$$

Thus $\rho^{23} < \gamma^{17}$, as required. The norm is -2, so $\beta\rho\gamma = 2$ and $\beta\rho^2 > 2$. Consequently

$$\frac{17}{40} < -\frac{\log \rho}{\log \beta} < \frac{1}{2}.$$

The gap-exponent assertion now follows by the same expansion as in Lemma 17.9; no prime hypothesis was used in that expansion.

Let T_n be the power sum of the three roots. It satisfies

$$\begin{aligned} T_0 = 3, \qquad T_1 = a, \qquad T_2 = a^2 + 2b, \\ T_n = aT_{n-1} + bT_{n-2} - 2T_{n-3} \qquad (n \geq 3). \end{aligned} \tag{236}$$

For every odd positive n, $T_n = \beta^n - \rho^n + \gamma^n$ and $0 < \rho^n - \gamma^n < 1$, so $T_n = \lfloor \beta^n \rfloor$. Put $p_r = T_{3^r}$ for $r \geq 0$. These are positive integers. Modulo 2, recurrence (236) and its initial values show that every positive T_n is odd. Modulo 3, the cubing identity for three variables gives $T_{3n} \equiv T_n^3 \pmod 3$. Thus $p_r \equiv a \equiv -1 \pmod 3$. We have proved

$$p_r \equiv 5 \pmod 6 \qquad (r \geq 0). \tag{237}$$

Strict nesting holds at every index here, not only eventually. For $n = 3^r$, the exact algebraic identity is

$$p_{r+1} - p_r^3 = 3(\beta^n - \rho^n)(\beta^n + \gamma^n)(\rho^n - \gamma^n) > 0. \tag{238}$$

The upper inequality follows from $p_{r+1} < \beta^{3n} < (p_r + 1)^3$. The asymptotic proved above also gives $p_{r+1} - p_r^3 = O((p_r^3)^\kappa)$ with $\kappa < 21/40$.

Apply Theorem 17.14 to

$$S = \{n \geq 1 : \gcd(n, 6) = 1\}, \qquad A = \beta^{1/3}, \qquad P_k = p_{k-1}.$$

This indexing gives $P_k = \left\lfloor A^{3^k} \right\rfloor$ for every $k \geq 1$, and $P_1 = p_0 = a \geq 2$. Define explicitly

$$S_* = (S \cap [a, \infty)) \setminus \bigcup_{r \geq 0} [p_r^3, p_{r+1}). \tag{239}$$

The base A is algebraic because β is algebraic, and the thinning theorem proves that A is the exact minimum for this set. For an interval of length L, the original set S has $L/3 + O(1)$ elements, by counting the residue classes 1 and 5 modulo 6. The proof of the thinning theorem removes only $O(x^\kappa)$ elements from $[x, x + x^{21/40}]$. Its count is therefore

$$\frac{x^{21/40}}{3} + O(1) + O(x^\kappa) = \left(\frac{1}{3} + o(1)\right) x^{21/40}.$$

Finally $2^8 \equiv -1 \pmod{257}$, so $2^{40} + 1 \equiv 0 \pmod{257}$ and $a > 257$. The first classical label is composite. □

This example combines the specified cubic root pattern, exact odd trace floors, a sub-BHP gap exponent, elementary coprimality to 6, an algebraic attained minimum, and richness on every BHP-scale window. It does not combine these properties with prime outputs. Its explicitly composite first label prevents an interpretation as a Mills counterexample.

17.8 Exact verification and reproducibility

The numerical certificates are retained in the original source archive in the `certificates/` directory. All four listed programs use only Python 3 and its standard library.

1. `adaptive_case_and_eisenstein_verify.py` uses the quotient ring R_d and integer binary powering. It checks (207)–(210), the divisibility $36 \mid h$, the numerical primality assertions, and (212). Its finite polynomial and gcd tests are sanity checks, not proofs of the universal theorems.
2. `audit_adaptive_matrix_verify.py` independently uses the 3 by 3 matrix (206). It checks the same returns, the field-interpretation determinant, and the negative-clock return (213). Its exhaustive primality routine tests the possible divisors of the form $6j \pm 1$, separately from the quotient-ring program.
3. `independent_audit.py` supplies a further independent exact arithmetic check of the adaptive witnesses.
4. `classical_minimality_portable_verify.py` checks the rational root brackets and inequalities for (234), the initial trace residues and nesting, the polynomial identity (238), and the algebra of the corridor exponents and width ratio. It includes a regression check for the indispensable hypothesis $P_1 \geq 2$. These are not primality or transcendence tests.

For the ring arithmetic, multiply triples as polynomials of degree at most 4, then reduce the terms of degrees 4 and 3, in that order, using $x^3 = ax^2 + 4x - 4$. All coefficients are reduced modulo d. Binary powering starts with $v = 1$ and successively squares the base, multiplying it into v whenever the current binary digit of the exponent is 1. The invariant is the usual exact product of the already processed powers. This computes the claimed identities in logarithmically many ring multiplications. The matrix verifier performs ordinary modular matrix products instead, so the two implementations do not share the quotient-ring multiplication routine.

Exhaustive trial division is a deterministic primality proof: a composite integer N has a prime divisor at most $\sqrt{N}$. After 2 and 3 have been checked, every possible prime divisor larger than 3 is of the form $6j \pm 1$. No probable-prime result is accepted as a proof. The proper-divisor conclusions themselves do not require that the recurring divisors be prime.

From the artifact directory the checks can be repeated with

```
python3 certificates/adaptive_case_and_eisenstein_verify.py
python3 certificates/audit_adaptive_matrix_verify.py
python3 certificates/independent_audit.py
python3 certificates/classical_minimality_portable_verify.py
sha256sum -c SHA256SUMS.txt
```

The accompanying successful logs and available JSON reports record these finite checks. The all-k, r Eisenstein theorem, all-m, k consecutive-coprimality theorem, fixed-depth asymptotics, and general thinning theorem are proved in the text, rather than inferred from finite samples.

17.9 Remaining questions

The arithmetic certificates settle the two specified coefficients, not the whole thin family $a = 7^m$. A universal adaptive approach would still need a theorem guaranteeing a suitable divisor and return, or another compositeness mechanism, for each remaining coefficient. The irreducibility theorem rules out a particular polynomial-composition shortcut; it does not prove that such a universal argument is impossible.

For the actual classical minimum, a skipped-level prime is useful only when every prescribed earlier floor is prime. The width calculation identifies the resulting one-step scale exactly. Possible additional inputs include a prime inside a competing-child interval, a correlation theorem excluding an entire candidate sequence of exceptional prime-free intervals, or a direct arithmetic obstruction to the candidate trace orbit. None of those inputs is proved here.

The altered-set theorem establishes the stability of a specified short-interval lower bound under thinning. It does not establish an algebraic all-prime tower, does not replace the full prime set by an equivalent one, and does not preclude a proof using further arithmetic information. No Riemann Hypothesis, Density Hypothesis, Artin-type conjecture, or presumed prime pattern on a prescribed orbit is used in the results proved here.

18 Projective return criteria and finite horizon limitations

The finite-ring examples admit a common projective formulation. This section gives recurrence criteria, finite-horizon obstructions, and the precise behavior of witness orders under powering; its projective exponent also strengthens the density theorem.

18.1 The clocks and the exact floor bridge

For an integer $a > 8$, put

$$f_a(X) = X^3 - aX^2 - 4X + 4, \qquad n_+(k) = 3^k, \quad n_-(k) = 2 \cdot 3^k - 1 \quad (k \geq 1).$$

The names classical and negative refer to these two exponent sequences; both have positive odd exponents. Write $P_n(A)$ for the formal Newton power-sum polynomial of $X^3 - AX^2 - 4X + 4$. Its recurrence is

$$P_0 = 3, \quad P_1 = A, \quad P_2 = A^2 + 8, \qquad P_n = AP_{n-1} + 4P_{n-2} - 4P_{n-3} \quad (n \geq 3). \tag{240}$$

In particular $P_n \in \mathbb{Z}[A]$, monic of degree n for $n \geq 1$.

Lemma 18.1. *The roots of f_a are $-\rho, \sigma, \beta_a$, where $0 < \sigma < \rho < 1$ and $a < \beta_a < a + 1$. The polynomial is irreducible, and at every positive odd exponent*

$$\lfloor \beta_a^n \rfloor = P_n(a), \qquad 0 < \beta_a^n - P_n(a) = \rho^n - \sigma^n < 1.$$

These floors strictly increase with the positive odd exponent. If $a > 2^{41}$, then additionally

$$\rho < \tfrac{1}{2}\beta_a^{-17/40}, \qquad \rho < \sigma^{17/23}, \qquad 4 < \beta_a^{3/20}. \tag{241}$$

Proof. The values at $-1, 0, 1, a, a + 1$ are respectively $7 - a, 4, 1 - a, 4(1 - a), (a - 1)^2$. Their signs put one root in each of $(-1, 0), (0, 1), (a, a + 1)$. No root is integral, so the rational-root test proves irreducibility. Their sum gives $\rho - \sigma = \beta_a - a > 0$. The odd power sum proves the floor identity. Since $\beta_a > 8$, $\beta_a^{n+2} - \beta_a^n > 1$, which proves strict increase. The equation at $-\rho$ gives $a\rho^2 = 4 + 4\rho - \rho^3 < 8$. With $\beta_a < 2a$,

$$(2\rho)^{40}\beta_a^{17} < (32/a)^{20}(2a)^{17} = 2^{117}/a^3 < 1.$$

The norm identity $\beta_a\rho\sigma = 4$ shows that $\rho < \sigma^{17/23}$ is equivalent to $\rho^{40}\beta_a^{17} < 4^{17}$, implied by the preceding bound. Finally $4^{20} < a^3 < \beta_a^3$. □

The bounds (241) are of the kind arising in the conditional Pisot reduction for the classical Mills minimum [4, Theorem 1.7]. Satisfying them does not make a cubic an all-prime tower or a least constant. Our main arithmetic family is $a = 7^m$, $m \geq 2$, whose integer part is already composite. Prime-leading examples will occur separately in the bounded-witness and finite-prime limitations.

For a modulus $D \geq 2$, let

$$R_D = (\mathbb{Z}/D\mathbb{Z})[X]/(f_a), \qquad x = X \bmod f_a.$$

Multiplication trace is the linear function

$$\tau(c_0 + c_1 x + c_2 x^2) = 3c_0 + ac_1 + (a^2 + 8)c_2 \pmod{D}. \tag{242}$$

It satisfies $\tau(x^n) = P_n(a) \pmod{D}$, by the common recurrence and initial values. Multiplication by x has matrix

$$M_a = \begin{pmatrix} 0 & 0 & -4 \\ 1 & 0 & 4 \\ 0 & 1 & a \end{pmatrix} \pmod{D}. \tag{243}$$

This interpretation includes repeated factors and composite moduli.

18.2 A positive projective recurrence mechanism

18.2.1 Every zero seed recurs at an irreducible reduction

If D is odd, x is a unit because the constant term is 4. Let Q_D be its order modulo scalar units:

$$Q_D = \operatorname{ord}([x] \in R_D^*/(\mathbb{Z}/D\mathbb{Z})^*).$$

Equivalently, Q_D is the least positive integer such that x^{Q_D} is a scalar unit. Multiplication by such a scalar preserves trace zero in both directions. Thus trace-zero status depends only on the exponent modulo Q_D, although the trace value itself need not.

Theorem 18.2 (Irreducible reduction)**.** *Let p be an odd prime and suppose f_a is irreducible modulo p. For either clock, if $p \mid P_{n_\pm(k_0)}(a)$ at any original index $k_0 \geq 1$, then the same prime divides infinitely many terms on that clock. For the real root, these yield infinitely many composite floors, with at most a finite initial exception for properness.*

Proof. The quotient is $\mathbb{F}_{p^3}$. Put $S = p^2 + p + 1$. Its field norm gives $x^S = -4 \in \mathbb{F}_p^*$, so $Q_p \mid S$. If $p \equiv 1 \pmod 3$, then $v_3(S) = 1$; otherwise, including $p = 3$, $v_3(S) = 0$. Write $S = 3^v B$ with $3 \nmid B$, and use $\operatorname{ord}_1(3) = 1$. For $h = \operatorname{ord}_B(3)$ and $k_0 \geq 1$, both

$$3^{k_0}(3^h - 1), \qquad 2 \cdot 3^{k_0}(3^h - 1)$$

are divisible by S. Hence the corresponding clock states at $k_0 + h$ and k_0 differ by a scalar unit. Their traces vanish together. Repetition gives every $k_0 + th$, $t \geq 0$. Lemma 18.1 makes the growing floors eventually larger than p. □

This proof needs a scalar return, not equality of the full ring states. For example, if $O = \operatorname{ord}(x)$ and $d = \operatorname{ord}_p(-4)$, the norm gives $d = O/\gcd(O, S)$. When $p \equiv 1 \pmod 3$ and $v_3(d) > 0$, this forces $v_3(O) = v_3(d) + 1$, which can exceed one. Those extra powers of 3 lie in scalar information and disappear projectively. Irreducibility cannot simply be omitted; an exact reducible transient example is given in Section 18.5.

18.2.2 Hits without factoring a scheduled integer trace

The following trace-zero geometry is classical. Trace-zero cosets in $\mathbb{F}_{p^3}^*/\mathbb{F}_p^*$ form the planar Singer difference set; its norm-one polynomial description is given explicitly in [33, Section 1.2 and Corollary 1.2]. We prove the needed root count directly and make no novelty claim for that ingredient.

Theorem 18.3 (A finite auxiliary-prime test)**.** *Suppose $p \geq 5$ is prime, $S = p^2 + p + 1$ is prime, $\mathrm{ord}_S(3) = S - 1$, and f_a is irreducible modulo p. Each clock has exactly $p+1$ trace-zero phases modulo $S - 1$. Consequently each clock has infinitely many composite floors for every integer $a > 8$ in that coefficient residue class. The two sets of phases need not intersect.*

Proof. The projective order of x divides the prime S and is not 1, since an irreducible cubic root is not scalar. Thus it equals S. The homomorphism $z \mapsto z^{p-1}$ has kernel $\mathbb{F}_p^*$, so $y = x^{p-1}$ generates the norm-one subgroup of order S. For every integer n,

$$\frac{\mathrm{Tr}(x^n)}{x^n} = 1 + y^n + y^{(p+1)n}. \tag{244}$$

The polynomial $F(Z) = Z^{p+1} + Z + 1$ has exactly $p + 1$ distinct roots, all in that subgroup. Indeed, a root z is nonzero; raising its equation to the pth power and multiplying by z gives $z^{p^2+p+1} = -z^{p+1} - z = 1$. These roots belong to $\mathbb{F}_{p^3}$ because $S \mid p^3 - 1$. Also $F'(Z) = Z^p + 1$; a common root of F and F' would give $1 = 0$. The roots are therefore distinct.

As y generates the group, exactly $p + 1$ classes $n \pmod S$ give trace zero. Class 0 does not, because $\mathrm{Tr}(1) = 3 \neq 0$. Class -1 does not either: in this cubic $\mathrm{Tr}(x^{-1}) = (-4)/(-4) = 1$. Over one index period $S - 1$, 3^k runs through every nonzero residue modulo S. The values $2 \cdot 3^k - 1$ run through every residue except -1, since S is odd. Each orbit therefore visits all $p + 1$ zero classes. Projective periodicity proves the asserted phase counts and recurrence. Only a finite prefix can fail to exceed p. □

For a general irreducible cubic with nonzero norm, the same argument still guarantees hits on both clocks under the two assumptions on S: the negative clock omits at most one of the $p + 1 > 1$ zero classes. The exact negative-clock count above uses the nonzero inverse trace of our specific cubic. The hypotheses are finite, checkable conditions; we do not assume or prove infinitely many primes satisfying them. In particular no primitive-root conjecture is used.

18.2.3 The exact same-prime simultaneous count

Theorem 18.4 (Common phases)**.** *Under the hypotheses of Theorem 18.3, the number of phases modulo $S - 1$ where both clocks have zero trace modulo p is 2 if a is a nonzero square modulo p, 0 if it is a nonsquare, and 1 if $a \equiv 0 \pmod p$.*

Proof. Put $z = x^{3^k}$. The conditions are exactly $\mathrm{Tr}(z) = 0$ and $\mathrm{Tr}(z^2/x) = 0$, with no reindexing. The trace-zero subspace has basis

$$v = 3x - a, \qquad w = 3x^2 - (a^2 + 8),$$

since $p \neq 3$ and the x, x^2 coefficients show independence. For $z = uv + tw$, its second condition is

$$Au^2 + 2But + Ct^2 = 0,$$

where direct trace calculation gives

$$A = a(a - 9), \quad B = a(a^2 - 3a + 8), \quad C = a^4 + 3a^3 + 16a^2 + 60a - 44.$$

These formulas follow from $\mathrm{Tr}(x^{-1}) = 1$ and the first four power sums. Their quadratic discriminant, up to the square factor 4, is

$$B^2 - AC = 36a(a^3 + a^2 + 18a - 11) = \frac{9}{4}a\Delta(f_a). \tag{245}$$

For an irreducible cubic over an odd finite field, $\Delta(f_a)$ is a nonzero square: Frobenius cyclically permutes the roots x, x^p, x^{p^2}, an even permutation, so it fixes their nonzero Vandermonde product. Its square is the discriminant. For $a \neq 0$ the form is nondegenerate. The elementary binary quadratic criterion, including a possible root at infinity, gives two projective zeros exactly for square a, and none otherwise. Indeed, when $A \neq 0$, multiplying the form by A gives $(Au + Bt)^2 - (B^2 - AC)t^2$; when $A = 0$, nondegeneracy forces $B \neq 0$ and the form factors as $t(2Bu + Ct)$ with two distinct lines. For $a = 0$ the form is $-44t^2$, nonzero because $p \geq 5$ and $p = 11$ would give $\Delta(f_0) = -176 = 0$, contradicting irreducibility. It has one projective zero.

Finally, $[x]$ generates the projective group of prime order S, and primitive 3 makes $[x^{3^k}]$ visit every projective point except the scalar identity exactly once per period. The omitted point has trace $3 \neq 0$, so every projective zero of the form is visited. The two equations scale by a nonzero scalar and its square, respectively, so this projective count is legitimate. □

The zero count concerns this particular auxiliary prime, not all possible divisors or the complete sets of composite indices.

18.3 The prime 173 and complete coefficient residue cells

Take $a = 7^{12}$. Modulo $p = 173$, its value is 81. Exhaustive evaluation shows that f_{81} has no root in $\mathbb{F}_{173}$, hence is irreducible. The integers 173 and $S = 30103$ are prime by trial division. Moreover

$$S - 1 = 30102 = 2 \cdot 3 \cdot 29 \cdot 173, \qquad 3^{30102} \equiv 1 \pmod{30103},$$
$$3^{30102/r} \not\equiv 1 \pmod{30103} \qquad (r = 2, 3, 29, 173).$$

Thus Theorem 18.3 applies without factoring any integer trace. Exact ring or companion-matrix calculations give the states in Table 16. Each tuple denotes coefficients in $1, x, x^2$. The

Original index k	Classical state	Negative state	Verified zero traces
318	$(143, 48, 33)$	—	Classical
421	—	$(99, 114, 107)$	Negative
14733	$(95, 89, 11)$	$(20, 140, 31)$	Both
25911	$(18, 43, 50)$	$(3, 142, 157)$	Both

Table 16: Exact states modulo 173 at their original clock indices.

complete scan has 174 zero phases for each clock; their intersection in $1 \leq k \leq 30102$ is exactly $\{14733, 25911\}$.

Corollary 18.5. *For $a = 7^{12}$, both actual floors are composite at every*

$$k = 14733 + 30102t \quad (t \geq 0),$$

and also at every $k = 25911 + 30102t$.

Proof. The exact identities $x^{30103} = -4$ and $3^{30102} \equiv 1 \pmod{30103}$ propagate each displayed zero by nonzero scalar multiplication. Every scheduled floor exceeds $a^3 - 1 > 173$, so properness holds already at the displayed seeds. □

The scalar -4 has order 43 modulo 173. Full-state order therefore retains information unnecessary for divisibility of its trace. The much larger previously available return period is not invalidated; this certificate simply supplies a shorter one.

The order of 7 modulo 173 is 172. The cubic is irreducible at $a = 7^m \pmod{173}$ precisely for the following 60 residue classes:

$$\begin{gathered} 2,3,4,5,6,11,12,13,16,17,18,20,24,27,28,34,36,39,46,50, \\ 53,54,57,58,64,65,67,70,71,74,81,83,87,90,94,101,102,103, \\ 104,106,107,110,113,115,121,124,126,129,136,137,138,140,142, \\ 144,152,153,154,160,166,170 \pmod{172}. \end{gathered} \tag{246}$$

For every $m \geq 2$ in these classes, both clocks have infinitely many composite floors. The phases can depend on the coefficient cell. The common phases in Corollary 18.5 extend specifically to $m \equiv 12 \pmod{172}$, not to all 60 cells. The list is a complete finite residue calculation, not a sample extrapolated to all m. Since 7 has order 172, it is a nonsquare modulo 173. Exactly 36 listed classes are even:

$$\begin{gathered} 2,4,6,12,16,18,20,24,28,34,36,46,50,54,58,64,70,74, \\ 90,94,102,104,106,110,124,126,136,138,140,142,144,152, \\ 154,160,166,170 \pmod{172}. \end{gathered}$$

By Theorem 18.4, each of these classes has two same-prime simultaneous phases modulo 30102. All indicated floors are already proper multiples, since $7^m \geq 49$ for $m \geq 2$. The other 24 irreducible classes have no common zero phase for this prime, but still have 174 zero phases on each clock. This does not rule out simultaneous composite floors using other primes.

18.4 Exact criteria for an arbitrary fixed modulus

The positive results above are sufficient conditions. We now give a complete local decision criterion for a fixed divisor, distinguishing transient states, recurrent states, and trace outputs.

18.4.1 Nilpotent and unit components

Lemma 18.6. *For R_D and x as above, there are an exact ordinary-power preperiod $\nu \geq 0$ and an exact eventual period $O \geq 1$ such that $x^{n+O} = x^n$ for $n \geq \nu$. They arise from a decomposition into a nilpotent component and a unit component, and satisfy*

$$\nu \leq 3 \max_{p^e \| D} e, \qquad O \leq D^3.$$

Use $\nu = 0$ if the nilpotent component is absent, and $O = 1$ if the unit component is the zero ring.

Proof. First work modulo p^e. The descending images $R \supseteq xR \supseteq x^2R \supseteq \cdots$ are additive p-groups inside a group of order p^{3e}. Each strict inclusion reduces cardinality by at least p. Once two consecutive images agree, all subsequent ones agree. Stabilization therefore occurs by $N = 3e$. Let $I = x^N R$ be a stable image. Multiplication by x is bijective on I, so multiplication by x^N is also bijective there. For every $r \in R$, choose $i \in I$ with $x^N i = x^N r$. This proves

$$R = \ker(x^N \cdot) \oplus I.$$

The intersection is zero by bijectivity on I. Both summands are ideals, and their products vanish, since elements of I have the form $x^N z$. Splitting 1 gives complementary idempotents

and thus a direct product of rings. On the first factor x is nilpotent; on the second it is a unit. Combine these decompositions by CRT. Take ν to be the nilpotence index on the combined nilpotent factor and O the order on the unit factor. The bounds follow from $N \le 3 \max e$ and the total ring size D^3. These are exact preperiod and period: on a nilpotent factor an equality $x^{n+h} = x^n$ implies $x^n = 0$, because $1 - x^h$ is a unit; on the unit factor equality is equivalent to $O \mid h$. □

Theorem 18.7 (Complete eventual zero criterion)**.** *Write $O = 3^s M$, with $3 \nmid M$, and use $\mathrm{ord}_1(3) = 1$. The exact eventual state periods in the original clock index are*

$$L_+ = \mathrm{ord}_M(3), \qquad L_- = \mathrm{ord}_{M/\gcd(M,2)}(3). \tag{247}$$

A clock state at $k \ge 1$ is recurrent exactly when $n_\pm(k) \ge \nu$ and $k \ge s$. Choose any $K \ge 1$ satisfying both conditions, and set, for its clock,

$$e_j = \nu + \big((n_\pm(K + j) - \nu) \bmod O\big), \qquad Z_D = \{0 \le j < L_\pm : \tau(x^{e_j}) = 0 \pmod D\}.$$

Then D divides infinitely many scheduled traces if and only if $Z_D \neq \varnothing$. More precisely, after K the zero indices are exactly the union of the classes $k \equiv K + j \pmod{L_\pm}$ for $j \in Z_D$.

Proof. Once the nilpotent component has vanished, equality after an index increment h is equivalent to divisibility by O of, respectively,

$$3^k(3^h - 1), \qquad 2 \cdot 3^k(3^h - 1).$$

Their 3-adic valuation is exactly k, since $3 \nmid 3^h - 1$. Thus no unit-state return is possible for $k < s$. For $k \ge s$, the remaining divisibility is exactly $M \mid 3^h - 1$ in the classical case and $M/\gcd(M, 2) \mid 3^h - 1$ in the negative case. This proves (247) and the recurrent-state threshold, using the last observation of Lemma 18.6 for the nilpotent part. Reduction of every exponent to the interval $[\nu, \nu + O)$ gives the stated states x^{e_j}. A finite eventual cycle has infinitely many zero outputs exactly when it has one. □

The trace-output period can be smaller than the state period. A transient zero need not recur, and two isolated zero traces do not by themselves establish an infinite return. The theorem checks the whole eventual cycle. Powers such as 3^{K+j} are evaluated modulo O; no enormous integer trace is required. By Lemma 18.1, an infinite zero set gives infinitely many proper composite floors for this fixed divisor. Finitely many zero indices do not.

For odd coefficients all positive traces are odd by (240), so useful divisors are odd and x is a unit. For even a, all positive traces are even, and the scheduled floors are already composite. The nonunit discussion ensures completeness but is not a missing obstacle in the odd-coefficient case.

18.4.2 The sharper projective criterion and uniform exponents

For odd D, write $Q_D = 3^q N$, with $3 \nmid N$. The same cyclic-group proof applied modulo scalar units gives exact projective preperiod q and periods

$$J_+ = \mathrm{ord}_N(3), \qquad J_- = \mathrm{ord}_{N/\gcd(N,2)}(3). \tag{248}$$

For $K \ge \max(1, q)$, replace Z_D by

$$W_D = \{0 \le j < J_\pm : \tau(x^{n_\pm(K+j) \bmod Q_D}) = 0 \pmod D\}.$$

It gives the same necessary-and-sufficient infinite-divisibility criterion and exactly all eventual zero phases. This can be much shorter than a full-state computation. The zero pattern may still have a smaller period than its projective state. For composite D, nonempty zero sets at its prime-power factors are not enough: their index phases must be simultaneously compatible. The generalized CRT requires pairwise agreement modulo the gcds of the phase periods, after the finite thresholds have passed.

Lemma 18.8 (A uniform unit exponent)**.** *For $p \geq 3$ and $e \geq 1$, every unit in a cubic algebra $(\mathbb{Z}/p^e\mathbb{Z})[X]/(f)$, with f monic, is killed by*

$$E_{p,e} = p^e(p^2-1)(p^2+p+1).$$

For $p \neq 3$,

$$v_3(E_{p,e}) = \begin{cases} v_3(p+1), & p \equiv 2 \pmod 3, \\ v_3(p-1)+1, & p \equiv 1 \pmod 3, \end{cases}$$

and $v_3(E_{3,e}) = e$.

Proof. Modulo p, the reduced algebra is a product of fields whose degrees sum to at most 3. Their unit orders divide $H = (p^2-1)(p^2+p+1)$. Thus a unit u has $u^H = 1+z$, with z in the nilradical. That ideal has cube zero in a cubic algebra, so $(1+z)^p = 1$ for $p \geq 3$. On lifting, the remaining factor $1+pY$ is killed by p^{e-1}, by repeated binomial expansion. This proves the exponent, without separability. The valuations follow by reducing the factors modulo 3; if $p \equiv 1 \pmod 3$, $p^2+p+1 = 3(1+3t+3t^2)$ for $p = 1+3t$. □

A seed divisible by p^e at $k_0 \geq v_3(E_{p,e})$ therefore has a full-state return. This is sufficient, not necessary. It explains the safe seeds $k_0 \geq 1$ for $p \equiv 2, 5 \pmod 9$ and $k_0 \geq 2$ for $p \equiv 4, 7 \pmod 9$, while the projective theorem can return earlier. For a fixed odd composite divisor, take the lcm of these exponents over its prime-power factors.

18.4.3 Exact coefficient-cell encoding

For a fixed $p \geq 5$, one can compute all recurrent-witness coefficient cells without factoring a trace. Work in

$$B = \mathbb{F}_p[A]/(A^p - A), \qquad R_B = B[X]/(X^3 - AX^2 - 4X + 4).$$

The algebra B is the product of its p coefficient cells. Using the uniform exponent, start at $K \geq \max(1, v_3(E_{p,1}))$ and take a valid eventual period L for the chosen clock. Let $F_j(A)$ be the trace function at index $K+j$, represented modulo $A^p - A$, for $0 \leq j < L$. Then

$$H_{p,\pm}(A) = \gcd\left(A^p - A, \prod_{j=0}^{L-1} F_j(A)\right) \tag{249}$$

has a residue a as a root exactly when p divides infinitely many scheduled traces there. Indeed, evaluation in each field cell makes the product zero exactly when some recurrent phase is zero. The separate gcds with F_j record its phases. Reducing after every operation keeps representatives of degree below p. For $a = 7^m$ and $p \neq 7$, evaluation on the finite subgroup generated by 7 gives a complete exponent-residue table. Backwards iteration on the prime-to-3 cyclic part is also valid, using the inverse of 3 modulo its order, but does not place an arbitrary guessed zero state on the desired orbit. Theorem 18.3 supplies that coverage in its special setting. None of these finite local decisions proves the existence of a useful modulus for every coefficient.

18.5 Finite horizons and unbounded witness primes

The next obstruction permits the witness prime to be arbitrarily large. It is a standard resultant-specialization and congruence-avoidance argument, applied to these particular trace polynomials. Put $C_k(A) = P_{3^k}(A)$, including $C_0(A) = A$.

Lemma 18.9. *For $k \geq 1$, C_k is Eisenstein at 3. Hence the polynomials $C_0, C_1, \ldots$ are pairwise relatively prime over $\mathbb{Q}[A]$.*

Proof. For the formal roots α_i, put $Q_n(A) = \sum_{i<j}(\alpha_i\alpha_j)^n \in \mathbb{Z}[A]$. Newton's cubing identity is

$$P_{3n} = P_n^3 - 3P_nQ_n + 3(-4)^n.$$

It gives $C_k(A) \equiv A^{3^k} \pmod 3$. At $A = 0$, $C_0(0) = 0$ and every $C_k(0)$ is divisible by 3; the same identity gives $C_{k+1}(0) \equiv 6 \pmod 9$. Together with monicity this proves Eisenstein's criterion. The irreducibles C_k have distinct degrees 3^k (including $C_0 = A$), so they are pairwise coprime. □

Theorem 18.10 (Pairwise coprime finite prefixes)**.** *For every $N \geq 1$ there is an integer L_N divisible by 12 such that, for every positive multiple m of L_N,*

$$\gcd(C_i(7^m), C_j(7^m)) = 1 \qquad (0 \leq i < j \leq N).$$

There are also infinitely many prime coefficients $a > 2^{41}$ with this same pairwise-coprimality property. These prime coefficients are a separate family, not powers of 7.

Proof. Let S_N be the finite set of prime divisors of

$$R_N = \prod_{0\leq i<j\leq N} |\operatorname{Res}_A(C_i, C_j)|.$$

Every factor is a nonzero integer by Lemma 18.9. If a prime divides both specialized values $C_i(a), C_j(a)$, their reductions share the root $a \pmod p$, so their Sylvester determinant vanishes modulo p. Thus that prime belongs to S_N. At $A = 1$ there is the integer factorization

$$X^3 - X^2 - 4X + 4 = (X - 1)(X - 2)(X + 2), \qquad P_n(1) = 1 \quad (n \text{ positive odd}).$$

Choose

$$L_N = \operatorname{lcm}\big(12, \{\operatorname{ord}_p(7) : p \in S_N,\ p \neq 7\}\big).$$

For every positive multiple m of L_N, all selected traces are $1 \pmod p$ for $p \in S_N \setminus \{7\}$. Prime 7 requires care: it divides $C_0(7^m)$, but no later classical trace. In the coefficient-0 quotient modulo 7,

$$u_1 = x^3 = (3, 4, 0), \quad u_2 = u_1^3 = (2, 0, 4), \quad u_2^3 = u_1.$$

Their traces under $\tau(c) = 3c_0 + 8c_2$ are 2 and 3. Determinism gives these nonzero values for every $k \geq 1$. Thus 7 cannot divide two distinct selected traces. All possible common prime divisors have been excluded.

For the prime-coefficient assertion set $Q_N = 2\prod_{p\in S_N} p$. Every $a \equiv 1 \pmod{Q_N}$ avoids all possible common primes. There are infinitely many primes in this particular residue class, by the elementary cyclotomic lemma proved next. Take them larger than 2^{41} and apply Lemma 18.1. □

Lemma 18.11 (Prime supply in the residue class one)**.** *For every integer $Q \geq 2$, infinitely many primes satisfy $q \equiv 1 \pmod Q$.*

Proof. Given a finite set of primes S, choose a sufficiently large positive integer M divisible by $Q\prod_{p\in S}p$. The monic cyclotomic polynomial Φ_Q has positive degree and constant term 1, so $\Phi_Q(M) > 1$. A prime divisor q of this value divides neither M nor Q, and is not in S, since the value is 1 modulo every prime dividing M. Over $\mathbb{F}_q$, the factorization $X^Q - 1 = \prod_{d|Q}\Phi_d(X)$ is squarefree because $q \nmid Q$. The residue M is a root of Φ_Q, so its multiplicative order is exactly Q. A smaller order $d \mid Q$ would make it a root of another factor of this squarefree product. Thus $Q \mid q-1$. Since the finite forbidden set was arbitrary, the supply is infinite. □

This standard argument is only for residue class 1; it is not a proof of the full theorem for arbitrary reduced progressions.

Corollary 18.12 (No simultaneous uniform seed and gap bound)**.** *Given $B \geq 0$ and $H \geq 1$, infinitely many positive m divisible by 12 admit no prime p, seed $0 \leq k \leq B$, and gap $1 \leq h \leq H$ satisfying $p \mid C_k(7^m)$ and $p \mid C_{k+h}(7^m)$.*

Proof. Apply Theorem 18.10 with $N = B + H$. □

There is no bound on p here. Even an adaptive recurring-prime certificate cannot cover every such coefficient while bounding both its first zero index and its return gap uniformly. This does not bound all compositeness algorithms. Pairwise coprime values may all be composite with unrelated factors. Two trace zeros are only a necessary prerequisite for a two-hit recurring certificate, not a sufficient state-return proof.

For a reproducible finite illustration, the six absolute resultants at $N = 3$ factor as

(i,j)	$\lvert\operatorname{Res}(C_i, C_j)\rvert$
$(0,1)$	$2^2 \cdot 3$
$(0,2)$	$2^6 \cdot 3 \cdot 13$
$(0,3)$	$2^{18} \cdot 3 \cdot 5 \cdot 53 \cdot 73$
$(1,2)$	$2^{18} \cdot 3^3$
$(1,3)$	$2^{56} \cdot 3^3 \cdot 5 \cdot 11 \cdot 727 \cdot 13789$
$(2,3)$	$2^{162} \cdot 3^9$.

They give $L_3 = 216885240$ and $Q_3 = 332776921680060$. The certificate checks determinants, factorizations, and orders, without constructing 7^{L_3}. The all-horizon result is the proof above, not an extrapolation from these six computations.

18.5.1 A divisor that occurs only in a transient

For $m \equiv 3 \pmod{16}$, $a = 7^m \equiv 3 \pmod{17}$, and

$$P_3(7^3) = 40357711 = 17 \cdot 2373983.$$

Modulo 17, $f_3 = (X-5)(X^2+2X+6)$, the quadratic discriminant 14 is a nonsquare, and the cubic discriminant is $6 \neq 0$. The classical states under cubing are

k	0	1	2	3	4	5	6	7	8	9	10
$\tau(u_k)$	3	0	12	11	5	6	12	11	5	6	12.

The exact states at $k = 2, \ldots, 9$ are, in order,

$$(10,11,4), (15,0,8), (8,5,5), (12,7,14),$$
$$(13,8,14), (16,16,0), (3,10,11), (8,11,12),$$

and $u_{10} = u_2$; also $u_1 = (13,4,3)$. Thus 17 divides the classical clock exactly at original index 1 and never again. This is a finite exact cycle proof, not a scan extrapolated past its end. It shows why merely factoring a scheduled term does not suffice. Its reducible cubic does not contradict Theorem 18.2, and this example is not confined to $12 \mid m$.

18.6 Projective orders along the actual powering orbit

We now allow the coefficients to move, but only along one prescribed cubic powering orbit. No coefficient-density statement is transferred to that orbit. Write

$$f_0(X) = X^3 - aX^2 + bX - c, \qquad c \neq 0,$$

and let f_r have root multiset $\alpha_i^{3^r}$, where the α_i are the roots of f_0, with multiplicities. For a totally real cubic Pisot number, these are still its minimal polynomials: odd powers preserve distinct real conjugates; a power cannot be rational since their absolute values differ. With $f_r = X^3 - a_r X^2 + b_r X - c_r$, Newton's identities give

$$(a, b, c) \longmapsto (a^3 - 3ab + 3c,\ b^3 - 3abc + 3c^2,\ c^3). \tag{250}$$

The second expression follows by applying the same cubing identity to the three pair products, whose sum, pair sum, and product are b, ac, c^2. In the real negative-norm notation $b = -B, c = -q$, this is

$$(a, B, q) \longmapsto (a^3 + 3aB - 3q,\ B^3 + 3aBq - 3q^2,\ q^3). \tag{251}$$

18.6.1 New polynomial discriminant divisors and nilpotent kernels

Let Δ_r be the polynomial discriminant. Factoring the pairwise differences of cubes gives

$$\Delta_{r+1} = \Delta_r J_r^2, \qquad J_r = a_r^2 b_r^2 - b_r^3 - a_r^3 c_r. \tag{252}$$

Indeed, elementary symmetric expansion identifies J_r with

$$\prod_{i<j} (\alpha_i^{2\cdot 3^r} + \alpha_i^{3^r}\alpha_j^{3^r} + \alpha_j^{2\cdot 3^r}).$$

A new polynomial-discriminant prime need not ramify in the fixed number field: powered roots can collide modulo that prime or the power basis can define a nonmaximal order. We include these collisions rather than excluding every prime dividing Δ_r.

Fix a prime $\ell \nmid c$ and put

$$R_0 = \mathbb{F}_\ell[X]/(f_0), \quad x = X \bmod f_0, \qquad R_r = \mathbb{F}_\ell[Y]/(f_r), \quad y = Y \bmod f_r.$$

There is a homomorphism

$$\psi_r : R_r \longrightarrow R_0, \qquad y \longmapsto x^{3^r}, \tag{253}$$

with image $A_r = \mathbb{F}_\ell[x^{3^r}]$. To justify it without separability, multiplication by x has characteristic polynomial f_0; its 3^rth power has characteristic polynomial f_r, including multiplicities. Cayley–Hamilton gives the required relation. The image need not equal R_0.

Lemma 18.13. *The kernel of (253) is a nilpotent ideal. The induced map $R_r^* \to A_r^*$ is surjective, with an ℓ-group kernel. The same is true after quotienting by the embedded scalar group $\mathbb{F}_\ell^*$.*

Proof. Extend scalars to a finite splitting field. If a polynomial $h(y)$ is in the kernel, evaluation at each original root gives $h(\alpha_i^{3^r}) = 0$. These powered roots are precisely the geometric root support of f_r. Thus h is divisible by every distinct irreducible factor of f_r, and its class is in the nilradical. In a cubic quotient this ideal has cube zero, so the kernel is nilpotent, even with original repeated roots or new collisions. To lift a unit from A_r, lift it and its inverse arbitrarily; their product is $1 + n$ for a kernel element n. This is invertible by a finite geometric series, proving surjectivity. The kernel on units is $1 + \ker \psi_r$, an ℓ-group since the ideal has cardinality a power of ℓ. If a unit maps to a scalar, dividing by that scalar puts it in this same kernel. This proves the projective assertion as well. □

Let $Q_{\ell,r}$ be the order of y in $R_r^*/\mathbb{F}_\ell^*$, with x for $r = 0$.

Theorem 18.14 (Exact projective transfer)**.** *For every $r \geq 0$, the quotient*

$$\frac{Q_{\ell,r}}{Q_{\ell,0}/\gcd(Q_{\ell,0}, 3^r)} \tag{254}$$

is a nonnegative integral power of ℓ. It is at most ℓ if $\ell \geq 3$ and at most 4 if $\ell = 2$. In particular, for $\ell \neq 3$,

$$v_3(Q_{\ell,r}) = \max\{v_3(Q_{\ell,0}) - r, 0\}. \tag{255}$$

Moreover, for every polynomial h,

$$\tau_r(h(y)) = \tau_0(h(x^{3^r})), \tag{256}$$

where both traces are regular multiplication traces in their cubic algebras.

Proof. Equality of an element of A_r to a scalar is the same equality inside R_0. The projective order of x^{3^r} in A_r is therefore $Q_{\ell,0}/\gcd(Q_{\ell,0}, 3^r)$. Restrict the surjection in Lemma 18.13 to the cyclic subgroup generated by $[y]$. Its order divided by its image order is the order of a subgroup of an ℓ-group, proving (254). If $n^3 = 0$, then $(1+n)^\ell = 1$ for $\ell \geq 3$, and $(1+n)^4 = 1$ for $\ell = 2$. These bounds apply to that cyclic kernel. Taking 3-adic valuations proves (255). Finally both sides of (256) sum $h(\alpha_i^{3^r})$ over the same three-root multiset, including multiplicities. No general invariance of regular trace under arbitrary quotient maps is asserted. □

Over $\mathbb{F}_7$, $f_0 = X^3 - 1$ has $Q_{7,0} = 3$, whereas after cubing $f_1 = (Y-1)^3$ has $Q_{7,1} = 7$. The image of y is 1, but its nilpotent thickening contributes the extra factor 7. Over $\mathbb{F}_3$ the same construction has $Q_{3,0} = Q_{3,1} = 3$, so (255) fails in characteristic 3. The exclusion is essential.

18.6.2 The original seed index does not reset

Suppose $\ell \neq 3$ and a classical seed in the powered algebra has trace zero at relative index $s \geq 0$. Trace transfer says

$$\tau_r(y^{3^s}) = \mathrm{Tr}(\beta^{3^{r+s}}) \pmod{\ell}.$$

A scalar return after some positive index increment h exists if and only if

$$Q_{\ell,r} \mid 3^s(3^h - 1) \text{ for some } h \geq 1 \quad \Longleftrightarrow \quad v_3(Q_{\ell,r}) \leq s.$$

The reverse implication uses an order of 3 modulo the prime-to-3 part of $Q_{\ell,r}$. The transfer theorem gives the exact equivalence

$$v_3(Q_{\ell,r}) \leq s \quad \Longleftrightarrow \quad v_3(Q_{\ell,0}) \leq r + s. \tag{257}$$

Thus the powered-state scalar certificate is the original certificate at its original total index, not a newly reset threshold. Actual periods may acquire prime-to-3 factors, as the $\mathbb{F}_7$ example shows. Non-scalar trace-zero returns can also exist; the displayed criterion characterizes scalar returns only. More general finite-state zero sequences still transfer exactly to the same original classical tail. For the negative clock, powering gives exponent $3^r(2 \cdot 3^s - 1)$, not in general $2 \cdot 3^{r+s} - 1$; no identical negative-clock reindexing is claimed.

18.6.3 First collisions and a factorization-free gcd test

Assume $\Delta_0 \neq 0$. Let $\ell \nmid 6c\Delta_0$ be prime, and let $r \geq 1$ be its first collision stage, so $\ell \mid \Delta_r$ but $\ell \nmid \Delta_{r-1}$. Some ratio of two distinct original reduced roots then has exact order 3^r. The original reduction is separable.

Proposition 18.15. *The original splitting type imposes the following restrictions:*

(i) *In type* $1+1+1$, $\ell \equiv 1 \pmod{3^r}$.

(ii) *In type* $1+2$, $\ell \equiv -1 \pmod{3^r}$ *and* $v_3(Q_{\ell,0}) = r$.

(iii) *In type* 3, *necessarily* $r = 1$ *and* $\ell \equiv 1 \pmod 3$*; no classical trace* T_{3^j} *with* $j \geq 1$ *vanishes modulo* ℓ.

Proof. In the split case the ratio belongs to $\mathbb{F}_\ell^*$, proving (i). In the irreducible case its norm is 1, so its order divides $\ell^2+\ell+1$, whose 3-part is at most 3. A collision is therefore possible only as in (iii). Frobenius cyclically permutes all three roots, so a collision of two powered roots makes all three coincide. Every subsequent classical trace is $3t$ with $t \neq 0$, hence is nonzero. Since the collision is at stage 1, this proves the assertion for every $j \geq 1$.

For mixed type, let $\alpha \in \mathbb{F}_\ell^*$ be the linear root and λ, λ^ℓ the quadratic pair. If the colliding ratio is λ/λ^ℓ, its order divides $\ell+1$. If it is λ/α, a pure 3-power-order element outside $\mathbb{F}_\ell$ is possible only for $\ell \equiv -1 \pmod 3$: when $\ell \equiv 1 \pmod 3$, the entire 3-primary subgroup of $\mathbb{F}_{\ell^2}^*$ lies in $\mathbb{F}_\ell^*$, a contradiction. For $\ell \equiv -1 \pmod 3$, the 3-part lies in $\ell+1$; in either case $3^r \mid \ell+1$. The projective group of $\mathbb{F}_\ell \times \mathbb{F}_{\ell^2}$ identifies with $\mathbb{F}_{\ell^2}^*$ by dividing the second component by the first. Thus $Q_{\ell,0} = \operatorname{ord}(y)$ for $y = \lambda/\alpha$. A cross-pair collision has $\operatorname{ord}(y) = 3^r$. A quadratic-pair collision has $\operatorname{ord}(y^{1-\ell}) = 3^r$; since $3 \nmid 1-\ell$, this also gives $v_3(\operatorname{ord}(y)) = r$. □

Corollary 18.16 (An adaptive gcd certificate)**.** *If for some* $j \geq 0$ *a prime* ℓ *satisfies*

$$\ell \nmid 6c\Delta_0, \quad \ell \equiv 2 \pmod 3, \quad \ell \mid \gcd(T_{3^j}, \Delta_j),$$

then it divides infinitely many classical traces. For a growing Pisot trace sequence whose late terms equal its floors, this gives infinitely many composite floors.

Proof. Its first collision stage $r \leq j$ must have mixed type by Proposition 18.15. Hence $v_3(Q_{\ell,0}) = r \leq j$, which is exactly the scalar-return threshold (257) in the original algebra. The seed trace is zero, so the returns are zero and eventually exceed the fixed prime. □

This admits a factorization-free sufficient test. Starting with the positive integer $\gcd(T_{3^j}, \Delta_j)$, repeatedly divide by its gcd with $|6c\Delta_0|$ until it is coprime to that integer. Call the remaining integer H_j. If $H_j \equiv 2 \pmod 3$, then some prime factor of H_j is $2 \pmod 3$ and the corollary applies. This procedure needs gcd computations, not a factorization of the large trace. It is not necessary: an even number of such prime factors can give $H_j \equiv 1 \pmod 3$. If a late classical trace sequence consists of increasing primes $p_j \equiv -1 \pmod 3$, the same argument implies

$$\gcd(p_j, \Delta_j) = 1 \qquad \text{for every sufficiently large } j.$$

Indeed, these prime values eventually avoid the fixed factors of $6c\Delta_0$; a violation would force the same prime to divide infinitely many strictly larger prime values. Neither this restriction nor the gcd test proves that a witness ever exists for a remaining candidate.

18.7 The invariant boundary and finite witness sets

The surface $a = 1, B = q$ is invariant under (251):

$$(1, q, q) \longmapsto (1, q^3, q^3).$$

Its polynomial factors as $(X - 1)(X^2 - q)$, so every positive odd trace equals 1 as an integer identity, including when reduced factors happen to coincide.

Proposition 18.17. *For any fixed finite prime set $\mathcal{P}$, infinitely many prime coefficients $a > 2^{41}$ give cubics $X^3 - aX^2 - 4X + 4$ satisfying (241) whose actual powering coefficient triples obey*

$$(a_r, B_r, q_r) \equiv (1, 4^{3^r}, 4^{3^r}) \pmod{\ell} \qquad (\ell \in \mathcal{P},\ r \geq 0).$$

No prime in $\mathcal{P}$ divides any positive odd trace at any powered stage.

Proof. Put $L = \operatorname{lcm}(24, \prod_{\ell \in \mathcal{P}} \ell)$. Lemma 18.11 supplies infinitely many prime $a \equiv 1 \pmod{L}$; take them above the stated threshold. Lemma 18.1 supplies the irreducible real Pisot roots, prime initial floor, and strict shape. Induction under the invariant coefficient map proves the displayed congruence. At each stage the odd traces are therefore 1 (mod ℓ) by the factorization. □

Letting the lower coefficients change along a prescribed orbit therefore does not make a fixed finite divisor list universal. The quantifiers are "for every finite list, infinitely many prime-leading parameters escape it", not the existence of one parameter escaping all primes. Nothing here says that the later traces are prime. The scalar transfer theorem also does not eliminate any candidate by itself. A useful remaining task is to find, for a given candidate, an actual $\ell \nmid 3c$ and index j with

$$\ell \mid T_{3^j}, \qquad v_3(Q_{\ell,0}) \leq j,$$

or a recurrent trace-zero state not explained by scalar return. An ambient coefficient-density theorem does not supply this statement on one prescribed moving triple.

18.8 Exact verification and scope

All numerical certificates are in `certificates/`; the packaged programs use only Python's standard library.

1. `fixed_modulus_projective_verify.py` checks full unit and nonunit state criteria on finite grids, projective criteria, coefficient witness polynomials, auxiliary primalities and orders, and the complete $p = 173$ phase and coefficient-cell tables.
2. `positive_projective_independent_verify.py` independently uses companion matrices for the $p = 173$ certificate, including scalar returns, all 174 phases on each clock, the two common phases, and all 60 irreducible coefficient cells.
3. `simultaneous_projective_quadratic_verify.py` checks (245) coefficientwise, reconstructs the quadratic form in irreducible cells, and compares projective zero counts with complete small-prime clock scans.
4. `bounded_witness_verify.py` computes the six $N = 3$ Sylvester determinants by exact fraction-free elimination, verifies their factorizations and orders, and checks the 7-cycle and the transient-only 17-orbit.

5. `powering_orbit_portable_verify.py` uses exact sparse polynomial arithmetic for the coefficient and discriminant identities, and finite quotient algebras for projective transfer, repeated-factor examples, matched traces, first collisions, and the invariant boundary. It is a standard-library transcription of the symbolic checks, not a probabilistic replacement.

6. `powering_orbit_independent_verify.py` independently checks powering transfer and first collisions with companion matrices, including a late mixed example whose return period changes.

Triple multiplication uses the defining monic cubic to reduce terms of degree 4 and 3, in that order. Binary powering then evaluates huge exponents in logarithmically many ring multiplications. The independent matrix implementation uses ordinary 3 by 3 modular products instead. Primality assertions use exhaustive trial division; order certificates check the exponent and its quotients by every distinct prime factor. No probable-prime screen is accepted as a proof. Return identities, not a finite numerical extrapolation, prove the infinite progressions.

From this directory run

```
python3 certificates/fixed_modulus_projective_verify.py
python3 certificates/positive_projective_independent_verify.py
python3 certificates/simultaneous_projective_quadratic_verify.py
python3 certificates/bounded_witness_verify.py
python3 certificates/powering_orbit_portable_verify.py
python3 certificates/powering_orbit_independent_verify.py
sha256sum -c SHA256SUMS.txt
```

The logs and JSON files record the finite calculations. The universal local criteria, projective theorems, resultant obstruction, and powering transfer rest on the proofs in this paper, not those grids.

The positive mechanism does not require factoring a scheduled integer trace once a suitable auxiliary prime and irreducible cell have been found. It does not prove an infinite supply of such primes, nor that every coefficient meets one. The bounded-horizon obstruction leaves unbounded adaptive indices and periods entirely possible. The same-prime simultaneous count is not a statement about all actual composite indices. No all-m closure, prime pattern on a prescribed orbit, global minimum result, or Mills transcendence theorem is proved. The finite-field trace geometry, finite-module decomposition, resultant avoidance, cyclotomic prime supply, and Cayley–Hamilton methods are standard tools; their use here is not a historical-priority claim.

19 Coset coverage and output prime order bounds

Coset coverage removes an unnecessary primitive-root hypothesis in several explicit families. The subsequent order bounds concern actual hypothetical output primes and must not be inferred from arbitrary local models.

19.1 Two clocks and the exact floor bridge

For an integer $a > 8$, define

$$f_a(X) = X^3 - aX^2 - 4X + 4, \qquad n_+(k) = 3^k, \quad n_-(k) = 2 \cdot 3^k - 1 \quad (k \geq 1).$$

The signs label the classical and negative clocks; both exponents are positive and odd. Their Newton power sums are the integer polynomials

$$T_0 = 3, \quad T_1 = a, \quad T_2 = a^2 + 8, \qquad T_n = aT_{n-1} + 4T_{n-2} - 4T_{n-3} \quad (n \geq 3). \tag{258}$$

We write $T_n(a)$ when the coefficient must be displayed.

Lemma 19.1. *The roots of f_a are $-w, v, \beta$ with $0 < v < w < 1$ and $a < \beta < a + 1$. The polynomial is irreducible over $\mathbb{Q}$. For every positive odd n,*

$$\lfloor \beta^n \rfloor = T_n(a), \qquad 0 < \beta^n - T_n(a) = w^n - v^n < 1.$$

These floors strictly increase with the odd exponent and tend to infinity.

Proof. The values of f_a at $-1, 0, 1, a, a + 1$ are respectively $7 - a, 4, 1 - a, 4(1 - a), (a - 1)^2$. The signs place a root in each of $(-1, 0), (0, 1), (a, a + 1)$. There are no integer roots, so the rational-root test proves irreducibility. The sum of the roots gives $w - v = \beta - a > 0$. Taking odd power sums proves the floor identity. Since $\beta > 8$, one has $\beta^{n+2} - \beta^n > 1$, proving strict increase. □

A recurring congruence $p \mid T_{n_\pm(k)}(a)$ therefore gives infinitely many composite *actual floors*, once the trace exceeds p. No assumption of prime initial floors is needed for that implication. The thin family $a = 7^m$, $m \geq 2$, is used below for explicit tests; its leading coefficient is not prime, and we do not present it as an all-prime Mills candidate.

19.2 Complete coset coverage beyond a primitive-root hypothesis

19.2.1 Projective trace-zero classes

Let $p \geq 5$ be prime and put $S = p^2 + p + 1$. The quotient

$$\mathbb{F}_{p^3}^* / \mathbb{F}_p^*$$

is cyclic of order S. Choose $x \in \mathbb{F}_{p^3}^*$ whose projective class has order S, and define

$$D = \{n \in \mathbb{Z}/S\mathbb{Z} : \mathrm{Tr}_{\mathbb{F}_{p^3}/\mathbb{F}_p}(x^n) = 0\}.$$

A scalar multiple of x^n has the same zero/nonzero trace status. The trace is a nonzero linear map, since $\mathrm{Tr}(1) = 3 \neq 0$. Its kernel is a plane, so

$$|D| = (p^2 - 1)/(p - 1) = p + 1, \qquad 0 \notin D, \qquad pD = D. \tag{259}$$

Every nonscalar γ is cx^u for some $c \in \mathbb{F}_p^*$ and $u \neq 0$ in $\mathbb{Z}/S\mathbb{Z}$, and

$$\mathrm{Tr}(\gamma^n) = 0 \quad \Longleftrightarrow \quad un \in D. \tag{260}$$

Because $[\mathbb{F}_{p^3} : \mathbb{F}_p] = 3$ is prime, nonscalar elements are exactly those having irreducible cubic minimal polynomial. These elementary projective trace facts are the standard Singer construction; see [33, §1.2 and Corollary 1.2].

Suppose now that S is prime. Necessarily $p \equiv 2 \pmod 3$, and every nonscalar element has projective order S. Set

$$G = (\mathbb{Z}/S\mathbb{Z})^*, \quad H = \langle 3 \rangle \subseteq G, \quad h = \mathrm{ord}_S(3), \quad A_- = (2H - 1) \setminus \{0\}.$$

For subsets of G, products and inverses below are taken in G.

Theorem 19.2 (Exact coverage criterion)**.** *Define $W_+ = DH^{-1}$ and $W_- = DA_-^{-1}$. For $\gamma = cx^u$ with $u \in G$:*

$$\exists k \geq 1 : \mathrm{Tr}(\gamma^{3^k}) = 0 \quad \Longleftrightarrow \quad u \in W_+,$$
$$\exists k \geq 1 : \mathrm{Tr}(\gamma^{2 \cdot 3^k - 1}) = 0 \quad \Longleftrightarrow \quad u \in W_-.$$

Every hit recurs with index gap h. Thus universal classical coverage is exactly the condition that D meet every multiplicative H-coset; universal negative-clock coverage is exactly $W_- = G$.

Proof. The phases $k = 1, \dots, h$ run through all $3^k \in H$. Apply (260). An exponent zero modulo S has scalar nonzero trace, so it is correctly omitted from A_-. Replacing k by $k + h$ changes each exponent by a multiple of S, hence changes its field value only by a nonzero scalar. This preserves trace zero. Finally, $DH^{-1} = G$ is equivalent to meeting every H-coset. □

Changing the chosen projective generator rescales D and relabels u; coverage is unaffected. Importantly, W_- is a separate finite set: it is not determined merely by the multiplicative coset counts for D.

19.2.2 Exact certificates at 17, 59, and 89

Theorem 19.3 (Three complete finite certificates)**.** *At $p = 17$, every irreducible cubic element has a recurring classical trace zero. At each of $p = 59$ and $p = 89$, every irreducible cubic element has recurring trace zeros on* both clocks separately*. In all three cases* 3 *is not primitive modulo $S = p^2 + p + 1$. At $p = 17$, exactly* 39 *of the* 306 *projective generators have no negative-clock zero.*

Exact finite verification and its algebraic reduction. Use a root of f_{a_0} as reference element, with the data in Table 17. Trial division proves that both p and S are prime. Testing the p possible roots proves each reference cubic irreducible. Exact quotient-ring powering gives $x^S = -4$, while x is nonscalar. Thus it has projective order S.

Enumerating x^n for $0 \le n < S$ gives all of D, not a sample. Enumerating H gives all coset counts and all elements of $W_\pm$. The indicated cardinalities are exact; Theorem 19.2 then proves every assertion. The portable verifier `frobenius_coset_coverage_verify.py` contains these complete loops; its JSON retains D, all cosets, and every negative miss at 17. For example, the class $u = 3$ is one of those misses. □

p	S	a_0	h	H-coset counts	$\lvert W_+\rvert$	$\lvert W_-\rvert$
17	307	5	34	2, 2, 2, 2, 2, 2, 2, 2, 2	306	267
59	3541	0	1770	33, 27	3540	3540
89	8011	0	2670	39, 24, 27	8010	8010

Table 17: Complete coverage, despite a proper subgroup $H = \langle 3 \rangle$.

For precision, coset representatives in increasing least-unused order are

$$\begin{array}{l|l} p = 17 & 1, 2, 4, 5, 7, 10, 14, 20, 28 \\ p = 59 & 1, 2 \\ p = 89 & 1, 7, 13. \end{array}$$

Here is the finite algorithm underlying the table; all operations in its first step are in the displayed cubic quotient:

```
D = {n in range(S) : trace(x**n) == 0}
H = {3**k mod S : 0 <= k < ord_S(3)}
Aminus = {2*t-1 mod S : t in H} - {0}
Wplus  = {d * inverse(t) mod S : d in D, t in H}
Wminus = {d * inverse(t) mod S : d in D, t in Aminus}
```

Quotient multiplication is reduced using $x^3 = a_0x^2 + 4x - 4$, and

$$\tau(c_0 + c_1x + c_2x^2) = 3c_0 + a_0c_1 + (a_0^2 + 8)c_2 \pmod p. \tag{261}$$

Thus the certificate requires neither an enormous integer trace nor its factorization. Universal assertions here exhaust *all nonidentity projective classes*, not only coefficients in f_a.

Corollary 19.4. *Let $a > 8$ be an integer. If f_a is irreducible modulo 59 or 89, then both sequences $\lfloor \beta^{n_+(k)} \rfloor$ and $\lfloor \beta^{n_-(k)} \rfloor$ contain infinitely many composite terms. If it is irreducible modulo 17, the same holds for the classical clock.*

Proof. Apply Theorem 19.3 to the residue class of X, then Lemma 19.1. A fixed divisor is proper for all sufficiently large indices in its recurring phase. □

The four irreducible cells of f_a modulo 17 are $a \in \{5, 6, 9, 14\}$. A separate complete check shows that all four also have negative-clock hits. This restricted-family fact does not remove the 39 generic projective misses in Theorem 19.3.

The complete irreducible coefficient cells of the thin family are

$$\begin{aligned} p = 17: &\quad m \equiv 6, 11, 13, 15 \pmod{16}, \\ p = 59: &\quad m \equiv 4, 5, 6, 12, 17, 18, 22, 23, 24, 26 \pmod{29}, \\ p = 89: &\quad m \equiv 6, 11, 12, 15, 16, 17, 19, 22, 25, 30, 31, 32, \\ &\qquad\qquad 36, 42, 46, 52, 55, 58, 59, 60, 62, 72, 74, 80, 81 \pmod{88}. \end{aligned}$$

The moduli are the exact respective orders of 7. For every $m \geq 2$ in one of these cells, both clocks have infinitely many composite floors separately. At 17 this uses the additional fixed-family checks, rather than the false generic negative-coverage assertion.

Corollary 19.5 (An explicit simultaneous powering class)**.** *For every $m \geq 2$ with $m \equiv 60 \pmod{88}$, both floors associated with $a = 7^m$ are composite at every*

$$k = 49 + 2670t \quad \text{and every } k = 179 + 2670t, \qquad t \in \mathbb{Z}_{\geq 0}.$$

Proof. The order of 7 modulo 89 is 88, so $a \equiv 73$ throughout this class. The reduction f_{73} is irreducible. At $k = 49$, direct quotient powering gives the respective states

$$x^{3^{49}} = (22, 48, 33), \qquad x^{2 \cdot 3^{49} - 1} = (20, 35, 41).$$

Both have trace zero by (261). The complete common zero-phase set is $\{49, 179\}$ modulo 2670, verified by enumerating one full projective period; $x^{8011} = -4$ and $\mathrm{ord}_{8011}(3) = 2670$ prove recurrence. Every positive m in the class is at least 60, so $a > 89$. Since $n_\pm(k) \geq 3$, the floors are already larger than $a > 89$ at both displayed seeds. Hence the divisor 89 is proper from $t = 0$ onward. □

The coefficient $m = 60$ illustrates a concrete gain over two earlier finite certificates in this research sequence: it is divisible by 12, and it is not an irreducible cell for the earlier auxiliary prime 173. Indeed $7^{60} \equiv 22 \pmod{173}$ and $f_{22}(161) \equiv 0 \pmod{173}$. This comparison concerns only those certificates, not a historical claim that this coefficient was previously unknown. Corollary 19.5 is a separately verified simultaneous class, not a theorem that every irreducible coefficient at 89 has a common zero phase.

19.2.3 Short certificates for $a = 7^{12}$; no common-phase inference

At $p = 59$, the coefficient is $a \equiv 9$, and f_9 is irreducible. The complete 1770-phase cycle contains 33 classical zero phases and 31 negative zero phases. The first positive phases and their states in $(1, x, x^2)$ are

clock	k	$x^{n_\pm(k)}$
$+$	36	$(26, 42, 32)$
$-$	19	$(31, 42, 2)$.

The trace functional (261) vanishes in each case. Consequently each clock has infinitely many composite floors, respectively on $k \equiv 36 \pmod{1770}$ and $k \equiv 19 \pmod{1770}$.

The complete phase sets have *empty intersection*. This is stronger than noticing that the two selected progressions are disjoint: there is no same-prime common zero phase at 59 for this coefficient. Although 9 is a square, a common-phase theorem requiring a primitive-3 orbit cannot be applied here. Other primes may still witness common composite indices.

An even smaller separate certificate occurs at $p = 11$, where $a \equiv 5$, $x^{133} = 7$, and $\mathrm{ord}_{133}(3) = 18$:

$$\begin{array}{c|c|l} + & k \equiv 16 \pmod{18} & (10,5,6) \text{ at } k = 16 \\ - & k \equiv 1 \pmod{18} & (5,8,7) \text{ at } k = 1. \end{array}$$

These are the complete zero phases at this prime. They too are separate, not a simultaneous certificate. The positive assertions above do not claim that finitely many auxiliary primes cover all a, or all 7^m.

19.3 What Frobenius enlargement and irreducibility do not supply

19.3.1 The full-group shortcut is exactly the primitive-root case

Frobenius invariance $pD = D$ immediately gives a sufficient classical criterion:

$$\langle 3, p\rangle = G \quad \Longrightarrow \quad W_+ = G. \tag{262}$$

Indeed, for $d \in D$ and $u \in G$, write $d/u = 3^k p^j$ and apply an inverse Frobenius power. Then $u3^k \in D$; adding a period makes k positive. This reasoning does not apply to the affine negative exponent: $p(2t-1) = 2pt - p$, not $2pt - 1$.

Proposition 19.6 (Cubic reciprocity limitation)**.** *If $p \geq 5$ and $S = p^2 + p + 1$ are prime, then*

$$\langle 3, p\rangle = (\mathbb{Z}/S\mathbb{Z})^* \quad \Longleftrightarrow \quad 3 \text{ is primitive modulo } S.$$

Proof. We first record the cubic supplement in the needed convention:

$$3^{(S-1)/3} \equiv p^{(p+1)/3} \pmod{S}. \tag{263}$$

This is a specialization of the standard supplements stated in [34, p. 291, equations (1.1)–(1.2)]. To see the specialization, let $\omega^2 + \omega + 1 = 0$ and take the primary Eisenstein prime $\pi = -1 - (p+1)\omega$, of norm S. Modulo π one has $\omega = p$. The supplements give

$$\left(\frac{1-\omega}{\pi}\right)_3 = 1, \qquad \left(\frac{\omega}{\pi}\right)_3 = \omega^{-(p+1)/3}.$$

Use $3 = -\omega^2(1-\omega)^2$ and the fact that -1 is a cube. The cubic symbol of 3 is $\omega^{(p+1)/3}$; its defining Euler criterion gives (263).

Set $H = \langle 3\rangle$ and $N = S - 1 = p(p+1)$. The residue p has order exactly 3 modulo S. If $\langle 3, p\rangle = G$ but $H \neq G$, then $[G : H] = 3$ and $p \notin H$. In the cyclic group G, an order of H divisible by 3 would force H to contain the unique subgroup of order 3, including p. Thus $|H| = N/3$ is prime to 3, and $v_3(N) = 1$. The left side of (263) is then 1, whereas its right side is not 1, since $3 \nmid (p+1)/3$. This contradiction proves the nontrivial direction; the reverse is immediate. □

Frobenius can enlarge a proper subgroup without filling G; $p = 17$ is an example. Proposition 19.6 does not restrict the complete D-coset coverage in the preceding coset-coverage subsection. That coverage is the reason the positive certificates go beyond the full-group shortcut.

19.3.2 Composite projective group order and missing strata

For an irreducible cubic element γ, let Q be its projective order and put $y = \gamma^{p-1}$. Then $\mathrm{ord}(y) = Q \mid S$, and

$$\frac{\mathrm{Tr}(\gamma^n)}{\gamma^n} = 1 + y^n + (y^n)^{p+1}. \tag{264}$$

Write $Q = 3^e N$, $3 \nmid N$. One has $e \le 1$: when $p \equiv 1 \pmod 3$, $S \equiv 3 \pmod 9$, and otherwise $3 \nmid S$. For every classical index $k \ge 1$, y^{3^k} has exact order N. It follows that

$$\gcd(Z^{p+1} + Z + 1, \Phi_N(Z)) = 1 \text{ in } \mathbb{F}_p[Z] \quad \Longrightarrow \quad \text{no classical zero in this stratum.} \tag{265}$$

If this gcd is nonconstant and $\langle 3, p\rangle = (\mathbb{Z}/N\mathbb{Z})^*$, there is conversely a classical zero: the group acts transitively on the primitive Nth roots, and trace zero is invariant under Frobenius. This converse is sufficient, not necessary. Smaller exact coset coverage may suffice. For $N = 1$, the gcd in (265) is 1, since $1 + 1 + 1 \neq 0$.

The general exact criterion remains (260), now allowing nonunit u. If $3 \nmid S$, enumerate an exponent period $\mathrm{ord}_S(3)$; if $3 \mid S$, start at $k = 1$ and use $\mathrm{ord}_{S/3}(3)$. Both clocks are periodic modulo S from that point. One must not invert a nonunit exponent when S is composite.

At $p = 11$, $S = 133 = 7 \cdot 19$. The reference f_5 has full projective order and

$$D = \{2, 5, 16, 18, 22, 43, 50, 55, 65, 73, 74, 109\}.$$

Every member is coprime to 133. Thus every trace-zero projective point has full order 133; the order-7 and order-19 strata contain none. Both missing strata actually occur in the family:

a mod 11	scalar identity	projective order
2	$x^7 = 8$	7
4	$x^{19} = 2$	19.

Both cubics are irreducible. Enumerating one full projective period shows that 11 divides *no ordinary power trace* in either cell. For $a = 7^m$, these are respectively $m \equiv 3$ and 6 (mod 10). This diagnoses an existence gap, not an obstruction to using other primes.

For contrast, irreducibility does guarantee recurrence of a clock zero that already exists. Indeed, if $Q = 3^e N$ with $e \le 1$, take $L = \mathrm{ord}_N(3)$, with $L = 1$ for $N = 1$. For $k \ge 1$,

$$n_+(k + L) - n_+(k) = 3^k(3^L - 1), \qquad n_-(k + L) - n_-(k) = 2 \cdot 3^k(3^L - 1)$$

are multiples of Q. The two corresponding states are scalar multiples. Existence and recurrence are different assertions.

19.4 Actual output primes: norm and cyclotomic lower bounds

This section concerns a *different, conditional* setting. Let β be a fixed cubic Pisot number, with minimal polynomial

$$f(X) = X^3 - aX^2 + bX - c, \qquad c \neq 0, \quad C = |c|, \quad \Delta = \mathrm{disc}(f) \neq 0.$$

Write its other conjugates as β_2, β_3. Then

$$1 \le C = |\beta\beta_2\beta_3| < \beta. \tag{266}$$

Assume that, from some index onward,

$$p_k = \mathrm{Tr}(\beta^{3^k}) = \left\lfloor \beta^{3^k} \right\rfloor \tag{267}$$

are strictly increasing primes. We consider only the residue branch $p_k \equiv -1 \pmod 3$, and primes not dividing $6c\Delta$. The latter exclusion is automatic for sufficiently late increasing outputs. We do not assert that all hypothetical candidates lie in this branch.

19.4.1 Two independent components of projective order

For a good output prime, let Q_k be the order of the class of X in

$$(\mathbb{F}_{p_k}[X]/(f))^*/\mathbb{F}_{p_k}^*.$$

Write $Q_k = 3^{w_k} q_k$, with $3 \nmid q_k$. If $w_k \le k$, a positive index gap L with $3^L \equiv 1 \pmod{q_k}$ would make $X^{3^{k+L}}$ a nonzero scalar multiple of X^{3^k}. The trace is zero modulo p_k at the first exponent because its integer value is p_k. It would then be zero at every such later exponent, contradicting the increasing prime tail. Consequently

$$w_k \ge k + 1. \tag{268}$$

An irreducible reduction has projective order dividing $p_k^2 + p_k + 1$, whose 3-part is at most 3. A completely split reduction has order dividing $p_k - 1$, prime to 3 in the stated branch. Thus every such sufficiently late reduction is of mixed type $(1, 2)$.

Let $p \equiv -1 \pmod 3$ be any good mixed prime, with linear root $\alpha \in \mathbb{F}_p^*$ and quadratic roots λ, λ^p. Dividing all components by α identifies the projective order with

$$Q = \operatorname{ord}(\lambda/\alpha) = 3^w q, \qquad 3 \nmid q.$$

Its norm component is

$$\operatorname{Norm}_{\mathbb{F}_{p^2}/\mathbb{F}_p}(\lambda/\alpha) = c/\alpha^3, \qquad g = \operatorname{ord}_{\mathbb{F}_p^*}(c/\alpha^3).$$

Since $g \mid Q$ and $g \mid p - 1$, one has

$$3 \nmid g, \qquad g \mid q. \tag{269}$$

The large 3-part in (268) therefore cannot account for this separate norm-component order.

Theorem 19.7 (Elementary output-prime bound)**.** *For any good mixed prime $p \equiv -1 \pmod 3$ in the fixed setting above,*

$$g > \frac{\log p - \log 8}{3 \log \beta + 2 \log C}.$$

For an actual output in (267), once $\beta^{3^k} > 2$,

$$g_k > \frac{3^k \log \beta - \log 16}{3 \log \beta + 2 \log C}. \tag{270}$$

If the indices in the stated residue branch are infinite, then along those indices

$$\liminf \frac{q_k}{3^k} \ge \frac{\log \beta}{3 \log \beta + 2 \log C}, \tag{271}$$

$$\liminf \frac{Q_k}{9^k} \ge \frac{3 \log \beta}{3 \log \beta + 2 \log C}. \tag{272}$$

Limits here are along the relevant indices tending to infinity.

Proof. The order relation gives $\alpha^{3g} = c^g$ modulo p, so p divides

$$R_g = \operatorname{Norm}_{\mathbb{Q}(\beta)/\mathbb{Q}}(\beta^{3g} - c^g) \in \mathbb{Z}.$$

This integer is nonzero: $\beta^3 > C$ and $|\beta_i|^3 < 1 \le C$ for the two small conjugates. Triangle inequalities and (266) give

$$0 < |R_g| \le (\beta^{3g} + C^g) \prod_{i=2}^{3} (|\beta_i|^{3g} + C^g) < 8(\beta^3 C^2)^g.$$

Taking logarithms proves the first bound. For an actual output, (267) implies $p_k > \beta^{3^k} - 1 > \beta^{3^k}/2$, proving (270). Finally $q_k \geq g_k$ and $Q_k \geq 3^{k+1} g_k$ by (268)–(269). Divide and take lower limits. □

This is the standard elementary norm-height argument for an order of size at least a constant times $\log p$. The substitution of the growing *output equality* $p_k = T_{3^k}$ is essential: it cannot be made for an arbitrary fixed prime divisor of a large trace. For checking the norm, if U_n denotes the pair-product power sum, then

$$R_g = c^g(c^g T_{3g} - U_{3g}), \qquad R_1 = c(a^3 c - b^3). \tag{273}$$

Expansion of the three factors proves this: the first and last terms cancel because $\mathrm{Norm}(\beta^{3g}) = c^{3g}$.

If the small conjugates have moduli $\sigma < \rho \leq \beta^{-17/40}$, then $C = \beta\rho\sigma < \beta^{3/20}$. Thus the candidate-specific constants in (271) and (272) are strictly larger than $10/33$ and $10/11$, respectively. These root-size assumptions are compatible with the cubic totally real shape appearing in the algebraic reduction of Saito [4, Theorem 1.7]; here they are explicit hypotheses, not an additional reduction theorem.

19.4.2 Exact order gives a totient bound

Let $\Phi_j(X, Y) = Y^{\varphi(j)}\Phi_j(X/Y)$ be the homogeneous cyclotomic polynomial, with $\Phi_1(X, Y) = X - Y$. Since α^3/c has exact order g and $p \nmid cg$, separability of $Z^g - 1$ modulo p shows that

$$p \mid N_g, \qquad N_g = \mathrm{Norm}_{\mathbb{Q}(\beta)/\mathbb{Q}}\, \Phi_g(\beta^3, c) \neq 0. \tag{274}$$

Indeed, $f(X)$ and $\Phi_g(X^3, c)$ share the root α modulo p. No conjugate ratio β_i^3/c has modulus 1, so the norm cannot vanish.

Lemma 19.8. *For $0 < r < 1$, put $F(r) = r/(1-r)^2$. If $|z| = r$, then, uniformly for $j \geq 1$,*

$$\bigl|\log|\Phi_j(z)|\bigr| \leq F(r).$$

Proof. The Möbius product expresses the logarithm as a signed sum of $\log|1 - z^d|$ over $d \mid j$. Each absolute summand is at most

$$-\log(1 - r^d) \leq \frac{r^d}{1 - r^d} \leq \frac{r^d}{1 - r}.$$

Summing over all positive integers instead of divisors gives $F(r)$. For $j = 1$, the sign in $\Phi_1(z) = -(1 - z)$ has no effect. □

Define the fixed constant

$$E = F(C/\beta^3) + F(|\beta_2|^3/C) + F(|\beta_3|^3/C).$$

Reciprocity in absolute value, $|\Phi_j(z)| = |z|^{\varphi(j)}|\Phi_j(1/z)|$ for $|z| > 1$, and Lemma 19.8 yield

$$0 < |N_g| \leq e^E(\beta^3 C^2)^{\varphi(g)}.$$

Homogenization contributes $C^{3\varphi(g)}$, while the dominant ratio contributes $(\beta^3/C)^{\varphi(g)}$; this explains the exact base. For an actual output with $M = 3^k$, we obtain

$$\varphi(g) > \frac{M\log\beta - \log 2 - E}{3\log\beta + 2\log C}. \tag{275}$$

This improves an order bound to a bound on its totient. It is still only a necessary condition.

19.4.3 Cube norms and a unit-norm integer refinement

Suppose $c = d^3$ for a nonzero integer d, and put $D = |d| = C^{1/3}$. Cubing is an automorphism of $\mathbb{F}_p^*$ because $p \equiv -1 \pmod 3$. Consequently

$$g = \operatorname{ord}((d/\alpha)^3) = \operatorname{ord}(d/\alpha) = \operatorname{ord}(\alpha/d).$$

The smaller norm certificates are

$$p \mid \operatorname{Norm}(\beta^g - d^g), \qquad p \mid \operatorname{Norm} \Phi_g(\beta, d).$$

They are nonzero since $\beta > D$ and $|\beta_i| < 1 \le D$. The same triangle and cyclotomic estimates, now with

$$E_d = F(D/\beta) + F(|\beta_2|/D) + F(|\beta_3|/D),$$

give at an actual output

$$\begin{aligned} g &> \frac{M \log \beta - \log 16}{\log \beta + (2/3) \log C}, \\ \varphi(g) &> \frac{M \log \beta - \log 2 - E_d}{\log \beta + (2/3) \log C}. \end{aligned} \tag{276}$$

The relevant norm bounds are $8(\beta D^2)^g$ and $e^{E_d}(\beta D^2)^{\varphi(g)}$.

Corollary 19.9 (Unit norms, under explicit small-root bounds)**.** *Assume $c = \varepsilon \in \{1, -1\}$, $\beta \ge 16$, and $|\beta_2|, |\beta_3| \le \beta^{-17/40}$. At each good mixed output of (267) in the stated residue branch, with $k \ge 1$,*

$$\varphi(g_k) \ge 3^k + 1, \qquad g_k \ge 3^k + 2, \qquad Q_k \ge 3 \cdot 9^k + 6 \cdot 3^k.$$

Proof. Take $d = \varepsilon$ in (276). The small moduli are below $1/3$, since $2^{17} > 3^{10}$. As F is increasing,

$$E_\varepsilon \le F(1/16) + 2F(1/3) = \frac{16}{225} + \frac{3}{2} = \frac{707}{450} < 2 \le \log 8.$$

For the last inequality, $\log 2 \ge 2/3$ follows, for example, by integrating or differentiating $\log x - 2(x-1)/(x+1)$ for $x \ge 1$. Hence $E_\varepsilon + \log 2 < \log \beta$. With $M = 3^k$, (276) gives $\varphi(g) > M - 1$, so $\varphi(g) \ge M$. The bound excludes $g \le 2$; for $g > 2$, its totient is even, whereas M is odd. Thus $\varphi(g) \ge M + 1$ and $g \ge \varphi(g) + 1 \ge M + 2$. Finally $Q \ge 3^{k+1} g = 3Mg$ by (268)–(269). □

All indices are relative to the *fixed chosen* Pisot number, and the prime trace-floor tail must already have begun. Choosing a later cubic power may arrange $\beta \ge 16$, but shifts the clock. In particular, replacing β_0 by β_0^3 makes its norm a cube, yet the new certificate $\Phi_g(\beta_0^3, \operatorname{Norm}(\beta_0))$ is exactly the old general certificate. Forgetting the index shift does not create an extra improvement. Unit norms already are cubes at the original chosen power.

19.4.4 Why a local model is not an actual output model

For fixed $k \ge 1$, choose $p \equiv \pm 1 \pmod{3^{k+1}}$ and an element ζ of order 3^{k+1} in $\mathbb{F}_p^*$ in the plus case, or in the norm-one subgroup of $\mathbb{F}_{p^2}^*$ in the minus case. The root multiset

$$1, \zeta, \zeta^{-1}$$

has norm 1, trace zero at exponent 3^k, and trace 3 at every later classical exponent. Its projective 3-height is exactly $k + 1$. The plus case is split; the minus case is mixed. These are

legitimate finite-field examples where a seed zero and the required height do not force a later zero.

In the mixed case the norm-component order is only $g = 1$. For a *fixed* irreducible Pisot candidate, however, this forces $p \mid R_1 = c(a^3c - b^3)$, a fixed nonzero integer. The displayed linear root 1 also forces $p \mid f(1)$. It cannot model arbitrarily late increasing output primes of that candidate. Likewise, lifting coefficients modulo a fixed prime usually produces a large trace divisible by that prime, not an equality between the trace and that prime.

None of the lower bounds above contradicts $Q \mid p^2 - 1$. An actual output is exponential in 3^k, leaving far more room than the required order scale. No inert output, recurring divisor, or transcendence conclusion has been deduced from these necessary conditions.

19.5 Ordinary divisors and sparse clocks are different conditions

For a fixed coefficient and one clock, a sufficient inert-witness strategy would require

$$\exists p\ \exists k \geq 1: \quad f_a \text{ is irreducible modulo } p, \quad p \mid T_{n_{\pm}(k)}(a). \tag{277}$$

As proved in the preceding projective-return argument, such a zero recurs. The two clocks can use different witnesses; (277) for both does not itself give a common index progression. Chebotarev can provide the irreducibility condition, while a prime-factor theorem can provide divisors. Their intersection on a fixed sparse clock is additional arithmetic input.

19.5.1 The even powering family is not algebraically degenerate

Proposition 19.10. *For every even $m \geq 2$, the polynomial f_{7^m} has Galois group S_3, and its three roots are multiplicatively independent. For every $m \geq 2$, the ordinary trace recurrence is nondegenerate of exact order 3, as is its odd-index subsequence.*

Proof. Irreducibility follows from Lemma 19.1. The discriminant is

$$\Delta(a) = 16D(a), \qquad D(a) = a^3 + a^2 + 18a - 11.$$

For even m, write $a = b^2$ with $b = 7^{m/2}$ odd. If $m \geq 4$, then $b \geq 49$, and the consecutive integers

$$L = b^3 + (b-1)/2, \qquad U = L + 1 = b^3 + (b+1)/2$$

satisfy

$$4(D(b^2) - L^2) = 4b^3 + 71b^2 + 2b - 45 > 0,$$
$$4(U^2 - D(b^2)) = 4b^3 - 71b^2 + 2b + 45 > 0.$$

For the second inequality, $4b - 71 > 0$ already at $b \geq 18$. At $m = 2$, direct arithmetic gives

$$347^2 = 120409 < D(49) = 120921 < 121104 = 348^2.$$

Thus the discriminant is nonsquare and the irreducible cubic has group S_3.

The root magnitudes β, w, v are distinct, so no quotient of distinct roots is a root of unity. The three exponential coefficients in T_n are nonzero, giving a nondegenerate recurrence of exact order 3. The same reasoning applies to

$$T_{2r+1} = \beta(\beta^2)^r - w(w^2)^r + v(v^2)^r.$$

For multiplicative independence in the S_3 case, let $\prod \alpha_i^{e_i} = 1$ with integer e_i. Apply a Galois transposition and divide the two relations. Then $(\alpha_i/\alpha_j)^{e_i - e_j} = 1$, forcing $e_i = e_j$. All exponents are equal, and the norm -4 gives $(-4)^{e_1} = 1$, hence $e_1 = 0$. □

19.5.2 An inert odd-trace divisor that misses every classical phase

Proposition 19.11. *For every $m \geq 2$ with $m \equiv 4 \pmod{12}$, the cubic f_{7^m} is irreducible modulo 13, and $13 \mid T_{61}(7^m)$, but*

$$13 \nmid T_{3^k}(7^m) \qquad \text{for every } k \geq 1.$$

Proof. The coefficient is $a \equiv 9 \pmod{13}$. In $\mathbb{F}_{13}[x]/(x^3 - 9x^2 - 4x + 4)$, testing all 13 residues proves irreducibility. Direct quotient-ring powering gives

$$x^{61} = (11, 7, 9), \quad \tau(x^{61}) = 0, \qquad x^{183} = 9.$$

Thus the ordinary divisible exponent is odd, not an even exponent omitted by the floor bridge. The ten successive cubing states are given in Table 18. All traces are nonzero. The next state is $u_{11} = (1, 12, 1) = 3u_1$. Cubing preserves nonzero scalar proportionality, so by induction every later phase is a nonzero scalar multiple of one of the ten listed states. This proves the nondivisibility for all k. □

k	x^{3^k}	τ	k	x^{3^k}	τ
1	$(9, 4, 9)$	6	6	$(5, 1, 0)$	11
2	$(11, 4, 1)$	2	7	$(4, 1, 11)$	12
3	$(1, 9, 1)$	4	8	$(1, 8, 1)$	8
4	$(7, 0, 12)$	10	9	$(11, 6, 3)$	3
5	$(6, 8, 11)$	3	10	$(7, 7, 7)$	5

Table 18: Complete projective classical phases for $p = 13$, $a = 9$.

Equivalently, after removing the factor 3 from $183 = 3 \cdot 61$, the classical residual orbit is

$$\langle 3 \rangle \bmod 61 = \{1, 3, 9, 27, 20, 60, 58, 52, 34, 41\}.$$

The trace-zero classes among exponents divisible by 3, divided by 3, are $18, 51, 53$. They miss this orbit. The inertness of a prime and an ordinary odd trace divisor therefore do not suffice for the clock intersection in (277).

19.5.3 A primitive ordinary divisor need not be inert

At $a = 7^4 = 2401$, the integer

$$p = T_3(a) = 13841316001$$

is prime, as certified by exhaustive trial division in the portable verifier. The preceding ordinary terms are $T_0 = 3$, $T_1 = 2401$, and $T_2 = 5764809$, all positive and smaller than p. Thus p is primitive in the first-occurrence sense for this ordinary trace sequence. Nevertheless, modulo p,

$$f_a(X) = (X - 1977685529)(X^2 + 1977683128X - 6128330130). \tag{278}$$

The quadratic discriminant is a nonresidue; equivalently $\Delta(a)^{(p-1)/2} \equiv -1 \pmod{p}$. Hence the splitting type is $(1, 2)$, not (3). This does not disprove a hypothetical eventual inert-primitive-divisor theorem. It shows that primitivity alone is not the required inertness condition, and a prime seed is of course not a composite term at that seed.

19.6 No rational iteration in the trace coordinate

Theorem 19.12. *For any fixed $a > 8$, put $C_k = T_{3^k}(a)$ and $B_k = T_{2\cdot 3^k-1}(a)$. For every fixed rational function $F \in \mathbb{R}(X)$, each equality*

$$C_{k+1} = F(C_k), \qquad B_{k+1} = F(B_k)$$

holds for at most finitely many indices. In particular neither trace clock is eventually a scalar rational-function orbit in its own trace coordinate.

Proof. For the classical clock, write $n = 3^k$, $t = \beta^n$, $u = w^n$, $z = v^n$. Then $C_k = t - u + z$ and the exact identity

$$D_k := C_{k+1} - C_k^3 = 3(u - z)(t - u)(t + z) \tag{279}$$

gives

$$D_k \sim 3(\beta^2 w)^n, \qquad D_k/C_k^2 \longrightarrow 0, \qquad D_k/C_k \longrightarrow +\infty.$$

The final limit uses $\beta w = 4/v > 1$. If a fixed rational function matched infinitely many pairs, then on that unbounded subsequence $F(C_k)/C_k^3 \to 1$. Its Laurent expansion at infinity must be

$$F(X) = X^3 + b_2X^2 + b_1X + O(1).$$

The first defect ratio forces $b_2 = 0$; the second contradicts the finite limit b_1. This proves the classical assertion.

For the negative clock put $n = 2 \cdot 3^k - 1$, whose successor is $3n + 2$. The corresponding identity is

$$\begin{aligned} B_{k+1} - \beta^2 B_k^3 = {} & 3\beta^2(u - z)(t - u)(t + z) \\ & + (\beta^2 - w^2)u^3 + (v^2 - \beta^2)z^3. \end{aligned}$$

The last two terms are $O(w^{3n})$, so the difference is asymptotic to $3\beta^2(\beta^2 w)^n$. Dividing by B_k^2 and B_k again gives 0 and $+\infty$. A putative F must have leading term $\beta^2 X^3$, and the same Laurent-expansion contradiction follows. □

The proof extends to any cubic algebraic integer with conjugates $\beta, -\rho, \gamma$ satisfying

$$\beta > 1, \qquad 0 < |\gamma| < \rho < 1.$$

For odd n, set $u = \rho^n$ and $z = \gamma^n$, allowing $\gamma < 0$. Then $u - z \sim \rho^n$, while the nonzero integral norm gives $\beta\rho = |\operatorname{Norm}(\beta)|/|\gamma| > 1$. Both arguments are unchanged.

The conclusion is deliberately coordinate-specific. Classical conjugates still evolve by coordinatewise cubing, or by a polynomial map in three symmetric coordinates. Higher-dimensional dynamics and extra auxiliary coordinates remain available. Nor does the theorem forbid interpolation through finitely many pairs.

The scheduled sequences also cannot be fixed-order constant-coefficient linear recurrences in k: such a recurrence has growth at most $AR^k k^d$, whereas these traces grow as β^{3^k} or $\beta^{2\cdot 3^k-1}$. Ordinary-index recurrence results must therefore be applied with their actual indexing hypotheses.

19.7 Applicable literature, reproducibility, and the remaining gap

19.7.1 A prime-factor theorem that does transfer

For each fixed $a = 7^m$, the ordinary recurrence is nondegenerate with three nonzero roots by Proposition 19.10. Evertse's Corollary 3 [35, p. 230] implies that its greatest prime factor tends to infinity; the precise formulation is also recorded in [36, equation (2), p. 342]. Consequently

$$P^+(T_{3^k}(a)) \longrightarrow \infty, \qquad P^+(T_{2\cdot 3^k-1}(a)) \longrightarrow \infty.$$

This is an application of a published theorem, not a new general prime-factor result. It says that eventually each term has a factor outside any fixed finite prime set. It supplies no splitting-type control and no primitive divisor at every index. In particular it does not by itself establish (277).

The classical primitive-divisor theorem for Lucas and Lehmer sequences [37, Theorem 1.4] cannot be applied simply by naming our trace a recurrence: even its odd-index subsequence has exact order 3, and no factor identity reducing it to the relevant two-root form has been established here. Roskam's higher-order positive-density result [38] assumes a generalized Artin conjecture and concerns unrestricted ordinary indices. For fixed subgroup index h, Roskam also proves ordinary-index divisibility outside finitely many inert reduction primes whose root-generated subgroup has that index; the positive-density supply of such primes is conditional, and neither statement locates an index on our clocks. Järviniemi's Theorems 1.1–1.2 [39] assume GRH; in the cubic case the structural theorem provides ordinary-term divisors with mixed splitting type, and its Section 8 distinguishes the irreducible-reduction difficulty. Algebraic genericity alone does not remove these hypotheses or impose either clock.

Arithmetic-dynamics primitive-divisor results such as [40] concern iterates of a fixed rational map with a suitable periodic target. Theorem 19.12 rules out the direct scalar trace-coordinate interpretation. It does not rule out encoding traces as reduced numerators of another rational orbit with varying denominators, nor a higher-dimensional reduction. No such additional representation is established here. These are bounded applicability checks, not an exhaustive literature or historical-priority claim.

19.7.2 Portable exact verification

The original source archive contains six Python 3 programs requiring only the standard library, each with its exact JSON output and text log:

1. `frobenius_coset_coverage_verify.py`: primality by exhaustive division, irreducible reductions, complete D-sets and coverage sets, all displayed short phases and missing strata. A finite consistency scan of (263) is not substituted for its reciprocity proof.
2. `frobenius_coset_matrix_audit.py`: an independent companion-matrix recomputation of the coverage sets, using the preceding JSON only for comparison, not its quotient-ring implementation.
3. `mixed_output_prime_norm_verify.py`: 720 exact norm identities, finite-field norm/projective orders, and the local roots-of-unity models. Numerical samples do not prove the height bounds.
4. `cyclotomic_norm_height_verify.py`: homogeneous polynomial normalizations, 27 actual finite-field exact-order norm certificates, and rational checks of the unit-bound constants and parity rounding.

5. `inert_bridge_verify.py`: the complete $p = 13$ projective-phase obstruction, the primitive-but-mixed prime, and polynomial identities in the Galois and rational-dynamics arguments.

6. `inert_bridge_independent_verify.py`: independent companion-matrix returns and an Eratosthenes-sieve trial-divisor certification of the primitive-but-mixed prime.

Run each program from the `certificates` directory. The universal statements about norms, asymptotics, and all even m are proved in the text, not extrapolated from finite scans. Checksums identify the delivered sources and outputs.

19.7.3 Scope and the next arithmetic problem

The positive coverage certificates supply usable auxiliary primes whenever the fixed cubic is irreducible at one of the listed primes. They prove neither that every coefficient meets one of these cells nor that a suitable prime exists universally. The missing step for the inert-witness strategy is an intersection theorem between irreducible reductions of a fixed cubic and divisors occurring on its fixed sparse clock.

The conditional output-prime bounds are different: they use equality to a growing prime output and exclude overly small norm-component orders. They leave ample admissible group order and yield no contradiction. No coefficient-density theorem is transferred to a prescribed powering orbit, no separate-phase statement is promoted to a common-phase statement, and no universal Mills or Pisot exclusion is claimed.

Part V

Actual cubic candidates and analytic prime interfaces

20 Both residue signs and all norm depth

The positive residue branch is genuinely present and requires its own analysis. The generalized two-sign depth theorem and positive-branch resolvent statements below supplement, rather than erase, the negative-branch constraints in the preceding section.

20.1 The classical minimum and the two genuinely remaining signs

Let ξ denote the least real number greater than 1 for which $\left\lfloor \xi^{3^j} \right\rfloor$ is prime for every $j \geq 1$. We do not assume that ξ is algebraic. We study consequences *if it is*. Saito's reduction [4, Theorem 1.7 and Section 10] supplies a fixed power $\beta = \xi^{3^{j_0}}$ that is a totally real cubic Pisot number with conjugates

$$\beta, -\rho, \gamma, \qquad \beta > 1, \quad 0 < \sigma := |\gamma| < \rho < 1, \quad \rho \leq \beta^{-17/40}. \tag{280}$$

The theorem has additional conjugate inequalities; only those displayed are needed for the basic statements here. Write the irreducible minimal polynomial as

$$f(X) = X^3 - aX^2 + bX - c, \qquad a, b, c \in \mathbb{Z}, \quad c \neq 0, \qquad \Delta = \operatorname{disc}(f) \neq 0.$$

At clock index $k \geq 0$ put $M = 3^k$ and

$$\begin{aligned} A_k &= \operatorname{Tr}(\beta^M), \\ B_k &= \sum_{i<j} (\beta_i \beta_j)^M, \qquad C_k = c^M, \end{aligned}$$

where $\beta_1 = \beta$, $\beta_2 = -\rho$, $\beta_3 = \gamma$. All three are integers. The exact coefficient-cubing recurrence is

$$\begin{aligned} A_{k+1} &= A_k^3 - 3A_k B_k + 3C_k, \\ B_{k+1} &= B_k^3 - 3A_k B_k C_k + 3C_k^2, \qquad (A_0, B_0, C_0) = (a, b, c). \\ C_{k+1} &= C_k^3, \end{aligned} \tag{281}$$

For all sufficiently large k, the shadow $\rho^M - \gamma^M$ is positive and below 1. Hence

$$p_k := A_k = \left\lfloor \beta^{3^k} \right\rfloor \tag{282}$$

is prime, and these outputs strictly increase. All later arithmetic arguments require only the fixed cubic, (280), and this prime trace-floor tail; they make no additional use of minimality. Indices are always relative to this chosen fixed power β.

20.1.1 What the source arguments actually say

Saito's theorem specifies the real conjugate shape and sizes, not the residue $p_k \pmod 3$. The supplied manuscript [6] also retains both signs in Proposition `mil:prop:cubic-split-three`.

Its theorem `f13:thm:r2-splitbranch` explicitly treats a *positive-sign* norm-minus-one unit branch and, under an additional exact-depth hypothesis, forces complete splitting rather than excluding that branch. The subsequent discussion leaves those split outputs open.

By contrast, the formula $a \equiv 2 \pmod 3$ at `ft:eq:cubic-a-two` occurs in the nonreal-conjugate argument. That argument uses infinitely recurring floor shifts of both values 0 and 1. In the surviving real shape, the floor equals the trace eventually, so that premise is unavailable. A negative real conjugate is not a negative trace residue. These are scope distinctions, not evidence that an algebraic all-prime candidate exists in either branch.

We now prove the common arithmetic statements directly, so the subsequent bounds do not depend on a citation to an unstated sign restriction.

20.2 Original projective order and trace-limit rigidity

Call a prime p *good* if $p \nmid 6c\Delta$. Every sufficiently late increasing output is good. The reduced cubic is then separable with nonzero roots $\alpha_1, \alpha_2, \alpha_3$ in a finite field. Let Q_p be the order of the subgroup generated by all ratios α_i/α_j. Equivalently, it is the order of the residue of X in the unit group of the original cubic quotient modulo nonzero scalar units. Write

$$Q_p = 3^{w_p} q_p, \qquad 3 \nmid q_p.$$

This definition uses the original separable algebra, not a possibly nonreduced polynomial obtained after powering.

Lemma 20.1 (Scalar return)**.** *For every sufficiently late good output $p = p_k$ of (282),*

$$w_p \geq k + 1.$$

Such an output cannot have irreducible cubic reduction. Its reduction is split or mixed, and

$$v_3(p^2 - 1) \geq k + 1.$$

Proof. If $w_p \leq k$, choose $L \geq 1$ with $3^L \equiv 1 \pmod{q_p}$; use $L = 1$ if $q_p = 1$. Then Q_p divides $3^k(3^{jL} - 1)$ for every $j \geq 0$. All residual roots at exponent 3^{k+jL} are a common nonzero scalar multiple of their values at 3^k. Their sum is zero because $p = A_k$. Thus p divides every A_{k+jL}, contradicting the strictly increasing prime tail.

For an irreducible cubic reduction, the projective order divides $p^2 + p + 1$, whose 3-adic valuation is at most 1. This is incompatible with $w_p \geq k + 1$ at $k \geq 1$. In the split case $Q_p \mid p - 1$; in the mixed case $Q_p \mid p^2 - 1$. This proves the remaining assertion. □

Proposition 20.2 (Both signs and the initial coefficient congruence)**.** *There is a fixed $\varepsilon \in \{1, -1\}$ such that*

$$p_k \equiv \varepsilon \pmod{3^{k+1}}$$

for every sufficiently late output. Moreover

$$a \equiv \varepsilon \pmod 3, \qquad c \equiv ab \pmod 3, \qquad f(X) \equiv (X - \varepsilon)(X^2 + b) \pmod 3. \tag{283}$$

The argument does not choose between the two values of ε.

Proof. The recurrence gives $A_k \equiv a \pmod 3$ for every k. An increasing prime tail excludes $a \equiv 0 \pmod 3$. Let $\varepsilon \in \{1, -1\}$ represent this fixed residue. Lemma 20.1 implies that the factor $p_k - \varepsilon$ is divisible by 3^{k+1}, since the other factor of $p_k^2 - 1$ is prime to 3.

For completeness, the standard trace congruence follows directly from (281). Denote its three-coordinate polynomial map by F. For every integral vector z, $F(z) \equiv z \pmod 3$. If $u - v \in 3^r\mathbb{Z}^3$ with $r \geq 1$, expansion of each cube and of the terms with coefficient 3 gives $F(u) - F(v) \in 3^{r+1}\mathbb{Z}^3$. Induction therefore yields

$$A_{k+1} - A_k \in 3^{k+1}\mathbb{Z}.$$

The 3-adic limit is ε by the output congruence, and $A_1 \equiv \varepsilon \pmod 9$. But $A_1 = a^3 - 3ab + 3c$ and $a^3 \equiv \varepsilon \pmod 9$. Hence $c \equiv ab \pmod 3$. Expanding the residual factorization proves (283). □

For $\varepsilon = -1$, completely split outputs are impossible because $3 \nmid p - 1$, so only mixed type remains. The norm-component bounds for that branch in the preceding research round have precisely this hypothesis. In the next subsection we focus on

$$\varepsilon = 1, \qquad a \equiv 1 \pmod 3, \qquad c \equiv b \pmod 3. \tag{284}$$

Neither the negative-branch mixed classification nor its prime-to-3 norm-component interpretation may be transferred without modification.

20.3 The positive branch: exact splitting and a cyclotomic sum

Fix a sufficiently late good output $p = p_k \equiv 1 \pmod 3$. Write

$$t = v_3(p-1), \quad Q_p = 3^w q, \quad 3 \nmid q, \quad M = 3^k, \quad v = w - k \geq 1.$$

Proposition 20.3. *Both split and mixed reduction remain unexcluded by the current necessary conditions, and in either case*

$$k + 1 \leq w \leq t.$$

In mixed type, c is a cube modulo p if and only if the normalized quadratic root is a cube in $\mathbb{F}_{p^2}^*$. If c is a cube, then $w \leq t - 1$ and $t \geq k + 2$; if it is not, then $w = t$ exactly.

Proof. In split type $Q_p \mid p - 1$. In mixed type write the roots as $\alpha \in \mathbb{F}_p^*$ and $\lambda, \lambda^p \in \mathbb{F}_{p^2}^*$, and put $y = \lambda/\alpha$. Then $Q_p = \operatorname{ord}(y) \mid p^2 - 1$. Since $p \equiv 1 \pmod 3$, one has $v_3(p^2 - 1) = t$. The lower bound is Lemma 20.1.

The field norm of y is c/α^3. A generator of $\mathbb{F}_{p^2}^*$ maps under the norm to a generator of $\mathbb{F}_p^*$. Thus the exponent of y is divisible by 3 if and only if its norm is a cube, which is equivalent to c being a cube. The group $\mathbb{F}_{p^2}^*$ has 3-part 3^t; its cubes have 3-part at most 3^{t-1}, and a noncube has order with 3-part exactly 3^t. □

The cube improvement is specific to mixed type. It is automatic for unit norms $c = \pm 1$ or a fixed integral cube norm, but not for an arbitrary nonunit output. Since inert type is already excluded, the Legendre symbol (Δ/p) now distinguishes the surviving types: $+1$ means complete splitting and -1 means mixed splitting. This is a classification at actual outputs, not a fixed congruence criterion for complete splitting of every prime.

Lemma 20.4 (A small nonzero pair coefficient)**.** *For all odd $M = 3^k$, the integer B_k is negative. At every sufficiently late output,*

$$0 < |B_k| < p_k. \tag{285}$$

Proof. If $\gamma > 0$, then

$$B_k = -(\beta\rho)^M + (\beta\sigma)^M - (\rho\sigma)^M < 0.$$

If $\gamma < 0$, then

$$B_k = -(\beta\rho)^M - (\beta\sigma)^M + (\rho\sigma)^M < 0.$$

The inequalities follow from $\rho > \sigma$ and $\beta > \sigma$. Also

$$|B_k| \le \beta^M(\rho^M + \sigma^M) + (\rho\sigma)^M, \qquad p_k > \beta^M - 1,$$

so $|B_k|/p_k \to 0$. □

Theorem 20.5 (Nonvanishing cyclotomic-sum bound)**.** *Set*

$$N = Q_p/3^k = 3^v q.$$

For every sufficiently late positive-residue good output,

$$p^{f_p} \le 3^{\varphi(N)} = 3^{\varphi(3^v)\varphi(q)}, \qquad f_p = \begin{cases} 1, & \textit{split type}, \\ 2, & \textit{mixed type}. \end{cases} \tag{286}$$

The residue degrees stated here are exact.

Proof. Let $Y_i = \alpha_i^M$. Their sum is zero modulo p, and their ratio subgroup has exact order $Q_p/\gcd(Q_p, M) = N$. Choose a primitive Nth root ξ in the finite splitting field and write the normalized roots as $1, \xi^A, \xi^B$. For a complex primitive Nth root ζ_N, put

$$Z = 1 + \zeta_N^A + \zeta_N^B.$$

We first prove that $Z \neq 0$. If two unit complex numbers sum to -1, then $|1 + \zeta_N^A| = 1$ forces $\zeta_N^{2A} + \zeta_N^A + 1 = 0$, and the other term is its square. The normalized roots would be the three cube roots of unity. Reducing modulo $p \neq 3$ would then give $\sum_{i<j} Y_i Y_j = 0$ modulo p, contradicting (285). The nonzero second coefficient is essential: trace zero alone would not exclude this equilateral case.

Evaluation $\zeta_N \mapsto \xi$ gives a prime ideal above p in $\mathbb{Z}[\zeta_N]$ at which Z vanishes. Because $p \nmid N$, its residue degree is $\mathrm{ord}_N(p)$. The norm of that prime ideal is $p^{\mathrm{ord}_N(p)}$, so this integer divides the nonzero rational norm of Z. Every complex conjugate of Z has absolute value at most 3. Therefore

$$p^{\mathrm{ord}_N(p)} \le \left|\mathrm{Norm}_{\mathbb{Q}(\zeta_N)/\mathbb{Q}}(Z)\right| \le 3^{\varphi(N)}.$$

In split type $N \mid p - 1$, giving degree 1. In mixed type normalize by the linear root, so $N = \mathrm{ord}(y^M)$ for $y = \lambda/\alpha$. If $y^M \in \mathbb{F}_p$, then $(\lambda/\lambda^p)^M = 1$. That ratio has order dividing $p+1$, which is prime to 3. Raising to $M = 3^k$ cannot kill it, so $\lambda = \lambda^p$, contrary to the irreducible quadratic factor. Thus $N \mid p^2 - 1$ but $N \nmid p - 1$, giving degree exactly 2. □

The powered roots are geometric elements of the original separable algebra. Possible collisions in a separately defined powered polynomial do not alter their exact ratio-subgroup order used above. No equality between a nonreduced polynomial quotient and its image has been assumed.

20.4 Multiplicative independence and strict candidate bounds

Lemma 20.6 (Multiplicative relations among totally real cubic roots)**.** *Let z_1, z_2, z_3 be the roots of a monic irreducible totally real cubic in $\mathbb{Z}[X]$, and put $C = |z_1 z_2 z_3| \geq 1$. If $C > 1$, its three roots are multiplicatively independent modulo roots of unity. If $C = 1$, every multiplicative relation modulo roots of unity has exponent vector in $\mathbb{Z}(1,1,1)$. In particular every pair of distinct roots is multiplicatively independent modulo roots of unity, in either case.*

Proof. For $\ell_i = \log|z_i|$, apply a Galois three-cycle to a relation $\prod_i z_i^{e_i}$ equal to a root of unity. Taking absolute values and real logarithms makes (e_1, e_2, e_3) orthogonal to the three cyclic permutations of (ℓ_1, ℓ_2, ℓ_3). The corresponding circulant determinant is, up to sign,

$$\log C \left(\ell_1^2 + \ell_2^2 + \ell_3^2 - \ell_1\ell_2 - \ell_2\ell_3 - \ell_3\ell_1\right).$$

The second factor is positive, since three distinct real roots cannot all have the same absolute value. If $C > 1$, the matrix is invertible. If $C = 1$, its rank is two and its kernel is the line spanned by $(1,1,1)$. This gives the assertions; a vector supported on at most two coordinates cannot be a nonzero constant vector. □

Corollary 20.7 (Independence in the entire explicit family)**.** *For every integer $a > 8$, the three roots of $X^3 - aX^2 - 4X + 4$ are multiplicatively independent modulo roots of unity. In particular no parity restriction on m is needed when $a = 7^m$, $m \geq 2$.*

Proof. The interval/root argument proves irreducibility and total reality, and the absolute norm is 4. Apply Lemma 20.6. □

Corollary 20.8 (Strict shadows for the conditional classical candidate)**.** *Suppose an irreducible totally real cubic Pisot number has conjugates $\beta, -\rho, \gamma$, with $\beta > 1$ and $0 < \sigma = |\gamma| < \rho < 1$, and satisfies the Saito inequalities*

$$\rho \leq \beta^{-17/40}, \qquad \rho \leq \sigma^{17/23}.$$

Then both inequalities are strict. Moreover

$$\frac{1}{2} < \kappa := \frac{2 + \log\rho/\log\beta}{3} < \frac{21}{40}.$$

Consequently the mixed-positive anti-norm constant $3\log\beta/\log(\beta^2\rho)$ is strictly greater than $40/21$.

Proof. Equality in the first inequality would give $\beta^{17}(-\rho)^{40} = 1$, a forbidden two-root relation. Equality in the second would imply $(-\rho)^{46}\gamma^{-34} = 1$, again forbidden; the even exponents remove all sign issues. For the lower endpoint, the nonzero integral norm gives $1 \leq |\operatorname{Norm}(\beta)| = \beta\rho\sigma < \beta\rho^2$. Thus $\rho > \beta^{-1/2}$. Substituting this and the strict upper shadow bound into the definition of κ proves its two strict endpoints. Finally, writing $u = -\log\rho/\log\beta$ gives $17/40 < u < 1/2$ and the anti-norm constant is $3/(2-u) > 40/21$. □

20.5 A two-sign depth theorem for arbitrary totally real cubics

The positivity of the dominant root and the Pisot inequalities are not needed for the exact-depth argument. This observation both broadens that argument and closes its sign restriction.

Theorem 20.9 (Effective two-sign depth for totally real cubics)**.** *Let*

$$f(X) = X^3 - aX^2 + bX - c \in \mathbb{Z}[X]$$

be irreducible and totally real, with $c \neq 0$. Let $\varepsilon \in \{1, -1\}$ and assume

$$a \equiv \varepsilon \pmod 3, \qquad c \equiv \varepsilon b \pmod 3.$$

For the roots $\beta_1, \beta_2, \beta_3$ put $A_k = \sum_i \beta_i^{3^k}$. The following finite positive integral offsets exist and can be computed by terminating modular searches.

1. *If $b \equiv 0 \pmod 3$, put $s = v_3(f(\varepsilon))$. Then*

$$v_3(A_k - \varepsilon) = k + s \qquad (k \geq \max\{3, s-1\}).$$

2. *If $b \equiv 2 \pmod 3$, there is one offset $s \geq 1$ such that*

$$v_3(A_k - \varepsilon) = k + s \qquad (k \geq s).$$

3. *If $b \equiv 1 \pmod 3$, there are offsets $s_0, s_1 \geq 1$ such that*

$$v_3(A_k - \varepsilon) = k + s_j \quad (k \equiv j \pmod 2,\ k \geq s_j - 1).$$

Only the third case uses the proved Baker–Brumer theorem.

Proof. Set

$$g(X) = \varepsilon f(\varepsilon X) = X^3 - \varepsilon a X^2 + bX - \varepsilon c.$$

It is again a monic irreducible totally real cubic. Its roots are $\varepsilon\beta_i$, its trace at the odd exponent 3^k is εA_k, and its coefficients satisfy $a' = \varepsilon a \equiv 1$ and $c' = \varepsilon c \equiv b$ modulo 3. Also $v_3(\varepsilon A_k - 1) = v_3(A_k - \varepsilon)$ and $v_3(g(1)) = v_3(f(\varepsilon))$. We therefore prove the assertion for g and suppress the primes. All local valuations below are normalized by $v_3(3) = 1$.

If $b \equiv c \equiv 0$, Hensel factorization of $g \equiv (X-1)X^2$ gives a root $\alpha \in 1 + 3\mathbb{Z}_3$; the other roots have valuations at least $1/2$. The latter assertion follows by comparing valuations in their monic quadratic, whose other coefficients are divisible by 3. Put $L = \log \alpha$. Irreducibility implies $\alpha \neq 1$, so

$$v_3(L) = v_3(\alpha - 1) = v_3(g(1)) = s \geq 1.$$

For $k \geq 3$, the two nonunit powers have valuations at least $3^k/2 \geq 2k+2$, while the exponential remainder also has valuation at least $2k + 2$. Consequently

$$A_k - 1 = 3^k L + R_k, \qquad v_3(R_k) \geq 2k + 2.$$

For $k \geq s - 1$ the leading term has strictly smaller valuation.

If $b \equiv c \equiv 2$, then $g \equiv (X-1)^2(X+1)$. Let $\eta \in -1+3\mathbb{Z}_3$ be the simple-root lift, and let u, v be the other roots. They satisfy $v_3(u-1), v_3(v-1) \geq 1/2$. The logarithms $U = \log u, V = \log v$ have valuations at least $1/2$. Cubing first gives $v_3(u^3 - 1), v_3(v^3 - 1) \geq 3/2$, so logarithm and exponential are inverse after cubing, even in the ramified quadratic case. With $L = \log(-\eta)$ and $k \geq 1$, this gives

$$A_k - 1 = 3^k \lambda + R_k, \qquad \lambda = U + V - L = \log c - 2\log(-\eta) \in 3\mathbb{Z}_3, \qquad v_3(R_k) \geq 2k + 1.$$

Here $v_3(n!) \leq (n-1)/2$ bounds the exponential remainders. If $\lambda = 0$, then the rational local unit c/η^2, congruent to -1 modulo 3, lies in the logarithm kernel. It must equal -1, so

$\eta^2 = -c$, contrary to the cubic irreducibility of g. Thus $s = v_3(\lambda) \geq 1$ is finite, and $k \geq s$ makes the leading valuation strictly below the error valuation.

If $b \equiv c \equiv 1$, the separable reduction $(X-1)(X^2+1)$ splits in the unramified quadratic field $K = \mathbb{Q}_3(i)$. Label the roots

$$\alpha = e^L, \qquad u = ie^U, \qquad v = -ie^V, \qquad L, U, V \in 3\mathcal{O}_K.$$

Local conjugation fixes L, swaps U, V, and sends i to $-i$. Thus

$$\lambda_j = L + (-1)^j i(U - V) \in 3\mathbb{Z}_3, \qquad A_k - 1 = 3^k \lambda_{k \bmod 2} + R_k, \qquad v_3(R_k) \geq 2k + 2.$$

It remains to prove both λ_j nonzero, without a Pisot hypothesis. By Lemma 20.6, the three global roots are multiplicatively independent modulo roots of unity when $C = |c| > 1$, and every pair is independent when $C = 1$.

All three local roots are units. The kernel of their local logarithm consists of roots of unity, so the preceding argument gives rational linear independence of L, U, V when $C > 1$. When $C = 1$, U, V are rationally independent and $L + U + V = \log c = 0$. The Baker–Brumer theorem [41, Theorem 2.1] promotes these rational independence statements to independence over the algebraic numbers. It excludes $\lambda_j = 0$ in the first case; in the second it excludes

$$(-1 + (-1)^j i)U + (-1 - (-1)^j i)V = 0.$$

Hence $s_j = v_3(\lambda_j) \geq 1$ is finite. The leading valuation $k + s_j$ is strictly smaller than $2k + 2$ for $k \geq s_j - 1$.

Finally all offsets are effectively recognizable using the integral coefficient-cubing recurrence modulo powers of 3. In the second case, search $k \geq 1$ until $A_k - 1$ is nonzero modulo 3^{2k+1}; its valuation d gives $s = d - k$. In the third case search each parity until the corresponding residue is nonzero modulo 3^{2k+2} and set $s_j = d - k$. In the first case use $v_3(g(1))$ directly. The proved nonvanishing establishes termination, without asserting a uniform practical running time. □

20.5.1 One cyclotomic order bound for both output signs

Retain a fixed cubic Pisot candidate with real conjugates $\beta, -\rho, \gamma$, where $0 < |\gamma| < \rho < 1$, and suppose that $A_k = \lfloor \beta^{3^k} \rfloor = p_k$ is an increasing prime tail. For a good output $p = p_k \nmid 6c \operatorname{disc}(f)$ let $Q_p = 3^w q$ be the projective order, with $3 \nmid q$. The scalar-return argument gives $w \geq k + 1$, excludes irreducible reduction, and yields a fixed sign ε with $p_k \equiv \varepsilon \pmod{3^{k+1}}$ and $a \equiv \varepsilon$, $c \equiv \varepsilon b \pmod 3$. Use Theorem 20.9 and denote its offsets by s_j, repeating a common offset on both parities where appropriate.

Corollary 20.10 (Unified bounded excess and prime-to-3 growth)**.** *At every sufficiently late good output of parity j, set $N = Q_p/3^k = 3^{w-k}q$ and $F_j = \varphi(3^{s_j})$. Then*

$$1 \leq w - k \leq s_j, \qquad p^{f_p} \leq 3^{\varphi(N)}, \qquad \varphi(q) \geq \frac{f_p \log p}{F_j \log 3} > \frac{f_p(3^k \log \beta - \log 2)}{F_j \log 3},$$

where $f_p = 1$ for split outputs and $f_p = 2$ for mixed outputs. If $\varepsilon = -1$, every such output is mixed. In particular both signs satisfy $\varphi(q) \gg_f 3^k$ and $Q_p \gg_f 9^k$.

Proof. The exact depth gives $v_3(p - \varepsilon) = k + s_j$. As Q_p divides $p^2 - 1$, this bounds w above. Let Y_i be the residual roots raised to 3^k. Their ratio group has order N and their sum is zero. Normalize to $1, \xi^A, \xi^B$, with ξ a primitive Nth root, and lift to $Z = 1 + \zeta_N^A + \zeta_N^B$ in characteristic zero. The second powered coefficient B_k satisfies $0 < |B_k| < p$ for all sufficiently

late outputs. If $Z = 0$, its three unit terms form an equilateral triple, whose second coefficient vanishes after reduction; this contradicts $p \nmid B_k$. Thus $Z \neq 0$ and

$$p^{\mathrm{ord}_N(p)} \leq |\mathrm{Norm}_{\mathbb{Q}(\zeta_N)/\mathbb{Q}}(Z)| \leq 3^{\varphi(N)}.$$

The residue degree is one for split outputs. In mixed type put $y = \lambda/\alpha$, using the linear root α and one quadratic root λ. When $p \equiv 1 \pmod 3$, the nontrivial anti-norm ratio λ/λ^p has order dividing $p+1$, prime to 3; its 3^kth power remains nontrivial. Hence $y^{3^k} \notin \mathbb{F}_p$. When $p \equiv -1 \pmod 3$, the order N of y^{3^k} has a nontrivial 3-part because $w > k$, whereas $\mathbb{F}_p^*$ has order prime to 3. Again $y^{3^k} \notin \mathbb{F}_p$. Thus the residue degree is exactly two in either mixed branch. Finally $\varphi(N) \leq F_j\varphi(q)$, $p > \beta^{3^k}/2$ eventually, and $Q_p \geq 3^{k+1}q \geq 3^{k+1}\varphi(q)$. □

These results are necessary conditions for a hypothetical prime tail. They do not force an inert output or a composite term, and the available order $p^2 - 1$ remains far larger than the required 9^k scale.

Corollary 20.11 (Prime-to-3 growth for every positive candidate)**.** *For a fixed candidate satisfying the positive prime tail, use the same constant s on both parities in the first two cases of Theorem 20.9, and put $F_j = \varphi(3^{s_j}) = 2 \cdot 3^{s_j-1}$. At every sufficiently late good output of parity j,*

$$\varphi(q) \geq \frac{f_p \log p_k}{F_j \log 3} > \frac{f_p(3^k \log \beta - \log 2)}{F_j \log 3}, \tag{287}$$

where $f_p = 1$ in split type and $f_p = 2$ in mixed type. If c is a cube modulo a mixed output, then necessarily $s_j \geq 2$ and

$$\varphi(q) \geq \frac{2 \log p_k}{\varphi(3^{s_j-1}) \log 3} = \frac{6 \log p_k}{F_j \log 3}. \tag{288}$$

In particular, for every fixed surviving positive-residue candidate,

$$\varphi(q) \gg_\beta 3^k, \qquad Q_{p_k} \gg_\beta 9^k.$$

Proof. The depth theorem gives $t = k + s_j$. Proposition 20.3 implies $1 \leq v = w - k \leq s_j$, so $\varphi(3^v) \leq F_j$. Insert this in Theorem 20.5; then use $p_k > \beta^{3^k}/2$ eventually. In the mixed cube case, $w \leq t - 1$ gives $1 \leq v \leq s_j - 1$, hence $s_j \geq 2$ and the improved denominator. Finally $Q_{p_k} \geq 3^{k+1}q \geq 3^{k+1}\varphi(q)$. □

The improvement by a factor of six in (288), relative to the split bound with the same s_j, consists of the exact residue-degree factor 2 and the smaller 3-primary cyclotomic degree factor 3. For unit norms it is automatic in mixed type. If $s_j = 1$, mixed type with a cube norm is excluded on that parity, but complete splitting is not. For arbitrary nonunits the cube condition remains an additional condition at the particular output prime.

20.6 A separate anti-norm bound for every mixed positive output

The next condition does not need unit norm or a bounded excess 3-adic depth. Retain the mixed roots of Proposition 20.3 and define

$$h = \mathrm{ord}(\lambda/\lambda^p).$$

Then $h > 1$, $h \mid p+1$, and $h \mid Q_p$. Since $p \equiv 1 \pmod 3$,

$$3 \nmid h, \qquad h \mid q. \tag{289}$$

This is the anti-norm component. The norm-component order used in the negative residue branch need not be prime to 3 in the present branch.

For $j \geq 1$, let $\Phi_j(X, Y)$ be the homogeneous cyclotomic polynomial, and put $F(r) = r/(1-r)^2$ for $0 < r < 1$. The elementary estimate

$$|\log|\Phi_j(z)|| \leq F(|z|) \quad (0 < |z| < 1) \tag{290}$$

is uniform in j. Indeed, the Möbius product gives a signed sum of $\log|1 - z^d|$, each bounded in absolute value by $-\log(1 - |z|^d) \leq |z|^d/(1 - |z|)$. Summing over all positive d proves the claim. Reciprocity gives the corresponding estimate outside the unit circle.

Theorem 20.12 (Anti-norm height constraint)**.** *Define*

$$E_{\text{pair}} = F(\rho/\beta) + F(\sigma/\beta) + F(\sigma/\rho).$$

For every sufficiently late good mixed positive output,

$$\varphi(h) > \frac{3^k \log\beta - \log 2 - E_{\text{pair}}}{\log(\beta^2\rho)}. \tag{291}$$

The same lower bound holds for $\varphi(q)$. If there are infinitely many such outputs, then along them

$$\liminf \frac{Q_{p_k}}{9^k} \geq \frac{3\log\beta}{\log(\beta^2\rho)}. \tag{292}$$

Under the strict size inequality $\rho < \beta^{-17/40}$, the last candidate-specific constant is strictly greater than $40/21$. The weak inequality in (280) alone gives at least $40/21$.

Proof. For $h > 1$, $\Phi_h(X, Y)$ is symmetric. Hence

$$K_h = \prod_{i<j} \Phi_h(\beta_i, \beta_j)$$

is a rational algebraic integer, and therefore an integer. It is nonzero because the three real conjugates have distinct absolute values, so no pair ratio is a root of unity. The exact order of λ/λ^p and $p \nmid h$ give $\Phi_h(\lambda, \lambda^p) = 0$ modulo p; thus $p \mid K_h$. Applying (290) to the three ratios of smaller to larger absolute value yields

$$0 < |K_h| \leq e^{E_{\text{pair}}}(\beta^2\rho)^{\varphi(h)}.$$

The base exceeds 1, since the nonzero integral norm gives $\beta\rho = |c|/\sigma > 1$. The actual output satisfies $p_k > \beta^{3^k}/2$ eventually, proving (291). If $h \mid q$, the prime factor formula for Euler's function gives $\varphi(h) \leq \varphi(q)$. Finally $Q_{p_k} \geq 3^{k+1}q \geq 3^{k+1}\varphi(h)$, proving (292). Write $u = -\log\rho/\log\beta$; the constant is $3/(2-u)$, which is at least $40/21$ when $u \geq 17/40$, and strictly larger under the strict inequality. □

20.7 Powered resolvents preserve the splitting character

This section is a limitation of a proposed test, not an additional exclusion. Restrict to a totally real cubic Pisot *unit*, so $c \in \{1, -1\}$ and $C_j = c$ for every odd powering. Let $z_i = \beta_i^{3^j}$ at stage j. All these powers remain cubic: a rational value would have equal conjugates, impossible when exactly one conjugate has modulus greater than 1.

20.7.1 Three fields and an exact oriented covariance

Choose an ordering of the roots and write

$$\delta_j = \prod_{i<\ell} (\beta_i^{3^j} - \beta_\ell^{3^j}), \quad \Delta_j = \delta_j^2, \quad I_j = \delta_j/\delta_0 \in \mathbb{Z}.$$

The ratio is a symmetric algebraic integer, because each difference quotient is a polynomial with integral coefficients; hence it is an integer. In particular $\Delta_j = \Delta_0 I_j^2$. Let $\omega^2 + \omega + 1 = 0$ and $\eta_0 = (\omega - \omega^2)\delta_0$, so $\eta_0^2 = -3\Delta_0$. Distinguish

$$K = \mathbb{Q}(\sqrt{\Delta_0}), \qquad K' = \mathbb{Q}(\sqrt{-3\Delta_0}), \qquad E = \mathbb{Q}(\omega, \delta_0).$$

The quadratic resolvent is K, which may be $\mathbb{Q}$ in the cyclic case. The Cardano radicands lie in K', whereas E contains μ_3 and is the appropriate Kummer base. A cubic reciprocity formula for Eisenstein integers cannot simply be applied to an arbitrary element of K'.

Define the Fourier resolvents and two rational coefficients by

$$\begin{aligned} U_j &= z_1 + \omega z_2 + \omega^2 z_3, \quad & V_j &= z_1 + \omega^2 z_2 + \omega z_3, \\ P_j &= A_j^2 - 3B_j, & H_j &= 2A_j^3 - 9A_jB_j + 27c. \end{aligned}$$

Expansion gives

$$U_jV_j = P_j, \qquad U_j^3 + V_j^3 = H_j, \qquad U_j^3 - V_j^3 = 3\eta_0 I_j.$$

Here $P_j = \frac{1}{2}\sum_{i<\ell}(z_i - z_\ell)^2 > 0$, since the roots are distinct and real. Thus neither resolvent vanishes. Put

$$R_j^+ = (H_j + 3\eta_0 I_j)/2 = U_j^3, \qquad R_j^- = (H_j - 3\eta_0 I_j)/2 = V_j^3, \qquad R_j^+ R_j^- = P_j^3.$$

The weighted-square identity $\sum_i \omega^{i-1} z_i^2 = (2A_jU_j + V_j^2)/3$, together with $z_i^3 = A_j z_i^2 - B_j z_i + c$, yields

$$U_{j+1} = D_j U_j, \qquad D_j = \frac{2A_j^2 - 3B_j}{3} + \frac{A_j R_j^-}{3P_j} \in K'^*. \tag{293}$$

Consequently, with $\mathcal{C}_k = \prod_{0\le j<k} D_j$,

$$R_k^+ = R_0^+ \mathcal{C}_k^3, \qquad \mathcal{C}_k \in K'^*. \tag{294}$$

This is equality of *oriented* cube classes, not merely equality up to inversion or equality of splitting fields. Reversing the initial orientation consistently swaps the two classes. Standard Kummer classification over a base containing μ_3 is described in [42, Propositions 5.27–5.28]; the explicit multiplier here is proved by (293), not inferred from that classification.

20.7.2 The powered test is the original splitting test

Let $p \nmid 6\Delta_0 P_0$, assume $p \equiv 1 \pmod 3$ and $(\Delta_0/p) = 1$, and choose a degree-one prime of E over p. Then ω and δ_0 reduce to $\mathbb{F}_p$. If r_0 is the nonzero reduction of R_0^+, one has the exact test

$$f \text{ splits completely modulo } p \quad \Longleftrightarrow \quad r_0^{(p-1)/3} = 1. \tag{295}$$

If f splits, $U_0 \in \mathbb{F}_p$ proves that r_0 is a cube. Conversely, choose $u \in \mathbb{F}_p^*$ with $u^3 = r_0$ and set $v = P_0/u$. Then $v^3 = R_0^-$, and the inverse Fourier expressions

$$\frac{A_0 + \omega^{-\ell} u + \omega^\ell v}{3} \qquad (\ell = 0, 1, 2)$$

are all three roots of f in $\mathbb{F}_p$. They are distinct since $p \nmid \Delta_0$.

If also $p \nmid P_k$, both $R_0^{\pm}$ and $R_k^{\pm}$ are local units, by their products P_0^3 and P_k^3. Equation (294) therefore makes $\mathcal{C}_k$ a local unit, even if an intermediate formula for D_j has a denominator divisible by p. It follows that

$$(R_k^+)^{(p-1)/3} = (R_0^+)^{(p-1)/3} \pmod{\mathfrak{p}}$$

at the chosen prime $\mathfrak{p}$ of degree one. The powered character is exactly the original character, not a second independent condition. In the S_3 case this remains nonabelian splitting information over the base containing the quadratic resolvent and μ_3, not generally a fixed rational congruence criterion. Cyclic cubic fields are different: their abelian splitting may be described by conductor congruences.

At actual outputs the apparent degeneracy $P_k = 0 \pmod{p}$ disappears. There is even an integral unit-cubic nonvanishing fact independent of the real shape: $B_j \neq 0$ for every $j \geq 1$. For if the second recurrence in (281) were zero, it would force $3 \mid B_{j-1}$; its right side would then be $3 \pmod 9$, a contradiction. For our real candidate, Lemma 20.4 further gives $0 < |B_k| < p_k$ eventually. Thus at $p = A_k$,

$$P_k \equiv -3B_k \not\equiv 0 \pmod{p}, \qquad R_k^+ \equiv (27c + 3\eta_0 I_k)/2 \pmod{\mathfrak{p}}.$$

The latter nonzero trace-zero radicand has the same cube character as R_0^+. An additional restriction on that *fixed* character along the actual prime outputs would still be needed for an exclusion.

There is also a precise one-step index statement. Direct expansion gives

$$J_j := I_{j+1}/I_j = -A_j^3 c + A_j^2 B_j^2 - B_j^3 = \prod_{i<\ell} (z_i^2 + z_i z_\ell + z_\ell^2).$$

Hence $\Delta_{j+1} = \Delta_j J_j^2$. At a sufficiently late actual output $p = A_k$, one has $J_k \equiv -B_k^3 \neq 0 \pmod{p}$, so

$$v_p(\Delta_{k+1}) = v_p(\Delta_k).$$

This output cannot first enter the powering discriminant at the very next step. Earlier or later collisions are not excluded; this is a statement about polynomial discriminants, not an assertion of field ramification.

20.7.3 One split actual prime seed, followed by a composite trace

The following finite diagnostic prevents an overstrong individual-prime inference. Take

$$f(X) = X^3 - 3529X^2 - X + 1, \qquad c = -1.$$

Both 3529 and

$$p = A_1 = 43949615473$$

are prime, certified by exhaustive trial division, with the latter trial bound 209641. Its three distinct roots modulo p are

$$36438575596, \qquad 33092793291, \qquad 18367865588.$$

The complete projective order of this triple is

$$Q = 21974807736 = (p-1)/2, \qquad v_3(Q) = 3 \geq 1 + 1, \qquad p - 1 = 2^4 \cdot 3^3 \cdot 7^2 \cdot 2076229.$$

The last factor is prime. Exact ratio exponent tests for every prime factor of Q certify that it is the full projective order, not only an upper bound. The seed thus satisfies the required projective-depth inequality at $k = 1$.

The real roots lie in $(-1, 0), (0, 1), (3529, 3530)$; none is integral, so the cubic is irreducible. Their sum makes the negative small root have larger modulus. Consequently every positive odd floor is its trace. Furthermore

$$f(-1/50) = -48951/125000 < 0, \qquad 50^{40} > 3530^{17}.$$

Thus its negative small modulus satisfies $w < 1/50 < \beta^{-17/40}$. Since $\beta w v = 1$, this also gives $w < v^{17/23}$. These are strict Pisot shadow bounds, not merely a freely prescribed finite-field root configuration. Nevertheless the next classical trace is composite:

$$\begin{aligned} A_2 = T_9 &= 848917016345518666992366631739 29 \\ &= 991 \cdot 856626656251784729558392161 19. \end{aligned}$$

This is one prime seed, not an all-prime tail, a minimum, or a counterexample to any necessary-condition theorem. It shows only that unit norm, strict shadow, complete splitting, and the individual seed projective-depth condition are compatible.

20.8 Reproducibility and the current arithmetic gap

20.8.1 Portable verification

The original source archive supplies five Python 3 programs using only the standard library, each with JSON output and a text log:

1. `positive_branch_verify.py`: seven mixed square-prime cyclotomic norm certificates, four split certificates, 79 exactly certified totally real Pisot examples, and 1355 modular depth regressions across all three residue cases.
2. `positive_branch_independent_verify.py`: an independent implementation checking 540 mixed cube-equivalence cases, 30 mixed norm certificates, 18 split certificates, unit-depth examples, and an exact anti-norm integer at an isolated actual prime trace.
3. `positive_allnorm_depth_independent_verify.py`: 49 certified Pisot inputs and 1225 companion-matrix/Newton comparisons, including modular offset extraction and unequal parity offsets. No logarithm approximation is used to guess an offset.
4. `powered_resolvent_verify.py`: five formal polynomial identities, all 162 relevant modulo-9 inputs for the nonvanishing lemma, and exact primality, splitting, order, shadow, covariance, and next-composite certificates for the displayed seed.
5. `powered_resolvent_independent_verify.py`: an independent exact-grid verification of the formal identities, with a degree bound justifying polynomial interpolation, and a separate sieved trial-divisor certification of the prime seed.

These are exact finite regressions or certificates, not samples from which the universal depth theorem is inferred. The global logarithmic nonvanishing and terminating-search assertions are proved in the depth theorem above. The source programs contain no probable-prime proof oracle and require neither a computer algebra package nor network access. Checksums bind the delivered files.

20.8.2 What the partial results can be used for

For a specified totally real cubic Pisot candidate with a prime tail of either residue sign, the exact depth algorithm computes the offsets that limit its possible projective 3-part. The cyclotomic-sum bound then forces a prime-to-3 order component of exponential size in the clock index; for a mixed output, the independent anti-norm certificate can give another candidate-specific bound. A concrete proposed output or local model that violates any one of these necessary conditions is ruled out.

The vanishing-sum proof also identifies a useful constraint that is absent from arbitrary local root configurations: the actual second powered coefficient is a nonzero integer strictly smaller than the prime output. The resolvent calculation, in contrast, prevents counting the same splitting character twice as if powering supplied an independent obstruction. These are usable tests and method limitations, not only formal changes of notation.

20.8.3 What remains unresolved

The available finite-field order can still be as large as $p^2 - 1$, while p itself grows exponentially in 3^k. The lower bounds here therefore leave ample room and produce no asymptotic contradiction. They do not force an inert output, a particular discriminant character, a recurrent divisor, or a composite future term for every candidate. The single composite successor in the resolvent subsection is an exact isolated example, not a universal mechanism.

Both residue signs remain logically distinct. The negative-branch bounds from the preceding round do not cover the positive branch; the positive-branch refinements here complement the two-sign depth theorem, including nonunit norms. No density statement for varying coefficients is transferred to this fixed prime trace orbit, and Chebotarev is not invoked to select prime divisors of fixed trace values. The all-norm depth theorem uses a proved logarithm theorem, but neither it nor the cyclotomic bounds settle transcendence of the classical Mills minimum. This paper also makes no closure claim for the different schedule $2 \cdot 3^k - 1$.

21 Three analytic routes and their exact missing hypotheses

The final analytic comparison uses the preceding arithmetic structure to test three specific interfaces. Each positive deduction is kept distinct from the missing assertion needed to reach every fixed candidate.

21.1 Conclusion and scope

The main targets remain open in this investigation. The three routes do produce rigorous information, but none supplies the missing statement for every surviving fixed candidate. The distinction between a finite configuration, a prescribed local orbit, and a single global minimum is essential.

1. **Tao–Ziegler prime patterns.** Finite cubic trace shapes can occur after a common additive translation. Such translated prime patterns can coexist with recurring composite *unshifted* Pisot floors. There are exact finite obstructions to identifying the translation with a new Pisot trace sequence.
2. **Finite-field sum–product.** The subgroup of exponents is a valid input to BKT/BGK. Separate additive Fourier bounds do not control its correlation with the particular Singer

trace-zero set. The phase-blind estimates studied here have a precise obstruction; this is not a prohibition on every possible sum–product argument.

3. **Gafni–Tao short intervals.** A candidate gap forces a block of exceptional starting points. The currently inspected envelope is too weak to exclude that block. A stronger exponent or a theorem forcing many disjoint bad intervals at a common scale would be sufficient, but neither is established here.

The first two routes use the explicit family

$$f_a(X) = X^3 - aX^2 - 4X + 4, \qquad a \in \mathbb{Z}, \quad a > 8. \tag{296}$$

The third route concerns a *conditional reduction for the actual classical Mills minimum*; its gap hypotheses are stated separately in Section 21.5. We do not identify an arbitrary member of (296) with that minimum.

The proofs below are deductions from the cited theorems and elementary algebra, together with exact finite certificates. No claim of literature priority is made. Analytic infinitude statements are not inferred from the finite computations.

21.2 Trace normalization and actual floors

Write $T_n(a)$ for the sum of the nth powers of the three roots of f_a. Newton's identities give

$$\begin{aligned} T_0 = 3, \quad T_1 = a, \quad T_2 = a^2 + 8, \\ T_n = aT_{n-1} + 4T_{n-2} - 4T_{n-3} \qquad (n \geq 3). \end{aligned} \tag{297}$$

For $n \geq 1$, T_n is a monic integer polynomial of degree n.

Lemma 21.1. *For $a > 8$, the roots of f_a are $-w, v, \beta_a$, where $0 < v < w < 1$ and $a < \beta_a < a + 1$. The polynomial is irreducible over $\mathbb{Q}$, and for every positive odd integer n,*

$$\lfloor \beta_a^n \rfloor = T_n(a), \qquad 0 < \beta_a^n - T_n(a) = w^n - v^n < 1.$$

Moreover, $T_n(1) = 1$ for every positive odd n.

Proof. At $-1, 0, 1, a, a + 1$, the values of f_a are respectively $7 - a, 4, 1 - a, 4(1 - a), (a - 1)^2$. They locate the three roots in the stated intervals. Their sum gives $w - v = \beta_a - a > 0$. None of the roots is an integer, so the rational-root test proves irreducibility. The identity for odd powers proves the floor formula. Finally, $f_1 = (X - 1)(X^2 - 4)$, so its odd power sums are $1 + 2^n + (-2)^n = 1$. □

We use the two positive odd exponent clocks

$$n_+(k) = 3^k, \qquad n_-(k) = 2 \cdot 3^k - 1 \qquad (k \geq 1).$$

A recurring congruence $\ell \mid T_{n_\pm(k)}(a)$ produces infinitely many composite actual floors once the trace exceeds ℓ.

21.3 Tao Ziegler patterns and finite rigidity

21.3.1 The common-zero theorem gives translated trace shapes

The common-zero polynomial progression theorem of Tao–Ziegler [26] applies to any finite family of integer polynomials vanishing at zero. It allows unequal degrees and supplies infinitely many simultaneous prime translates, with the polynomial parameter at most any prescribed positive power of the translating parameter.

Proposition 21.2. *Let I be a finite nonempty set of positive odd integers and let B be any real lower bound. There are infinitely many pairs of positive integers (x, m) such that, for a fixed integer $L \geq \max(8, B)$ and*

$$a = 1 + Lm > B, \qquad c = x - 1 \geq 0,$$

all the numbers

$$T_n(a) + c = \lfloor \beta_a^n \rfloor + c \qquad (n \in I)$$

are prime. If $d = \max I$, these pairs can be chosen with $m \leq x^\varepsilon$ for any fixed $0 < \varepsilon < 1/d$. Along an unbounded subsequence of such pairs,

$$c \longrightarrow \infty, \qquad \max_{n \in I} |T_n(a)| = o(x), \qquad \frac{T_n(a) + c}{x} \longrightarrow 1. \tag{298}$$

Proof. Use $P_n(m) = T_n(1 + Lm) - 1$. Lemma 21.1 gives $P_n(0) = 0$, and the cited theorem makes $x + P_n(m)$ prime for every $n \in I$. Bounded x permits only finitely many $m \leq x^\varepsilon$, so infinitely many pairs force an unbounded subsequence of x. Polynomial growth gives $T_n(1 + Lm) = O_{I,L}(x^{\varepsilon d}) = o(x)$. □

This is a common translation of an entire finite vector, not unrelated coordinatewise shifts. Nevertheless, in this application the translation dominates every actual trace. It does not provide $c = 0$, and choosing L large does not produce one fixed coefficient that survives all increasing prefix lengths.

The unequal-constant theorem in Tao–Ziegler's later paper [43, Theorems 1.2 and 1.4, Corollary 1.5] has a different hypothesis: every pairwise polynomial difference must have the maximal degree. A raw tuple of three or more distinct trace degrees fails it, since two members below the maximal degree have a lower-degree difference. Subtracting a common polynomial does not change this defect.

21.3.2 A valid prime pair inside an obstructed family

Proposition 21.3. *For infinitely many distinct integers $a \equiv 43 \pmod{186}$, $a \geq 229$, there are primes q_1, q_2 with*

$$q_2 - q_1 = T_3(a) - a = a^3 + 11a - 12.$$

At each such coefficient, the actual floors are composite on the recurring index classes

$$\left\lfloor \beta_a^{3^k} \right\rfloor \quad \text{for } k \equiv 1 \pmod 4,$$
$$\left\lfloor \beta_a^{2 \cdot 3^k - 1} \right\rfloor \quad \text{for } k \equiv 3 \pmod 4.$$

Proof. Put $a(m) = 43 + 186m$ and apply the 2016 theorem to $P_1(m) = a(m)$ and $P_2(m) = T_3(a(m)) = a(m)^3 + 12a(m) - 12$. Their degree-three coefficients are 0 and 186^3, so the required degree condition holds.

We check local admissibility. For a prime $\ell \nmid 186$, choose m modulo ℓ with $a(m) = 1$; then $P_1 = P_2 = 1$, and any $x \not\equiv -1 \pmod{\ell}$ works. At $\ell = 2$, both values are odd; at $\ell = 3$, both are 1, so $x = 0$ works in either case. At $\ell = 31$, at most two residues of x are forbidden. Hence the singular series is positive and the theorem gives prime pairs $x + P_1(m), x + P_2(m)$.

The coefficients $a(m)$ are unbounded, not merely the translations: Theorem 1.4 provides a positive asymptotic weighted count of order NM with M near $N^{1/3}$. A fixed finite set of m contributes only $O(N \log^2 N) = o(NM)$. Prime powers can be removed as in the cited Corollary 1.5. Thus infinitely many distinct m remain.

For compositeness, $a \equiv 12 \pmod{31}$ and

$$f_a(X) \equiv (X+6)^2(X+7) \pmod{31}.$$

For odd n, this gives

$$T_n(a) \equiv 2(-6)^n + (-7)^n = -6^n(2-4^n) \pmod{31}.$$

Since $\mathrm{ord}_{31}(4) = 5$ and $4^3 \equiv 2 \pmod{31}$, the trace vanishes exactly when $n \equiv 3 \pmod{5}$. The two exponent clocks yield the displayed classes of k. Their traces exceed 31 and equal the actual floors by Lemma 21.1. □

For a small exact illustration, take $a = 229, c = 42$. Then

$$a + c = 271, \qquad T_3(a) + c = 12{,}011{,}767$$

are prime, whereas

$$T_3(a) = 12{,}011{,}725 = 5^2 \cdot 11 \cdot 31 \cdot 1409.$$

The theorem, not this numerical example, proves infinitude. The example shows concretely why translated prime geometry can occur in an already obstructed unshifted family.

21.3.3 Seven odd trace coordinates force trivial translation

Lemma 21.4. *A nonzero real exponential sum $F(t) = \sum_{j=1}^{s} b_j e^{\lambda_j t}$, with distinct real λ_j and nonzero real b_j, has at most $s-1$ distinct real zeros.*

Proof. Induct on s. Divide by $e^{\lambda_1 t}$ and differentiate. The derivative has $s-1$ exponential terms. If the original sum had s distinct zeros, Rolle's theorem would give at least $s-1$ derivative zeros, contradicting the induction hypothesis. □

Theorem 21.5. *Let β, γ be totally real cubic Pisot numbers and $c \in \mathbb{R}$. If*

$$\mathrm{Tr}(\gamma^n) = \mathrm{Tr}(\beta^n) + c$$

at seven distinct positive odd integers n, then $\beta = \gamma$ and $c = 0$. In particular, seven distinct exponents from either of the actual clocks suffice.

Proof. Let β_i, γ_i be the real conjugates. At odd integers, $z^n = \mathrm{sgn}(z)|z|^n$. Thus the assumed equalities are zeros of the real function

$$F(t) = \sum_{i=1}^{3} \mathrm{sgn}(\gamma_i)|\gamma_i|^t - \sum_{i=1}^{3} \mathrm{sgn}(\beta_i)|\beta_i|^t - c.$$

Merge equal positive bases and remove zero coefficients. At most seven distinct exponential terms remain, including the constant base 1. Lemma 21.4 forces F to vanish identically. No Pisot conjugate has magnitude 1, so the coefficient at that base is $-c$, whence $c = 0$. If $\beta \neq \gamma$, the larger dominant root supplies a unique largest base greater than 1 with nonzero coefficient, a contradiction. □

No common norm, common polynomial family, or consecutive-index assumption is used. The theorem does not address cubics with nonreal conjugates or algebraic numbers of unbounded degree. Nor does this trace statement alone rule out every realization of an arbitrary translated finite vector as floors of a non-Pisot base.

21.4 BKT and BGK in the exponent field

21.4.1 The subgroup is genuine after the correct change of field

Let $p \geq 5$ be prime and suppose

$$Q = p^2 + p + 1$$

is also prime. The group $\mathbb{F}_{p^3}^*/\mathbb{F}_p^*$ is cyclic of order Q. Choose x generating it and define

$$D = \{d \in \mathbb{Z}/Q\mathbb{Z} : \mathrm{Tr}_{\mathbb{F}_{p^3}/\mathbb{F}_p}(x^d) = 0\}.$$

The trace-zero plane gives $|D| = p + 1$, $0 \notin D$, and $pD = D$. For every $r \neq 0$ there is exactly one ordered representation $r = d - e$ with $d, e \in D$: the two distinct trace planes meet in one projective point. This is the standard Singer difference-set construction; see [33].

Write the unnormalized additive Fourier transform as

$$\widehat{A}(r) = \sum_{a \in A} e^{-2\pi i r a/Q}.$$

The difference-set identity gives

$$|\widehat{D}(r)| = \sqrt{p} \qquad (r \neq 0). \tag{299}$$

Now set

$$H = \langle 3 \rangle \subset \mathbb{F}_Q^*, \qquad h = |H|, \qquad m = (Q-1)/h.$$

For a nonscalar element $\gamma = cx^u$, the classical local question is

$$N(u) = |D \cap uH| > 0. \tag{300}$$

Within its stated size range, BKT's prime-field sum–product theorem [44, Theorem 1.1] can be applied to H; here $HH = H$ supplies multiplicative closure. BGK's subgroup exponential-sum theorem [45, Theorem 6] legitimately yields, for $h \geq Q^\delta$,

$$\max_{r \neq 0} |\widehat{H}(r)| \leq hQ^{-\eta(\delta)}, \qquad \eta(\delta) = \exp(-C_1/\delta^{C_2}).$$

The BGK paper is by Bourgain, Glibichuk and Konyagin; Tao is a coauthor of the precursor BKT paper. Crucially, H is a subgroup in the *exponent field* $\mathbb{F}_Q$. The set $\{x^{3^k}\}$ is not in general a multiplicative subgroup of $\mathbb{F}_{p^3}^*$, and we do not apply the subgroup theorem to that set.

21.4.2 The exact limitation of phase-blind Fourier bounds

Fourier inversion and Parseval give

$$N(u) = \frac{(p+1)h}{Q} + \frac{1}{Q} \sum_{r \neq 0} \widehat{D}(r) \overline{\widehat{H}(ur)}, \tag{301}$$

$$\sum_{r \neq 0} |\widehat{H}(r)|^2 = h(Q-h). \tag{302}$$

Proposition 21.6. *The Cauchy–Schwarz estimate from (299)–(302) is*

$$\left| N(u) - \frac{(p+1)h}{Q} \right| \leq \frac{\sqrt{ph(Q-1)(Q-h)}}{Q}.$$

Its main term exceeds its displayed error if and only if $h > p^2$. Consequently it does not certify positivity for any proper subgroup H.

Proof. The estimate is Cauchy–Schwarz. Positivity of the resulting lower bound is equivalent to $(p+1)^2h > p(Q-1)(Q-h)$. Since $(p+1)^2 + p(Q-1) = (p+1)Q$, this simplifies to $h > p^2$. A proper subgroup satisfies $h \le (Q-1)/2 < p^2$. □

The next statement is stronger, but still concerns only a specified bounding method, not the actual error or every sum–product method.

Theorem 21.7. *For every proper subgroup $H \subset \mathbb{F}_Q^*$ and $p \ge 5$, the exact termwise triangle-inequality allowance*

$$E_{\mathrm{tri}} = \frac{\sqrt{p}}{Q} \sum_{r \neq 0} |\widehat{H}(r)|$$

is larger than the main term $\mu = (p+1)h/Q$ in (301). Thus even knowing every individual amplitude $|\widehat{H}(r)|$ exactly would not make that phase-blind bound certify a hit.

Proof. Expanding 1_H in the m multiplicative characters trivial on H and using the usual Gauss-sum magnitude gives

$$\max_{r \neq 0} |\widehat{H}(r)| \le \frac{1 + (m-1)\sqrt{Q}}{m} \le \sqrt{Q}.$$

Together with Parseval, this implies

$$\sum_{r \neq 0} |\widehat{H}(r)| \ge \frac{h(Q-h)}{\sqrt{Q}}, \qquad \frac{E_{\mathrm{tri}}}{\mu} \ge \frac{\sqrt{p}(Q-h)}{(p+1)\sqrt{Q}}.$$

For a proper subgroup $Q - h > Q/2$. Also $pQ > 4(p+1)^2$ for $p \ge 5$: the difference is $p^3 - 3p^2 - 7p - 4 > 0$. The claimed ratio is therefore strictly greater than 1. □

Directly inserting a BGK supremum gives an error of order $\sqrt{p}\, hQ^{-\eta}$. Since the main term is of order $hQ^{-1/2}$, this would require $\eta > 3/4$ asymptotically. For a proper subgroup of size Q^δ, Parseval forces $\max|\widehat{H}| \ge \sqrt{h/2}$, limiting any relative saving to $\eta \le \delta/2 + o(1) \le 1/2 + o(1)$. The issue is therefore not a modest improvement to the BGK constant: one needs information about the *mixed phases* in (301), or a different structural argument.

21.4.3 An exact local counterexample at 71

Proposition 21.8. *At $p = 71$, $Q = 5113$ is prime and $H = \langle 3 \rangle$ has order $213 > p$. For the Singer set formed from a root of $f_0(X) = X^3 - 4X + 4$ over $\mathbb{F}_{71}$, ten of the 24 multiplicative H-cosets miss D. In particular $D \cap 13H = \varnothing$.*

Inside the original family, f_{17} is irreducible modulo 71, but $\mathrm{Tr}(x^{3^k}) \neq 0$ for every $k \ge 0$. Thus prime projective order, irreducibility, and $|H| > p$ are not sufficient for a classical local hit.

Exact certificate. Trial division proves that 71 and 5113 are prime. Root testing proves f_0 and f_{17} irreducible over $\mathbb{F}_{71}$. In the quotient by f_0, exact powering gives $x^{5113} = -4$; the nonscalar projective class therefore has order 5113. The trace recurrence enumerates all 72 elements of D, and quotient-ring multiplication checks the same traces independently. All 5112 ordered nonzero differences occur exactly once.

The complete coset intersection multiset is

$$\underbrace{0, \ldots, 0}_{10}, \quad \underbrace{3, \ldots, 3}_{6}, \quad \underbrace{6, \ldots, 6}_{6}, \quad 9, 9.$$

The least-unused missing representatives are $13, 19, 22, 26, 43, 46, 50, 59, 67, 89$. Order checks show $\mathrm{ord}_{5113}(3) = 213$.

For f_{17}, all 213 traces on the projective cubing clock are nonzero, and the state after 213 steps is $43x$. Subsequent cycles differ by nonzero scalars, so no trace zero can appear. The independent companion-matrix certificate reproduces these computations. □

This is a counterexample to a proposed *local sufficient condition*, not to the global composite-floor target. The same f_{17} has negative-clock zeros at $k = 77, 79, 211 \pmod{213}$ at this auxiliary prime, and another auxiliary prime may work on its classical clock. Also $D - D = \mathbb{F}_Q$ in this example: maximal additive expansion itself does not imply the prescribed multiplicative-coset hit.

21.4.4 A positive statement for almost all additive translates

Proposition 21.9. *For any fixed $a \in \mathbb{F}_Q^*$, put $R_a(b) = |D \cap (b + aH)|$. Then*

$$\sum_b R_a(b) = (p+1)h,$$
$$\sum_b R_a(b)(R_a(b)-1) = h(h-1),$$
$$\sum_b R_a(b)^2 = h^2 + ph.$$

In particular,

$$\#\{b : R_a(b) = 0\} \le \frac{p(Q-h)}{h+p}. \tag{303}$$

Proof. The first identity counts pairs in $D \times H$. For two distinct elements of H, the perfect-difference property provides exactly one corresponding ordered pair in D with their a-scaled difference. This gives the second identity, and adding the first gives the third. Cauchy–Schwarz on the support of R_a yields

$$\#\{b : R_a(b) > 0\} \ge \frac{(p+1)^2 h}{h+p}.$$

Subtract from Q and use $(p+1)^2 = Q + p$. □

If $h/p \to \infty$, the fraction of missing additive translates tends to zero. But the classical clock prescribes $b = 0$, and the negative clock prescribes the coupled values $a = 2u, b = -u$; neither grants a freely random translation. At $p = 71, a = 13$, exactly 262 of the 5113 shifts miss, including $b = 0$, whereas (303) gives the valid upper bound 1225.

For the classical clock, the missing input can also be formulated using multiplicative characters. Let $H^\perp$ be the m characters trivial on H, and set $T_D(\chi) = \sum_{d \in D} \chi(d)$. Then

$$N(u) = \frac{1}{m}\left(p + 1 + \sum_{\substack{\chi \in H^\perp \\ \chi \ne 1}} \overline{\chi(u)} T_D(\chi)\right).$$

Thus $(m-1)\max_{\chi \ne 1} |T_D(\chi)| < p+1$ would suffice. A hypothetical uniform bound $|T_D(\chi)| \le C\sqrt{p}$ would give hits in a range roughly $h > Cp^{3/2}$. No such bound is proved here. The identity $pD = D$ only removes characters with $\chi(p) \ne 1$; equivalently it allows H to be enlarged to $\langle 3, p \rangle$ without changing whether a coset hits.

Even a successful local correlation theorem would still need a prime-supply argument ensuring the required irreducibility, order, and scheduled-orbit hypotheses for every fixed coefficient.

21.5 Gafni Tao exceptional intervals

21.5.1 A precise conditional candidate interface

Gafni–Tao [46, Definition 1.1 and Theorem 1.3] study the exceptional starting points for the prime number theorem in short intervals. For fixed $\delta > 0$ and θ, let

$$E_\delta(X,\theta) = \left\{ x \in [X, 2X] : \left| \sum_{x < n \le x + x^\theta} \Lambda(n) - x^\theta \right| \ge \delta x^\theta \right\}.$$

Their estimates bound its Lebesgue measure by $X^{\mu(\theta)+\varepsilon}$, with constants and onset depending on fixed parameters. The explicit recipe uses zero-density and zero-additive-energy inputs. These are asymptotic exponents, not numerical constants at a prescribed finite height.

For clarity, the candidate-dependent hypotheses used here are stated explicitly. Assume a fixed candidate supplies late prime traces $p_r = \mathrm{Tr}(\beta^{3^r})$ and centers $x_r = p_r^3$, with

$$p_{r+1} \text{ is the least prime above } x_r,\ G_r = p_{r+1} - x_r \sim 3x_r^\kappa, \qquad \tfrac{1}{2} < \kappa < \tfrac{21}{40}, \qquad x_{r+1} \sim x_r^3. \tag{304}$$

These are the inputs at the present conditional classical-minimum reduction. We do not assert (304) for an arbitrary cubic Pisot number or for every member of (296).

Proposition 21.10. *Under (304), fix $1/2 < \theta < \kappa$. At arbitrarily large scales $X \asymp x_r$,*

$$|E_{1/2}(X,\theta)| \gg X^\kappa.$$

Consequently an exceptional-set exponent strictly below κ at such a θ would exclude this candidate.

Proof. The interval (x_r, p_{r+1}) is prime-free. Choose, for instance, $X = 2x_r/3$, so this interval lies inside $[X, 2X]$ away from its endpoints for large r. Every starting point in this gap at distance at least $(2X)^\theta$ from its right endpoint has a prime-free window $(x, x + x^\theta]$.

The total contribution of proper prime powers below $3X$ is $O(X^{1/2}\log^2 X) = o(X^\theta)$. Thus every such window is $1/2$-exceptional for sufficiently large r. The set of its starts has measure at least $G_r - O(X^\theta) \gg X^\kappa$. If an upper exponent is $\mu(\theta) < \kappa$, choose $\varepsilon < \kappa - \mu(\theta)$ to obtain a contradiction. □

21.5.2 What the displayed numerical envelope does not prove

Let $U(\theta)$ denote the upper-bound envelope produced by the displayed majorants and optimization in [46, Tables 1–2 and Theorem 1.3]. The following calculation is about this *recipe*, not a lower bound for the unknown true exceptional exponent $\mu(\theta)$.

At $\sigma = 7/10$, the displayed inputs are

$$\widetilde{A} = \frac{30}{13}, \qquad \widetilde{A}^* = \frac{235}{39}.$$

For $1/2 < \theta < 21/40$, this point is eligible since $30/13 > 1/(1-\theta)$. Its two objective values are

$$u_2 = \frac{9(1-\theta)}{13} + \frac{2}{5}, \qquad u_4 = \frac{47(1-\theta)}{26} - \frac{1}{5}.$$

Here $u_4 < u_2$, so the supremum furnished by that recipe obeys

$$U(\theta) \geq u_4 \geq \frac{137}{208} > \frac{21}{40} > \kappa, \qquad \frac{137}{208} - \frac{21}{40} = \frac{139}{1040}. \tag{305}$$

Thus those specific inputs do not yield the sufficient exponent in Proposition 21.10. This does not prove $\mu(\theta) \geq 137/208$, nor exclude improvements to the zero-density input, moment bounds, sieve treatment, or candidate-specific analysis. At the inspected σ, an improved recipe would have to remove its eligibility or lower the relevant objective; reindexing the same candidate does neither.

21.5.3 The missing packing theorem

If N disjoint prime-free intervals in $[X, 2X]$ each have length at least cX^κ, their interior start-sets give, by the same argument,

$$NX^\kappa \ll |E_{1/2}(X, \theta)|.$$

An upper estimate $\ll X^{U(\theta)+\varepsilon}$ then implies

$$N \ll X^{U(\theta)-\kappa+\varepsilon}. \tag{306}$$

A theorem forcing one candidate to generate

$$N \geq X^{U(\theta)-\kappa+\eta}$$

disjoint bad intervals at arbitrarily large common scales, for fixed $\eta > 0$, would contradict (306) after choosing $\varepsilon < \eta$. The current hypotheses provide no such theorem. The lacunarity $x_{r+1} \sim x_r^3$ instead gives at most one late center in any $[X, 2X]$. Replacing β by β^{3^j} merely reindexes the same centers.

Even the favorable endpoint in (305) leaves a required positive-power multiplicity gap of at least 139/1040. Prime patterns at unrelated coefficients or translations do not manufacture disjoint *prime-free* intervals for this one candidate.

21.5.4 A sparse deletion diagnostic

For $\theta > 1/2$, replacing the von Mangoldt sum by the weighted prime sum changes it by a uniformly negligible proper-prime-power contribution. Suppose primes are deleted in intervals J_r around x_r, of length $O(x_r^\kappa)$, with $x_{r+1} \sim x_r^3$. For a fixed range $[X, 2X]$, only $O(1)$ late deletion intervals can affect its windows. All affected starting points lie in neighborhoods of total measure

$$O\big(X^{\max(\theta,\kappa)}\big).$$

Outside these neighborhoods the original and retained weighted prime sums are identical. Hence an exceptional-set estimate $O(X^U)$ survives these deletions whenever $U > \max(\theta, \kappa)$, with the usual harmless adjustments of constants and tolerance.

Conditionally on an all-prime Pisot tower, deleting competing-child primes while retaining the tower can make its base the minimum for a *modified subset of primes*. The inspected exceptional envelope is stable under those sparse deletions. This is neither an actual Mills counterexample nor a construction of an algebraic all-prime tower. It identifies information absent from the exceptional-measure bound alone.

21.6 Verification and the remaining statements

The portable scripts in the accompanying `certificates/` directory use the Python standard library only. They check:

- the odd trace normalization, modular repeated-root identity, both index classes, and exact prime-pair illustration;
- the Singer difference sets, subgroup orders, complete coset counts, and the $p = 71$ counterexample, including an independent companion-matrix reconstruction;
- the exact translated-coset moments and finite shift counts;
- the rational numerical comparison in (305).

They do not numerically prove an analytic infinitude theorem, the real-exponential zero-count lemma, or a universal prime-supply claim. Independent proof reviews and exact regressions were used before packaging.

The remaining requirements are specific. The polynomial-pattern route would need control of the unshifted, fixed-candidate slice rather than free translations. The local finite-field route would need an adequate mixed-correlation statement and a suitable auxiliary prime for every candidate. The exceptional-interval route would need a sufficiently small exponent or candidate-driven same-scale packing. None of those requirements is met here. Conversely, the phase-blind obstruction and the numerical-envelope obstruction are scoped results, not assertions that all future uses of the cited methods are impossible.

22 Final status and the remaining arithmetic problem

The proved restrictions and infinite subfamilies do not constitute a proof of the universal cubic Pisot recurring-composite statement. They also do not prove the transcendence of the classical Mills minimum $\xi(3^k)$ or of $\xi(2 \cdot 3^k - 1)$. The surviving candidate space has not been reduced to a finite list. Both residue signs must be respected in the classical analysis, and reindexing a powering orbit does not reset an original seed index or preserve a fixed coefficient line.

A completion by recurrent divisors would require a theorem that assigns to every fixed surviving candidate an auxiliary prime with an admissible trace-zero phase on the required sparse clock. Neither ordinary divisors of a recurrence, nor density-one coefficient results, nor finitely many successful tests supply this universal assertion. A completion by prime intervals would require a genuinely stronger candidate-specific interval input or a packing theorem at common scales. A completion by finite-field analysis must control the relevant mixed correlation rather than only separate Fourier magnitudes. These are precise sufficient directions, not claims that all possible proof methods have been classified.

A Source coverage and reproducibility

The twenty source packages are represented below. The first package contains proof notes and certificates rather than a standalone PDF. Copied predecessor sources in later packages are dependencies, not additional papers. The initial positive divisibility exclusions $28s$ and $403s$ are subsumed by the later complete positive-shift elimination in Section 3; the initial large negative cutoff is strengthened by the later negative-shift results. Such obsolete restrictions are not counted as additional final theorems.

Source	Mathematical contribution	Location
1	Scaling consequences local minima and normalized tails	2
2	Positive shifts and elimination of the exceptional unit candidates	3
3	Negative shifts and integral fixed point rigidity	5
4	The multiplier thirteen and a residual branch exclusion	6
5	Prime cells height bounds and Pell families	8
6	Odd multiplier extensions	7
7	Finite prime forests and explicit continuation criteria	9
8	Nonunit cubic families and fixed modulus limitations	11
9	Full prefix positive extensions and cube rescaling	4
10	Density on fixed norm lines	12
11	Shifted prime patterns and compactness criteria	10
12	Prime coefficients and simultaneous composite floors	13
13	Signature controlled density and quantitative exceptional sets	14
14	The thin prime power family	15
15	A prime seed outlier and certified returns	16
16	Adaptive returns and ancestral minimality	17

Source	Mathematical contribution	Location
17	Projective return criteria and finite horizon limitations	18
18	Coset coverage and output prime order bounds	19
19	Both residue signs and all norm depth	20
20	Three analytic routes and their exact missing hypotheses	21

The standalone LaTeX source contains the complete typesetting input; no external bibliography, figure, or input file is needed to compile it. Portable certificate procedures printed in the text reproduce the core finite algebra. Commands and relative certificate paths quoted in the verification sections refer to the corresponding extracted original archive, not to additional files required by this standalone source. The original archives also retain their more extensive machine-readable logs, prime certificates, and checker files. Claims whose verification relies on a stated analytic theorem retain that theorem as an explicit mathematical input rather than being described as computationally certified.

A.1 Independent checks of the strengthened statements

The scalar-exponent identity was independently checked in 55 222 monic cubic quotient algebras with nonzero constant term at the primes $5, 7, 11, 13, 17, 19, 23, 31$. Each check used exact polynomial reduction and repeated squaring, including reducible and repeated-root cases. An additional implementation checked 14 754 cubic reductions and 7 264 returned clock states. These are consistency checks for the universal proof, not an exhaustive verification over all primes. The two-sign depth formulas were also tested on 536 irreducible totally real cubics at indices 0 through 7 using integer coefficient cubing and exact 3-adic valuations. These finite tests do not replace the nonvanishing arguments or establish uniform practical bounds for the modular searches. The previously supplied source archives remain the record for the older computational certificates; the consolidation is not represented as a rerun of every original search.

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